

Daisy Program Reference Manual

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Part I

The Framework

Chapter 1

Introduction

Welcome to the Daisy Program Reference Manual. This manual provides you with a reference to the Daisy software. That is, it describes the command line version of Daisy, and the format of the setup files. As a start, let's look at what this manual isn't.

- It is not an introduction to the Daisy simulation model itself. See [Hansen, 2002], [Hansen et al., 1990] and [Hansen et al., 1991] instead.
- It is not a reference manual for the mathematical or physical aspects of the various models in Daisy. The models are listed together with their parameters, but not described in any detail. Some models are explained in [Hansen, 2002], [Hansen et al., 1990] and [Hansen et al., 1991].
- It is not a tutorial for the Daisy software. The manual is intended as a reference for people who already know the system. A tutorial can be found in [Abrahamsen, 2011].
- It is not a programmers manual. See [Abrahamsen, 1997] for how to add new models to Daisy, or [Abrahamsen, 1998] for a description of how to add Daisy to larger systems (not currently supported). An overview of the program design can be found in [Abrahamsen and Hansen, 2000].

Enough negative statements, what this manual *does* provide is a complete reference to all components and models provided by Daisy at the time the manual was generated, as well as all the parameters, state variables and log variables supported by these models. These are collected in part II of the manual. The reason why we can state with absolute confidence that the listing is complete, is that it is generated automatically from the code. Which means, if Daisy accepts a parameter in the setup files, it will also be listed in the manual. It also means that the components, models and parameters cannot be organized into logical groups with helpful descriptions of each. Instead they are presented in alphabetical order. This manual is generated from the version of Daisy listed at page 629.

Part I of the manual is written by humans though. Here, we will try to describe enough of the concepts, format, and nomenclature to be able to use part II efficiently.

Chapter 2

Concepts

In order to use the Daisy model, as well as this manual, you must already possess a fairly good understanding of the agronomic and physical processes that are modelled by Daisy. However, when describing the organizing and principles behind the Daisy software, we have to introduce some new concepts, which are used throughout this manual. This section is somewhat abstract and dry. The best way to read it might be to read it once, without worrying too much about understanding it, then get some experience with the Daisy program and the setup files, and then read this section again.

2.1 Components, Models, and Parameterizations

The two most significant concepts behind the organization of the Daisy, are *components* and *models*. Typically, a component corresponds to some physical process, and a model is a particular way to describe (or simulate) the process. For example, water movement in the unsaturated zone is a process, while Richards Equation is a particular way to describe that process. In Daisy, water transport is a component (described in chapter 98), while Richards Equation is a model (described in section 98.3).

Just about everything in Daisy are either components or models, not just what you naturally think of as physical processes. For example, crops (chapter 29), management (chapter 15), and even log files (chapter 52) are all components, each having several different models. Which leads us to the next issue, *fixed* vs. *library* components.

The general component/model framework makes it very easy to add new models to existing Daisy components. All that is required is a C++ file with an implementation of the model. No other code has to be changed. To activate the model, the user just selects it in the setup file. However, for some components we only support a single model. This makes them slightly simpler, for example the user will not have to specify a model in the setup file, as there can only be one. The cost is less flexibility. We call the first kind of components, which can (theoretically) can have many different models, for *library components*. Each library component has its own chapter in part II, with a section for each model of that component. The second less flexible kind of components is called *fixed components*. All the fixed components are described in chapter 107.

2.1.1 Parameterizations

Library components have another important feature, which fixed components lack. They allow the user to save standard parameterizations of the model in text files, and refer to these parameterizations by name. Daisy is distributed with a large number of such standard parameterizations. The most obvious example where they are useful is parameterization of crop models. The default crop model (29.1) is sufficiently generic to be able to describe crops ranging from grass to potatoes, but it takes lot of parameters to do so. This is why Daisy comes with already parameterized versions of the model, with names like “Grass” and “Potato”.

All in all, this gives a three layered hierarchy, with library components in top. For each library component, there can be one or more models implemented in C++. For each model, there can be one or more named parameterizations.

2.2 Model Attributes

A parameterization can set *parameters* (as the name implies) for the model, as well as initial values for the models *state variables*. The parameters are constant during the simulation, while state variables can change as the simulation progresses. A typical parameter could be the hydraulic conductivity for saturated soil, or the maximum height for a crop. Typical state variables could then be the water content in the soil, or the current height of a crop.

2.2.1 Optional and Default Parameters

In general, all the parameters and state variables must be specified before a given model can be used in the simulation. There are two exceptions to that rule though.

First, some parameters and state variables are marked *optional*. This means the model will be able to infer the value of one variable (or parameter) from the values of other variables or parameters. For example, the ‘default’ chemical model (section 23.17) has both an ‘M’ state variable, indicating the total chemical content in the soil, and a ‘C’ state variable, indicating the chemical concentration in the soil water. Given one of these, and information about the amount of water in the soil and the sorption characteristics, the ‘default’ chemical component is able to calculate the other assuming equilibrium.

The other exception is parameters or state variables which have a *default value*. Many models have parameters that the typical user will rarely change, because the default value is well established, but where the parameter is still provided, for those people who want to fiddle with the model. For state variables, there can also be reasonable values provided by default. For example the ‘crops’ vegetation model (section 100.2) will by default start with an empty list of crops, i.e. there will be no crops in the field at the start of the simulation. This is what most people will expect.

2.2.2 Partial Parameterizations

One can say that the model itself can provide a *partial parameterization* of itself this way, i.e. a parameterization of some, but not necessarily all, parameters in the model. The user can also create partial parameterizations, which is a third example where you don’t need to specify all the parameters and state variables. A partial parameterization cannot be used directly in the simulation without specifying the missing parameters and state variables, but can be used as a *base* for other *specialized* parameterizations. For example, the standard fertilizer library in the Daisy

distribution includes a partial parameterization named ‘slurry’. This parameterization is used as the base for two specialized parameterizations, namely ‘pig_slurry’ and ‘cattle_slurry’, and contains the parameters that are common for both kinds of slurry. With partial parameterizations, the user can build a hierarchy of parameterizations with the most general on top, and the most specialized (complete) parameterizations at the bottom.

2.2.3 Log Variables

A model can also support a second kind of variables, namely *log variables*. Log variables are computed during the simulation, but are not part of the model state, i.e. they are not stored between time steps. This also means that it makes no sense to initialize them, so they cannot appear in a parameterization of the model. However, log variables and state variables can be written to output files (or other external media) by the various ‘log’ components (chapter 52). Thus, we have parameters and state variables in the input files, and state variables and log variables in the output files.

In the Daisy vocabulary, parameters, state variables, and log variables are three different *categories* of model *attributes*. One can think of the attributes in general as the users mean to interact with the model, and each of the three categories as specifications of how the particular attribute can be used, for input, for output, or for both.

2.3 Attribute Types

Beside having a category, attributes also have a *type* and a *size*.

A type is the set of legal values. For example, an attribute whose type is ‘Integer’ (see section 4.1.3) cannot hold a string (see section 4.1.5) value. This prevents you from setting the attribute ‘NoOfIntervals’ to ‘"Gnat"' in the input file. For log variables, the type will give you an idea of what kind of values to expect in the output files. The various types are described in section 4.1.

The attribute size specifies the number of values to expect. If the attribute is a *singleton*, only a single value can appear. If the attribute is a *variable*, any number of values can appear. The attribute can also have an integer size, which means exactly that number of values should appear. There are also four additional symbolic sizes, *canopy cells*, *canopy edges*, *soil cells*, and *soil edges*, which means the variable must hold a number of values corresponding to the discretization of the canopy or soil, respectively. A *cell* is an interval, area, or volume, depending on whether we have a 1D, 2D, or 3D model, respectively. A cell sized variable usually hold the content or sometimes sink of the interval, area or volume of the cell. An *edge* is the connection between two neighbouring cells, and edge variables usually hold fluxes between cells.

Each attribute has an associated description, which hopefully will give some idea of what the attribute is used for. Attributes with the type ‘number’ (see section 4.1.4) will also have a unit, so you for example will know whether the length should be given in millimeter or kilometer. This can make a difference. Trust me on that one.

2.3.1 Ordered Attributes

Finally, some attributes may be *ordered* within the model. This order is used by the ‘file’ parser model (section 61.1) for parsing the setup files. Normally, you will have to specify both the name and the value of the attributes in the setup files. However,

ordered attributes are identified by the order they appear in the setup file, so you will not have to specify their names. In fact, you are not allowed to specify their names. Ordered attributes are typically used for “obvious” parameters, for example the ‘file’ parser model has a single parameter, ‘where’, the name of the file to parse. That parameter is ordered, so you can write

```
(input file "filename.dai")          ; Correct.
```

instead of

```
(input file (where "filename.dai")) ; Wrong.
```

as you would have had to write, if the ‘where’ attribute had been unordered.

Chapter 3

Nomenclature

Having covered the concepts, it is now possible to explain the nomenclature used in part II of the manual. As already explained, each library component has its own chapter, while all the fixed components are collected in chapter 107. All the library component chapters start with a short explanation of the purpose of that component.

Each model (including those models that are part of a fixed component) has its own section. Here, we will explain how to read the content of these sections. This is best done with an example. Now, keep a finger here, and find section 15.12 (that's on page 80). Section 15.12 is a description of the 'fertilize' model, which is part of the action component. You specify this model when you want the manager to fertilize the field.

3.1 Name and Description

The name of the section is 'fertilize', which is the name you have to specify in the setup file when you want to refer to the model. The text in the start tells you that this is a specialization of 'fertilize.base', followed by a brief description of the model and what it does.

3.2 Sample

The next element in the section is a *sample*, which shows the format recognized by the file parser model when it reads the setup files. The sample starts with '<' and ends with '>', these should not be included in the setup files, they are only there to show the start and end of the sample. The sample text written in a non-slanted typewriter font should be included directly in the setup files, while the text written in slanted text should be replaced with the parameter (or state variable) values, as listed after the sample.

All parameters and state variables will be listed in the sample. Log variables will not be listed, since they cannot be set in input files.

Ordered attributes will be listed first, in order, since that is how they must appear in the setup files. In the example (that's section 15.12, your finger should still be there!), *am* is such an ordered attribute. You will note that the text *am* only occurs once in the sample, while the texts *to* and *from* both occur twice. This is because you only have to specify the value for the 'am' component in the setup files, since it is ordered, while you have to specify both the name and value for the unordered 'to' and 'from' attributes.

For the unordered attributes, it doesn't matter what order they appear in the setup files. In the sample, they will be shown in alphabetic order, 'from' will appear before 'to'. However, there is no reason (except good taste) why you shouldn't set the 'to' parameter before the 'from' parameter in the setup files.

All three attributes ('am', 'from', and 'to') are singletons (reread section 2.3 if you have forgotten what that means). If any of the attributes had been sequences, they would have been followed by the string '...'. This does not mean you should write '...' in the setup files, but that you can write any number of values at that point, separated by white space. Section 15.4 has an example of that.

If an unordered attribute has a simple value, it will be printed directly in the sample.

3.3 The Attribute List

After the sample comes a list with information about each attribute in the model. The parameters and state variables are listed in the same order as in the sample, that is, first the ordered attributes (in the specified order), and then the rest of the parameters and state variables in alphabetic order. After that, the log variables are listed, if any, in their own section. The 'fertilize' management action model doesn't have any log variables, so look at the default bioclimate model (section 19.2) instead. It has plenty.

The documentation for each attribute is divided in three parts, which are explained below.

3.3.1 The Type Line

The first line (the line with the bullet) contains the name and the type (section 2.3, remember?) of the attribute. The name is written in slanted *italic*, to match the way it is written in the sample (section 3.2). Two types are treated specially: Numbers have a unit, which is written in bold in square brackets; Components have their name written in bold, and a reference to the chapter (or section, for fixed components) in the manual where they are described. Other types (such as integers and strings) have a reference to the subsection of section 4.1 where they are explained.

3.3.2 The Category Line

The second line contains the following information:

1. Whether the attribute is optional. If so, the line will start with the string 'Optional'.
2. The category of the attribute, that is, whether it is a parameter or a state variable. The category line is not included for log variables.
3. The default value, if any. If the value is something simple (like a number or a very short string), the value will be shown on the line. Otherwise, the manual will just state that the attribute has a default value, but not what it is.

Sometimes the category line is left out. It is never included for log variables, since their category is obvious from context, and they never have default values. The category line is also left out for components, unless the component is optional or has a default value. This is because components can contain both state and log variables, as well as parameters. So the category of a component is not a very useful concept. If a component does have a default value or is optional, the category line

will be printed. However, instead of ‘parameter’ or ‘state variable’, the attribute will be categorized as a ‘component’.

3.3.3 The Textual Description

On the third line (or second, if the category line is missing) there is a brief textual description of the attribute. The description may continue on the following lines.

3.4 Submodels

Sometimes it is useful to combine attributes in a model into groups. One example of this is the ‘cond’ management action model (section 15.4), where we use the ‘clauses’ group to emphasize the strong association between each condition, and the list of actions to perform when the condition is true. Another example is the ‘default’ crop model (section 29.1), where there simply are so many attributes that grouping them logically becomes a necessity.

If you have looked at the two examples, you will see the attribute groups are categorized and typed as ‘submodel’ in the manual. After the textual description of the submodel follows a sample illustrating the formatting of the submodel attributes. The submodel sample follows the same conventions as the model sample, described in section 3.2. After the sample follows a nested attribute list, documenting attributes of the submodel in the exact same way as the attributes of the main model are documented, as described in section 3.3. You can even have submodels nested inside submodels, ad infinitum.

Chapter 4

The file parser model

This section describes Daisy setup files. A Daisy setup file is read by the ‘file’ model of the ‘parser’ component, hence the strange name.

4.1 Types and Formatting

There are a few general rules for the formatting accepted by the ‘file’ parser.

1. Elements are separated by white space.
2. Grouping is done with parentheses. You can always (but need not) have white space before and after parentheses.
3. White space is line feeds, spaces, and tabs. You can have any number (greater than one) of these anywhere you can have white space.
4. Comments start with the character ‘;’ and ends at the end of the line. Comments are legal anywhere you can have white space.
5. All names (attribute names, component names, model names, and parameterization names) follow the same conventions as string values, described in section 4.1.5.

In this section, we will describe how each type of attribute values should be formatted for each of the attribute types. The formatting does also depend slightly on whether the attribute is a singleton or a sequence, and whether the attribute is ordered or not, so we will cover that subject first.

4.1.1 Order and Size

Booleans, integers, numbers, and strings have a common format though. If *val* denotes the attribute value as described in sections 4.1.2, 4.1.3, 4.1.4, and 4.1.5, and *name* is the attribute name, then the format for the attributes in the setup files depends on whether they are ordered or not, and whether they are singletons or sequences, as illustrated in table 4.1.

	singleton	sequence
ordered	<i>val</i>	<i>(val ...)</i>
unordered	<i>(name val)</i>	<i>(name val ...)</i>

Table 4.1: Format for booleans, integers, numbers, and strings.

PLF's, submodels and fixed components are different though. Unlike the simpler types described above, these already contain several parts that are white space separated. Therefore, it is necessary to mark their start and end with parentheses except for unordered singletons, when there can be no ambiguity. Again, *name* is the attribute name, and *value* is the attribute value in the format described in sections 4.1.6 and 4.1.7.

For sequences, you can use the the special keyword `&old` anywhere in the specified value, and it will be replaced with the original value for the attribute. A typical example of its use is with the 'path' top level parameter, where you may add your own library files to the ones bundled with Daisy.

```
(path "C:/MyDaisyLib" &old)
```

	singleton	sequence
ordered	<i>(val)</i>	<i>((val)...) </i>
unordered	<i>(name val)</i>	<i>(name (val)...) </i>

Table 4.2: Format for time values, PLF's, submodels, and fixed components.

In general, library component values are written using the formatting conventions in table 4.2. However, there are two exceptions.

1. If you just specify the name of a complete parameterization, you do not need the extra parentheses, and can use the format conventions from table 4.1.
2. If there are no unordered attributes, then you do not need to put parentheses around the last attribute in the order. The action sequence is the last attribute in the 'clauses' submodel in the 'cond' action model.

4.1.2 Booleans

The simplest attribute type is booleans. A Boolean value can be either 'false' or 'true', spelled exactly that way, with small letters.

4.1.3 Integers

Integers should start with a non-zero digit, and be followed by a sequence of digits. Negative numbers should start with a '-' sign, followed immediately with the digits. No white space is allowed. The only integer which should start with '0' is the integer zero itself, don't use leading zeros for padding. Use space instead. Otherwise, Daisy might misinterpret the integer on some systems.

4.1.4 Numbers

The general format for writing numbers is

<sign><integer-part>.<fraction>e<sign><exponent>

All of these can be left out, except for the integer part. There can be no white space between any of the parts.

After the number, you may optionally specify the unit inside square brackets, like this [cm]. Fractions and other unitless numbers are specified as []. Daisy will check that the specified unit is also the unit it expects, or a simple conversion thereof (like [m] instead of [cm]). If the unit is listed as "unknown" in this manual, Daisy will be unable to check or convert it. It can still be specified for informational purposes, but the string must start with a question mark, like this [?cm]. The

known units are listed in chapter 96, more units can be defined with the ‘defunit’ command.

For lists of numbers, the unit applies to all the number before it.

In some contexts, numbers may be replaced by simple arithmetic expressions, using the operations defined in chapter 59.

4.1.5 Strings

Daisy accepts three different string formats. The first format is called ‘identifier’. Identifiers must start with a letter or a special characters (one of = < > _ + - * /) and can be followed by letters, digits or special characters only. They are conventionally used for attribute names.

The second format is called ‘quoted strings’, and are most often used for attribute values. Quoted strings start and begin with a double quote (“”). Any characters in between the two double quote signs are part of the string, with these exceptions:

- To include a “ in the string, you must write \”.
- To include a \, you must write \\.
- If you put a backslash as the last character on the line, neither it, nor the following newline character will not be part of the string.
- A \n will be replaced by a newline character.

It is an error to put any other character after backslash.

The third format is intended for units. It begins with [and ends with]. The string consists of the text between the two delimiters, and can contain any character but].

For some string values, a sequence of the form ‘\${var}’ will be replaced with the value of parameter *var*, which must be a string, integer, or number. To include a ‘\$’ in such a string value, you must write ‘\$\$’. See section 4.3 for more information.

Case is always significant.

File Names

Under some operating systems, file names are conventionally written like this:

```
C:\daisy\lib\log.dai
```

However, since \ is treated specially in Daisy setup files, you will have to double all the backslashed, like this:

```
(input file "C:\\daisy\\sample\\test.dai")
```

You can instead use ordinary slashes in file names:

```
(input file "C:/daisy/sample/test.dai")
```

4.1.6 PLF

PLF is the name used in Daisy for piecewise linear functions. In the setup files, such a function is specified as a sequence of (*x y*) pairs, where the *x* value must be monotonically increasing. The value of the PLF for any point in the *x* axis is then calculated by linear interpolation between the two closest points. The value *before* the first specified point will be the same as the value *at* the first specified point. Similarly, The value *after* the last specified point will be the same as the value *at* the last specified point.

An example might help. If ‘f’ is a unordered singleton PLF, we can specify its value like this:

```
(f (1.0 1.0) (3.0 2.0) (5.0 4.0))
```

We then get the values in table 4.3.

x	$f(x)$
0.0	1.0
1.0	1.0
2.0	1.5
3.0	2.0
4.0	3.0
5.0	4.0
6.0	4.0

Table 4.3: Interpolated values for f

4.1.7 Submodels and Fixed Components

A submodel or fixed component is basically a set of attribute names, with a value associated to each value. Some of the attributes are ordered, when specifying the value of a submodel you must specify the values for all the ordered attributes separated by white space, followed by the values for each of those unordered attributes you wish to set. The unordered attributes can, as the name implies, come in any order. How to write the attributes in the file depends on whether they are ordered or not, their type, and whether they are sequences or singletons. This is all described in section 4.1, in particular 4.1.1.

The fixed components are found in chapter 107, and the submodels are found with their parent models throughout part II. In the description of each submodel or fixed component, there is a *sample* showing the attributes you can specify, and how they should be placed. The sample is explained in section 3.2.

Submodels and fixed components are special compared to the other attribute values, in that they do not replace the original value. Instead, the original value will be amended with the individual attributes you specify. If you don't specify the value for a particular attribute, the original value will be used.

4.1.8 Library Components

A library component value should always start with the name of model or parameterization. If you want to overwrite some of the attribute values of the model or parameterization you just specified, you should write all the values of all the ordered attributes immediately after the name of the model or parameterization, separated by white space, and then the unordered attributes, just like you would for a submodel or fixed component (see section 4.1.7).

Each section of each chapter in part II describes a specific model. Part of that description is a *sample*, which shows how to specify that model in a setup file, and how to set the parameters for that model.

If you want to make partial modifications to the original value of the library component, you can specify the special keyword “original” instead of a name of a model or parameterization, followed by a list of those attribute names and values you want to overwrite.

4.2 Declaring new parameters

You can declare new parameters most places where you can specify parameter values. Once you have declared a new parameter, you can set it just like the build-in parameters. The syntax for declaring a new parameter is

```
(declare par [size] type "doc")
```

Here *par* is the name of the new parameter we want to declare, [*size*] is either missing, in which case the new parameter is a singleton, [], in which case the new parameter is a sequence with an arbitrary length, or *size* is an integer, in which case the sequence must hold exactly that number of values. *type* is either a name of a component, in which case the parameter must hold a parameterization of a model within that component, or **String**, **Integer**, or **Number**, in which case the value must be a primitive of that type. For **Number**, you must also specify a unit afterwards. *doc* is a non-optional description of the new parameter.

4.3 Referring to parameters

Instead of a parameter value, you can specify the name of another parameter.

```
(par1 $par2)
```

Here we specify the value of the parameter *par1* should be set to the value of the parameter *par2*. What makes this interesting is that *par2* does not have to be declared or have a value at the point in the setup file where you write this. The value of *par2* is only used at the time where the simulation starts, and must then be defined and have a value in an outer scope. What this means is best explained with an example. If you set the value of *clay* in a horizon to *\$MyClay*, then at the time where you start the simulation, Daisy will look for a parameter named *MyClay* in the horizon itself. If none is found, it will look for the parameter in the column where the horizon is used. If not declared there either, it will look for it in the Daisy program where the column is used. If it is still not found, it look for the parameter at the top level. And if that doesn't help, Daisy will give an error message.

Note that this substitution works on basis of a complete parameter value. You cannot substitute a single number in a PLF, nor single number in a sequence. The only exception in strings, where the sequence *\$par2* anywhere in the string is substituted with the value of *par2*.

Also note than some parameters doesn't accept this kind of substitution at all. Only way to find out is to try.

4.4 Toplevel

A Daisy setup file consist of definitions of parameterizations and a small number of commands. The complete list of commands accepted can be found in 107.30. The two most important are 'input' and 'run'. The following short setup file demonstrates all three.

```
;;; test-refman.dai -- Sample file using the Daisy libraries.

(input file "dk-soil.dai")
(input file "dk-management.dai")

(defprogram RefSim Daisy
  "Simulation for use in reference manual."
  (weather default "dk-taastrup.dwf")
  (column Askov)
```

```

(manager "SBarley w. catch crop")
(time 1986 12 1 1)
(stop 1988 4 1 1)
(output harvest))

(run RefSim)

;;; test-refman.dai ends here.

```

The format will be described in more details in section 4.1, here we will concentrate on the three commands being used. The first is ‘input’. It directs Daisy to read commands from somewhere else. It takes a ‘parser’ component (see chapter 61, page 335) argument. The ‘file’ model (see section 61.1, page 335) will read the commands from the specified name, in the same format we are currently describing. Typically, these files contains standard parameterizations.

The next command is ‘defprogram’. It should be read as *define program*, or define a new parameterization of the ‘program’ component. This is followed by the name of the new parameterization, and the name of an existing parameterization or model to use as basis for the new parameterization. That is, if some attribute is not specified for the new parameterization, assume the value from old parameterization. Here, the new parameterization is called ‘RefSim’, and the original model to base it on is called ‘Daisy’.

In general, the format for library component values described in section 4.1.8 is also used when defining named parameterizations. In the setup file, you write:

```
(defcomponent name value)
```

where *component* is the name of the component you want to define a named parameterization for, *name* is the name of the new parameterization you want to define, and *value* is a library component value as described in this section (4.1.8).

The last command is called ‘run’. It simply runs the specified program, ‘RefSim’. So, what happened here? Daisy, the program you start from Unix or Windows, actually itself contains different programs. A full list can be found in chapter 66, page 353. The main program is itself called ‘Daisy’, and will run a single Daisy simulation. We just created a parameterization containing all the mandatory parameters for a Daisy simulation, and called it ‘RefSim’. For a complete list of Daisy parameters, see section 66.2, page 353.

For compatibility with older version of Daisy (the Unix or Windows program) you can also specify the Daisy (the Daisy program) attributes directly on top level. In that case, the “run” command is not necessary.

Chapter 5

Log File Generation

The value of the ‘output’ attribute should be a sequence of log component parameterizations. Each parameterization should specify some state or log variables to be output during the simulation.

The ‘log.dai’ file contains a number of standard log parameterizations, for example ‘Crop Production’, which will write information about the development and production of any crops in the field to a file named ‘crop_prod.dlf’ located in the directory specified by the ‘directory’ attribute, or, by default, the directory where daisy was started. This file will contain a header with some meta-information about the simulation, a line of dashes, and then a large number of tab separated columns, with each column describing one aspect of the crop development (e.g. the stem nitrogen content), and each row a specific time step. The two first rows are special, the first will contain a label identifying the column (e.g. NStem), and the second row the unit used for the numbers in that column (e.g. kg N/ha).

You can obviously also define your own parameterizations. If you wish to do so, the ‘table’ log model (see section 52.5) is far the most useful. In the example, we specify two standard log parameterizations (‘Crop Production’ and ‘Harvest’) and one home-made parameterization (‘Pond’).

```
(deflog Pond table
  (where "pond.dlf")
  (when daily)
  (entries (number (path time year))
            (number (path time month))
            (number (path time mday))
            (number (path column "*" Surface pond)
                    (dimension "mm"))))
```

The ‘table’ log model has a number of parameters, most of these have default values. The three listed here are ‘where’, which specifies the name of the file to log the results in, ‘when’, which specifies how often to log the results, and ‘entries’, which specifies what to log. The ‘entries’ attribute is a sequence of ‘select’ (see chapter 80) models, where each entry specifies one column in the log file. The first entry listed will correspond to the first column in the log file.

Each log entry model has a large number of attributes, allowing you to accumulate results, extract values from the soil array, specify the tag and unit printed in the two first rows of the column, and more. Most of these have default values, and only the fourth (and last) log entry submodel specifies one of these attributes, namely the unit string to print in the second row.

The most important log entry parameter is ‘path’, which specifies the state or log variable we want to include in the log file. The value is a sequence of attribute

names, which should be read as follows: Start with the Daisy model (section 66.2), and select an attribute in that model. That should be the first name listed in the path sequence.

- If the attribute is a time value, as in the three first log entries, you list one of ‘year’, ‘month’, ‘mday’, or ‘hour’ next in the path. The table log model will then log the selected part of the time value.
- If the selected attribute is a component sequence, as in the fourth log entry in the example, you should list the name of the model or parameterization you want to log next in the path list. In this case ‘JB1_Andeby’ would do, but instead we use the special value ‘*’, which will match *all* entries in the sequence. However, their values will be accumulated into a single number in the log file, which is why we say most of the standard log models don’t work well with multiple columns.
- After listing the name of the model or parameterization in the sequence to log, we must list an attribute of the model. In this case we list the ‘Surface’ fixed component.
- As long as the selected attribute itself is a submodel or fixed component, we keep listing a name of an attribute in *that* model next.
- In this case, it ends with the ‘pond’ attribute of the ‘Surface’ fixed component. The value of the ‘pond’ attribute is a single number, which will be logged in the fourth column of the ‘pond.dlf’ log file.

That is the general strategy for writing log file entry paths. Keep listing names of nested attributes, until you reach a simple value, which will then be logged.

Chapter 6

Error Messages

Sometimes the simulation will fail to run for some reason or another. In these cases, Daisy will attempt to give an error message, which hopefully will give some indication of what went wrong. There are zillions of situations where Daisy might give some error message, in the following we describe some of them.

6.1 Parser errors

If the format of the setup files is wrong, for example if there are missing parentheses or unknown attributes, Daisy will give you information about which file the error was found, the line number, the column number, and what the problem was.

```
test.dai:15:6: Unknown attribute 'kurt'
```

This is the “best” kind of errors, since it will lead you directly to the place where the error occurred.

If you run Daisy with ‘M-x compile RET’ inside the Emacs text editor, you can go directly to the place the error occurred by clicking on the error message with the mouse.

6.2 Completeness checks

After parsing the setup files, Daisy will check that all the required parameters and state variables are defined, and make some very simple consistency checks. Errors during this phase looks as follows:

```
* Daisy
time missing
```

6.3 Consistency checks

After checking that all the required attributes are there, Daisy will create the simulation objects, and then perform a more throughout consistency check. Errors found during these phases will be reported like the one below.

```
* column
** Andeby
*** SoilHeat
You have 20 intervals but 60 T values
```

6.4 Runtime exceptions

Now Daisy is ready to run the simulation. However, some errors are too hard to find before the simulation has started. If Daisy finds one of those during the simulation, it will throw an exception, which normally causes the program to exit.

```
Exception: Cannot restrict already restricted field
```

6.5 Assertion failures

During debugging, a lot of ‘assert’ statements have been added to the code. Each of these statements will check if a certain condition is true, and if not, exit the program with a message like

```
field.C:71: failed assertion 'false'
```

A failed assertion message will always print the file name, the line number, and the assertion that failed, but the format may vary between different platforms.

The difference between assertion failures and runtime exceptions is that the runtime exceptions are normally due to an error in the setup files, while the assertion may also be caused by bugs in the program. The printed information is a great help for debugging those. So failed assertions should be reported to us.

6.6 Program errors

Sometimes we can get errors that aren’t even caught by the assertions, they can manifest themselves as “Bus error” or “Segmentation violation” errors, or weird pop-up windows under win32. Or even worse, as bogus numbers in the log files. These can be due to program errors, hardware errors, too little memory, or errors in the operating system. They can also be caused by errors in the setup files, but in this case it is also a program error, as such errors should be caught by one of the mechanisms above.

Part II

The Components

Chapter 7

ABAProduction

The 'ABAProduction' component calculates the prod of ABA in soil.

Used by root component ABAProd (see 72, page 401) .

7.1 none

No ABA production.

Used by root component ABAProd (see 72, page 401) .

7.2 root

ABA production based on production in roots.

The assumptions are that that each length of root will produce ABA with a rate that depends solely on the water pressure in that cell, and that all the ABA will be included in the water uptake.

< root (expr *expr*) >

- *expr*: **number** component (see chapter 59)

Expression to evaluate to ABA production per root length [g/cm/h]. The symbol 'h' will be bound to the water pressure [cm].

7.3 soil

ABA production based on soil location.

< soil (expr *expr*) >

- *expr*: **number** component (see chapter 59)

Expression to evaluate to ABA uptake [g/cm³/h]. The symbol 'h' will be bound to the water pressure [cm]. The symbol 'L' will be bound to the root density [cm/cm³]. The symbol 'S' will be bound to the water uptake [cm³/cm³/h].

7.4 uptake

ABA production based on concentration in water uptake.

The assumption is water uptake from roots in specific region of the soil comes with a specific ABA concentration, which depends solely on the water pressure in that region.

< uptake (expr *expr*) >

- *expr*: **number** component (see chapter 59)
Expression to evaluate to ABA concentration in water uptake [g/cm³]. The symbol 'h' will be bound to the water pressure [cm].

Chapter 8

AOM

A single Added Organic Matter pool.

Used by Harvesting @ Root (see 107.7, page 593) , and Bioincorporation @ AOM (see 107.11, page 603) .

```
< component (N N ...)
              (C C ...)
              (heat_factor heat_factor)
              (water_factor water_factor)
              (turnover turnover)
              (efficiency efficiency ...)
              (C_per_N C_per_N ...)
              (fractions fractions ...)
              (initial_C_per_N initial_C_per_N)
              (initial_fraction initial_fraction)
              (top_C 0 [g C/cm2])
              (top_N 0 [g N/cm2]) >
```

- *N*: number [g N/cm³] soil cells
Optional state variable
Nitrogen in each soil interval.
- *C*: number [g C/cm³] soil cells
Optional state variable
Carbon in each soil interval.
- *heat_factor*: plf [dg C → <none>]
Optional parameter
Heat factor. If empty, use default from 'OrganicMatter'.
- *water_factor*: plf [cm → <none>]
Optional parameter
Water potential factor. If empty, use default from 'OrganicMatter'.
- *turnover*: **rate** component (see chapter 69)

Fraction converted to other pools each hour.

- *efficiency*: number [<fraction>] sequence
Parameter
The efficiency this pool can be digested by each of the SMB pools.

- *C_per_N*: number $[(\text{g C}/\text{cm}^3)/(\text{g N}/\text{cm}^3)]$ soil cells
Optional state variable
The carbon/nitrogen ratio.
- *fractions*: number [**<fraction>**] sequence
Parameter
How this pool is divided into other pools. The first numbers corresponds to each of the SMB pools, the next number to the SOM buffer, and any remaining numbers to each of the DOM pools. The length of the sequence should thus be the number of SMB pools plus 1 plus optionally the number of DOM pools.
- *initial_C_per_N*: number $[\text{g C}/\text{g N}]$
Optional state variable
The initial C/N ratio when this pool is created. Negative numbers mean unspecified.
- *initial_fraction*: number [**<fraction>**]
Optional parameter
The initial fraction of the total available carbon allocated to this pool for AOM. One pool should be left unspecified.
- *top_C*: number $[\text{g C}/\text{cm}^2]$
State variable (default 0)
Carbon on top of soil.
- *top_N*: number $[\text{g N}/\text{cm}^2]$
State variable (default 0)
Nitrogen on top of soil.

8.1 AOM-DIRECT

Third AOM pool of already decomposed material.

```
< (turnover turnover) ; Default rate value.
  (efficiency 1)
  (fractions 0 0 1) >
```

8.2 AOM-FAST

Fast AOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003]

Used by crop simple root_am (see 29.47, page 178) , am root om (see 17.4, page 105) , vegetation permanent litter_am (see 100.3, page 545) , vegetation afforestation root_am (see 100.1, page 542) , Harvesting @ Root (see 107.7, page 593) , and Bioincorporation @ AOM (see 107.11, page 603) .

```
< (turnover turnover) ; Default rate value.
  (efficiency 0.5 0.5)
  (fractions 0 1 0) >
```

8.3 CROP-FAST

A ‘AOM-FAST’ model (see 8.2, page 54) build into Daisy.

Parameterization used for fast pool of some crop residuals. See also [Bruun et al., 2003]

```
< (turnover turnover) ; Default rate value. >
```

8.4 AOM-SLOW

Slow AOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003]

Used by crop simple root_am (see 29.47, page 178) , am root om (see 17.4, page 105) , vegetation permanent litter_am (see 100.3, page 545) , vegetation afforestation root_am (see 100.1, page 542) , and Harvesting @ Root (see 107.7, page 593) .

```
< (turnover turnover) ; Default rate value.
    (efficiency 0.5 0.5)
    (C_per_N 90)
    (fractions 0 1 0)
    (initial_fraction 0.8 [<fraction>]) >
```

8.5 AOM-SLOW-BIOINCORPORATION

A ‘AOM-SLOW’ model (see 8.4, page 55) build into Daisy.

Lower C/N ration for bioincorporated matter. See also [Bruun et al., 2003]

Used by Bioincorporation @ AOM (see 107.11, page 603) .

```
< (C_per_N 60) >
```

8.6 AOM-SLOW-OLD

A ‘AOM-SLOW’ model (see 8.4, page 55) build into Daisy.

Original parameterization of the slow AOM pool. See also [Müller et al., 1997]

```
< (fractions 0.5 0.5 0) >
```

8.7 CROP-SLOW

A ‘AOM-SLOW’ model (see 8.4, page 55) build into Daisy.

Parameterization used for slow pool of some crop residuals. See also [Bruun et al., 2003]

```
< (turnover turnover) ; Default rate value.
    (C_per_N 100) >
```

8.8 Ryegrass-SLOW

A ‘CROP-SLOW’ model (see 8.7, page 55) defined in ‘ryegrass.dai’.

Slow AOM pool modified for ryegrass. See also [Bruun et al., 2003]

8.9 Wclover-SLOW

A ‘CROP-SLOW’ model (see 8.7, page 55) defined in ‘wclover.dai’.

Slow AOM pool modified for Wclover. See also [Bruun et al., 2003]

Chapter 9

AWI

Estimate area of air-water interface.

< component >

Log Variables

- *area*: number [cm^2/cm^3] soil cells
Air-Water interface area.

9.1 Brusseau2023

Estimate AWI area from water and median particle size (Equation 5). See also [Brusseau, 2023]

Used by column default AWI (see 26.1, page 153) .

9.2 Costanza2008x

Estimate AWI area from water and median particle size (Equation 1 and 2). See also [Costanza-Robinson et al., 2008]

Chapter 10

ClayOM

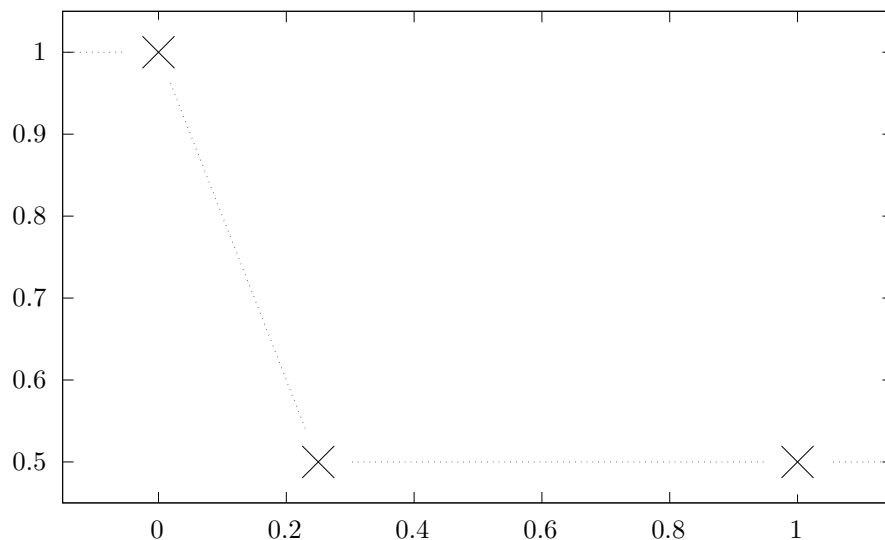
Find the effect of clay on organic matter processing.

10.1 biomod

Clay influence on organic matter from BIOMOD project. All SMB pools are affected, but not the SOM pools. Additionally, the ration between maintenance and turnover is also clay dependent. See also [Petersen et al., 2005a, Petersen et al., 2005b, Pedersen et al., 2007]

```
< biomod (factor factor) ; Has default value.  
  (a a)  
  (E_SMB E_SMB)  
  (f_SMB1 f_SMB1) >
```

- *factor*: plf [**<fraction>** → **<none>**]
Parameter (has default value with 3 points)



```
(factor (0 [] 1 []) (0.25 [] 0.5 []) (1 [] 0.5 []))
```

Parameter description:

Function of clay content, multiplied to the maintenance and turnover rates of the SMB pools.

- *a*: number (dimensionless)
Parameter
Maintenance parameter.
- *E_{SMB}*: number [<fraction>]
Parameter
SMB efficiency in processing organic matter. Note that you must set the 'efficiency' parameter for all OM pools to this number for the BIOMOD clay response model to work correctly.
- *f_{SMB1}*: number [<fraction>]
Parameter
Fraction of AOM pools goind to SMB1. Only the fraction of AOM going to a SMB pool count, so this is really a fraction of the fraction coing to the SMB pools. Note that you must set the 'fraction' parameter of all AOM pools to reflect this for the BIOMOD clay response model to work correctly.

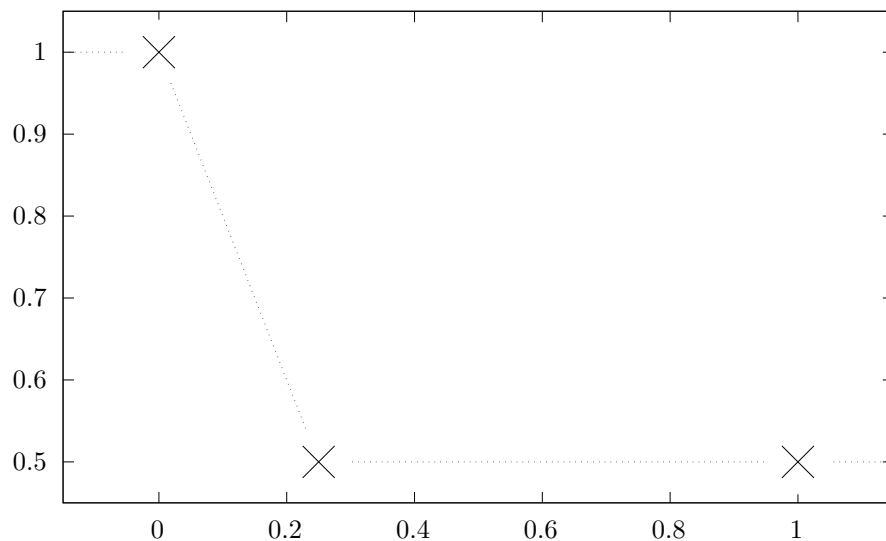
10.2 old

Traditional clay influence on organic matter.

Used by organic default ClayOM (see 60.1, page 327) .

< old (factor *factor*) ; Has default value. >

- *factor*: plf [<fraction> → <none>]
Parameter (has default value with 3 points)



```
(factor (0 [] 1 []) (0.25 [] 0.5 []) (1 [] 0.5 []))
```

Parameter description:

Function of clay content, multiplied to the maintenance and turnover rates of SMB1 and all SOM pools.

Chapter 11

MV_Crop

Description of a crop for use by the MARKVAND model.

11.1 default

Standard MARKVAND crop model.

```
< default  (S_F S_F ...)
            (A_F A_F ...)
            (L_gv L_gv)
            (L_ge L_ge)
            (L_gx L_gx)
            (L_gm L_gm)
            (L_ym L_ym)
            (S_Le S_Le)
            (S_Lx S_Lx)
            (S_Lr S_Lr)
            (S_Lm S_Lm)
            (z_0 z_0)
            (z_v z_v)
            (z_xA z_xA)
            (z_m z_m)
            (c_r c_r) >
```

- *S_F*: number [**dg C d**] sequence
Parameter
Temperature sum for each phase.
- *A_F*: number [**<fraction>**] sequence
Parameter
Allowable water deficit for each phase before irrigation.
- *L_gv*: number (dimensionless)
Parameter
Green leaf area index at emergence / growth start.
- *L_ge*: number (dimensionless)
Parameter
Green leaf area index at the time where growth rate become exponential.
- *L_gx*: number (dimensionless)
Parameter
Maximum green leaf area index.

- L_{gm} : number (dimensionless)
Parameter
Green leaf area index at maturity.
- L_{ym} : number (dimensionless)
Parameter
Yellow leaf area index at maturity.
- S_{Le} : number [dg C d]
Parameter
Temperature sum when green LAI growth turn exponential.
- S_{Lx} : number [dg C d]
Parameter
Temperature sum maximum green LAI.
- S_{Lr} : number [dg C d]
Parameter
Temperature sum for start of yellow leaves.
- S_{Lm} : number [dg C d]
Parameter
Temperature sum at maturity.
- z_0 : number [mm]
Parameter
Root depth before emergence (growth start).
- z_v : number [mm]
Parameter
Root depth at emergence (growth start).
- z_{xA} : number [mm]
Parameter
Maximum root depth for this crop.
- z_m : number [mm]
Parameter
Root depth at maturity.
- c_r : number [mm/d]
Parameter
Root penetration rate.

Chapter 12

MV_Soil

Description of a soil for use by the MARKVAND model.

12.1 default

Standard MARKVAND soil model.

```
< default  (z_o z_o)
            (z_xJ z_xJ)
            (Theta_fo Theta_fo)
            (Theta_wo Theta_wo)
            (Theta_fu Theta_fu)
            (Theta_wu Theta_wu)
            (C_e C_e)
            (c_e c_e)
            (c_T c_T)
            (k_qr k_qr)
            (k_qb k_qb) >
```

- *z_o*: number [**mm**]
Parameter
Depth of top soil.
- *z_xJ*: number [**mm**]
Parameter
Max rooting depth.
- *Theta_fo*: number [<**fraction**>]
Parameter
Field capacity, topsoil.
- *Theta_wo*: number [<**fraction**>]
Parameter
Wilting point, topsoil.
- *Theta_fu*: number [<**fraction**>]
Parameter
Field capacity, subsoil.
- *Theta_wu*: number [<**fraction**>]
Parameter
Wilting point, subsoil.

- C_e : number [**mm**]
Parameter
Capacity of evaporation reservoir.
- c_e : number [<**fraction**>]
Parameter
Basic evaporation factor.
- c_T : number [**mm**]
Parameter
Transpiration constant.
- k_{qr} : number (dimensionless)
Parameter
Drainage constant root zone.
- k_{qb} : number (dimensionless)
Parameter
Drainage constant subsone.

Chapter 13

SMB

A single Soil MicroBiological pool.

```
< component  (N N ...)
              (C C ...)
              (heat_factor heat_factor)
              (water_factor water_factor)
              (turnover turnover)
              (efficiency efficiency ...)
              (C_per_N C_per_N ...)
              (fractions fractions ...)
              (initial_C_per_N initial_C_per_N)
              (maintenance maintenance) >
```

- *N*: number [g N/cm³] soil cells
Optional state variable
Nitrogen in each soil interval.
- *C*: number [g C/cm³] soil cells
Optional state variable
Carbon in each soil interval.
- *heat_factor*: plf [dg C → <none>]
Optional parameter
Heat factor. If empty, use default from 'OrganicMatter'.
- *water_factor*: plf [cm → <none>]
Optional parameter
Water potential factor. If empty, use default from 'OrganicMatter'.
- *turnover*: **rate** component (see chapter 69)

Fraction converted to other pools each hour.

- *efficiency*: number [<fraction>] sequence
Parameter
The efficiency this pool can be digested by each of the SMB pools.
- *C_per_N*: number [(g C/cm³)/(g N/cm³)] soil cells
Optional state variable
The carbon/nitrogen ratio.
- *fractions*: number [<fraction>] sequence
Parameter

How this pool is divided into other pools. The first numbers corresponds to each of the SMB pools, the next numbers corresponds to the SOM pools, and the last numbers to each of the DOM pools. The length of the sequence should thus be the number of SMB pools plus the number of SOM pools plus the number of DOM pools.

- *initial_C_per_N*: number [g C/g N]
Optional state variable
The initial C/N ratio when this pool is created. Negative numbers mean unspecified.
- *maintenance*: number [h⁻¹]
Parameter
The fraction used for staying alive each hour.

13.1 SMB-FAST

Fast SMB pool parameterization.

Used by organic SOM2025 smb (see 60.3, page 334) .

```
< (turnover turnover) ; Default rate value.
  (efficiency 0.6 0.6)
  (C_per_N 6.7)
  (fractions 0 0.6 0 0.4 0)
  (maintenance 0.000416667 [h-1]) >
```

13.2 SMB-FAST-2000

A ‘SMB-FAST’ model (see 13.1, page 66) build into Daisy.

Used by organic SOM2000 smb (see 60.2, page 334) .

```
< (fractions 0 0.4 0 0.6 0) >
```

13.3 SMB-SLOW

Slow SMB pool parameterization.

Used by organic SOM2000 smb (see 60.2, page 334) , and organic SOM2025 smb (see 60.3, page 334) .

```
< (turnover turnover) ; Default rate value.
  (efficiency 0.6 0.6)
  (C_per_N 6.7)
  (fractions 0 0.6 0 0.4 0)
  (maintenance 7.5e-05 [h-1]) >
```

Chapter 14

SOM

A single Soil Organic Matter pool.

```
< component (N N ...)
              (C C ...)
              (heat_factor heat_factor)
              (water_factor water_factor)
              (turnover turnover)
              (efficiency efficiency ...)
              (C_per_N C_per_N ...)
              (fractions fractions ...)
              (initial_C_per_N initial_C_per_N) >
```

- *N*: number [g N/cm³] soil cells
Optional state variable
Nitrogen in each soil interval.
- *C*: number [g C/cm³] soil cells
Optional state variable
Carbon in each soil interval.
- *heat_factor*: plf [dg C → <none>]
Optional parameter
Heat factor. If empty, use default from 'OrganicMatter'.
- *water_factor*: plf [cm → <none>]
Optional parameter
Water potential factor. If empty, use default from 'OrganicMatter'.
- *turnover*: **rate** component (see chapter 69)

Fraction converted to other pools each hour.
- *efficiency*: number [<fraction>] sequence
Parameter
The efficiency this pool can be digested by each of the SMB pools.
- *C_per_N*: number [(g C/cm³)/(g N/cm³)] soil cells
Optional state variable
The carbon/nitrogen ratio.
- *fractions*: number [<fraction>] sequence
Parameter

How this pool is divided into other pools. The first numbers corresponds to each of the SMB pools, the next numbers corresponds to the SOM pools, and the last numbers to each of the DOM pools. The length of the sequence should thus be the number of SMB pools plus the number of SOM pools plus the number of DOM pools.

- *initial_C_per_N*: number [g C/g N]
Optional state variable
The initial C/N ratio when this pool is created. Negative numbers mean unspecified.

14.1 SOM-FAST

Fast SOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003]

Used by organic default som (see 60.1, page 327) .

```
< (turnover turnover)           ; Default rate value.
  (efficiency 0.5 0.5)
  (fractions 0.7 0 0.3 0 0) >
```

14.2 SOM-FAST-OLD

A ‘SOM-FAST’ model (see 14.1, page 68) build into Daisy.

Original parameterization of the fast SOM pool. See also [Müller et al., 1997]

```
< (fractions 0.9 0 0.1 0 0) >
```

14.3 SOM-INERT

Inert SOM pool parameterization.

Used by organic default som (see 60.1, page 327) .

```
< (turnover turnover)           ; Default rate value.
  (efficiency 0.5 0.5)
  (fractions 0 0 0 0 1) >
```

14.4 SOM-SLOW

Slow SOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003]

Used by organic default som (see 60.1, page 327) .

```
< (turnover turnover)           ; Default rate value.
  (efficiency 0.4 0.4)
  (fractions 1 0 0 0 0) >
```

14.5 SOM-SLOW-OLD

A ‘SOM-SLOW’ model (see 14.4, page 68) build into Daisy.

Original parameterization of the slow SOM pool. See also [Müller et al., 1997]

```
< (turnover turnover)           ; Default rate value. >
```

Chapter 15

action

The 'action' component represents management on different abstraction levels, from a single tillage operation to strategies of how to manage a farm. Typically, but not necessarily, the high level management strategies are build by combining low level management operations.

15.1 BBCH

Management according to external BBCH scale.

```
< BBCH (crop crop)  
      (file file)  
      (from from)  
      (to to)  
      (from_start true)  
      (to_start false)  
      (do do)  
      (when when) >
```

- *crop*: string (see section 4.1.5)
Parameter
Name of crop to use BBCH number for.
- *file*: string (see section 4.1.5)
Parameter
Name of file where data is found.
- *from*: integer
Parameter
BBCH at beginning of interval.
- *to*: integer
Parameter
BBCH at end of interval.
- *from_start*: boolean (see section 4.1.2)
Parameter (default true)
If true, the interval begins when the crop enters the specified BBCH. Otherwise, the interval begins when the crop leaves the specified BBCH.
- *to_start*: boolean (see section 4.1.2)
Parameter (default false)

If true, the interval ends when the crop enters the specified BBCH. Otherwise, the interval ends when the crop leaves the specified BBCH.

- *do*: **action** component (see chapter 15)
Action to perform.
- *when*: **condition** component (see chapter 28)
When to perform the action. It will be performed the first time the condition is true within the specified BBCH interval, or at the end of the interval, whichever comes first. Use 'true' to perform the action at the start of the interval, or 'false' to perform it at the end.

15.2 assert

Assert that condition is true, if not, stop the simulation.

```
< assert condition
      (message message) ; Has default value. >
```

- *condition*: **condition** component (see chapter 28)
Condition to check.
- *message*: string (see section 4.1.5)
Parameter (has default value with length 32)

```
(message "Required condition not fulfilled")
```

Parameter description:
Error message to give iff assertion fails.

15.3 at

Do action at specific time.

```
< at year month day
      (hour 8)
      (do do) >
```

- *year*: integer
Parameter
Perform action this year.
- *month*: integer
Parameter
Perform action this month.
- *day*: integer
Parameter
Perform action this day in the month.
- *hour*: integer
Parameter (default 8)
Perform action this hour.
- *do*: **action** component (see chapter 15)
Action to perform at specified time..

15.4 cond

Perform the actions associated with the first true condition in the list.

```
< cond  clauses...  >
```

- *clauses*: submodel (see section 4.1.7) sequence
Each clause consist of a condition and a sequence of actions. The first clause whose condition is true, will have its actions activated.

```
<  condition actions...  >
```

- *condition*: **condition** component (see chapter 28)
Condition for performing the actions.
- *actions*: **action** component (see chapter 15) sequence
Actions to perform when condition is meet.

15.5 crop

Manage a specific crop or multicrop.

```
< crop  (spray)
        (secondary secondary)
        (tillage)
        (primary primary) ; Has partial value.
        (harvest_annual harvest_annual)
        (harvest_perennial harvest_perennial)
        (fertilize_at)
        (fertilize_at_index 0)
        (fertilize_incorporate false)
        (tillage_index 0)
        (spray_index 0)
        (irrigation irrigation)
        (irrigation_rest irrigation_rest)
        (irrigation_year irrigation_year)
        (irrigation_delay irrigation_delay) >
```

- *spray*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
List of chemicals to apply.

```
<  month day name amount  >
```

- *month*: integer
Parameter
Month in the year.
- *day*: integer
Parameter
Day in the month.
- *name*: string (see section 4.1.5)
Parameter
Name of chemichal to spray.
- *amount*: number [g/ha]
Parameter
Amount of chemichal to spray.

- *secondary*: submodel (see section 4.1.7)
 Optional submodel
 Secondary crop, if any.

```
< (crop crop)
    (date date)      ; Has partial value.
    (done false) >
```

 - *crop*: **crop** component (see chapter 29)
 Crop to sow.
 - *date*: submodel (see section 4.1.7)
 Submodel (has partially specified default value)

```
(date <missing month> <missing day>)
```

 Parameter description:
 Date to sow.

```
< month day
    (hour 8) >
```

 - * *month*: integer
 Parameter
 Month in the year.
 - * *day*: integer
 Parameter
 Day in the month.
 - * *hour*: integer
 Parameter (default 8)
 Hour in the day.
 - *done*: boolean (see section 4.1.2)
 State variable (default false)
 True iff the crop has been sowed.
- *tillage*: submodel (see section 4.1.7) sequence
 Submodel (default: an empty sequence)
 List of tillage operations to apply.

```
< month day operation >
```

 - *month*: integer
 Parameter
 Month in the year.
 - *day*: integer
 Parameter
 Day in the month.
 - *operation*: **action** component (see chapter 15)
 Tillage operation.
- *primary*: submodel (see section 4.1.7)
 Submodel (has partially specified default value)

```
(primary )
```

 Parameter description:
 Primary crop.

```
< (crop crop)
    (date date)      ; Has partial value.
    (done false) >
```


- *crop*: **crop** component (see chapter 29)
Crop to sow.
- *date*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(date <missing month> <missing day>)

Parameter description:

Date to sow.

```
< month day
    (hour 8) >
```

* *month*: integer
Parameter
Month in the year.

* *day*: integer
Parameter
Day in the month.

* *hour*: integer
Parameter (default 8)
Hour in the day.

- *done*: boolean (see section 4.1.2)
State variable (default false)
True iff the crop has been sowed.

- *harvest_annual*: submodel (see section 4.1.7)

Optional submodel

Harvest parameters for annual crops.

```
< (done false)
    (loss loss)
    (remove_residuals remove_residuals)
    (latest latest) ; Has partial value. >
```

- *done*: boolean (see section 4.1.2)
State variable (default false)
True iff the crop has been sowed.

- *loss*: number [<fraction>]
Parameter
Fraction lost during harvest.

- *remove_residuals*: boolean (see section 4.1.2)
Parameter
Remove residuals at harvest.

- *latest*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(latest <missing month> <missing day>)

Parameter description:

Latest harvest date. If the crop is ripe before this date, it will be harvested at that point.

```
< month day
    (hour 8) >
```

* *month*: integer
Parameter
Month in the year.

- * *day*: integer
Parameter
Day in the month.
 - * *hour*: integer
Parameter (default 8)
Hour in the day.
- *harvest_perennial*: submodel (see section 4.1.7)
Optional submodel
Harvest conditions for perennial crops.


```

      < (DS DS)
        (fertilize fertilize ...)
        (end end) ; Has partial value.
        (DM DM)
        (seasons seasons)
        (year_of_last_harvest year_of_last_harvest)
        (fertilize_index 0)
        (fertilize_rest fertilize_rest ...)
        (fertilize_rest_index 0)
        (fertilize_year fertilize_year) >
      
```

 - *DS*: number (dimensionless)
Parameter
Development stage at or above which to initiate harvest.
 - *fertilize*: **am** component (see chapter 17) sequence
Optional component
Fertilizer applications after harvest first season. First season is defined as the year where the first harvest occurs.
 - *end*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(end <missing month> <missing day>)

Parameter description:
End management this date the last season.


```

          < month day
            (hour 8) >
          
```

 - * *month*: integer
Parameter
Month in the year.
 - * *day*: integer
Parameter
Day in the month.
 - * *hour*: integer
Parameter (default 8)
Hour in the day.
 - *DM*: number [**kg DM/ha**]
Parameter
Dry matter at or above which to initiate harvest.
 - *seasons*: integer
Parameter
Number of seasons to harvest crop. The crop will be harvested whenever the specified DS or DM are reached. The first season is the year the crop management starts.

- *year_of_last_harvest*: integer
Optional state variable
Year of last season.
- *fertilize_index*: integer
State variable (default 0)
Next entry in 'fertilize' to execute.
- *fertilize_rest*: **am** component (see chapter 17) sequence
Optional component
Fertilizer applications after harvest remaining seasons. If missing, use the same fertilizer as first season.
- *fertilize_rest_index*: integer
State variable (default 0)
Next entry in 'fertilize_rest' to execute.
- *fertilize_year*: integer
Optional state variable
Year last fertilization was applied.
- *fertilize_at*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Fertilizer application by date.

```
< month day what >
```

 - *month*: integer
Parameter
Month in the year.
 - *day*: integer
Parameter
Day in the month.
 - *what*: **am** component (see chapter 17)
Fertilizer to apply.
- *fertilize_at_index*: integer
State variable (default 0)
Next entry in 'fertilize_at' to execute.
- *fertilize_incorporate*: boolean (see section 4.1.2)
Parameter (default false)
Incorporate organic fertilizer in plowing zone.
- *tillage_index*: integer
State variable (default 0)
Next entry in 'tillage' to execute.
- *spray_index*: integer
State variable (default 0)
Next entry in 'spray' to execute.
- *irrigation*: submodel (see section 4.1.7)
Optional submodel
Irrigation model for first season. If missing, don't irrigate.

```
< (flux 2 [mm/h])
  (potential potential)
  (from from)           ; Has partial value.
  (to to)               ; Has partial value.
  (amount amount) >
```

- *flux*: number [**mm/h**]
Parameter (default 2)
Water application speed.
- *potential*: number [**cm**]
Parameter
Soil potential at which to irrigate.
- *from*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(from <missing month> <missing day>)

Parameter description:
Start of irrigation period.
 - < *month day*
 (hour 8) >
 - * *month*: integer
 Parameter
 Month in the year.
 - * *day*: integer
 Parameter
 Day in the month.
 - * *hour*: integer
 Parameter (default 8)
 Hour in the day.
- *to*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(to <missing month> <missing day>)

Parameter description:
End of irrigation period.
 - < *month day*
 (hour 8) >
 - * *month*: integer
 Parameter
 Month in the year.
 - * *day*: integer
 Parameter
 Day in the month.
 - * *hour*: integer
 Parameter (default 8)
 Hour in the day.
- *amount*: number [**mm**]
Parameter
Amount of water to apply on irrigation.
- *irrigation_rest*: submodel (see section 4.1.7)
Optional submodel
Irrigation model for remaining seasons. If missing, use the same model as first season.
 - < (flux 2 [mm/h])
 (potential *potential*)
 (from *from*) ; Has partial value.
 (to *to*) ; Has partial value.
 (amount *amount*) >

- *flux*: number [**mm/h**]
Parameter (default 2)
Water application speed.
- *potential*: number [**cm**]
Parameter
Soil potential at which to irrigate.
- *from*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(from <missing month> <missing day>)

Parameter description:
Start of irrigation period.
 - < *month day*
(hour 8) >
 - * *month*: integer
Parameter
Month in the year.
 - * *day*: integer
Parameter
Day in the month.
 - * *hour*: integer
Parameter (default 8)
Hour in the day.
- *to*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(to <missing month> <missing day>)

Parameter description:
End of irrigation period.
 - < *month day*
(hour 8) >
 - * *month*: integer
Parameter
Month in the year.
 - * *day*: integer
Parameter
Day in the month.
 - * *hour*: integer
Parameter (default 8)
Hour in the day.
- *amount*: number [**mm**]
Parameter
Amount of water to apply on irrigation.
- *irrigation_year*: integer
Optional state variable
Year management started. Negative number means it hasn't started yet.
- *irrigation_delay*: integer
Optional state variable
Hours we test for irrigation again. This is set at each irrigation, to avoid multiple applications.

15.6 emerge

Force a crop to emerge.

```
< emerge  crop  >
```

- *crop*: string (see section 4.1.5)
Parameter (default 'all')
Name of the crop to emerge. If you specify 'all', all crops will emerge. If there are no crop on the field with the specified name, nothing will happen.

15.7 error

Write a error message to the user.

```
< error  message  >
```

- *message*: string (see section 4.1.5)
Parameter
Error message to give.

15.8 extern

Select an external scope, and perform action.

```
< extern  scope action  >
```

- *scope*: **scopesel** component (see chapter 77)
Scope to evaluate expressions in.
- *action*: **action** component (see chapter 15)
Action to perform if the condition is false.

15.9 extern_fertigation

Continues irrigation with mineral nitrogen mix.

If the nitrogen amount is non-zero, it will be applied in the irrigation water if available, and otherwise be spread on the soil surface.

```
< extern_fertigation  (NO3 NO3)
                      (NH4 NH4)
                      (volume volume)           ; Default box value.
                      (surface surface)
                      (scope scope)               ; Default null value.
                      (from -10 [cm])
                      (to to)
                      (overhead overhead)
                      (subsoil subsoil) >
```

- *NO3*: **number** component (see chapter 59)
Amount of NO3 in irrigation.
- *NH4*: **number** component (see chapter 59)
Amount of NH4 in irrigation.
- *volume*: **volume** component (see chapter 102)
Component (default 'box')
Soil volume to add irrigation.

- *surface*: **number** component (see chapter 59)
Amount of surface irrigation applied.
- *scope*: **scopesel** component (see chapter 77)
Component (default 'null')
Scope to evaluate expressions in.
- *from*: number [**cm**]
Parameter (default -10)
Height where you want to start the incorporation (a negative number).
- *to*: number [**cm**]
Parameter
Height where you want to end the incorporation (a negative number).
- *overhead*: **number** component (see chapter 59)
Amount of overhead irrigation applied.
- *subsoil*: **number** component (see chapter 59)
Amount of subsoil irrigation applied.

15.10 extern_subsoil

Subsoil irrigation controlled externally.

```
< extern_subsoil  (flux flux)
                  (volume volume)                      ; Default box value.
                  (scope scope)
                  (from from)
                  (to to)
                  (constituents constituents ...) >
```

- *flux*: **number** component (see chapter 59)
Amount of irrigation applied.
- *volume*: **volume** component (see chapter 102)
Component (default 'box')
Soil volume to add irrigation.
- *scope*: **scopesel** component (see chapter 77)
Scope to evaluate expressions in.
- *from*: number [**cm**]
Optional parameter
Height where you want to start the incorporation (a negative number). OB-SOLETE: Use (volume box (top FROM)) instead.
- *to*: number [**cm**]
Optional parameter
Height where you want to end the incorporation (a negative number). OB-SOLETE: Use (volume box (bottom TO)) instead.
- *constituents*: string (see section 4.1.5) sequence
Parameter
List of solutes to add to the irrigation water. The values are taken from the external scope, dimensions must be convertible to ppm.

15.11 fertilize_base

Shared parameters for all fertilize actions.

```
< fertilize_base (am am)
    (equivalent_weight equivalent_weight)
    (minimum_weight 0 [kg N/ha])
    (precision precision)
    (second_year_compensation false) >
```

- *am*: **am** component (see chapter 17)
The fertilizer you want to apply.
- *equivalent_weight*: number [kg N/ha]
Optional parameter
When fertilizing with organic matter, you may let Daisy calculate the amount of dry matter that corresponds to the specified amount of nitrogen. This requires that the fertilizer has specified the 'first_year_utilization' parameter, but not the 'weight' parameter.
- *minimum_weight*: number [kg N/ha]
Parameter (default 0)
Minimum amount of nitrogen to fertilize with.
- *precision*: submodel (see section 4.1.7)
Optional submodel
Let the amount of fertilizer depend on the inorganic nitrogen in the soil. The amount of fertilizer will be the specified 'target', minus the amount already present in the soil zone between 'from' and 'to'.


```
< target
    (from 0 [cm])
    (to -100 [cm]) >
```

 - *target*: number [kg N/ha]
Parameter
How much N you want.
 - *from*: number [cm]
Parameter (default 0)
Height where you want to start measuring (a negative number).
 - *to*: number [cm]
Parameter (default -100)
Height where you want to end measuring (a negative number).
- *second_year_compensation*: boolean (see section 4.1.2)
Parameter (default false)
Compensate for the second year effect of previous fertilizations. The second year effect is solely governed by the 'second_year_utilization' organic fertilizer parameter. The second year effect does not fade with time, but is zeroed once you fertilize with this flag set.

15.12 fertilize

A 'fertilize_base' model (see 15.11, page 80) build into Daisy.

Apply fertilizer to the soil surface.


```
< fertilize  am
              (from 0 [cm])
              (to 0 [cm])
              ;; Shared parameters are described in section 15.11.
              (equivalent_weight equivalent_weight)
              (minimum_weight 0 [kg N/ha])
              (precision precision)
              (second_year_compensation false) >
```

- *am*: **am** component (see chapter 17)
The fertilizer you want to apply.
- *from*: number [**cm**]
Parameter (default 0)
Height where you want to start the incorporation (a negative number) OBSOLETE: Use 'fertilize.incorporate' instead.
- *to*: number [**cm**]
Parameter (default 0)
Height where you want to end the incorporation (a negative number) OBSOLETE: Use 'fertilize.incorporate' instead.

15.13 incorporate_fertilizer

A 'fertilize_base' model (see 15.11, page 80) build into Daisy.
Incorporate fertilizer.

```
< incorporate_fertilizer  am
                          (volume volume)
                          ;; Shared parameters are described in section 15.11.
                          (equivalent_weight equivalent_weight)
                          (minimum_weight 0 [kg N/ha])
                          (precision precision)
                          (second_year_compensation false) >
```

- *am*: **am** component (see chapter 17)
The fertilizer you want to apply.
- *volume*: **volume** component (see chapter 102)
Soil volume to incorporate fertilizer in.

15.14 harvest_base

Common parameters for harvest operations.

```
< harvest_base  crop
                 (combine false)
                 (stub 0 [cm])
                 (stem 1 [<fraction>])
                 (leaf 1 [<fraction>])
                 (sorg 1 [<fraction>]) >
```

- *crop*: string (see section 4.1.5)
Parameter (default 'all')
Name of the crop to harvest or cut. If you specify 'all', all crops will be harvested. If there are no crop on the field with the specified name, nothing will happen.

- *combine*: boolean (see section 4.1.2)
Parameter (default false)
Set this to 'true' in order to combine all crop parts into stem in the harvest log files. This is mostly useful for silage.
- *stub*: number [**cm**]
Parameter (default 0)
Leave stem and leafs below this height on the field.
- *stem*: number [<**fraction**>]
Parameter (default 1)
Fraction of stem (above stub) to harvest.
- *leaf*: number [<**fraction**>]
Parameter (default 1)
Fraction of leafs (above stub) to harvest.
- *sorg*: number [<**fraction**>]
Parameter (default 1)
Fraction of storage organ to harvest.

15.15 cut

A 'harvest_base' model (see 15.14, page 81) build into Daisy.

Cut a crop.

15.16 harvest

A 'harvest_base' model (see 15.14, page 81) build into Daisy.

Harvest a crop.

15.17 if

If the condition is true, perform the first action, otherwise perform the second action.

```
< if  if then else >
```

- *if*: **condition** component (see chapter 28)
Condition determining which action to perform.
- *then*: **action** component (see chapter 15)
Action to perform if the condition is true.
- *else*: **action** component (see chapter 15)
Component (default 'nil')
Action to perform if the condition is false.

15.18 irrigate_base

Shared parameter for irrigate actions.

```

< irrigate_base  flux
                  (solute)
                  (hours hours)
                  (days 0)
                  (remaining_time remaining_time)
                  (temperature temperature) >

```

- *flux*: **number** component (see chapter 59)
Amount of irrigation applied.
- *solute*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Solutes in irrigation water.

```

      <  name value  >

      — name: string (see section 4.1.5)
        Parameter
        Name of chemical.

      — value: number [ppm]
        Parameter
        Value for chemical.

```
- *hours*: integer
Optional parameter
Irrigate this number of hours. By default, irrigate 1 hour if days is 0, and 0 hours plus the specified number of days else.
- *days*: integer
Parameter (default 0)
Irrigate this number of days.
- *remaining_time*: number [**h**]
Optional state variable
Irrigate this number of hours. Setting this overrides the 'days' and 'hours' parameters.
- *temperature*: number [**dg C**]
Optional parameter
Temperature of irrigation (default: air temperature).

15.19 irrigate_overhead

A 'irrigate_base' model (see 15.18, page 82) build into Daisy.
Irrigate the field from above.

15.20 irrigate_subsoil

A 'irrigate_base' model (see 15.18, page 82) build into Daisy.
Irrigate the field directly into the soil. Currently, the 'temperature' parameter is ignored.

```

< irrigate_subsoil  flux
                    (volume volume)                                ; Default box value.
                    (from from)
                    (to to)
                    ;; Shared parameters are described in section 15.18.
                    (solute)
                    (hours hours)
                    (days 0)
                    (remaining_time remaining_time)
                    (temperature temperature) >

```

- *flux*: **number** component (see chapter 59)
Amount of irrigation applied.
- *volume*: **volume** component (see chapter 102)
Component (default ‘box’)
Soil volume to add irrigation.
- *from*: number [**cm**]
Optional parameter
Height where you want to start the incorporation (a negative number). OBSOLETE: Use (volume box (top FROM)) instead.
- *to*: number [**cm**]
Optional parameter
Height where you want to end the incorporation (a negative number). OBSOLETE: Use (volume box (bottom TO)) instead.

15.21 irrigate_surface

A ‘irrigate_base’ model (see 15.18, page 82) build into Daisy.

Irrigate the field directly on the soil surface, bypassing the canopy.

15.22 markvand

Irrigate the field according to MARKVAND scheduling.

```

< markvand  (soil soil)
            (solute)
            (flux 2 [mm/h])
            (dt dt)
            (T_sum T_sum)
            (map map ...)
            (V_I 0 [mm])
            (V_r V_r)
            (V_e V_e)
            (C_u 0 [mm])
            (V_u 0 [mm])
            (V_b V_b) >

```

- *soil*: **MV_Soil** component (see chapter 12)
Soil type to schedule irrigation on.
- *solute*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Solutes in irrigation water.

< *name value* >

- *name*: string (see section 4.1.5)
Parameter
Name of chemical.
- *value*: number [**ppm**]
Parameter
Value for chemical.

- *flux*: number [**mm/h**]
Parameter (default 2)
Water application speed.
- *dt*: number [**d**]
Optional state variable
Days since emergence.
- *T_{sum}*: number [**dg C d**]
Optional state variable
Temperature sum since emergence.
- *map*: submodel (see section 4.1.7) sequence
Map of Daisy crop names into MARKVAND crop descriptions.

< *Daisy MARKVAND* >

- *Daisy*: string (see section 4.1.5)
Parameter
Name of Daisy crop.
- *MARKVAND*: **MV_Crop** component (see chapter 11)
MARKVAND crop description.
- *V_I*: number [**mm**]
Optional state variable (default 0)
Amount of water interceptor by leaves.
- *V_r*: number [**mm**]
Optional state variable
Amount of available water in root zone. By default, the reservoir will be full at plant emergence.
- *V_e*: number [**mm**]
Optional state variable
Amount of available water in top soil reservoir. This is the water that can be extracted by soil evaporation. Included in 'V_r'. By default, the reservoir will be full at plant emergence.
- *C_u*: number [**mm**]
Optional state variable (default 0)
Capacity of available water in upper root zone.
- *V_u*: number [**mm**]
Optional state variable (default 0)
Amount of available water in upper root zone. Included in 'V_r'.
- *V_b*: number [**mm**]
Optional state variable
Amount of water between current and max root depth. By default, the reservoir will be full at plant emergence.

15.23 message

Write a message to the user.

```
< message message >
```

- *message*: string (see section 4.1.5)
Parameter
Message to give to the user.

15.24 mix

Mix soil content down to the specified depth. The effect is that nitrogen, water, temperature and such are averaged in the interval.

```
< mix depth
    (random_roughness 0.012 [m])
    (penetration 1 [<fraction>])
    (surface_loose surface_loose) >
```

- *depth*: number [**cm**]
Parameter
How far down to mix the soil (a negative number).
- *random_roughness*: number [**m**]
Parameter (default 0.012)
Random roughness generated by tillage operation.
- *penetration*: number [<**fraction**>]
Parameter (default 1)
Fraction of organic matter on surface that are incorporated in the soil by this operation.
- *surface_loose*: number [<**fraction**>]
Optional parameter
Fraction of surface being loosened by operation. By default, this is equal to 'penetration'.

15.25 disk_harrowing

A 'mix' model (see 15.24, page 86) defined in 'tillage.dai'.

15.26 rotavation

A 'mix' model (see 15.24, page 86) defined in 'tillage.dai'.

15.27 seed_bed_preparation

A 'mix' model (see 15.24, page 86) defined in 'tillage.dai'.

15.28 stubble_cultivation

A 'mix' model (see 15.24, page 86) defined in 'tillage.dai'.

15.29 nil

This action does nothing, always done.

Used by action if else (see 15.17, page 82) .

15.30 panic

Write a error message to the user and stop the simulation.

```
< panic message >
```

- *message*: string (see section 4.1.5)
Parameter
Error message to give.

15.31 pluck

Pluck a crop. Unlike the 'harvest' operation, this allows you to pluck selected parts of the above ground dry matter without killing the crop. It is intended for crops like tomatoes, that are harvested multiple times.

```
< pluck crop
      (stem 0 [<fraction>])
      (leaf 0 [<fraction>])
      (sorg 1 [<fraction>]) >
```

- *crop*: string (see section 4.1.5)
Parameter (default 'all')
Name of the crop to pluck. If you specify 'all', all crops will be plucked. If there are no crop on the field with the specified name, nothing will happen.
- *stem*: number [<fraction>]
Parameter (default 0)
Fraction of stem to pluck.
- *leaf*: number [<fraction>]
Parameter (default 0)
Fraction of leaves to pluck.
- *sorg*: number [<fraction>]
Parameter (default 1)
Fraction of storage organ to pluck.

15.32 progn

Perform all the specified actions in the sequence listed. All the actions will be performed in the same time step.

```
< progn actions... >
```

- *actions*: **action** component (see chapter 15) sequence
List of actions to perform.

15.33 remove_litter

Remove organic matter from soil surface.

15.34 `remove_solute`

Remove a specific chemical from the field.

```
< remove_solute  chemical  >
```

- *chemical*: string (see section 4.1.5)
Parameter
Name of chemical to remove.

15.35 `repeat`

Perform all of the specified action. When done, repeat the action. The action may take several timesteps.

```
< repeat  repeat
      (do do) >
```

- *repeat*: **action** component (see chapter 15)
Action to perform repeatedly.
- *do*: **action** component (see chapter 15)
Optional component
Action currently being performed.

15.36 `restore_SOM`

Restore the content of the soil organic matter pools.

15.37 `sequence`

Perform all the specified actions in the sequence listed. Each action is performed until done. At most one action can be performed at each time step.

```
< sequence  (do) >
```

- *do*: **action** component (see chapter 15) sequence
Component (default: an empty sequence)
Sequence of actions to perform.

15.38 `activity`

A ‘sequence’ model (see 15.37, page 88) build into Daisy.

15.39 `set_porosity`

Set the porosity of the horizon at the specified depth. To get useful results, you need to use a hydraulic model that supports this.

```
< set_porosity  (depth 0 [cm])
      (porosity porosity) >
```

- *depth*: number [**cm**]
Parameter (default 0)
A point in the horizon to modify.
- *porosity*: number [<**fraction**>]
Parameter
Non-solid fraction of soil.

15.40 set_surface_detention_capacity

Set amount of ponding the surface can retain.

```
< set_surface_detention_capacity height >
```

- *height*: number [mm]
Parameter
Max ponding height before runoff.

15.41 sorption_table

Print a sorption table for all chemicals.

```
< sorption_table (Theta 0.5 [<fraction>])
                 (factor 10 [])
                 (cell 0)
                 (start 1e-10 [g/cm3])
                 (intervals 10) >
```

- *Theta*: number [<fraction>]
Parameter (default 0.5)
Soil water content.
- *factor*: number (dimensionless)
Parameter (default 10)
Multiply C with this number for the next entry in the table. If zero, instead add start to C for the next entry in the table.
- *cell*: integer
Parameter (default 0)
Numeric cell to use for soil information.
- *start*: number [g/cm³]
Parameter (default 1e-10)
Lowest solute concentration in table.
- *intervals*: integer
Parameter (default 10)
Number of entries in the table.

15.42 sow_base

Shared sowing parameters.

```
< sow_base (seed seed)
           (row_position 0 [cm])
           (row_width 0 [cm])
           (plant_distance plant_distance)
           (plant_position plant_position) >
```

- *seed*: number [g w.w./m²]
Optional parameter
Amount of seed applied. By default, initial growth will be governed by 'typical' seed amounts.
- *row_position*: number [cm]
Parameter (default 0)
Position of plant row on x-axes. Specify zero to spread equally over the area (no rows).

- *row_width*: number [cm]
Parameter (default 0)
Distance between rows. Specify zero to spread equally over the area (no rows).
- *plant_distance*: number [cm]
Optional parameter
Distance between plants.

Setting this will overrule 'row_width'. The only purpose of this parameter is to provide the user with a more intuitive name for 'row_width' for the situation where you have a 2D simulation, where the x axis is parallel with the actual rows in the field, rather than orthogonal to the rows as is otherwise assumed by Daisy.

- *plant_position*: number [cm]
Optional parameter
Position of plant on x-axes.

Setting this will overrule 'row_position'. The only purpose of this parameter is to provide the user with a more intuitive name for 'row_position' for the situation where you have a 2D simulation, where the x axis is parallel with the actual rows in the field, rather than orthogonal to the rows as is otherwise assumed by Daisy.

15.43 sow

A 'sow_base' model (see 15.42, page 89) build into Daisy.

Sow a crop on the field.

```
< sow  crop
      ;; Shared parameters are described in section 15.42.
      (seed seed)
      (row_position 0 [cm])
      (row_width 0 [cm])
      (plant_distance plant_distance)
      (plant_position plant_position) >
```

- *crop*: **crop** component (see chapter 29)
Crop to sow.

15.44 plant

A 'sow' model (see 15.43, page 90) build into Daisy.

'plant' is another name for 'sow'

15.45 sow_name

A 'sow_base' model (see 15.42, page 89) build into Daisy.

Variant of 'sow' that takes the name of a crop. This name can be a reference parameter.

```
< sow_name  crop
      ;; Shared parameters are described in section 15.42.
      (seed seed)
      (row_position 0 [cm])
      (row_width 0 [cm])
      (plant_distance plant_distance)
      (plant_position plant_position) >
```

- *crop*: string (see section 4.1.5)
Parameter
Name of crop to sow.

15.46 sow_object

A 'sow_base' model (see 15.42, page 89) build into Daisy.

Variant of 'sow' that allows reference parameters. Unlike the normal 'sow', this action can only be called once per simulation.

```
< sow_object  crop
      ;; Shared parameters are described in section 15.42.
      (seed seed)
      (row_position 0 [cm])
      (row_width 0 [cm])
      (plant_distance plant_distance)
      (plant_position plant_position) >
```

- *crop*: **crop** component (see chapter 29)
Crop to sow.

15.47 spray

Spray a chemical (typically a pesticide) on the field.

```
< spray  chemical amount >
```

- *chemical*: string (see section 4.1.5)
Parameter
Name of pesticide to spray.
- *amount*: number [g/ha]
Parameter
Amount of pesticide to spray.

15.48 spray_surface

Spray a chemical (typically a pesticide) on the field below the canopy.

```
< spray_surface  chemical amount >
```

- *chemical*: string (see section 4.1.5)
Parameter
Name of pesticide to spray.
- *amount*: number [g/ha]
Parameter
Amount of pesticide to spray.

15.49 stop

Stop the simulation.

15.50 store_SOM

Store the content of the soil organic matter pools.

15.51 swap

Swap two soil layers. The top layer start at the surface and goes down to 'middle', and the second layer starts with 'middle' and goes down to 'depth'. After the operation, the content (such as heat, water, and organic matter) will be averaged in each layer, and the bottom layer will be placed on top of what used to be the top layer.

```
< swap (depth depth)
      (random_roughness 0.024 [m])
      (middle middle) >
```

- *depth*: number [cm]
Parameter
The end of the second layer to swap.
- *random_roughness*: number [m]
Parameter (default 0.024)
Random roughness generated by tillage operation.
- *middle*: number [cm]
Parameter
The end of the first layer and the start of the second layer to swap.

15.52 plowing

A 'swap' model (see 15.51, page 92) defined in 'tillage.dai'.

15.53 t

This action does nothing, never done.

15.54 table

Read management actions from a Daisy data file.

After the ddf header, the following column tags are recognized (with the dimension for the dimension line in square brackets).

Date [date]: The date for fertilization or irrigation.

Planting [date]: The content should be a date in yyyy-mm-dd format, where the operation specified by the 'sow' attribute will be performed.

Emerging [date]: The content should be a date in yyyy-mm-dd format, where the operation specified by the 'emerge' attribute will be performed.

Harvest [date]: The content should be a date in yyyy-mm-dd format, where the operation specified by the 'harvest' attribute will be performed.

Irrigate [mm]: The content should be an irrigation amount, that will be applied as overhead irrigation for the date specified in the 'Date' field. You can disable it with the 'enable_irrigation' attribute.

Fertilize [kg N/ha]: The content should be an amount of nitrogen fertilizer to be applied on the date specified in the 'Date' field. The fertilizer type will be either the one specified in the 'Fertilizer' column, or the 'fertilizer' attribute. You can disable it with the 'enable_fertilization' attribute.

Fertilizer [name]: The type of fertilizer to be applied.

```
< table (file file)
      (flux 2 [mm/h])
      (emerge emerge)
      (harvest harvest)
      (sow sow)
      (volume volume)           ; Default box value.
      (missing missing ...)     ; Has default value.
      (filter)
      (original original ...)
      (dim_line dim_line)
      (fertilizer fertilizer)
      (enable_irrigation true)
      (enable_fertilization true)
      (from -5 [cm])
      (to -25 [cm]) >
```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *flux*: number [mm/h]
Parameter (default 2)
Water application speed.
- *emerge*: **action** component (see chapter 15)
Optional component
Emerge action.
- *harvest*: **action** component (see chapter 15)
Optional component
Harvest action.
- *sow*: **action** component (see chapter 15)
Optional component
Sow action.
- *volume*: **volume** component (see chapter 102)
Component (default 'box')
Soil volume to add irrigation.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

 - *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
 - *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.
- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.
- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *fertilizer*: **am** component (see chapter 17)
Optional component
The fertilizer you want to apply.
- *enable_irrigation*: boolean (see section 4.1.2)
Parameter (default true)
Set this to false to ignore any irrigation information in the file.
- *enable_fertilization*: boolean (see section 4.1.2)
Parameter (default true)
Set this to false to ignore any fertilization information in the file.
- *from*: number [**cm**]
Parameter (default -5)
Height where you want to start the incorporation (a negative number). OBSOLETE: Use (volume box (top FROM)) instead.
- *to*: number [**cm**]
Parameter (default -25)
Height where you want to end the incorporation (a negative number). OBSOLETE: Use (volume box (bottom TO)) instead.

15.55 wait

Wait until the specified condition is true.

```
< wait condition >
```

- *condition*: **condition** component (see chapter 28)
Condition to wait for.

15.56 wait_mm_dd

Wait until a specific month and day in the year.

```
< wait_mm_dd  month day
                (hour 8) >
```

- *month*: integer
Parameter
Wait until this month.
- *day*: integer
Parameter
Wait until this day in the month.
- *hour*: integer
Parameter (default 8)
Wait until this hour.

15.57 wait_period

Waits the specified period.

```
< wait_period  (hours hours)
                (days days)
                (end_time end_time) >
```

- *hours*: integer
Parameter
Wait this number of hours.
- *days*: integer
Parameter
Wait this number of days.
- *end_time*: **Time** fixed component (see section 107.27)
Optional submodel
Wait until this date. Setting this overrides the 'days' and 'hours' parameters.

15.58 wait_days

A 'wait_period' model (see 15.57, page 95) build into Daisy.

Waits the specified number of days.

```
<  days
    (hours 0) >
```

15.59 wait_hours

A 'wait_period' model (see 15.57, page 95) build into Daisy.

Waits the specified number of hours.

```
<  hours
    (days 0) >
```

15.60 warning

Write a warning to the user.

```
< warning message >
```

- *message*: string (see section 4.1.5)
Parameter
Warning to give to the user.

15.61 while

Perform all the specified actions in the sequence listed, but in the same timestep.

The 'while' action is done when the first action in the list is done.

```
< while actions... >
```

- *actions*: **action** component (see chapter 15) sequence
List of actions to perform.

15.62 with-column

Perform actions on a specific column.

```
< "with-column" column actions... >
```

- *column*: string (see section 4.1.5)
Parameter
Name of column to perform actions on.
- *actions*: **action** component (see chapter 15) sequence
Actions to perform on the specified column.

Chapter 16

adsorption

This component describes the adsorption of a chemical to the soil, which among other things affects how large a fraction can be transported with the water.

16.1 Freundlich

$M = \rho K C_{ref} (C / C_{ref})^m + \Theta C$. Formula from FOCUS page 93. See also [Freundlich and Heller, 1939, FOCUS, 2000]

```
< Freundlich (m m)
      (K_clay K_clay)
      (K_OC K_OC)
      (C_ref C_ref) >
```

- *m*: number (dimensionless)
Parameter
Freundlich parameter (1/n). If not known, FOCUS recommend a value of 0.9 (page 93). See also [FOCUS, 2000]
- *K_{clay}*: number [**cm**³/g]
Optional parameter
Clay dependent distribution parameter. It is multiplied with the soil clay fraction to get the clay part of the 'K' factor. If 'K_OC' is specified, 'K_{clay}' defaults to 0.
- *K_{OC}*: number [**cm**³/g]
Optional parameter
Humus dependent distribution parameter. It is multiplied with the soil organic carbon fraction to get the carbon part of the 'K' factor. By default, 'K_OC' is equal to 'K_{clay}'.
- *C_{ref}*: number [g/**cm**³]
Parameter
Reference concentration for determining the other parameters. According to FOCUS, the usual value is 1 mg/L (page 93). See also [FOCUS, 2000]

16.2 Guo2020

Linear sorption with *K_{awi}* estimated from Equation 11. See also [Guo et al., 2020]

Used by chemical PFOS_Guo adsorption (see 23.25, page 142) , and chemical PFOA_Guo adsorption (see 23.23, page 142) .

```
< Guo2020 (K_clay K_clay)
           (K_OC K_OC)
           (K_d K_d)
           (a a)
           (b b) >
```

- *K_{clay}*: number [**cm³/g**]
Optional parameter
Clay dependent distribution parameter. It is multiplied with the soil clay fraction to get the clay part of the 'K_d' factor. If 'K_{OC}' is specified, 'K_{clay}' defaults to 0.
- *K_{OC}*: number [**cm³/g**]
Optional parameter
Humus dependent distribution parameter. It is multiplied with the soil organic carbon fraction to get the carbon part of the 'K_d' factor. By default, 'K_{OC}' is equal to 'K_{clay}'.
- *K_d*: number [**cm³/g**]
Optional parameter
Soil dependent distribution parameter. By default, it will be calculated from 'K_{OC}' and 'K_{clay}'.
- *a*: number [**mol/cm³**]
Parameter
Fitting parameter, Eq11.
- *b*: number (dimensionless)
Parameter
Fitting parameter, Eq11.

16.3 Langmuir

$M = \text{rho} \cdot (\text{my_max} \cdot C) / ((1/K) + C) + \text{Theta} \cdot C$

```
< Langmuir (K K)
           (my_max_clay my_max_clay)
           (my_max_OC my_max_OC) >
```

- *K*: number [**cm³/g**]
Parameter
Slope parameter.
- *my_max_clay*: number [**g/g**]
Optional parameter
Max adsorption capacity (clay). It is multiplied with the soil clay fraction to get the clay part of 'my_max'. If 'my_max_OC' is specified, 'my_max_clay' defaults to 0.
- *my_max_OC*: number [**g/g**]
Optional parameter
Max adsorption capacity (humus). It is multiplied with the soil organic carbon fraction to get the carbon part of 'my_max'. By default, 'my_max_OC' is equal to 'my_max_clay'.

16.4 Python

Adsorption defined in user specified Python module.

```
< Python (module module)
      (C_to_M C_to_M)
      (M_to_C M_to_C)
      (extra extra ...) >
```

- *module*: string (see section 4.1.5)
Parameter
Python module to find the functions.
- *C_to_M*: string (see section 4.1.5)
Parameter
Name of function to convert concentration to mass. C_to_M (C, *extra*) -> [g CHEMICAL/cm³ SPACE] total content in soil/water/air.
where *extra* is the extra parameters specified by 'extra'.
- *M_to_C*: string (see section 4.1.5)
Optional parameter
Name of the function to convert mass to concentration. M_to_C (M *extra*) -> [g CHEMICAL/cm³ WATER] concentration in soil water.
where *extra* is the extra parameters specified by 'extra'. By default this will use C_to_M and bisection between 0 and 1.
- *extra*: string (see section 4.1.5) sequence
Parameter
Extra input parameters to Python soil function.
Options include:
Theta_sat [cm³ W/cm³ V]: Saturated water per system volume. Theta [cm³ W/cm³ V]: Actual water per system volume. rho_b [g S/cm³]: Soil dry bulk density. f_OC [g OC/g SOIL]: organic carbon content in soil f.clay [g clay/g SOIL]: soil clay fraction d50 [um]: median particle diameter area_AWI [cm²/cm³]: air-water interface area molar_mass [g/mol]: molar mass of compound T [dg C]: soil temperature.

16.5 air

Henry's law, fixed ratio between concentration in air and water. See also [?]

```
< air (K_wa K_wa)
      (T_ref 25 [dg C])
      (B B) >
```

- *K_wa*: number (dimensionless)
Parameter
Ratio between concentration in air and in water. $K_{wa} = C_a / C_w$ where C_a is mass per volume air, and C_w is mass per volume water.
- *T_ref*: number [dg C]
Parameter (default 25)
Temperature at which K_{aw} is valid.
- *B*: number [K]
Parameter
Temperature effect coefficient.

16.6 full

Full adsorption. Used for non-solutes, fully adsorped in the soil.

Used by chemical solid adsorption (see 23.30, page 143) .

16.7 linear

$M = \rho K C + \Theta C$

Used by chemical Atrazine adsorption (see 23.8, page 138) , chemical Bentazon adsorption (see 23.9, page 138) , chemical IPU adsorption (see 23.10, page 138) , chemical MCPP adsorption (see 23.12, page 138) , chemical Pendimethalin adsorption (see 23.13, page 138) , chemical Ioxynil adsorption (see 23.11, page 138) , chemical 2,4-D adsorption (see 23.7, page 138) , chemical Heptachlor adsorption (see 23.16, page 139) , chemical DDT adsorption (see 23.15, page 138) , chemical PFOS adsorption (see 23.24, page 142) , chemical PFOA adsorption (see 23.22, page 142) , chemical PFHxA adsorption (see 23.21, page 142) , chemical PFPeA adsorption (see 23.26, page 142) , chemical PFBS adsorption (see 23.20, page 141) , and chemical PFBA adsorption (see 23.19, page 141) .

```
< linear (K_clay K_clay)
          (K_OC K_OC)
          (K_d K_d)
          (K_AWI 0 [cm3/cm2]) >
```

- *K_{clay}*: number [cm³/g]
Optional parameter
Clay dependent distribution parameter. It is multiplied with the soil clay fraction to get the clay part of the 'K_d' factor. If 'K_{OC}' is specified, 'K_{clay}' defaults to 0.
- *K_{OC}*: number [cm³/g]
Optional parameter
Humus dependent distribution parameter. It is multiplied with the soil organic carbon fraction to get the carbon part of the 'K_d' factor. By default, 'K_{OC}' is equal to 'K_{clay}'.
- *K_d*: number [cm³/g]
Optional parameter
Soil dependent distribution parameter. By default, it will be calculated from 'K_{OC}' and 'K_{clay}'.
- *K_{AWI}*: number [cm³/cm²]
Parameter (default 0)
Sorpton to air-water interface.

16.8 NH4

A 'linear' model (see 16.7, page 100) build into Daisy.

Adsorption of ammonium, fitted by Styczen for Daisy 7.

Used by chemical NH4 adsorption (see 23.37, page 143) .

```
< (K_clay 28 [cm3/g])
    (K_OC 213 [cm3/g]) >
```

16.9 *NH4_Hansen*

A ‘linear’ model (see 16.7, page 100) build into Daisy.

Adsorption of ammonium, traditional value.

```
< (K_clay 117.116 [cm3/g])
    (K_OC 117.116 [cm3/g]) >
```

16.10 *none*

No adsorption. Used for solutes that are not adsorped to the soil.

Used by chemical base adsorption (see 23.1, page 125) .

16.11 *vS_S*

Double langmuir description of *NH4-N* sorption to clay. See also [Van Schouwenburg and Schuffelen, 1963]

```
< vS_S (S_planar S_planar)
        (S_edge S_edge)
        (K_planar K_planar)
        (K_edge K_edge) >
```

- *S_planar*: number [g N/g clay]
Parameter
Absorption capacity of the clay planar sites.
- *S_edge*: number [g N/g clay]
Parameter
Absorption capacity of the clay edge sites.
- *K_planar*: number [g N/cm³]
Parameter
Half-saturation constant of the clay planer sites.
- *K_edge*: number [g N/cm³]
Parameter
Half-saturation constant of the clay edge sites.

16.12 *vS_S_Hansen*

A ‘vS_S’ model (see 16.11, page 101) build into Daisy.

. See also [Van Schouwenburg and Schuffelen, 1963]

```
< (S_planar 0.005964 [g N/g clay])
    (S_edge 0.000308 [g N/g clay])
    (K_planar 6.3e-05 [g N/cm3])
    (K_edge 1.372e-05 [g N/cm3]) >
```

16.13 *vS_S_Styczen*

A ‘vS_S’ model (see 16.11, page 101) build into Daisy.

Parameter update.. See also [Van Schouwenburg and Schuffelen, 1963]

```
< (S_planar 0.005964 [g N/g clay])
    (S_edge 0.0002801 [g N/g clay])
    (K_planar 0.0006338 [g N/cm3])
    (K_edge 1.369e-05 [g N/cm3]) >
```

16.14 vS_S_old

Buggy implementation included as vS_S before Daisy 6.47. Included here as reference. Note, linear sorption was used as default for NH4 as long as memory serve.

Chapter 17

am

The 'am' component describes various kinds of fertilizer and other added matter such as crop residues. In particular, it describes how they decompose.

17.1 base

Common attributes for all added organic matter models.

```
< base (name name)  
      (initialized false)  
      (creation creation)  
      (lock lock)  
      (om om ...) >
```

- *name*: string (see section 4.1.5)
Optional state variable
A name given to this AOM so you can identify it in for example log files.
- *initialized*: boolean (see section 4.1.2)
State variable (default false)
True if this AM has been initialized. It will usually be false in user setup files, but true in checkpoints.
- *creation*: **Time** fixed component (see section 107.27)
Optional submodel
Time this AM was created.
- *lock*: submodel (see section 4.1.7)
Optional submodel
This AM belongs to a still living plant

```
< (crop crop)  
  (part part) >
```

 - *crop*: string (see section 4.1.5)
State variable
Crop to which this am is locked
 - *part*: string (see section 4.1.5)
State variable
Crop part to which this am is locked
- *om*: **AOM** component (see chapter 8) sequence
Optional component
The individual AOM pools.

17.2 initial

A ‘base’ model (see 17.1, page 103) build into Daisy.

Initial added organic matter at the start of the simulation.

```
< initial (layers layers ...)
          ;; Shared parameters are described in section 17.1.
          (name name)
          (initialized false)
          (creation creation)
          (lock lock)
          (om om ...) >
```

- *layers*: submodel (see section 4.1.7) sequence
Carbon content in different soil layers. The carbon is assumed to be uniformly distributed in each layer.

```
< end weight >
```

- *end*: number [cm]
Parameter
Height where this layer ends (a negative number).
- *weight*: number [kg C/m²]
Parameter
Carbon in this layer.

17.3 organic

A ‘base’ model (see 17.1, page 103) build into Daisy.

Organic fertilizer, typically slurry or manure from animals.

```
< organic (weight 0 [Mg w.w./ha])
          (total_C_fraction total_C_fraction)
          (total_N_fraction total_N_fraction)
          (NH4_fraction 0 [<fraction>])
          (volatilization 0 [<fraction>])
          (first_year_utilization first_year_utilization)
          (second_year_utilization second_year_utilization)
          (dry_matter_fraction dry_matter_fraction)
          (NO3_fraction 0 [<fraction>])
          ;; Shared parameters are described in section 17.1.
          (name name)
          (initialized false)
          (creation creation)
          (lock lock)
          (om om ...) >
```

- *weight*: number [Mg w.w./ha]
Parameter (default 0)
Amount of fertilizer applied.
- *total_C_fraction*: number [<fraction>]
Parameter
Carbon fraction of dry matter.
- *total_N_fraction*: number [<fraction>]
Parameter
Nitrogen fraction of dry matter.

- *NH₄-fraction*: number [**<fraction>**]
Parameter (default 0)
Ammonium fraction of total N in fertilizer. The remaining nitrogen is assumed to be nitrate or organic.
- *volatilization*: number [**<fraction>**]
Parameter (default 0)
Fraction of NH₄ that evaporates on application.
- *first_year_utilization*: number [**<fraction>**]
Optional parameter
Estimated useful N fraction for the first year. In Denmark, this is governed by legalisation.
- *second_year_utilization*: number [**<fraction>**]
Optional parameter
Estimated useful N fraction for the second year. In Denmark, this is governed by legalisation.
- *dry_matter_fraction*: number [**<fraction>**]
Parameter
Dry matter fraction of total (wet) weight.
- *NO₃-fraction*: number [**<fraction>**]
Parameter (default 0)
Nitrate fraction of total N in fertilizer. The remaining nitrogen is assumed to be ammonium or organic.

17.4 root

A ‘base’ model (see 17.1, page 103) build into Daisy.

Initialization of old root remains.

Used by organic default am (see 60.1, page 327) .

```
< root (depth depth)
      (om om ...) ; Has default value.
      (dist 7 [cm])
      (weight 1.2 [Mg DM/ha])
      (total_C_fraction 0.4 [<fraction>])
      (total_N_fraction 0.01 [<fraction>])
      ;; Shared parameters are described in section 17.1.
      (name name)
      (initialized false)
      (creation creation)
      (lock lock) >
```

- *depth*: number [**cm**]
Optional parameter
How far down does the old root reach? (a negative number) By default, the soils maximal rooting depth will be used.
- *dist*: number [**cm**]
Parameter (default 7)
Distance to go down in order to decrease the root density to half the original.
- *weight*: number [**Mg DM/ha**]
Parameter (default 1.2)
Total weight of old root dry matter.

- *total_C_fraction*: number [**<fraction>**]
Parameter (default 0.4)
Carbon fraction of total root dry matter
- *total_N_fraction*: number [**<fraction>**]
Parameter (default 0.01)
Nitrogen fraction of total root dry matter

17.5 state

A ‘base’ model (see 17.1, page 103) build into Daisy.

Most AM models are only used for initialization, they will be converted to this generic model after creation, so this is what you will see in a checkpoint. This model contains a number (typically 2) of separate pools, each of which have their own turnover rate.

17.6 mineral

Mineral fertilizer.

```
< mineral (weight 0 [kg N/ha]
          (NH4_fraction NH4_fraction)
          (volatilization 0 [<fraction>]) >
```

- *weight*: number [**kg N/ha**]
Parameter (default 0)
Amount of fertilizer applied.
- *NH4_fraction*: number [**<fraction>**]
Parameter
Ammonium fraction of total N in fertilizer. The remaining nitrogen is assumed to be nitrate.
- *volatilization*: number [**<fraction>**]
Parameter (default 0)
Fraction of NH4 that evaporates on application.

Chapter 18

average

Find the average of two numbers.

18.1 arithmetic

Arithmetic average $'(a+b)/2'$.

Used by uz1d richards K_average (see 97.2, page 531) .

18.2 geometric

Geometric average $'\sqrt{a*b}'$.

18.3 harmonic

Harmonic average $'2ab/(a+b)'$.

Chapter 19

bioclimate

The 'bioclimate' component is responsible for distributing the water and energy provided by the weather component among the crops and soil for a given column.

19.1 base

Common parameters for all bioclimate parameterizations.

```
< base (cloudiness cloudiness)
      (difrad difrad)
      (ghf ghf)
      (net_radiation net_radiation)
      (pet pet)
      (raddist raddist)
      (svat svat)
      (deposition deposition ...)
      (Snow Snow) ; Has default value.
      (NoOfIntervals 30)
      (irrigation_subsoil_permanent 0 [mm/h])
      (canopy_water_storage 0 [mm])
      (litter_water_storage 0 [mm])
      (max_svat_iterations 100)
      (max_svat_absolute_difference 0.01 [mm/h])
      (maxTdiff 0.02 [K])
      (maxEdiff 1 [Pa])
      (min_sun_angle 0.0628319 [rad]) >
```

- *cloudiness*: **cloudiness** component (see chapter 25)
Cloudiness model.

- *difrad*: **difrad** component (see chapter 35)
Optional component
Diffuse radiation component.

By default, choose depending on available climate data.

If diffuse radiation is available in the climate data, Daisy will use these (the weather difrad model). Otherwise Daisy will use the DPF model.

- *ghf*: **ghf** component (see chapter 43)
Ground heat flux.
- *net_radiation*: **net_radiation** component (see chapter 57)
Net radiation.

- *pet*: **pet** component (see chapter 62)
Optional component
Potential Evapotranspiration component.

Some pet models provide answers for both dry and wet surface. For those, the wet answer will limit total evapotranspiration, while the dry answer will further limit transpiration.

The default model depends on available climate data.

If reference evaporation is available in the climate data, Daisy will use these (the weather pet model).

If vapor pressure and wind are available, it will use Penman-Monteith (PM).

If the timestep is larger than 12, and daily minimum and maximum temperature are available, Samani and Hargreaves (Hargreaves).

As a last resort, Makkink (makkink) will be used.

- *raddist*: **raddist** component (see chapter 67)
Radiation distribution model.
- *svat*: **svat** component (see chapter 88)
Soil Vegetation Atmosphere component.
- *deposition*: **deposition** component (see chapter 33) sequence
Deposition models.
- *Snow*: **Snow** fixed component (see section 107.24)
Submodel (has fully specified default value)
Surface snow pack.
- *NoOfIntervals*: integer
Parameter (default 30)
Number of vertical intervals in which we partition the canopy.
- *irrigation_subsoil_permanent*: number [**mm/h**]
State variable (default 0)
Long term irrigation below soil surface.
- *canopy_water_storage*: number [**mm**]
State variable (default 0)
Intercepted water on canopy.
- *litter_water_storage*: number [**mm**]
State variable (default 0)
Intercepted water on litter.
- *max_svat_iterations*: integer
Parameter (default 100)
Max number of svat iterations before giving up on convergence.
- *max_svat_absolute_difference*: number [**mm/h**]
Parameter (default 0.01)
Maximum absolute difference in svat ea values for convergence.
- *maxTdiff*: number [**K**]
Parameter (default 0.02)
Largest temperature difference for convergence.

- *maxEdiff*: number [**Pa**]
Parameter (default 1)
Largest humidity difference for convergence.
- *min_sun_angle*: number [**rad**]
Parameter (default 0.0628319)
Minimum sun angle above ground for some 'raddist' and 'svat' models.
The 'DPF' raddist model will zero radiation if the angle is below this, and the 'SSOC' svat model will revert to a one leaf description.

Log Variables

- *production_stress*: number (dimensionless)
SVAT module induced stress, -1 means use water stress.
- *Height*: number [**cm**] canopy boundaries
End points of canopy layers. First entry is top of canopy, last is soil surface.
- *cloudiness_index*: number [**<fraction>**]
Cloudiness index, 0 = no light, 1 = clear sky.
- *L_n*: number [**W/m²**]
The calculated net longwave radiation (positive downwards).
- *L_ia*: number [**W/m²**]
The calculated incoming longwave radiation (positive downwards).
- *epsilon_0*: number (dimensionless)
Atmospheric effective clearsky emmissivity (range 0-1).
- *black_body_radiation*: number [**W/m²**]
Radiation emitted by black bodies at current air temperature. Stefan-Boltzmann's law.
- *Rn*: number [**W/m²**]
Net radiation on field.
- *Rn_ref*: number [**W/m²**]
Net radiation on reference surface.
- *G*: number [**W/m²**]
Ground heat flux.
- *total_ep*: number [**mm/h**]
Potential evapotranspiration.
- *total_ea*: number [**mm/h**]
Actual evapotranspiration.
- *direct_rain*: number [**mm/h**]
Rain hitting surface directly. This includes rain hitting ponded water or litter, but excludes rain hitting canopy or snow, as well as snow and all forms for irrigation. The intended use is colloid generation.
- *irrigation_overhead*: number [**mm/h**]
Irrigation above canopy.
- *irrigation_overhead_temperature*: number [**dg C**]
Water temperature.

- *irrigation_surface*: number [**mm/h**]
Irrigation below canopy.
- *irrigation_surface_temperature*: number [**dg C**]
Water temperature.
- *irrigation_subsoil*: number [**mm/h**]
Irrigation below soil surface this hour.
- *irrigation_total*: number [**mm/h**]
Total irrigation above of below the soil surface.
- *tillage_water*: number [**mm/h**]
Water added to surface due to tillage operations.
- *snow_ep*: number [**mm/h**]
Potential snow evaporation.
- *snow_ea*: number [**mm/h**]
Actual snow evaporation.
- *snow_water_in*: number [**mm/h**]
Water entering snow pack.
- *snow_water_in_temperature*: number [**dg C**]
Temperature of water entering snow pack.
- *snow_water_out*: number [**mm/h**]
Water leaving snow pack
- *pond_water_to_snow*: number [**mm/h**]
Water leaving pond to snow pack
- *snow_water_out_temperature*: number [**dg C**]
Temperature of water leaving snow pack.
- *canopy_cover*: number [**<fraction>**]
Fraction of ground covered by canopy.
- *canopy_ep*: number [**mm/h**]
Potential canopy evaporation.
- *canopy_ea*: number [**mm/h**]
Actual canopy evaporation.
- *canopy_water_capacity*: number [**mm**]
Potential intercepted water on canopy.
- *canopy_water_temperature*: number [**dg C**]
Temperature of incoming water.
- *canopy_water_in*: number [**mm/h**]
Water entering canopy.
- *canopy_water_out*: number [**mm/h**]
Canopy drip throughfall.
- *canopy_water_bypass*: number [**mm/h**]
Water from above bypassing the canopy.

- *canopy_water_below*: number [mm/h]
Total water input below canopy.
- *litter_cover*: number [<fraction>]
Fraction of ground covered by litter.
- *litter_ep*: number [mm/h]
Potential evaporation litter.
- *litter_ea*: number [mm/h]
Actual litter evaporation.
- *litter_water_capacity*: number [mm]
Potential intercepted water on litter.
- *litter_water_temperature*: number [dg C]
Temperature of incoming water.
- *litter_water_in*: number [mm/h]
Water entering litter.
- *litter_water_out*: number [mm/h]
Litter drip throughfall.
- *litter_wash_off*: number [mm/h]
Water hitting but not entering litter.
- *litter_water_bypass*: number [mm/h]
Water bypassing litter.
- *litter_pond_up*: number [mm/h]
Water extracted from pond to litter.
- *litter_soil_up*: number [mm/h]
Water extracted from soil to litter.
- *pond_ep*: number [mm/h]
Potential evaporation from pond.
- *pond_ea*: number [mm/h]
Actual evaporation from pond.
- *soil_ep*: number [mm/h]
Potential exfiltration.
- *soil_ea*: number [mm/h]
Actual exfiltration.
- *svat_failed*: integer
'0' if the SVAT model found a solution, '1' otherwise.
- *crop_ep*: number [mm/h]
Potential transpiration. Transpiration under the assumption that the soil have an unlimited water supply. For a fully irrigated crop, this will be equal to the actual transpiration.
- *crop_ea_soil*: number [mm/h]
Soil limited transpiration. The part of the potential transpiration that the soil can supply.

- *crop_ea_svat*: number [**mm/h**]
Transpiration suggested by the SVAT module. Under stressed conditions, the soil, vegetation and atmosphere behave different than what was assumed when calculating the potential transpiration.
- *crop_ea*: number [**mm/h**]
Actual transpiration. This is the transpiration limited either by what the soil can deliver, or what the SVAT module requires.
- *CanopyTemperature*: number [**dg C**]
Actual canopy temperature.
- *SunLeafTemperature*: number [**dg C**]
Sunlit leaf temperature.
- *ShadowLeafTemperature*: number [**dg C**]
Shadow leaf temperature.
- *wind_speed_field*: number [**m/s**]
Wind speed in the field at reference height.
- *wind_speed_weather*: number [**m/s**]
Measured wind speed.
- *albedo*: number (dimensionless)
Reflection factor.
- *difrad0*: number [**W/m²**]
Diffuse radiation above canopy.
- *total_PAR*: number [**W/m²**] canopy boundaries
Total PAR between canopy layers.
- *sun_PAR*: number [**W/m²**] canopy boundaries
Sun PAR between canopy layers.
- *total_NIR*: number [**W/m²**] canopy boundaries
Total NIR between canopy layers.
- *sun_NIR*: number [**W/m²**] canopy boundaries
Sun NIR between canopy layers.
- *sun_LAI_fraction*: number [**<fraction>**] canopy intervals
Sunlit LAI in canopy layers.
- *absorbed_total_PAR_canopy*: number [**W/m²**]
Canopy absorbed PAR (sun+shade)
- *absorbed_total_NIR_canopy*: number [**W/m²**]
Canopy absorbed NIR (sun+shade)
- *absorbed_total_Long_canopy*: number [**W/m²**]
Canopy absorbed long wave radiation (sun+shade)
- *absorbed_total_PAR_soil*: number [**W/m²**]
Soil absorbed PAR (sun+shade)
- *absorbed_total_NIR_soil*: number [**W/m²**]
Soil absorbed NIR (sun+shade)

- *absorbed_total_Long_soil*: number [W/m^2]
Soil absorbed long wave radiation (sun+shade)
- *absorbed_sun_PAR_canopy*: number [W/m^2]
Canopy absorbed PAR on sunlit leaves
- *absorbed_sun_NIR_canopy*: number [W/m^2]
Canopy absorbed NIR on sunlit leaves
- *absorbed_sun_Long_canopy*: number [W/m^2]
Canopy absorbed long wave radiatio on sunlit leaves
- *absorbed_shadow_PAR_canopy*: number [W/m^2]
Canopy absorbed PAR on shadow leaves
- *absorbed_shadow_NIR_canopy*: number [W/m^2]
Canopy absorbed NIR on shadow leaves
- *absorbed_shadow_Long_canopy*: number [W/m^2]
Canopy absorbed long wave radiation on shadow leaves
- *incoming_Long_radiation*: number [W/m^2]
Incoming longwave radiation
- *incoming_PAR_radiation*: number [W/m^2]
Incoming PAR radiation
- *incoming_NIR_radiation*: number [W/m^2]
Incoming NIR radiation
- *incoming_Total_radiation*: number [W/m^2]
Incoming radiation, sum of shortwave and longwave
- *outgoing_Short_radiation*: number [W/m^2]
Outgoing shortware radiation.

19.2 default

A ‘base’ model (see 19.1, page 109) build into Daisy.

Use best models given available data.

Used by column default Bioclimate (see 26.1, page 153) .

```
< (cloudiness cloudiness)           ; Default Taastrup value.
  (ghf ghf)                         ; Default surface value.
  (net_radiation net_radiation)     ; Default brunt value.
  (raddist raddist)                 ; Default default value.
  (svat svat)                      ; Default none value.
  (deposition deposition ...)      ; Has default value. >
```

19.3 FAO56_daily

A ‘default’ model (see 19.2, page 115) build into Daisy.

Follow FAO56 for daily data. See also [Allen et al., 1998]

```
< (cloudiness cloudiness)           ; Default FAO56 value.
  (ghf ghf)                         ; Default none value.
  (pet pet)                         ; Default FAO_PM value. >
```

19.4 FAO56_hourly

A ‘default’ model (see 19.2, page 115) build into Daisy.

Follow FAO56 for hourly data. See also [Allen et al., 1998, Allen et al., 2006]

```
< (cloudiness cloudiness) ; Default FA056 value.  
    (ghf ghf) ; Default FA056 value.  
    (pet pet) ; Default FA0_PM_hourly value. >
```

19.5 SSOC

A ‘default’ model (see 19.2, page 115) build into Daisy.

The SSOC SVAT model.

```
< (raddist raddist) ; Default sun-shade value. >
```

Chapter 20

biopore

A single class of biopores.

```
< component (density density)  
            (height_start height_start)  
            (height_end height_end)  
            (diameter diameter) >
```

- *density*: **number** component (see chapter 59)
Biopore density [cm^{-2}] as a function of 'x' [cm].
- *height_start*: number [**cm**]
Parameter
Biopores starts at this depth (a negative number).
- *height_end*: number [**cm**]
Parameter
Biopores ends at this depth (a negative number).
- *diameter*: number [**cm**]
Parameter
Biopore diameter.

Log Variables

- *S*: number [**cm³/cm³/h**] soil cells
Total stream from matrix domain to biopore.
- *J*: submodel (see section 4.1.7) sequence
Solute flux between cells.

```
< name value... > Log Variables
```

 - *value*: number [**g/cm²/h**] soil edges
Value for chemical.
 - *name*: string (see section 4.1.5)
Name of chemical.
- *q*: number [**cm/h**] soil edges
Water flow in this biopore class.
- *S_chem*: submodel (see section 4.1.7) sequence
Matrix to biopore term for solutes.

```
< name value... > Log Variables
```

- *value*: number [$\text{g}/\text{cm}^3/\text{h}$] soil cells
Value for chemical.
- *name*: string (see section 4.1.5)
Name of chemical.
- *M2B*: number [$\text{cm}^3/\text{cm}^3/\text{h}$] soil cells
Stream from matrix domain to biopore. Never negative.
- *B2M*: number [$\text{cm}^3/\text{cm}^3/\text{h}$] soil cells
Stream from biopore to matrix domain. Never negative.
- *infiltration*: number [cm/h]
Surface infiltration.
- *solute_infiltration*: submodel (see section 4.1.7) sequence
Rate of solute infiltration through surface.

< *name value* > **Log Variables**

- *value*: number [$\text{g}/\text{cm}^2/\text{h}$]
Value for chemical.
- *name*: string (see section 4.1.5)
Name of chemical.

20.1 drain

Biopores that ends in the drain pipes.

```
< drain  (pipe_position pipe_position)
          ;; Shared parameters are described in chapter 20.
          (density density)
          (height_start height_start)
          (height_end height_end)
          (diameter diameter) >
```

- *pipe_position*: number [cm]
Optional parameter
Height pipes are placed in the soil (a negative number). By default, use the height specified for pipes in the column.

Log Variables

- *S_to_drain*: number [$\text{cm}^3/\text{cm}^3/\text{h}$] soil cells
Total stream from biopore to drain.
- *S_chem_to_drain*: submodel (see section 4.1.7) sequence
Biopore to drain term for solutes.

< *name value...* > **Log Variables**

- *value*: number [$\text{g}/\text{cm}^3/\text{h}$] soil cells
Value for chemical.
- *name*: string (see section 4.1.5)
Name of chemical.

20.2 matrix

Biopores that ends in the matrix.

```
< matrix (solute)
      (debug 0)
      (xplus xplus ...)
      (K_wall_relative K_wall_relative)
      (h_bottom h_bottom ...)
      (max_iterations 50)
      (max_absolute_difference 0.02 [cm])
      (max_relative_difference 0.001 [])
      (allow_upward_flow true)
      ;; Shared parameters are described in chapter 20.
      (density density)
      (height_start height_start)
      (height_end height_end)
      (diameter diameter) >
```

- *solute*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Chemical concentration in biopore intervals.

```
< name value... >
```

 - *name*: string (see section 4.1.5)
Optional state variable
Name of chemical.
 - *value*: number [g] sequence
Optional state variable
Value for chemical.
- *debug*: integer
Parameter (default 0)
Debug level. Increase value to get more debug message.
- *xplus*: number [cm] sequence
Optional parameter
Right side of each biopore interval. Water and chemical content is tracked individually for each interval. By default, use intervals as specified by the geometry.
- *K_wall_relative*: number (dimensionless)
Parameter
Relative conductivity of biopore wall compared to matrix.
- *h_bottom*: number [cm] sequence
Optional state variable
Pressure at the bottom of the biopores in each interval.
- *max_iterations*: integer
Parameter (default 50)
Maximum number of iterations when seeking convergence.
- *max_absolute_difference*: number [cm]
Parameter (default 0.02)
Maximum absolute difference in biopore content for convergence.

- *max_relative_difference*: number (dimensionless)
Parameter (default 0.001)
Maximum relative difference in biopore content for convergence.
- *allow_upward_flow*: boolean (see section 4.1.2)
Parameter (default true)
Allow water to enter from saturated soil at the bottom of the biopore. And leave in unsaturated soil above.

Log Variables

- *water*: number [cm^3]
Water content.
- *Theta*: number [cm^3/cm^3] soil cells
Water content in this biopore class.
- *M*: submodel (see section 4.1.7) sequence
Chemical content in soil cells.

$\langle \text{ name value... } \rangle$ **Log Variables**

 - *value*: number [g/cm^3] soil cells
Value for chemical.
 - *name*: string (see section 4.1.5)
Name of chemical.
- *iterations*: integer
Number of iterations used for finding a solution.
- *h3_soil*: number [cm] sequence
Pressure suggested by the soil for each interval.
- *z3_lowest*: number [cm] sequence
Depth of lowest unsaturated cell in each interval. Water may not enter the macropore below this depth.

Chapter 21

boolean

Generic representation of booleans.

Used by IntegerCondClause @ condition (see 107.28, page 624) , and Stringer-CondClause @ condition (see 107.29, page 624) .

21.1 false

Always false.

21.2 not

True if and only if the operand is not true.

< not *operands* ... >

- *operands*: **boolean** component (see chapter 21) array of length 1
The operand to check.

21.3 numbers

Base class for boolean expressions involving numbers.

< numbers *operands* ... >

- *operands*: **number** component (see chapter 59) sequence
List of operands to compare.

21.4 <

A ‘numbers’ model (see 21.3, page 121) build into Daisy.

True iff each operand is smaller than the next.

21.5 <=

A ‘numbers’ model (see 21.3, page 121) build into Daisy.

True iff each operand is smaller than or equal to the next.

21.6 >

A ‘numbers’ model (see 21.3, page 121) build into Daisy.
True iff each operand is larger than the next.

21.7 >=

A ‘numbers’ model (see 21.3, page 121) build into Daisy.
True iff each operand is at least as large as the next.

21.8 operands

Base class for boolean expressions involving multiple boolean operands.

< **operands** *operands*... >

- *operands*: **boolean** component (see chapter 21) sequence
List of operands to compare.

21.9 and

A ‘operands’ model (see 21.8, page 122) build into Daisy.
True if and only if all operands are true.

21.10 or

A ‘operands’ model (see 21.8, page 122) build into Daisy.
True if and only if any operand is true.

21.11 string-equal

True iff the supplied strings are identical.

< "string-equal" *values*... >

- *values*: string (see section 4.1.5) sequence
Parameter
Strings to compare.

21.12 true

Always true.

Used by source arithmetic valid (see 83.1, page 445) , program extract valid (see 66.14, page 361) , xysource arithmetic valid (see 105.1, page 569) , and xysource frequency valid (see 105.4, page 573) .

21.13 xor

True if and only if one operand is true, and one false.

< **xor** *operands*... >

- *operands*: **boolean** component (see chapter 21) array of length 2
The two operands to compare.

Chapter 22

bound

Specify one end of an interval boundary.

22.1 finite

Finite interval bound.

```
< finite bound >
```

- *bound*: number [**cm**]
Parameter
Interval bound to use.

22.2 full

Maximum value for the interval boundary.

22.3 none

No boundary specified.

22.4 state

Bound used for checkpoints.

```
< state (bound bound)
      (type type) >
```

- *bound*: number [**cm**]
Optional state variable
Interval bound to use. Only valid for the 'finite' type.
- *type*: string (see section 4.1.5)
State variable
Bound type

22.5 empty

A 'state' model (see 22.4, page 123) build into Daisy.

A 'state' model set to 'none'.

Used by volume box top (see 102.1, page 551) , and zone box top (see 106.1, page 581) .

< (type none) >

Chapter 23

chemical

This component should, for a specific chemical (typically a pesticide), provide a description of the properties of interest to Daisy.

23.1 base

Read chemical properties as normal Daisy parameters.

```

< base (adsorption adsorption) ; Default none value.
      (initial_C initial_C ...)
      (C C ...)
      (initial initial) ; Default initial_zero
      (molar_mass molar_mass)
      (solubility 1 [g/cm3])
      (solubility_infiltration_factor 1 [])
      (crop_uptake_reflection_factor 1 [<fraction>])
      (canopy_dissipation canopy_dissipation)
      (canopy_washoff_coefficient canopy_washoff_coefficient)
      (surface_decompose surface_decompose)
      (litter_decompose litter_decompose)
      (litter_washoff_coefficient 1 [<fraction>])
      (litter_diffusion litter_diffusion) ; Default rate value.
      (diffusion_coefficient diffusion_coefficient)
      (decompose decompose)
      (decompose_conc_factor decompose_conc_factor) ; Has default value.
      (decompose_lag_increment decompose_lag_increment) ; Has default value.
      (enable_surface_products false)
      (soil_affects_surface_decompose false)
      (drain_secondary false)
      (C_below C_below) ; Default zero_gradient
      (decompose_products)
      (snow_storage 0 [g/m2])
      (canopy_storage 0 [g/m2])
      (litter_storage 0 [g/m2])
      (surface_storage 0 [g/m2])
      (initial_C_plf initial_C_plf)
      (C_secondary C_secondary ...)
      (initial_C_secondary initial_C_secondary ...)
      (initial_C_secondary_plf initial_C_secondary_plf)
      (initial_C_primary initial_C_primary ...)
      (initial_C_primary_plf initial_C_primary_plf)
      (M M ...)
      (initial_M initial_M ...)
      (initial_M_plf initial_M_plf)
      (M_secondary M_secondary ...)
      (initial_M_secondary initial_M_secondary ...)
      (initial_M_secondary_plf initial_M_secondary_plf)
      (initial_M_primary initial_M_primary ...)
      (initial_M_primary_plf initial_M_primary_plf)
      (Ms Ms ...)
      (initial_Ms initial_Ms ...)
      (initial_Ms_plf initial_Ms_plf)
      (S_permanent)
      (lag lag ...) >

```

- *adsorption*: **adsorption** component (see chapter 16)
 Component (default 'none')
 Instant equilibrium between sorbed and solute phases.

Specify the equilibrium model here for chemicals where the sorbed and solute phases typically reaches equilibrium within a single timestep. Slower adsorption processes should be modelled as two chemicals, one with 'none' adsorption and one with 'full' adsorption, and an 'adsorption' reaction between them.

- *initial_C*: submodel (see section 4.1.7) sequence
Optional submodel
Initial value of the 'C' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'C' is unspecified.

```
< end value >
```

 - *end*: number [cm]
Parameter
End point of this layer (a negative number).
 - *value*: number [g/cm³]
Parameter
Concentration in water.
- *C*: number [g/cm³] soil cells
Optional state variable
Concentration in water.
- *initial*: **number** component (see chapter 59)
Component (default 'initial_zero')
Initial content (M) if otherwise unspecified. [g/cm³].
- *molar_mass*: number [g/mol]
Optional parameter
Molar mass of the chemical.
- *solubility*: number [g/cm³]
Parameter (default 1)
Maximal concentration in water at 20 dg C.
- *solubility_infiltration_factor*: number (dimensionless)
Parameter (default 1)
Adjustment for maximum concentration in infiltrated water.
- *crop_uptake_reflection_factor*: number [<fraction>]
Parameter (default 1)
How much of the chemical is reflected at crop uptake.
- *canopy_dissipation*: **rate** component (see chapter 69)
How fast does the chemical dissipate on canopy.
- *canopy_washoff_coefficient*: number [<fraction>]
Parameter
Fraction of the chemical that follows the water off the canopy.
- *surface_decompose*: **rate** component (see chapter 69)
Optional component
How fast does the chemical decompose on surface. Defaults to 'canopy_dissipation' rate.
- *litter_decompose*: **rate** component (see chapter 69)
Optional component
How fast does the chemical decompose on litter. Default to 'canopy_dissipation' rate.

- *litter_washoff_coefficient*: number [**<fraction>**]
Parameter (default 1)
Fraction of the chemical that follows the water off the litter.

- *litter_diffusion*: **rate** component (see chapter 69)
Component (default 'rate')

(*litter_diffusion* rate 0 [h⁻¹])

Parameter description:
How fast chemical diffuse to water passing on surface.

- *diffusion_coefficient*: number [cm²/s]
Parameter
Diffusion coefficient.
- *decompose*: **rate** component (see chapter 69)
How fast the chemical is being decomposed in the soil.
- *decompose_conc_factor*: plf [g/cm³ H₂O → **<none>**]
Parameter (has default value with 0 points)
Concentration development factor on decomposition.
- *decompose_lag_increment*: plf [g/cm³ → h⁻¹]
Parameter (has default value with 0 points)
Increment lag with the value of this PLF for the current concentration each hour. When lag in any cell reaches 1.0, decomposition begins. It can never be more than 1.0 or less than 0.0. By default, there is no lag.
- *enable_surface_products*: boolean (see section 4.1.2)
Parameter (default false)
True if metabolites of this chemical can be generated on the surface.
- *soil_affects_surface_decompose*: boolean (see section 4.1.2)
Parameter (default false)
True if soil conditions affect surface decomposition.
- *drain_secondary*: boolean (see section 4.1.2)
Parameter (default false)
Concentration in secondary soil water user for drainage. If you set this to true the concentration in the secondary domain is used for concentration in drain water. Otherwise, the average concentration is the matrix is used. Using the secondary domain is more physically correct, but also more likely to give unstable results.
- *C_below*: **number** component (see chapter 59)
Component (default 'zero_gradient')
Concentration below the layer of soil being examined. Use a negative number to indicate same concentration as in lowest cell.
- *decompose_products*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
List of products from decomposition.

< *fraction chemical* >

- *fraction*: number [**<fraction>**]
 Parameter
 Fraction of decomposed matter that become this chemical.
 If both chemicals have molar_mass specified, the fraction will be mole based, otherwise it will be mass based.
 - *chemical*: string (see section 4.1.5)
 Parameter
 Chemical product of decomposed matter.
 - *snow_storage*: number [**g/m²**]
 State variable (default 0)
 Stored in the snow pack.
 - *canopy_storage*: number [**g/m²**]
 State variable (default 0)
 Stored on the canopy.
 - *litter_storage*: number [**g/m²**]
 State variable (default 0)
 Stored in the litter (mulch, surface residuals).
 - *surface_storage*: number [**g/m²**]
 State variable (default 0)
 Stored on the soil surface. This includes the mixing layer, and constitute 'surface_solute' and 'surface_immobile'.
 - *initial_C_plf*: plf [**cm → g/cm³**]
 Optional parameter
 Initial value of the 'C' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'C' and 'initial_C' are unspecified.
 - *C_secondary*: number [**g/cm³**] soil cells
 Optional state variable
 Concentration in secondary domain.
 - *initial_C_secondary*: submodel (see section 4.1.7) sequence
 Optional submodel
 Initial value of the 'C_secondary' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'C_secondary' is unspecified.
- < end value >**
- *end*: number [**cm**]
 Parameter
 End point of this layer (a negative number).
 - *value*: number [**g/cm³**]
 Parameter
 Concentration in secondary domain.
 - *initial_C_secondary_plf*: plf [**cm → g/cm³**]
 Optional parameter
 Initial value of the 'C_secondary' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up.

The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'C_secondary' and 'initial_C_secondary' are unspecified.

- *initial_C_primary*: submodel (see section 4.1.7) sequence
Optional submodel
Initial value of the 'C_primary' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'C_primary' is unspecified.

$$< \text{end value} >$$
 - *end*: number [cm]
Parameter
End point of this layer (a negative number).
 - *value*: number [g/cm³]
Parameter
Concentration in primary domain.
- *initial_C_primary_plf*: plf [cm → g/cm³]
Optional parameter
Initial value of the 'C_primary' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'C_primary' and 'initial_C_primary' are unspecified.
- *M*: number [g/cm³] soil cells
Optional state variable
Total mass per volume water, soil, and air.
- *initial_M*: submodel (see section 4.1.7) sequence
Optional submodel
Initial value of the 'M' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'M' is unspecified.

$$< \text{end value} >$$
 - *end*: number [cm]
Parameter
End point of this layer (a negative number).
 - *value*: number [g/cm³]
Parameter
Total mass per volume water, soil, and air.
- *initial_M_plf*: plf [cm → g/cm³]
Optional parameter
Initial value of the 'M' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'M' and 'initial_M' are unspecified.
- *M_secondary*: number [g/cm³] soil cells
Optional state variable
Secondary domain mass per volume water, soil, and air.

- *initial_M_secondary*: submodel (see section 4.1.7) sequence
Optional submodel
Initial value of the 'M_secondary' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'M_secondary' is unspecified.

$$< \text{end value} >$$
 - *end*: number [cm]
Parameter
End point of this layer (a negative number).
 - *value*: number [g/cm³]
Parameter
Secondary domain mass per volume water, soil, and air.
- *initial_M_secondary_plf*: plf [cm → g/cm³]
Optional parameter
Initial value of the 'M_secondary' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'M_secondary' and 'initial_M_secondary' are unspecified.
- *initial_M_primary*: submodel (see section 4.1.7) sequence
Optional submodel
Initial value of the 'M_primary' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'M_primary' is unspecified.

$$< \text{end value} >$$
 - *end*: number [cm]
Parameter
End point of this layer (a negative number).
 - *value*: number [g/cm³]
Parameter
Primary domain mass per volume water, soil, and air.
- *initial_M_primary_plf*: plf [cm → g/cm³]
Optional parameter
Initial value of the 'M_primary' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'M_primary' and 'initial_M_primary' are unspecified.
- *Ms*: number [<fraction>] soil cells
Optional parameter
Mass in dry soil. This include all matter in both soil and water, relative to the dry matter weight. Only for initialization of the 'M' parameter.
- *initial_Ms*: submodel (see section 4.1.7) sequence
Optional submodel
Initial value of the 'Ms' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'Ms' is unspecified.

$$< \text{end value} >$$

- *end*: number [**cm**]
Parameter
End point of this layer (a negative number).
- *value*: number [**<fraction>**]
Parameter
Mass in dry soil. This include all matter in both soil and water, relative to the dry matter weight. Only for initialization of the 'M' parameter.
- *initial_Ms_plf*: plf [**cm** → **<fraction>**]
Optional parameter
Initial value of the 'Ms' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'Ms' and 'initial_Ms' are unspecified.
- *S_permanent*: number [**g/cm³/h**] soil cells
State variable (default: an empty sequence)
Permanent external source, e.g. subsoil irrigation.
- *lag*: number (dimensionless) soil cells
Optional state variable
This state variable grows with lag_increment (C) each hour. When it reached 1.0, decomposition begins.

Log Variables

- *spray*: number [**g/m²/h**]
Amount currently being applied.
- *spray_surface*: number [**g/m²/h**]
Amount currently being applied below canopy.
- *deposit*: number [**g/m²/h**]
Amount deposited from the atmosphere.
- *spray_overhead*: number [**g/m²/h**]
Amount currently being applied above canopy.
- *surface_tillage*: number [**g/m²/h**]
Amount removed from surface due to tillage operations.
- *litter_tillage*: number [**g/m²/h**]
Amount removed from litter due to tillage operations.
- *snow_in*: number [**g/m²/h**]
Entering snow pack.
- *snow_out*: number [**g/m²/h**]
Leaking from snow pack.
- *canopy_in*: number [**g/m²/h**]
Entering canopy.
- *canopy_dissipate*: number [**g/m²/h**]
Dissipating from canopy.
- *canopy_out*: number [**g/m²/h**]
Falling through or off the canopy.

- *canopy_transform*: number $[\text{g}/\text{m}^2/\text{h}]$
Added through transformation to canopy.
- *canopy_harvest*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount removed with crop harvest.
- *litter_in*: number $[\text{g}/\text{m}^2/\text{h}]$
Entering litter .
- *litter_decomposed*: number $[\text{g}/\text{m}^2/\text{h}]$
Decomposed from the litter.
- *litter_transform*: number $[\text{g}/\text{m}^2/\text{h}]$
Added through transformation in litter layer.
- *litter_out*: number $[\text{g}/\text{m}^2/\text{h}]$
Leaking from litter.
- *litter_leak*: number $[\text{g}/\text{m}^2/\text{h}]$
Leaking from bottom litter.
- *litter_diffuse*: number $[\text{g}/\text{m}^2/\text{h}]$
Diffusing from litter to water passing on surface.
- *surface_solute*: number $[\text{g}/\text{m}^2]$
Stored in the soil water of the mixing layer. This is part of 'surface_storage'.
- *surface_immobile*: number $[\text{g}/\text{m}^2]$
Bound to soil particles in the mixing layer. This is part of 'surface_storage'.
- *surface_in*: number $[\text{g}/\text{m}^2/\text{h}]$
Falling on the bare soil surface.
- *surface_runoff*: number $[\text{g}/\text{m}^2/\text{h}]$
Removed through lateral movement on the soil.
- *surface_decomposed*: number $[\text{g}/\text{m}^2/\text{h}]$
Decomposed from the surface.
- *surface_transform*: number $[\text{g}/\text{m}^2/\text{h}]$
Added through transformation to surface.
- *surface_mixture*: number $[\text{g}/\text{m}^2/\text{h}]$
Entering the soil through mixture with ponded water.
- *surface_out*: number $[\text{g}/\text{m}^2/\text{h}]$
Entering the soil with water infiltration.
- *surface_release*: number $[\text{fraction}]$
Fraction of available soil particles released as colloids this timestep. Only relevant for chemicals representing colloids.

The idea behind this is that reactions that generate colloids will set the value of this variable, and then reactions that convert immobile chemicals into colloid bound chemicals will use it. For this to work, the reactions that set the variable must be listed before the reactions that use it.

Note that the value is relative to the current timestep.
- *top_storage*: number $[\text{g}/\text{m}^2]$
Sum of above ground (surface, litter, snow, canopy) storage.

- *top_loss*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount lost from the system from the surface. This includes runoff, canopy dissipation and harvest, but not soil infiltration. It also includes the net loss through transformation, which can be negative.
- *C_primary*: number $[\text{g}/\text{cm}^3]$ soil cells
Concentration in primary domain.
- *M_primary*: number $[\text{g}/\text{cm}^3]$ soil cells
Primary domain mass per volume water, soil, and air.
- *M_error*: number $[\text{g}/\text{cm}^3]$ soil cells
Mass subtracted to avoid negative values.
- *M_tertiary*: number $[\text{g}/\text{cm}^3]$ soil cells
Mass in tertiary domain (biopores).
- *S_secondary*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Secondary matrix source term.
- *S_primary*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Primary matrix source term.
- *S_exchange*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Exchange from primary to secondary domain.
- *S_indirect_drain*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Added to the soil matrix from drains indirectly via biopores. This can be non-zero wherever there are drain connected biopores.
- *S_soil_drain*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Added to the soil matrix from drains directly, not via biopores. This is only non-zero in drain nodes.
- *S_p_drain*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Removed from the biopores to the drain. This is only non-zero in drain nodes.
- *S_B2M*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Biopores to matrix.
- *S_M2B*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Matrix to biopores (negative matrix source). Does not count flow to drain connected biopores.
- *S_external*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
External source, such as incorporated fertilizer.
- *S_root*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Source term (root uptake only, always negative).
- *S_decompose*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Source term for decompose, is never positive.
- *S_decompose_primary*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Source term for decompose in primary domain, is never positive.
- *S_decompose_secondary*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Source term for decompose in secondary domain, is never positive.

- *S_transform*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Source term for transformations other than sorption.
- *decompose_factor*: number (dimensionless) soil cells
Decompose factor due to soil conditions.
- *surface_decompose_factor*: number (dimensionless)
Decompose factor due to soil conditions near surface.
- *J_primary*: number $[\text{g}/\text{cm}^2/\text{h}]$ soil edges
Transportation in primary matrix water (positive up).
- *J_secondary*: number $[\text{g}/\text{cm}^2/\text{h}]$ soil edges
Transportation in secondary matrix water (positive up).
- *J_matrix*: number $[\text{g}/\text{cm}^2/\text{h}]$ soil edges
Transportation in matrix (positive up).
- *J_tertiary*: number $[\text{g}/\text{cm}^2/\text{h}]$ soil edges
Transportation in tertiary water (positive up).
- *tillage*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Changes during tillage.
- *dt*: number $[\text{h}]$
Suggested timestep length based on sink terms.
- *sink_cell*: integer
Cell with largest forward sink compared to available matter.

23.2 FOCUS

A ‘base’ model (see 23.1, page 125) build into Daisy.

Pesticides inspired by FOCUS surface water.

Depth, moisture, and heat effect according to the FOCUS Surface Water WG.

Water factor is zero below 0.5 Theta_wp , one above field capacity, and $((\text{Theta} - 0.5 \text{ Theta_wp}) / (\text{Theta_fc} - 0.5 \text{ Theta_wp}))^B$ otherwise. From MACRO 5.2.

Heat factor is zero below zero degrees, $0.2 T e^{(\alpha (5 - T_{\text{ref}}))}$ below 5 dg C, and $e^{(\alpha (T - T_{\text{ref}}))}$ above 5 dg C. From MACRO 5.0.

Depth factor is specified by ‘z’ and ‘z_factor’. See also [FOCUS, 2002, FOCUS, 2012]

```

< FOCUS (alpha 0.0948 [K-1])
(T_ref 20 [dg C])
(B 0.49 [])
(canopy_dissipation canopy_dissipation) ; Default halftime value.
(canopy_washoff_coefficient 0.075 [<fraction>])
(diffusion_coefficient 5e-06 [cm2/s])
(z -30 -60 -100)
(z_factor 1 0.5 0.3)
;; Shared parameters are described in section 23.1.
(adsorption adsorption) ; Default none value.
(initial_C initial_C ...)
(C C ...)
(initial initial) ; Default initial_zero value
(molar_mass molar_mass)
(solubility 1 [g/cm3])
(solubility_infiltration_factor 1 [])
(crop_uptake_reflection_factor 1 [<fraction>])
(surface_decompose surface_decompose)
(litter_decompose litter_decompose)
(litter_washoff_coefficient 1 [<fraction>])
(litter_diffusion litter_diffusion) ; Default rate value.
(decompose decompose)
(decompose_conc_factor decompose_conc_factor) ; Has default value.
(decompose_lag_increment decompose_lag_increment) ; Has default value.
(enable_surface_products false)
(soil_affects_surface_decompose false)
(drain_secondary false)
(C_below C_below) ; Default zero_gradient value
(decompose_products)
(snow_storage 0 [g/m2])
(canopy_storage 0 [g/m2])
(litter_storage 0 [g/m2])
(surface_storage 0 [g/m2])
(initial_C_plf initial_C_plf)
(C_secondary C_secondary ...)
(initial_C_secondary initial_C_secondary ...)
(initial_C_secondary_plf initial_C_secondary_plf)
(initial_C_primary initial_C_primary ...)
(initial_C_primary_plf initial_C_primary_plf)
(M M ...)
(initial_M initial_M ...)
(initial_M_plf initial_M_plf)
(M_secondary M_secondary ...)
(initial_M_secondary initial_M_secondary ...)
(initial_M_secondary_plf initial_M_secondary_plf)
(initial_M_primary initial_M_primary ...)
(initial_M_primary_plf initial_M_primary_plf)
(Ms Ms ...)
(initial_Ms initial_Ms ...)
(initial_Ms_plf initial_Ms_plf)
(S_permanent)
(lag lag ...) >

```

- *alpha*: number [K⁻¹]

Parameter (default 0.0948)
 Value description: Section 7.4.4 See also [FOCUS, 2012]
 Temperature effect parameter.

- *T_{ref}*: number [**dg C**]
 Parameter (default 20)
 Value description: Section 7.4.2 See also [FOCUS, 2012]
 Reference temperature.

- *B*: number (dimensionless)
 Parameter (default 0.49)
 Value description: Section 7.4.5 See also [FOCUS, 2012]
 Soil moisture effect parameter.

- *z*: number [**cm**] sequence
 Parameter (has default value with length 3)

(*z* -30 -60 -100 [cm])

Parameter description:
 End of each interval.

- *z_{factor}*: number (dimensionless) sequence
 Parameter (has default value with length 3)

(*z_{factor}* 1 0.5 0.3 [])

Parameter description:
 Factor to apply to decompose rate at each interval.

23.3 metabolite

A ‘FOCUS’ model (see 23.2, page 135) defined in ‘chemistry-base.dai’.
 Degraded pesticide. See also [FOCUS, 2002, FOCUS, 2012]

23.4 pesticide

A ‘FOCUS’ model (see 23.2, page 135) defined in ‘chemistry-base.dai’.
 This stuff protects plants. See also [FOCUS, 2002, FOCUS, 2012]

23.5 fungicide

A ‘pesticide’ model (see 23.4, page 137) defined in ‘chemistry-base.dai’.
 This stuff kills swamps. See also [FOCUS, 2002, FOCUS, 2012]

23.6 herbicide

A ‘pesticide’ model (see 23.4, page 137) defined in ‘chemistry-base.dai’.
 This stuff kills plants. See also [FOCUS, 2002, FOCUS, 2012]

23.7 2,4-D

A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’.
Selective herbicide, kill weeds but not crops. See also [FOCUS, 2002, FOCUS, 2012]
Used by chemistry pesticides trace (see 24.7, page 146) .

23.8 Atrazine

A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’.
Used by chemistry pesticides trace (see 24.7, page 146) .

23.9 Bentazon

A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’.
Used by chemistry pesticides trace (see 24.7, page 146) .

23.10 IPU

A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’.
Used by chemistry pesticides trace (see 24.7, page 146) .

23.11 Ioxynil

A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’.
Used by chemistry pesticides trace (see 24.7, page 146) .

23.12 MCPP

A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’.
Used by chemistry pesticides trace (see 24.7, page 146) .

23.13 Pendimethalin

A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’.
Used by chemistry pesticides trace (see 24.7, page 146) .

23.14 insecticide

A ‘pesticide’ model (see 23.4, page 137) defined in ‘chemistry-base.dai’.
This stuff kills insects. See also [FOCUS, 2002, FOCUS, 2012]

23.15 DDT

A ‘insecticide’ model (see 23.14, page 138) defined in ‘chemistry.dai’.
Used by chemistry pesticides trace (see 24.7, page 146) .

23.16 Heptachlor

A ‘insecticide’ model (see 23.14, page 138) defined in ‘chemistry.dai’.

Used by chemistry pesticides trace (see 24.7, page 146) .

23.17 default

A ‘base’ model (see 23.1, page 125) build into Daisy.

```

< default  (T_ref 10 [dg C])
            (decompose_heat_factor decompose_heat_factor) ; Has default value
            (decompose_water_factor decompose_water_factor) ; Has default value
            (decompose_SMB_pool 1)
            (decompose_SMB_KM 0 [g C/cm3])
            (SMB_ref SMB_ref)
            (decompose_CO2_factor decompose_CO2_factor) ; Has default value
            (decompose_depth_factor decompose_depth_factor) ; Has default value
            ;; Shared parameters are described in section 23.1.
            (adsorption adsorption) ; Default none value
            (initial_C initial_C ...)
            (C C ...)
            (initial initial) ; Default initial value
            (molar_mass molar_mass)
            (solubility 1 [g/cm3])
            (solubility_infiltration_factor 1 [])
            (crop_uptake_reflection_factor 1 [<fraction>])
            (canopy_dissipation canopy_dissipation)
            (canopy_washoff_coefficient canopy_washoff_coefficient)
            (surface_decompose surface_decompose)
            (litter_decompose litter_decompose)
            (litter_washoff_coefficient 1 [<fraction>])
            (litter_diffusion litter_diffusion) ; Default rate value
            (diffusion_coefficient diffusion_coefficient)
            (decompose decompose)
            (decompose_conc_factor decompose_conc_factor) ; Has default value
            (decompose_lag_increment decompose_lag_increment) ; Has default value
            (enable_surface_products false)
            (soil_affects_surface_decompose false)
            (drain_secondary false)
            (C_below C_below) ; Default zero gradient
            (decompose_products)
            (snow_storage 0 [g/m2])
            (canopy_storage 0 [g/m2])
            (litter_storage 0 [g/m2])
            (surface_storage 0 [g/m2])
            (initial_C_plf initial_C_plf)
            (C_secondary C_secondary ...)
            (initial_C_secondary initial_C_secondary ...)
            (initial_C_secondary_plf initial_C_secondary_plf)
            (initial_C_primary initial_C_primary ...)
            (initial_C_primary_plf initial_C_primary_plf)
            (M M ...)
            (initial_M initial_M ...)
            (initial_M_plf initial_M_plf)
            (M_secondary M_secondary ...)
            (initial_M_secondary initial_M_secondary ...)
            (initial_M_secondary_plf initial_M_secondary_plf)
            (initial_M_primary initial_M_primary ...)
            (initial_M_primary_plf initial_M_primary_plf)
            (Ms Ms ...)
            (initial_Ms initial_Ms ...)
            (initial_Ms_plf initial_Ms_plf)
            (S_permanent)
            (lag lag ...) >

```

- *T_ref*: number [dg C]
Parameter (default 10)
Reference temperature for decomposition. The heat factor on decomposition will be scaled so it is 1 at this temperature..
- *decompose_heat_factor*: plf [dg C $\rightarrow$ <none>]
Parameter (has default value with 0 points)
Heat factor on decomposition.
- *decompose_water_factor*: plf [cm $\rightarrow$ <none>]
Parameter (has default value with 0 points)
Water potential factor on decomposition.
- *decompose_SMB_pool*: integer
Parameter (default 1)
SMB pool for Michaelis-Menten kinetics. Use 0 for SMB1, 1 for SMB2, or -1 for all SMB pools.
- *decompose_SMB_KM*: number [g C/cm³]
Parameter (default 0)
Michaelis-Menten kinetics parameter. Decompose rate is modified by $C / (KM + C)$, where C is the carbon content in the pool specified by 'decompose_SMB_pool'.
- *SMB_ref*: number [g C/cm³]
Optional parameter
Reference SMB carbon for decomposition of mulch. The SMB factor for decomposition will be scaled so it is 1 at this amount of SMB carbon. By default, it will not be scaled.
- *decompose_CO2_factor*: plf [g CO₂-C/cm³/h $\rightarrow$ <none>]
Parameter (has default value with 0 points)
CO₂ development factor on decomposition.
- *decompose_depth_factor*: plf [cm $\rightarrow$ <none>]
Parameter (has default const value 1 [])
Depth influence on decomposition.

23.18 PFAS

A 'default' model (see 23.17, page 139) defined in 'PFAS.dai'.

23.19 PFBA

A 'PFAS' model (see 23.18, page 141) defined in 'PFAS.dai'.

Aqueous diffusion coefficient from Table 1 See also [Gauthier and Mabury, 2024]

Used by chemistry PFAS trace (see 24.4, page 145) .

23.20 PFBS

A 'PFAS' model (see 23.18, page 141) defined in 'PFAS.dai'.

Aqueous diffusion coefficient from Table 1 See also [Gauthier and Mabury, 2024]

Used by chemistry PFAS trace (see 24.4, page 145) .

23.21 PFHxA

A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’.
Aqueous diffusion coefficient from Table 1 See also [Gauthier and Mabury, 2024]
Used by chemistry PFAS trace (see 24.4, page 145) .

23.22 PFOA

A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’.
Aqueous diffusion coefficient from Table 1 See also [Gauthier and Mabury, 2024]
Used by chemistry PFAS trace (see 24.4, page 145) .

23.23 PFOA_Guo

A ‘PFOA’ model (see 23.22, page 142) defined in ‘PFAS.dai’.
Used by chemistry PFAS trace (see 24.4, page 145) .

23.24 PFOS

A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’.
Aqueous diffusion coefficient from Table 1 See also [Gauthier and Mabury, 2024]
Used by chemistry PFAS trace (see 24.4, page 145) .

23.25 PFOS_Guo

A ‘PFOS’ model (see 23.24, page 142) defined in ‘PFAS.dai’.
Used by chemistry PFAS trace (see 24.4, page 145) .

23.26 PFPeA

A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’.
Aqueous diffusion coefficient from Table 1 See also [Gauthier and Mabury, 2024]
Used by chemistry PFAS trace (see 24.4, page 145) .

23.27 Tracer

A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’.
No adsorption, plante uptake or degradation
Used by chemistry Tracer trace (see 24.5, page 145) .

23.28 classic

A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’.
For chemicals where we know no better. See also [FOCUS, 2002]

23.29 colloid

A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’.
Mobile colloids.
Used by chemistry colloid trace (see 24.6, page 146) .

23.30 solid

A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’.
Non-dissolvable chemicals

23.31 nutrient

A ‘base’ model (see 23.1, page 125) build into Daisy.

Plants eat this stuff.

```
< (canopy_dissipation canopy_dissipation)           ; Default rate value.
  (canopy_washoff_coefficient 1 [<fraction>])
  (decompose decompose)                             ; Default rate value. >
```

23.32 C

A ‘nutrient’ model (see 23.31, page 143) build into Daisy.
Carbon.

23.33 DOC

A ‘C’ model (see 23.32, page 143) build into Daisy.

Nitrate-N.

Used by chemistry DOM trace (see 24.2, page 145) .

```
< (litter_diffusion litter_diffusion)           ; Default rate value.
  (diffusion_coefficient 1.9e-06 [cm2/s]) >
```

23.34 N

A ‘nutrient’ model (see 23.31, page 143) build into Daisy.
Nitrogen.

23.35 DON

A ‘N’ model (see 23.34, page 143) build into Daisy.

Nitrate-N.

Used by chemistry DOM trace (see 24.2, page 145) .

```
< (diffusion_coefficient 1.9e-06 [cm2/s]) >
```

23.36 MIN

A ‘N’ model (see 23.34, page 143) build into Daisy.
Non-organic nitrogen.

23.37 NH4

A ‘MIN’ model (see 23.36, page 143) build into Daisy.

Ammonium-N.

Used by chemistry N trace (see 24.3, page 145) .

```

< (adsorption adsorption)                ; Default NH4 value.
  (initial initial)                        ; Default initial_NH4 value.
  (diffusion_coefficient 1.8e-05 [cm2/s]) >

```

23.38 NO3

A 'MIN' model (see 23.36, page 143) build into Daisy.

Nitrate-N.

Used by chemistry N trace (see 24.3, page 145) .

```

< (initial initial)                        ; Default initial_NO3 value.
  (diffusion_coefficient 2e-05 [cm2/s]) >

```


Chapter 24

chemistry

Pesticides and other chemicals.

24.1 default

Handle chemicals and reactions.

```
< default (reaction)
      (trace) >
```

- *reaction*: **reaction** component (see chapter 70) sequence
Component (default: an empty sequence)
List of chemical reactions you want to simulate.
- *trace*: **chemical** component (see chapter 23) sequence
Component (default: an empty sequence)
List of chemicals you want to trace in the simulation.

24.2 DOM

A ‘default’ model (see 24.1, page 145) build into Daisy.

Dissolved organic matter.

```
< (reaction reaction ...) ; Has default value.
  (trace trace ...) ; Has default value. >
```

24.3 N

A ‘default’ model (see 24.1, page 145) build into Daisy.

Inorganic nitrogen.

Used by chemistry nutrient combine (see 24.10, page 147) .

```
< (reaction reaction ...) ; Has default value.
  (trace trace ...) ; Has default value. >
```

24.4 PFAS

A ‘default’ model (see 24.1, page 145) defined in ‘PFAS.dai’.

24.5 Tracer

A ‘default’ model (see 24.1, page 145) defined in ‘chemistry.dai’.

24.6 colloid

A ‘default’ model (see 24.1, page 145) defined in ‘chemistry.dai’.

24.7 pesticides

A ‘default’ model (see 24.1, page 145) defined in ‘chemistry.dai’.
Trace all known pesticides.

24.8 multi

Handle multile chemistries.

```
< multi  (combine combine ...)
          (ignore)
          (max_sink_total 0.5 [])
          (max_sink_solute 0.9 [])
          (max_sink_secondary 1.5 [])
          (min_sink_total 0.01 []) >
```

- *combine*: **chemistry** component (see chapter 24) sequence
List of chemistry parameterizations you want to combine.
- *ignore*: string (see section 4.1.5) sequence
State variable (default: an empty sequence)
Don’t warn when spraying one of these chemicals. The first time an untraced chemical not on the list is sprayed on the field, Daisy will issue a warning and add the chemical to this list.
- *max_sink_total*: number (dimensionless)
Parameter (default 0.5)
Maximum allowed sink term as a fraction of total content.

If variable timesteps are enabled, Daisy will try to scale down the timestep in order to ensure that no more than this fraction of the total content is removed by drains or biopores within the timestep.
- *max_sink_solute*: number (dimensionless)
Parameter (default 0.9)
Maximum allowed sink term as a fraction of solute content.

If variable timesteps are enabled, Daisy will try to scale down the timestep in order to ensure that no more than this fraction of the solute content is removed by drains or biopores within the timestep.
- *max_sink_secondary*: number (dimensionless)
Parameter (default 1.5)
Maximum allowed sink term as a fraction of secondary domain content.

If variable timesteps are enabled, Daisy will try to scale down the timestep in order to ensure that no more than this fraction of the secondary domain content is removed by drains or biopores within the timestep. This should usually be above 1 to allow for the case where the secondary domain is emptied within a timestep.
- *min_sink_total*: number (dimensionless)
Parameter (default 0.01)

Always allow this fraction of total content to be removed by sink term.
 This overwrites all the 'max_sink' parameters.

Log Variables

- *trace*: **chemical** component (see chapter 23) sequence
 List of chemicals in nested chemistries.

24.9 none

A 'multi' model (see 24.8, page 146) build into Daisy.

No active chemistries.

```
< (combine) >
```

24.10 nutrient

A 'multi' model (see 24.8, page 146) build into Daisy.

Include 'N' chemistry so organic matter and plants will work.

Used by column default Chemistry (see 26.1, page 153) .

```
< (combine combine ...) ; Has default value. >
```


Chapter 25

cloudiness

Calculate cloudiness from meteorological data.

25.1 clear_sky

Calculate based on clear sky radiation: $a * Si / Si0 + 1 - a$ where Si is global radiation and $Si0$ is clear sky radiation.

```
< clear_sky (cloudiness cloudiness)  
  (a a)  
  (min_extraterrestrial_radiation 0 [W/m2])  
  (min_global_radiation 0 [W/m2])  
  (min_solar_elevation_angle 0.174533 [rad])  
  (min_radiation_ratio 0.3 [<fraction>])  
  (min_time_to_sunset 0 [h])  
  (min_time_from_sunrise min_time_from_sunrise) >
```

- *cloudiness*: number [<fraction>]
Optional state variable
Cloudiness.
- *a*: number (dimensionless)
Parameter
Cloudiness coefficient.
- *min_extraterrestrial_radiation*: number [W/m²]
Parameter (default 0)
Only calculate cloudiness when extraterrestrial radiation is above this. Otherwise, last value will be used.
- *min_global_radiation*: number [W/m²]
Parameter (default 0)
Only calculate cloudiness when global radiation is above this. Otherwise, last value will be used.
- *min_solar_elevation_angle*: number [rad]
Parameter (default 0.174533)
Value description: FAO56 suggest 2-3 hour before sunset, ASCE suggest 20 dg solar angle. See also [Allen et al., 2005]
Only calculate cloudiness when solar elevation angle is above this. Otherwise, last value will be used.

- *min_radiation_ratio*: number [**<fraction>**]
Parameter (default 0.3)
Lowest Si/Si0 ration for calculating cloudiness.
- *min_time_to_sunset*: number [**h**]
Parameter (default 0)
Don't calculate cloudiness after this time before sunset.
- *min_time_from_sunrise*: number [**h**]
Optional parameter
Don't calculate cloudiness less than this time after sunrise. By default, equal to 'min_time_to_sunset'.

Log Variables

- *clear_sky_radiation*: number [**W/m²**]
Global radiation with clear sky conditions.

25.2 ASCE

A 'clear_sky' model (see 25.1, page 149) build into Daisy.

Equation D.1 in the ASCE std ref evap report. See also [Allen et al., 2005]

```
< ASCE  (Kt 1 [<fraction>])
      ;; Shared parameters are described in section 25.1.
      (cloudiness cloudiness)
      (a a)
      (min_extraterrestrial_radiation 0 [W/m2])
      (min_global_radiation 0 [W/m2])
      (min_solar_elevation_angle 0.174533 [rad])
      (min_radiation_ratio 0.3 [<fraction>])
      (min_time_to_sunset 0 [h])
      (min_time_from_sunrise min_time_from_sunrise) >
```

- *Kt*: number [**<fraction>**]
Parameter (default 1)
Turbidity coefficient, 1 = clean air, <= 0.5 extremely unclean.

Log Variables

- *W*: number [**mm**]
Precipitable water in atmosphere.

25.3 Taastrup

A 'ASCE' model (see 25.2, page 150) build into Daisy.

ASCE calibrated for Taastrup, Denmark. See also [Kjaersgaard et al., 2007a]

Used by bioclimate default cloudiness (see 19.2, page 115) .

```
<  (a 1 [])
      (min_extraterrestrial_radiation 25 [W/m2])
      (min_solar_elevation_angle 0 [rad]) >
```

25.4 FAO56

A ‘clear_sky’ model (see 25.1, page 149) build into Daisy.

Function of station height, Eq 37, FAO56. See also [Allen et al., 1998]

Used by bioclimate FAO56_daily cloudiness (see 19.3, page 115) , and bioclimate FAO56_hourly cloudiness (see 19.4, page 116) .

```
< (a 1.35 []) >
```

25.5 Kjaersgaard

A ‘FAO56’ model (see 25.4, page 151) build into Daisy.

FAO56 calibrated for Taastrup, Denmark. See also [Kjaersgaard et al., 2007a]

```
< (a 1 [])
  (min_extraterrestrial_radiation 25 [W/m²])
  (min_solar_elevation_angle 0 [rad]) >
```

25.6 const

The cloudiness never changes.

```
< const cloudiness >
```

- *cloudiness*: number [<fraction>]
Parameter
The cloudiness never changes.

25.7 weather

Use value from weather file.

```
< weather >
```

Log Variables

- *cloudiness*: number [<fraction>]
Cloudiness. 1 represent clear sky, 0 total eclipse.

Chapter 26

column

A 'column' is an one-dimensional vertical description of the soil/crop/atmosphere system. The column component contains most of the other processes in Daisy as submodels.

```
< component   (area 1 [m2])  
              (location) >
```

- *area*: number [m²]
State variable (default 1)
Area covered by this column. When logging multiple columns, the values are weighted by relative area.
- *location*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Location of this column.

The meaning depends on the number of point in the sequence. 0 points: The column has no specific location. 1 point: The column has a location, but no specific area. 3 or more points: The column represents the area specified by a polygon with the specified corner points.

```
<   x y   >
```

- *x*: number (dimension not specified)
Parameter
X-Coordinate.
- *y*: number (dimension not specified)
Parameter
Y-Coordinate.

26.1 default

Hansen et.al. 1990. with generic movement in soil.

```

< default  (AWI AWI)                                ; Default Brusseau2023 value.
            (Irrigation Irrigation)                  ; Has default value.
            (Soil Soil)                                ; Has partial value.
            (SoilHeat SoilHeat)                      ; Has default value.
            (SoilWater SoilWater)                    ; Has default value.
            (weather weather)
            (scope scope)                              ; Default null value.
            (second_year_utilization 0 [kg N/ha])
            (Surface Surface)                        ; Default default value.
            (SoilpH SoilpH)                          ; Default neutral value.
            (Movement Movement)                    ; Default vertical value.
            (Vegetation Vegetation)                  ; Default crops value.
            (Litter Litter)                          ; Default none value.
            (Bioclimate Bioclimate)                  ; Default default value.
            (Drain Drain)                            ; Default none value.
            (Groundwater Groundwater)
            (Chemistry Chemistry)                    ; Default nutrient value.
            (OrganicMatter OrganicMatter)            ; Default SOM2025 value.
            (tillage_age tillage_age ...)
            ;; Shared parameters are described in chapter 26.
            (area 1 [m2])
            (location) >

```

- *AWI*: **AWI** component (see chapter 9)
Component (default ‘Brusseau2023’)
Air-water interface area.
- *Irrigation*: **Irrigation** fixed component (see section 107.10)
Submodel (has fully specified default value)
Active irrigation events.
- *Soil*: **Soil** fixed component (see section 107.21)
Submodel (has partially specified default value)
The numeric and physical soil properties.
- *SoilHeat*: **SoilHeat** fixed component (see section 107.22)
Submodel (has fully specified default value)
Soil heat capacity and transportation.
- *SoilWater*: **SoilWater** fixed component (see section 107.23)
Submodel (has fully specified default value)
Soil water content and transportation.
- *weather*: **weather** component (see chapter 103)
Optional component
Weather model for providing climate information during the simulation. If unspecified, used global weather.
- *scope*: **scopesel** component (see chapter 77)
Component (default ‘null’)
Scope to evaluate expressions in.
- *second_year_utilization*: number [kg N/ha]
State variable (default 0)
Estimated accumulated second year fertilizer effect.

- *Surface*: **surface** component (see chapter 87)
Component (default 'default')
The upper border of the soil.
- *SoilpH*: **daisy/soil/soil.h** component (see chapter 31)
Component (default 'neutral')
The soil pH.
- *Movement*: **movement** component (see chapter 56)
Component (default 'vertical')
Discretization and movement of water, heat and solutes in the soil.
- *Vegetation*: **vegetation** component (see chapter 100)
Component (default 'crops')
The crops on the field.
- *Litter*: **litter** component (see chapter 51)
Component (default 'none')
The litter layer below the canopy.
- *Bioclimate*: **bioclimate** component (see chapter 19)
Component (default 'default')
The water and energy distribution among the crops.
- *Drain*: **drain** component (see chapter 37)
Component (default 'none')
Drainage.
- *Groundwater*: **groundwater** component (see chapter 45)
The groundwater level.
- *Chemistry*: **chemistry** component (see chapter 24)
Component (default 'nutrient')
Chemical compounds in the system.
- *OrganicMatter*: **organic** component (see chapter 60)
Component (default 'SOM2025')
The organic matter in the soil and on the surface.
- *tillage_age*: number [**d**] soil cells
Optional state variable
Time since the latest tillage operation was performed. By default, the top 25 cm will have an initial tillage age of 100 days, while the soil below that will have an initial tillage age of 100 years. If you specify fewer values than there are soil cells, the last specified value will be used for the remaining cells.

Log Variables

- *first_year_utilization*: number [**kg N/ha/h**]
Estimated first year fertilizer effect.
- *yield_DM*: number [**g/m²/h**]
Amount of DM removed as yield this hour. This is the economic part of the storage organ, e.g. the grains.
- *yield_N*: number [**g/m²/h**]
Amount of nitrogen removed as yield this hour. This is the economic part of the storage organ, e.g. the grains.

- *harvest_DM*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of DM removed by harvest this hour.
- *harvest_N*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of nitrogen removed by harvest this hour.
- *harvest_C*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of carbon removed by harvest this hour.
- *residuals_DM*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of dry matter removed from crops to surface and soil this hour. This includes loss as harvest, as well as loss of old leaves and roots.
- *residuals_N_top*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of nitrogen removed from crops to soil this hour. This includes loss as harvest, as well as loss of old leaves.
- *residuals_C_top*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of carbon removed from crops to surface this hour. This includes loss as harvest, as well as loss of old leaves.
- *residuals_N_soil*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Amount of nitrogen removed from crops in soil this hour. This includes loss as harvest, as well as loss of old roots.
- *residuals_C_soil*: number $[\text{g}/\text{cm}^3/\text{h}]$ soil cells
Amount of carbon removed from crops in soil this hour. This includes loss as harvest, as well as loss of old roots.
- *residuals_N_root*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of nitrogen removed from crops to soil this hour. This includes loss as harvest, as well as loss of old roots.
- *residuals_C_root*: number $[\text{g}/\text{m}^2/\text{h}]$
Amount of carbon removed from crops to surface this hour. This includes loss as harvest, as well as loss of old roots.
- *surface_water*: number $[\text{mm}]$
Amount of water in the system above ground. This include ponded water, intercepted water and the snow pack.
- *seed_N*: number $[\text{kg N}/\text{ha}/\text{h}]$
Amount of nitrogen in seed applied this time step.
- *seed_C*: number $[\text{kg C}/\text{ha}/\text{h}]$
Amount of carbon in seed applied this time step.
- *applied_DM*: number $[\text{ton DM}/\text{ha}/\text{h}]$
Amount of dry matter applied this time step. This includes dry matter incorporated directly in the soil.

Chapter 27

condedge

Find the hydraulic conductivity between two cells.

27.1 arithmetic

Use the arithmetic average of the conductivity in the two cells.

Used by uzrect Mollerup `K_average` (see 99.1, page 537) .

27.2 geometric

Geometric average `'sqrt(a*b)'`.

27.3 harmonic

Use harmonic average of the conductivity of the two cells. This corresponds to using the average hydraulic resistance.

27.4 pressure

Pressure dependent average of the two cells. Use harmonic average of the conductivity of unsaturated cells. This corresponds to using the average hydraulic resistance. For saturated cells, water may stream into unsaturated neighbor cells with saturated conductivity if `'allow_sideways'` is true. For cells where pressure is above `'h_lim'`, water may stream downward to dryer cell with a conductivity corresponding to `'h_lim'`.

```
< pressure (allow_sideways true)
           (use_h_old true)
           (h_lim h_lim) >
```

- *allow_sideways*: boolean (see section 4.1.2)
Parameter (default true)
Allow water to flow fast from saturated cells to all neighbor cells. Not just the cell below.
- *use_h_old*: boolean (see section 4.1.2)
Parameter (default true)
Use pressure at the start of the small timestep for enabling fast flow. If false, use the pressure at end of the small timestep.

- *h_lim*: number [**cm**]
Parameter
Lower pressure limit for fast downward flow.

Chapter 28

condition

A 'condition' component tests the state of the simulation, like whether the water pressure in a specific depth is above a given threshold. Logic conditions like 'and' and 'or' can be used for testing whether multiple conditions are fulfilled simultaneously.

Used by select component when (see 80, page 427) .

28.1 BBCH_after

True after specified BBCH.

```
< BBCH_after (crop crop)  
              (file file)  
              (start start)  
              (bbch bbch) >
```

- *crop*: string (see section 4.1.5)
Parameter
Name of crop to use BBCH number for.
- *file*: string (see section 4.1.5)
Parameter
Name of file where data is found.
- *start*: boolean (see section 4.1.2)
Parameter
If true, the interval begins when the crop enters the specified BBCH. Otherwise, the interval begins when the crop leaves the specified BBCH.
- *bbch*: integer
Parameter
BBCH to test for.

28.2 TSum_above

Test if the temperature sum is above the specified value The temperature sum is the sum of the daily average air temperature since last reset. It is reset once a year. Days where the average is below 0 does not count in the sum.

```

< TSum_above  TSum_limit
               (check_hour 6)
               (reset_mday 1)
               (reset_month 3)
               (TSum_now TSum_now) >

```

- *TSum_limit*: number [dg C d]
Parameter
Temperature sum above which the condition becomes true.
- *check_hour*: integer
Parameter (default 6)
Hour in day to update TSum.
- *reset_mday*: integer
Parameter (default 1)
Day in month to reset TSum.
- *reset_month*: integer
Parameter (default 3)
Month in year to reset TSum.
- *TSum_now*: number [dg C d]
Optional state variable
Current temperature sum since last reset.

28.3 and

True iff all the listed conditions are true. The conditions are tested in the sequence listed, until a false is found, or the end of the list is reached.

```

< and  operands ... >

```

- *operands*: **condition** component (see chapter 28) sequence
Conditions to test.

28.4 trafficable

A ‘and’ model (see 28.3, page 160) defined in ‘tillage.dai’.

28.5 check

Test if a boolean expression is true.

```

< check  expr >

```

- *expr*: **boolean** component (see chapter 21)
Expression to evaluate.

28.6 crop_dm_over

True iff the crop has reached the specified amount of dry matter.

```

< crop_dm_over  crop_weight
                 (height 0 [cm]) >

```


- *crop*: string (see section 4.1.5)
Parameter
Name of crop on the field to test.
- *weight*: number [kg DM/ha]
Parameter
Amount of non-root dry-matter required for the condition to be true.
- *height*: number [cm]
Parameter (default 0)
Height above which we measure the DM weight.

28.7 crop_dm_sorg_over

True iff the storage organ has reached the specified amount of dry matter.

< crop_dm_sorg_over *crop weight* >

- *crop*: string (see section 4.1.5)
Parameter
Name of crop on the field to test.
- *weight*: number [kg DM/ha]
Parameter
Amount of non-root dry-matter required for the condition to be true.

28.8 crop_ds_after

True iff the crop has reached development stage 'ds'.

< crop_ds_after *crop ds* >

- *crop*: string (see section 4.1.5)
Parameter
Name of crop on the field to test. Specify "all" to use combined weight of all crops on the field in test.
- *ds*: number (dimensionless)
Parameter
Development stage [-1.0:2.0].

28.9 crop_stage_after

True iff the crop has reached development stage 'stage'.

< crop_stage_after *crop stage* >

- *crop*: string (see section 4.1.5)
Parameter
Name of crop on the field to test. Specify "all" to use combined weight of all crops on the field in test.
- *stage*: number (dimensionless)
Parameter
Crop specific phenological stage.

28.10 daily

True at the end of each day.

Used by log Crop when (see 52.16, page 299) .

28.11 daily_air_temperature_above

Test if the daily air is warmer than the specified temperature.

```
< daily_air_temperature_above temperature >
```

- *temperature*: number [**dg C**]
Parameter
Lowest air temperature for which the condition is true.

28.12 daily_precipitation_above

Test if the daily precipitation is warmer than the specified value.

```
< daily_precipitation_above precipitation >
```

- *precipitation*: number [**mm**]
Parameter
Lowest precipitation for which the condition is true.

28.13 every

Matches simulation with fixed time intervals.

```
< every (seconds 0)
        (minutes 0)
        (hours 0)
        (days 0)
        (microseconds 0)
        (next next) >
```

- *seconds*: integer
State variable (default 0)
Number of seconds.
- *minutes*: integer
State variable (default 0)
Number of minutes.
- *hours*: integer
State variable (default 0)
Number of hours.
- *days*: integer
State variable (default 0)
Number of days.
- *microseconds*: integer
State variable (default 0)
Number of microseconds.
- *next*: **Time** fixed component (see section 107.27)
Optional submodel
Time for next match.

28.14 **extern**

Test if a boolean expression is true, using *extern* log.

`< extern scope expr >`

- *scope*: **scopesel** component (see chapter 77)
Scope to evaluate expression in.
- *expr*: **boolean** component (see chapter 21)
Expression to evaluate.

28.15 **false**

Always false.

Used by program *Daisy* *stop_when* (see 66.2, page 353) .

28.16 **finished**

True iff the simulation has finished.

Used by log checkpoint when (see 52.1, page 281) .

28.17 **hour**

True, at the specified hour.

`< hour at >`

- *at*: integer
Parameter
Hour when the condition is true [0-23].

28.18 **hourly**

True at the end of each hour.

Used by log Field water when (see 52.8, page 289) , log Soil water when (see 52.10, page 293) , log Field nitrogen when (see 52.7, page 286) , log Soil nitrogen when (see 52.9, page 290) , log Field chemical when (see 52.13, page 295) , and log Soil chemical when (see 52.14, page 297) .

28.19 **if**

If the first condition is true, return the value of the second condition, else return the value of the third condition.

`< if if then else >`

- *if*: **condition** component (see chapter 28)
Condition to test for.
- *then*: **condition** component (see chapter 28)
Condition to use if the 'if' test was true.
- *else*: **condition** component (see chapter 28)
Condition to use if the 'if' test was false.

28.20 mday

True, at the specified day in the month.

```
< mday  at  >
```

- *at*: integer
Parameter
Day in the month when the condition is true [1-31].

28.21 microsecond

True, at the specified microsecond.

```
< microsecond  at  >
```

- *at*: integer
Parameter
Microsecond when the condition is true [0-999999].

28.22 minute

True, at the specified minute.

```
< minute  at  >
```

- *at*: integer
Parameter
Minute when the condition is true [0-59].

28.23 minutely

True at the end of each minute.

28.24 mm_dd_base

Conditions based on month and day.

```
< mm_dd_base  month day
                (second 0)
                (minute 0)
                (hour 8) >
```

- *month*: integer
Parameter
Month to test for.
- *day*: integer
Parameter
Day in the month to test for.
- *second*: integer
Parameter (default 0)
Second to test for.
- *minute*: integer
Parameter (default 0)
Minute to test for.

- *hour*: integer
Parameter (default 8)
Hour to test for.

28.25 after_mm_dd

A ‘mm_dd_base’ model (see 28.24, page 164) build into Daisy.
True after specific month, day and hour in the year.

28.26 before_mm_dd

A ‘mm_dd_base’ model (see 28.24, page 164) build into Daisy.
True before specific month, day and hour in the year.

28.27 mm_dd

A ‘mm_dd_base’ model (see 28.24, page 164) build into Daisy.
True a specific month, day and hour in the year.

28.28 month

True, at the specified month.

< month *at* >

- *at*: integer
Parameter
Month when the condition is true [1-12].

28.29 monthly

True at the end of each month.

28.30 not

True iff the operand is not true.

Used by condition trafficable operands (see 28.4, page 160) .

< not *operand* >

- *operand*: **condition** component (see chapter 28)
Condition to test.

28.31 or

True iff any of the listed conditions are true. The conditions are tested in the sequence listed, until a true is found, or the end of the list is reached.

< or *operands* ... >

- *operands*: **condition** component (see chapter 28) sequence
Conditions to test.

28.32 periodic

True if more than a specified walltime has passed since last time it was true.

Used by program Daisy print_time (see 66.2, page 353) .

< periodic *period* >

- *period*: integer
Parameter (default 1)
Number of walltime seconds between success.

28.33 running

True iff the simulation is still running.

28.34 second

True, at the specified second.

< second *at* >

- *at*: integer
Parameter
Second when the condition is true [0-59].

28.35 secondly

True at the end of each second.

28.36 soil_inorganic_N_above

Test if the soil contains more mineral nitrogen than the specified amount.

< soil_inorganic_N_above *amount*
(from 0 [cm])
(to *to*) >

- *amount*: number [kg N/ha]
Parameter
The soil should contain more inorganic nitrogen than this for the condition to be true.
- *from*: number [cm]
Parameter (default 0)
Top of interval to measure soil content in.
- *to*: number [cm]
Parameter
Bottom of interval to measure soil content in.

28.37 soil_temperature_above

Test if the soil is warmer than the specified temperature.

Used by condition trafficable operands (see 28.4, page 160) .

< soil_temperature_above (height *height*)
(temperature *temperature*) >

- *height*: number [**cm**]
Parameter
Soil depth in which to test the temperature.
- *temperature*: number [**dg C**]
Parameter
Lowest soil temperature for which the condition is true.

28.38 soil_water_content_above

Test if the soil contains more water than the specified amount.

```
< soil_water_content_above  water
                               (from 0 [cm])
                               (to to) >
```

- *water*: number [**mm**]
Parameter
The soil should contain more water than this for the condition to be true.
- *from*: number [**cm**]
Parameter (default 0)
Top of interval to measure soil water content in.
- *to*: number [**cm**]
Parameter
Bottom of interval to measure soil water content in.

28.39 soil_water_pressure_above

Test if the soil is wetter than the specified pressure potential.

```
< soil_water_pressure_above  (potential potential)
                               (height height) >
```

- *potential*: number [**cm**]
Parameter
The soil should be wetter than this for the condition to be true.
- *height*: number [**cm**]
Parameter
Depth at which to example the pressure potential.

28.40 time

Conditions based on a specific time.

```
< time  time  >
```

- *time*: **Time** fixed component (see section 107.27)
Submodel (has partially specified default value)
Fixed time to test for.

28.41 after

A ‘time’ model (see 28.40, page 167) build into Daisy.

True, iff the simulation time is after the specified time.

28.42 at

A ‘time’ model (see 28.40, page 167) build into Daisy.

True, iff the simulation time is at the specified time.

28.43 before

A ‘time’ model (see 28.40, page 167) build into Daisy.

True, iff the simulation time is before the specified time.

28.44 timestep

Add a timestep to a condition. It is true whenever ‘operand’ is true, but will let Daisy know what ‘timestep’ it represents. The timestep is used for the dimension in log files.

```
< timestep operand timestep >
```

- *operand*: **condition** component (see chapter 28)
Condition to use.
- *timestep*: string (see section 4.1.5)
Parameter
Timestep to use.

28.45 true

Always true.

Used by program Daisy activate_output (see 66.2, page 353) , and log select active (see 52.3, page 283) .

28.46 walltime

True if more than a specified walltime has passed since last time it was true.

```
< walltime (seconds 0)
           (minutes 0)
           (hours 0)
           (days 0)
           (microseconds 0) >
```

- *seconds*: integer
State variable (default 0)
Number of seconds.
- *minutes*: integer
State variable (default 0)
Number of minutes.
- *hours*: integer
State variable (default 0)
Number of hours.
- *days*: integer
State variable (default 0)
Number of days.

- *microseconds*: integer
State variable (default 0)
Number of microseconds.

28.47 weekly

True at the end of each week.

28.48 with

Test condition for specific column.

< with *where condition* >

- *where*: string (see section 4.1.5)
Parameter
Name of column to test condition on.
- *condition*: **condition** component (see chapter 28)
Condition to test on the specified column.

28.49 yday

True, at the specified julian day.

< yday *at* >

- *at*: integer
Parameter
Julian day when the condition is true [1-366].

28.50 year

True, at the specified year.

< year *at* >

- *at*: integer
Parameter
Year when the condition is true.

28.51 yearly

True at the end of each year.

Chapter 29

crop

The 'crop' component simulates a specific crop on the field, typically averaged over one square meter, not individual plants. Of particular interest is water and nitrogen uptake at different depths, and the vertical leaf area distribution, which are used for competition with other crops.

29.1 default

Standard Daisy crop model. Hansen, 1999.

```
< default  (CrpN CrpN)                                ; Has partial value.
            (Harvest Harvest)                          ; Has default value.
            (Seed Seed)                                ; Default LAI value.
            (Root Root)                                ; Default classic value.
            (Canopy Canopy)                            ; Has partial value.
            (Prod Prod)                                ; Has partial value.
            (last_time last_time)
            (sow_time sow_time)
            (emerge_time emerge_time)
            (flowering_time flowering_time)
            (ripe_time ripe_time)
            (Devel Devel)
            (CStage CStage)                            ; Default Daisy value.
            (Partit Partit)                            ; Has partial value.
            (Vernal Vernal)                            ; Default none value.
            (LeafPhot LeafPhot)                       ; Default GL value.
            (water_stress_effect water_stress_effect)
            (enable_N_stress enable_N_stress)
            (min_light_fraction 0 [<fraction>])
            (RubiscoN RubiscoN)                       ; Default default value. >
```

- *CrpN*: **CrpN** fixed component (see section 107.6)
Submodel (has partially specified default value)
Nitrogen parameters.
- *Harvest*: **Harvesting** fixed component (see section 107.7)
Submodel (has fully specified default value)
Harvest parameters.
- *Seed*: **seed** component (see chapter 79)
Component (default 'LAI')
Initial crop growth.

- *Root*: **root** component (see chapter 72)
Component (default ‘classic’)
Root system.
- *Canopy*: **CanopyStandard** fixed component (see section 107.5)
Submodel (has partially specified default value)
Canopy.
- *Prod*: **Production** fixed component (see section 107.9)
Submodel (has partially specified default value)
Production.
- *last_time*: **Time** fixed component (see section 107.27)
Optional submodel
The time of the previous timestep. Don’t set, calculated by Daisy.
- *sow_time*: **Time** fixed component (see section 107.27)
Optional submodel
The time the crop was sown. Don’t set, calculated by Daisy.
- *emerge_time*: **Time** fixed component (see section 107.27)
Optional submodel
The time the crop emerged. Don’t set, calculated by Daisy.
- *flowering_time*: **Time** fixed component (see section 107.27)
Optional submodel
The time the crop flowered. Don’t set, calculated by Daisy.
- *ripe_time*: **Time** fixed component (see section 107.27)
Optional submodel
The time the crop became ripe. Don’t set, calculated by Daisy.
- *Devel*: **phenology** component (see chapter 63)
Development and phenology.
- *CStage*: **cstage** component (see chapter 30)
Component (default ‘Daisy’)
Phenological messages.
- *Partit*: **Partition** fixed component (see section 107.8)
Submodel (has partially specified default value)
Assimilate partitioning.
- *Vernal*: **vernalization** component (see chapter 101)
Component (default ‘none’)
Vernalization.
- *LeafPhot*: **photosynthesis** component (see chapter 64)
Component (default ‘GL’)
Leaf photosynthesis. Note that if the selected radiation distribution model distinguishes between sunlit and shadow leaves, only the shadow leaves will be
- *water_stress_effect*: **wse** component (see chapter 104)
Optional component
Effect of water stress on production. By default, this will be ‘none’ iff the selected photosynthesis model does handle water stress implicitly, and ‘full’ otherwise.

- *enable_N_stress*: boolean (see section 4.1.2)
Optional parameter
Set this true to let nitrogen stress limit production. By default, it will be true iff the selected photosynthesis model does handle nitrogen stress implicitly.
- *min_light_fraction*: number [**<fraction>**]
Parameter (default 0)

When multiple crops are competing for light, this parameter specifies a minimum amount of the light this crop will receive. The idea is that the field has patches where one crop is dominating, as specified by this parameter, and in these patches the crop will not have to compete for light. The crop still needs LAI in order to catch the light though. Competition for water and nutrients are unaffected.

- *RubiscoN*: **rubiscoN** component (see chapter 74)
Component (default 'default')
Fraction of N in leaves that is photosynthetically active.

Log Variables

- *sunlit*: **photosynthesis** component (see chapter 64)
Leaf photosynthesis for sunlit leaves. This will be zero if the selected radiation distribution model does not distinguish between sunlit and shadow leaves.
- *reserved*: **photosynthesis** component (see chapter 64)
Leaf photosynthesis for reserved leaves. This is used for simulating "patches" in multi-crop systems, such as a clover-grass mixture. This is controlled by the 'min_light_fraction' parameter.

29.2 Beetroot

A 'default' model (see 29.1, page 171) defined in 'beetroot.dai'.

For AArhus County - Samsøe Project

29.3 Broccoli

A 'default' model (see 29.1, page 171) defined in 'broccoli.dai'.

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29.4 Broccoli - transplanted

A 'Broccoli' model (see 29.3, page 173) defined in 'broccoli.dai'.

29.5 Brussels sprouts

A 'default' model (see 29.1, page 171) defined in 'brusselssprouts.dai'.

For AArhus County - Samsøe Project

29.6 Brussels sprouts - transplanted

A 'Brussels sprouts' model (see 29.5, page 173) defined in 'brusselssprouts.dai'.

29.7 Celeriac

A ‘default’ model (see 29.1, page 171) defined in ‘celeriacy.dai’.
For AArhus County - Samsøe Project

29.8 Celeriac - transplanted

A ‘Celeriac’ model (see 29.7, page 174) defined in ‘celeriacy.dai’.

29.9 Fodder Beet

A ‘default’ model (see 29.1, page 171) defined in ‘fodderbeet.dai’.

29.10 Grass to grain

A ‘default’ model (see 29.1, page 171) defined in ‘grass.dai’.

29.11 Grass

A ‘Grass to grain’ model (see 29.10, page 174) defined in ‘grass.dai’.

29.12 Maize

A ‘default’ model (see 29.1, page 171) defined in ‘maize.dai’.
Roskilde 74-76 ukendt varietet

29.13 Ikuwala Maize

A ‘Maize’ model (see 29.12, page 174) defined in ‘maize.dai’.
SASA Project

29.14 Silage Maize

A ‘Maize’ model (see 29.12, page 174) defined in ‘maize.dai’.
LG11

29.15 Onion

A ‘default’ model (see 29.1, page 171) defined in ‘onion.dai’.
For AArhus County - Samsøe Project

29.16 Onion - planting of sets

A ‘Onion’ model (see 29.15, page 174) defined in ‘onion_plantingofsets.dai’.

29.17 Pea

A ‘default’ model (see 29.1, page 171) defined in ‘pea.dai’.

29.18 Green Pea

A ‘Pea’ model (see 29.17, page 174) defined in ‘pea.dai’.

29.19 Pioneer Maize

A ‘default’ model (see 29.1, page 171) defined in ‘maize.dai’.

29.20 Potato; Koege

A ‘default’ model (see 29.1, page 171) defined in ‘potato.dai’.
Koege Project

29.21 Potato; SCRI

A ‘default’ model (see 29.1, page 171) defined in ‘potato.dai’.
Potato parameterized by sha@kvl.dk – 2003. It is based on experimental data from SCRI from 1983 and 1984.

29.22 Potato; FertOrgaNic

A ‘Potato; SCRI’ model (see 29.21, page 175) defined in ‘potato.dai’.
These parameters represent common traits from various site specific calibrations performed as part of the FertOrgaNic project from 2003-2005. It is not recommended for direct use, but as a base for a variety specific calibration.
The calibration was performed by Charlotte Tofteng <cto@kvl.dk> in 2006.

29.23 Potato; Agria

A ‘Potato; FertOrgaNic’ model (see 29.22, page 175) defined in ‘potato.dai’.
Recalibration of Agria potato performed by Charlotte Tofteng <cto@kvl.dk> in 2006 based on experimental results from Czechia (2003-2005), Slovakia (2003) and Portugal (2003).

29.24 Potato; Folva

A ‘Potato; FertOrgaNic’ model (see 29.22, page 175) defined in ‘potato.dai’.
Recalibration of Agria potato performed by Tove Heidman in 2006 based on experimental results from DJF.

29.25 Potato

A ‘Potato; Folva’ model (see 29.24, page 175) defined in ‘potato.dai’.

29.26 Potato; Triada

A ‘Potato; FertOrgaNic’ model (see 29.22, page 175) defined in ‘potato.dai’.
Recalibration of Triada potato performed by Charlotte Tofteng <cto@kvl.dk> based on experimental results from Jadwisin, Poland 2003-2005.

29.27 Potato; SCRI - AArhus

A ‘default’ model (see 29.1, page 171) defined in ‘earlypotato.dai’.

Potato parameterized by sha@kvl.dk – 2003. It is based on experimental data from SCRI from 1983 and 1984.

29.28 Early potato

A ‘Potato; SCRI - AArhus’ model (see 29.27, page 176) defined in ‘earlypotato.dai’.

For AArhus County - Samsøe Project

29.29 Rye

A ‘default’ model (see 29.1, page 171) defined in ‘rye.dai’.

29.30 Ryegrass

A ‘default’ model (see 29.1, page 171) defined in ‘ryegrass.dai’.

ryegrass v.27 Henning Hoegh Jensen

29.31 Spring Barley

A ‘default’ model (see 29.1, page 171) defined in ‘sbarley.dai’.

29.32 Spring Barley; Foulum

A ‘Spring Barley’ model (see 29.31, page 176) defined in ‘sbarley.dai’.

RS-Model Projekt

29.33 Spring Rape

A ‘default’ model (see 29.1, page 171) defined in ‘srape.dai’.

29.34 Spring Wheat

A ‘default’ model (see 29.1, page 171) defined in ‘swheat.dai’.

Koege Project

29.35 Sugar Beet

A ‘default’ model (see 29.1, page 171) defined in ‘sugarbeet.dai’.

Neuenkirchen Project

29.36 Wclover

A ‘default’ model (see 29.1, page 171) defined in ‘wclover.dai’.

v.30 Developed by Henning Hoegh Jensen

29.37 White cabbage

A ‘default’ model (see 29.1, page 171) defined in ‘whitecabbage.dai’.
For Aarhus County - Samsøe Project

29.38 White cabbage - transplanted

A ‘White cabbage’ model (see 29.37, page 177) defined in ‘whitecabbage.dai’.

29.39 Early white cabbage - transplanted

A ‘White cabbage - transplanted’ model (see 29.38, page 177) defined in ‘early-whitecabbage.dai’.

29.40 Winter Barley

A ‘default’ model (see 29.1, page 171) defined in ‘wbarley.dai’.

29.41 Winter Barley; Foulum

A ‘Winter Barley’ model (see 29.40, page 177) defined in ‘wbarley.dai’.

29.42 Winter Barley; Koge

A ‘default’ model (see 29.1, page 171) defined in ‘wbarley.dai’.
Koege Project

29.43 Winter Rape

A ‘default’ model (see 29.1, page 171) defined in ‘wrape.dai’.

29.44 Winter Wheat

A ‘default’ model (see 29.1, page 171) defined in ‘wwheat.dai’.

29.45 Winter Wheat JG

A ‘Winter Wheat’ model (see 29.44, page 177) defined in ‘wwheat.dai’.
Mainly calibrated on the variety Benchmark See also [Gyldengren et al., 2020]

29.46 Winter Wheat; Foulum

A ‘Winter Wheat’ model (see 29.44, page 177) defined in ‘wwheat.dai’.
RS-Model Project

29.47 simple

Forced growth crop model.

```
< simple (Root Root) ; Default classic value.
      (Canopy Canopy) ; Has default value.
      (LAIvsTS LAIvsTS)
      (LAIvsDay LAIvsDay)
      (forced_LAI 0 [m2/m2])
      (height_max 80 [cm])
      (T_sum 0 [dg C d])
      (day 0 [d])
      (spring spring ...) ; Has default value.
      (spring_LAI 0.1 [m2/m2])
      (root_DM 2 [Mg DM/ha])
      (root_N 20 [kg N/ha])
      (root_am root.am ...) ; Has default value.
      (potential_N potential.N)
      (N_actual 0 [g N/m2])
      (N_b 10 [kg N/ha])
      (N_flowering 0.9 [<fraction>]) >
```

- *Root*: **root** component (see chapter 72)
Component (default ‘classic’)
Root system.
- *Canopy*: **CanopySimple** fixed component (see section 107.4)
Submodel (has fully specified default value)
Canopy.
- *LAIvsTS*: plf [**dg C d** → **m²/m²**]
Optional parameter
LAI as a function of T_sum
- *LAIvsDay*: plf [**d** → **m²/m²**]
Optional parameter
LAI as a function of number of days since sowing.
- *forced_LAI*: number [**m²/m²**]
State variable (default 0)
Minimum LAI, automatically cleared when exceeded by ‘LAIvsTS’.
- *height_max*: number [**cm**]
Parameter (default 80)
Maximum height of plant, reached when flowering.
- *T_sum*: number [**dg C d**]
State variable (default 0)
Temperature sum since sowing (or spring).
- *day*: number [**d**]
State variable (default 0)
Number of days since sowing (or spring).
- *spring*: integer array of length 2
Parameter (has default value with length 2)

```
(spring 3 1)
```

Parameter description:

Zero 'T_sum' at this month and day.

- *spring_LAI*: number [m^2/m^2]
Parameter (default 0.1)
Set 'forced_LAI' to this after spring clearence of 'T_sum'.
- *root_DM*: number [Mg DM/ha]
Parameter (default 2)
Fully developed root drymatter.
- *root_N*: number [kg N/ha]
Parameter (default 20)
Fully developed root N content.
- *root_am*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(root_am "AOM-SLOW"
         "AOM-FAST")
```

Parameter description:

Root AM parameters.

- *potential_N*: number [kg N/ha]
Parameter
Potential N content at harvest.
- *N_actual*: number [g N/m^2]
State variable (default 0)
N uptake until now.
- *N_b*: number [kg N/ha]
Parameter (default 10)
N uptake form parameter.
- *N_flowering*: number [<fraction>]
Parameter (default 0.9)
Fraction of potential N uptake reached at flowering.

Log Variables

- *N_demand*: number [g N/m^2]
Current potential N content.

Chapter 30

cstage

Crop phenological stage. The 'default' crop model uses a continuous development stage [-1:2] in order to track the phenological age of a crop. The 'cstage' library allows translation of the internal model to other phenological models.

30.1 table

A table mapping Daisy DS to other model.

```
< table (DS -1 [DS])  
      (table table ...)  
      (value value) >
```

- *DS*: number [DS]
State variable (default -1)
Last Daisy development stage.
- *table*: submodel (see section 4.1.7) sequence
List of Daisy and other model stages

```
< DS value message >
```

 - *DS*: number [DS]
Parameter
Daisy development stage where this stage starts.
 - *value*: number (dimension not specified)
Parameter
Numeric value of stage in other model.
 - *message*: string (see section 4.1.5)
Optional parameter
Message to display when entering this stage.
- *value*: number (dimension not specified)
State variable
Current stage in this phenological model.

30.2 BBCH

A 'table' model (see 30.1, page 181) build into Daisy.

Parent parameterization for all BBCH derived cropstage definitions.

```
< (value 0 [?]) >
```

30.3 Daisy

A ‘table’ model (see 30.1, page 181) build into Daisy.

DS 0, 1, and 2 have meaning.

Used by crop default CStage (see 29.1, page 171) .

```
< (table table ...) ; Has default value.  
  (value -1 [?]) >
```

Chapter 31

daisy/soil/soil.h

pH of soil.

< component >

Log Variables

- *pH*: number [**pH**] soil cells
Current pH of soil.

31.1 neutral

A soil with a constant pH of 7.

Used by column default SoilpH (see 26.1, page 153) .

31.2 year

A soil with a constant pH of 7.

< year (pH.depth_year *pH.depth_year* ...) >

- *pH.depth_year*: submodel (see section 4.1.7) sequence
pH as a function of depth and calendar year. If specified, Daisy will interpolate between depth and between years. By default, a pH of 7 is assumed.

< *year pH.depth* >

– *year*: integer

Parameter

Calendar year.

– *pH.depth*: plf [**cm** → **pH**]

Parameter

pH as a function of depth.

Chapter 32

denprod

Find products (gasses) produced by denitrification.

32.1 Parton1996

Find N₂O from denitrification based on WFPS, NO₃, and CO₂.

Used by reaction denitrification denprod (see 70.11, page 390) .

```
< Parton1996 (FR_NO3 FR_NO3) ; Default P96NO3 value.
              (FR_CO2 FR_CO2) ; Default P96CO2 value.
              (FR_WFPS FR_WFPS) ; Default P96WFPS value.
              (FR_CO2_depth 30 [cm]) >
```

- *FR_NO3*: function [**ug N/g** → **<none>**] (see chapter 42)
Parameter (default 'P96NO3')
NO₃ effect N₂O.
- *FR_CO2*: function [**kg C/ha/d** → **<none>**] (see chapter 42)
Parameter (default 'P96CO2')
CO₂ effect N₂O.
- *FR_WFPS*: function [**<fraction>** → **<none>**] (see chapter 42)
Parameter (default 'P96WFPS')
WFPS effect N₂O.
- *FR_CO2_depth*: number [**cm**]
Parameter (default 30)
Depth used for converting CO₂ from volume to area units.

Log Variables

- *N2O*: number [**g N/cm³/h**] sequence
Amount of N₂O-N generated by denitrification.

Chapter 33

deposition

Deposition of inorganic material from atmosphere.

< component >

Log Variables

- *deposit*: submodel (see section 4.1.7) sequence
Total atmospheric deposition.

< name value > **Log Variables**

- *value*: number [g/cm²/h]
Value for chemical.
- *name*: string (see section 4.1.5)
Name of chemical.

33.1 const

Constant deposition in time interval.

< const (before *before*) ; Has partial value.
(after *after*) ; Has partial value.
(wet)
(dry) >

- *before*: **Time** fixed component (see section 107.27)
Submodel (has partially specified default value)
Only deposit before or at this time.
- *after*: **Time** fixed component (see section 107.27)
Submodel (has partially specified default value)
Only deposit after this time.
- *wet*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Solutes in precipitation.

< name value >

- *name*: string (see section 4.1.5)
Parameter
Name of chemical.

- *value*: number [**ppm**]
Parameter
Value for chemical.
- *dry*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Dry deposition.

```
<  name  value  >
```

 - *name*: string (see section 4.1.5)
Parameter
Name of chemical.
 - *value*: number [**kg/ha/y**]
Parameter
Value for chemical.

33.2 weather

Rely solely on weather model for deposition.

Used by bioclimate default deposition (see 19.2, page 115) .

Chapter 34

depth

Depth below soil surface.

34.1 PLF

Linear interpolation of depth.

< PLF *table...* >

- *table*: submodel (see section 4.1.7) sequence
Height as a function of time. This is a list where each element has the form (TIME VALUE). The TIME entries must be increasing cronologically. The corresponding VALUE represents the value at that time. In order to find the depth for other times, linear interpolation between the entries in the list will be used.

< *time value* >

- *time*: **Time** fixed component (see section 107.27)
Submodel (has partially specified default value)
Time.
- *value*: number [**cm**]
Parameter
Depth.

34.2 const

Constant depth.

< const *value* >

- *value*: number [**cm**]
Parameter
Constant depth.

34.3 deep

A ‘const’ model (see 34.2, page 189) build into Daisy.

Way down.

Used by drain lateral pipe_outlet (see 37.1, page 197) , and movement rectangle pipe_outlet (see 56.2, page 307) .

< *value* >

34.4 extern

Look up depth in an scope.

```
< extern (value value)
        (initial_value initial_value) >
```

- *value*: **number** component (see chapter 59)
Expression that evaluates to a depth.
- *initial_value*: number [**cm**]
Optional parameter
Initial depth.

34.5 file

Linear interpolation of depth read from file.

```
< file file >
```

- *file*: string (see section 4.1.5)
Parameter
Name of file to read data from. The format of each line in the file is 'YEAR MONTH DAY HEIGHT', where HEIGHT should in cm above ground (i.e. a negative number). Linear interpolation is used between the datapoints.

34.6 season

Linear interpolation of depth within a year.

```
< season season >
```

- *season*: plf [**d** → **cm**]
Parameter
Depth as a function of Julian day. First and last entry must be identical.

34.7 table

Linear interpolation of depth read from file.

```
< table (file file)
        (offset 0 [cm])
        (missing missing ...) ; Has default value.
        (filter)
        (original original ...)
        (dim_line dim_line) >
```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *offset*: number [**cm**]
Parameter (default 0)
Add this number to the Level read from the table.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

(missing "" "00.00")

Parameter description:

List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

`< tag allowed... >`

- *tag*: string (see section 4.1.5)

Parameter

Name of column in Daisy log file to filter for.

- *allowed*: string (see section 4.1.5) sequence

Parameter

List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.

Chapter 35

difrad

The 'difrad' component should calculate the diffuse radiation from meteorological data.

35.1 BRL

Diffuse radiation calculation based on the model of Ridley et al., 2010. Last predictor (persistence) is not taken into account. See also [Ridley et al., 2010]

```
< BRL  (beta0 -2.4 [])  
        (beta1 2.77 [])  
        (beta2 0.033 [])  
        (beta3 0.054 [])  
        (beta4 1.028 []) >
```

- *beta0*: number (dimensionless)
Parameter (default -2.4)
1st parameter (constant) of BRL model.
- *beta1*: number (dimensionless)
Parameter (default 2.77)
2nd parameter (hourly clearness index) of BRL model.
- *beta2*: number (dimensionless)
Parameter (default 0.033)
3rd parameter (apparent solar time) of BRL model.
- *beta3*: number (dimensionless)
Parameter (default 0.054)
4th parameter (solar altitude) of BRL model.
- *beta4*: number (dimensionless)
Parameter (default 1.028)
5th parameter (daily clearness index) of BRL model.

35.2 DPF

Diffuse radiation calculated using the model of De Pury and Farquhar, 1997. See also [de Pury and Farquhar, 1997]

```
< DPF  (a 0.72 [])  
        (fa 0.426 [<fraction>]) >
```

- a : number (dimensionless)
Parameter (default 0.72)
Value description: Table 5 in cited paper.
Atmospheric transmission coefficient of PAR. Value around 0.6-0.9 depending on dust particles.
- fa : number [**<fraction>**]
Parameter (default 0.426)
Value description: Table 2 in cited paper.
Forward scattering coefficient of PAR in the atmosphere.

35.3 weather

Diffuse radiation using weather data.

Chapter 36

domsorp

Sorption and desorption of DOM to SOM.

36.1 default

Transformation between two soil chemicals.

```
< default (transform transform)  
          (dom_pool dom_pool)  
          (som_pool som_pool) >
```

- *transform*: **transform** component (see chapter 91)
Transformation process between DOM and SOM.
- *dom_pool*: integer
Parameter
Number of the DOM pool affected by the transformation.
- *som_pool*: integer
Parameter
Number of the SOM pool affected by the transformation.

Log Variables

- *S_C*: number [g C/cm³/h] soil cells
Carbon converted from DOM to SOM (may be negative).
- *S_N*: number [g N/cm³/h] soil cells
Carbon converted from DOM to SOM (may be negative).

Chapter 37

drain

Lateral transport of water.

37.1 lateral

Pipe drainage. See also [Hooghoudt, 1940]

```
< lateral (L 1800 [cm])
          (x x)
          (rad 3.5 [cm])
          (height height)
          (pipe_position -110 [cm])
          (eq_depth eq_depth) ; Default MolenWesseling value.
          (pipe_outlet pipe_outlet) ; Default deep value.
          (K_to_pipes K_to_pipes)
          (pipe_level pipe_level) >
```

- *L*: number [**cm**]
Parameter (default 1800)
Distance between pipes.
- *x*: number [**cm**]
Optional parameter
Horizontal distance to nearest pipe. By default, this is 1/2 L.
- *rad*: number [**cm**]
Parameter (default 3.5)
Inner radius of drain pipes.
- *height*: number [**cm**]
Optional state variable
Current groundwater level (a negative number).
- *pipe_position*: number [**cm**]
Parameter (default -110)
Height pipes are placed in the soil (a negative number).
- *eq_depth*: **draineqd** component (see chapter 38)
Component (default 'MolenWesseling')
Model for calculating equivalent depth for drains.

- *pipe_outlet*: **depth** component (see chapter 34)
Component (default 'deep')
Water table in drain pipe outlet.

By default this will be identical to 'pipe_position', meaning free flow of water through drains. Specifying a lower water level will not affect the simulation. Specifying a higher water level is functionally equivalent to temporarily increasing the 'pipe_position', lowering the ability of the drain system to drain the soil.

Currently, there is no possibility of water flowing from the pipe outlet to soil.
- *K_to_pipes*: number [**cm/h**]
Optional parameter
Horizontal conductivity in saturated soil. By default this is calculated from the horizontal conductivity and the anisotropy of the horizon.
- *pipe_level*: number [**cm**]
Optional state variable
Current effective pipe position (a negative number).

Log Variables

- *S*: number [**cm³/cm³/h**] soil cells
Pipe drainage.
- *DrainFlow*: number [**cm/h**]
Drain flow to pipes.
- *EqDrnFlow*: number [**cm/h**]
Equilibrium drain flow to pipes.

37.2 none

No lateral transport.

Used by column default Drain (see 26.1, page 153) .

Chapter 38

draineqd

Find the equilibrium drain depth for the Hooghoudt drainage model.

38.1 MolenWesseling

Equivalent depth calculated with model by van der Molen and Wesseling. See also [der Molen and Wesseling, 1991]

Used by drain lateral eq_depth (see 37.1, page 197) .

38.2 Moody

Equivalent depth calculated with model by Moody. See also [Moody, 1966]

38.3 Wesseling

Equivalent depth calculated with model by Wesseling. See also [Wesseling, 1973]

38.4 none

No modifications. See also [Hooghoudt, 1940]

Chapter 39

equilibrium

Find equilibrium between two soil chemicals.

39.1 Langmuir

```
A = (my_max B) / (K + B)
< Langmuir (K K)
(my_max my_max) >
```

- *K*: **number** component (see chapter 59)
Half saturation constant [g/cm³].
- *my_max*: **number** component (see chapter 59)
Max equilibrium capacity [g/cm³].

39.2 goal_A

Attempt to maintain A at at fixed level.

```
< goal_A (goal_A goal_A)
(A_solute A_solute)
(min_B min_B)
(B_solute B_solute)
(debug_cell -1) >
```

- *goal_A*: **number** component (see chapter 59)
The desired level of A [g/cm³].
- *A_solute*: boolean (see section 4.1.2)
Parameter
True iff 'goal_A' is in solute (mass per volume water). If false, the unit is assumed to be mass per volume space.
- *min_B*: **number** component (see chapter 59)
Do not convert B to A if B is smaller than this [g/cm³].
- *B_solute*: boolean (see section 4.1.2)
Parameter
True iff 'min_B' is in solute (mass per volume water). If false, the unit is assumed to be mass per volume space.

- *debug_cell*: integer
Parameter (default -1)
Print debug information for this cell. Set it to a negative number to disable it.

39.3 linear

$$A = K B$$

< linear (K K) >

- *K*: **number** component (see chapter 59)
The ratio A/B at equilibrium [].

Chapter 40

exchange

A named value to exchange with external models.

```
< component (name name) >
```

- *name*: string (see section 4.1.5)
Parameter
Name of value to exchange.

40.1 name

Exchange a string value.

```
< name (value value)  
    ;; Shared parameters are described in chapter 40.  
    (name name) >
```

- *value*: string (see section 4.1.5)
Parameter
Current value to exchange.

40.2 number

Exchange a numeric value.

```
< number (value value)  
        (dimension dimension)  
        ;; Shared parameters are described in chapter 40.  
        (name name) >
```

- *value*: number (dimension not specified)
Optional state variable
Current value to exchange.
- *dimension*: string (see section 4.1.5)
Parameter
Dimension of value to exchange.

Chapter 41

format

Text formatting component.

41.1 LaTeX

Format text as LaTeX.

Used by program document format (see 66.13, page 361) , and program docmodel format (see 66.12, page 361) .

Chapter 42

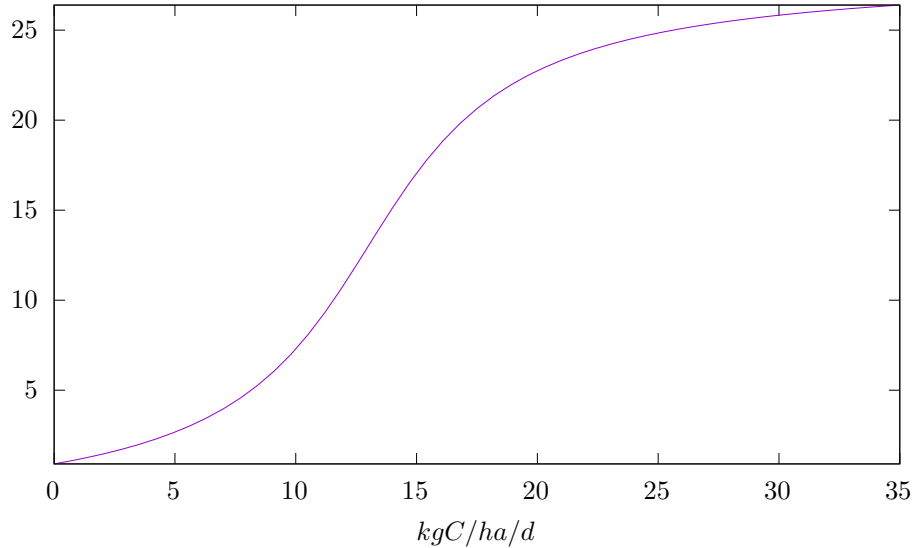
function

Pure function of one parameter.

42.1 P96CO2

A function [**kg C/ha/d** → **<none>**] build into Daisy.

$$13 + \frac{30.78 \arctan(\pi \cdot 0.07 \cdot (\text{CO2} - 13))}{\pi}$$

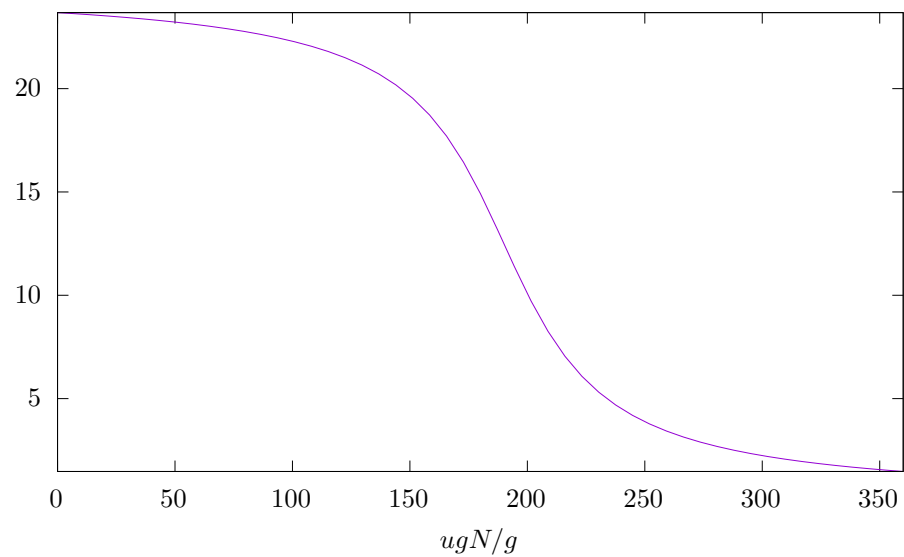


CO2 effect on N2/N2O ratio. Figure 5, bottom. See also [Parton et al., 1996]

42.2 P96NO3

A function [**ug N/g** → **<none>**] build into Daisy.

$$\left(1 - \left(0.5 + \frac{1 \arctan(\pi \cdot 0.01 \cdot (\text{NO3} - 190))}{\pi}\right)\right) 25$$

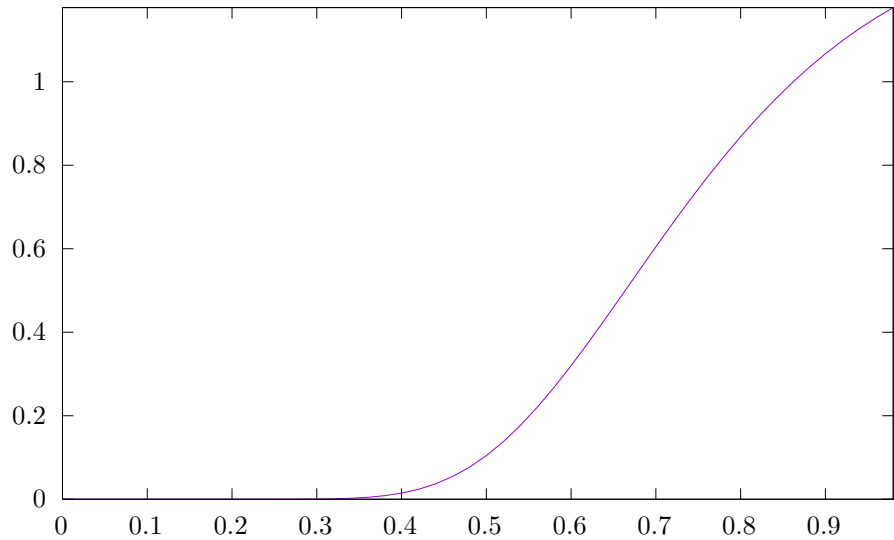


NO3 effect on N2/N2O ratio. Figure 5, middle. Note extra parentheses compared to source. See also [Parton et al., 1996]

42.3 P96WFPS

A function [`<fraction>` $\rightarrow$ `<none>`] build into Daisy.

$$\frac{1.4}{13\left(13\left(\frac{17}{2.2\text{ wfps}}\right)\right)}$$



Water filled pore space effect on N2/N2O ratio. Figure 5, top. Note extra parentheses compared to source. See also [Parton et al., 1996]

42.4 Python

A function [`?` $\rightarrow$ `?`] build into Daisy.

Call Python function.

```
< Python (name name)
      (module module) >
```

- *name*: string (see section 4.1.5)
Parameter
Name of the function.
- *module*: string (see section 4.1.5)
Parameter
Where to find the function.

42.5 *T_scale*

A base function [**dg C** → **<none>**] build into Daisy.

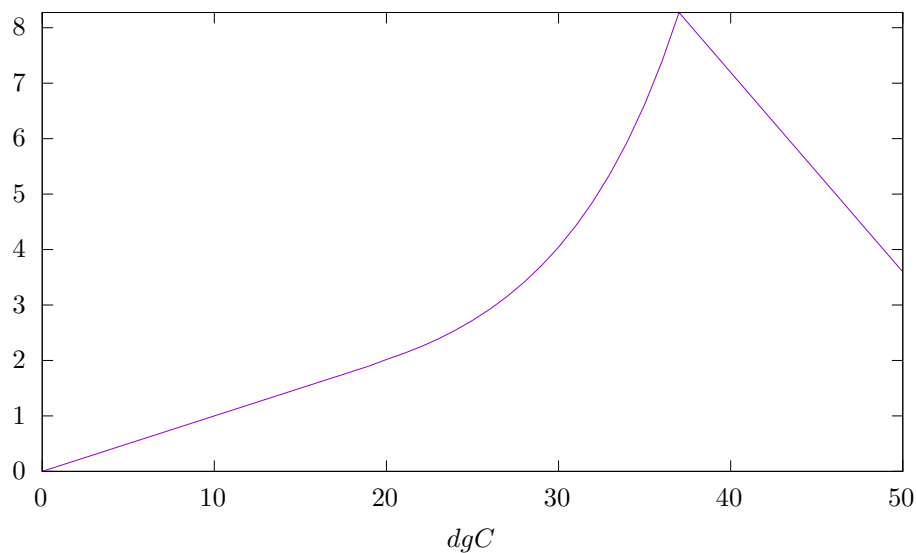
Scale to reference temperature.

```
< T.scale (ref ref) >
```

- *ref*: number [**dg C**]
Parameter
Temperature at which the function is one.

42.6 *T_min*

A ‘*T_scale*’ function [**dg C** → **<none>**] (see 42.5, page 209) build into Daisy.

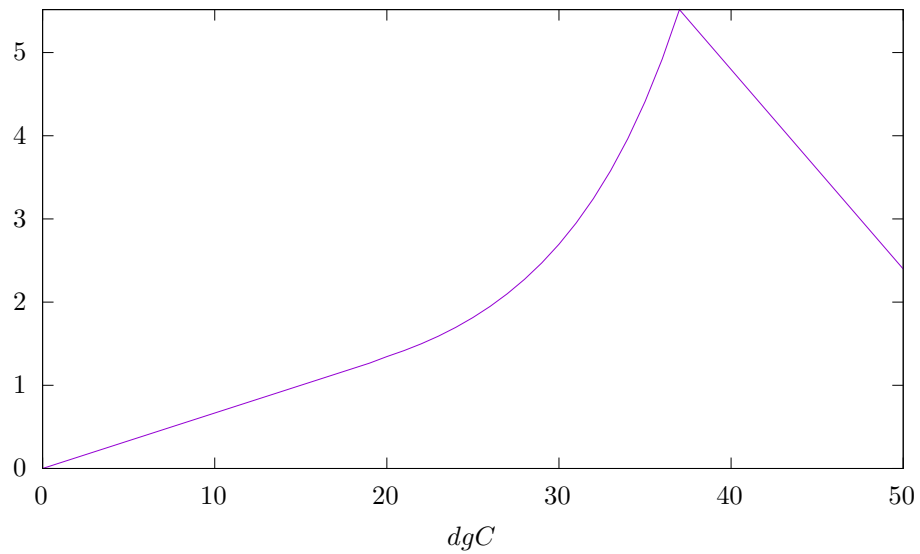


Default temperature function used for mineralization. Equation (6-13) in A10, with a linear decrease beginning at 37 dg C, down to 0 at 60 dg C, according to J.A. van Veen and M.J.Frissel. See also [Hansen et al., 1990]

```
< (ref 10 [dg C]) >
```

42.7 *T_min_15*

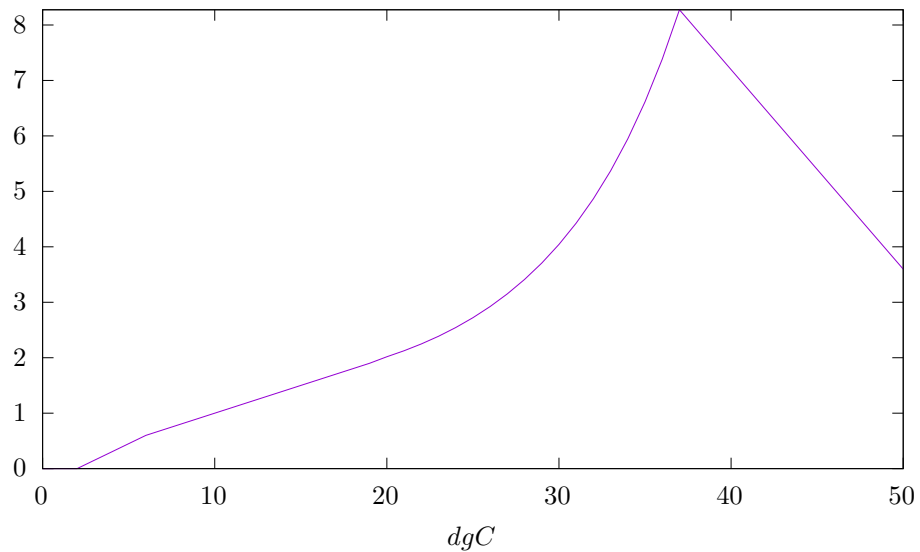
A ‘*T_min*’ function [**dg C** → **<none>**] (see 42.6, page 209) build into Daisy.



T_min normalized to 15 dg C. See also [Hansen et al., 1990]
 < (ref 15 [dg C]) >

42.8 T_nit

A 'T_scale' function [dg C → <none>] (see 42.5, page 209) build into Daisy.



Default temperature function used for nitrification. Equation (7-3) in A10, with a linear decrease beginning at 37 dg C, down to 0 at 60 dg C, according to J.A. van Veen and M.J.Frissel. See also [Hansen et al., 1990]

< (ref 10 [dg C]) >

42.9 const

A function [? → ?] build into Daisy.

Always return the same number.

< const value >

- *value*: number [**<user>**]
Parameter
The number to return.

42.10 plf

A function [$? \rightarrow ?$] build into Daisy.

A piecewise linear function.

< **plf** *plf* >

- *plf*: plf [**<user>** $\rightarrow$ **<user>**]
Parameter
The piecewise linear function.

Chapter 43

ghf

Ground heat flux.

43.1 FAO56

Equation 45 and 46. Ground heat flux is 10 % of Rn during day, and 50 % of Rn during night. See also [Allen et al., 1998]

Used by bioclimate FAO56_hourly ghf (see 19.4, page 116) .

43.2 const

The ground heat flux never changes.

< const G >

- G : number [W/m^2]
Parameter
The ground heat flux never changes.

43.3 none

A ‘const’ model (see 43.2, page 213) build into Daisy.

No ground heat flux.

Used by bioclimate FAO56_daily ghf (see 19.3, page 115) .

< G >

43.4 old

Based soil numeric cell 0 and 1. The model used in Daisy 6.08 and earlier, should not be used.

43.5 surface

Based on flux through soil surface. Calculated from previous timestep.

Used by bioclimate default ghf (see 19.2, page 115) .

43.6 weather

Use value from weather file.

Chapter 44

gnuplot

Plot a graph with gnuplot.

44.1 common

Common parameters.

```
< common  (extra)
           (where screen)
           (title title)
           (device device)
           (canvas canvas)
           (size size)
           (legend auto) >
```

- *extra*: string (see section 4.1.5) sequence
Parameter (default: an empty sequence)
List of extra gnuplot commands. The commands will be inserted right before the plot command. Note that if you have multiple plots in the same command file, The extra commands may affect the subsequence plots.
- *where*: string (see section 4.1.5)
Optional parameter (default 'screen')
File to store results in. By default, show them on a window. The format is determined from the file name extension: *.tex: LaTeX code with PostScript specials. *.eps: Encapsulated PostScript. *.pdf: Adobe PDF files. *.emf: Enhanced Metafile.

The special name 'screen' indicate that the data should be shown on the screen instead of being stored in a file.
- *title*: string (see section 4.1.5)
Optional parameter
Title of the plot, if any. Set it to an empty string to disable.
- *device*: string (see section 4.1.5)
Optional parameter
Output device. By default, this is derived from the file extension.
- *canvas*: string (see section 4.1.5)
Optional parameter
Canvas size. By default, this depend on the device.

- *size*: submodel (see section 4.1.7)
Optional submodel
Relative to size of plot. The standard size is 1.0, specify other numbers to scale accordingly.

< *x y* >

- *x*: number (dimensionless)
Parameter
Relative horizontal size of plot.
- *y*: number (dimensionless)
Parameter
Relative vertical size of plot.

- *legend*: string (see section 4.1.5)
Optional parameter (default 'auto')
Placement of legend. This can be one of the four corners, named by compass locations (nw, ne, sw, se) to get the legend inside the graph in that corner, 'below' to get the legend below the graph, 'outside' to get the legend to the right of the graph, or 'none' to avoid getting a legend at all.

The value 'auto' mean the legend will be places in the corner located farthest away from any data points. Note that datapoints outside the graph are ignored, and so are the lines connecting the datapoints. Thus, a line connecting two datapoints, one of them outside the graph, may cross the legend.

44.2 profile

A 'common' model (see 44.1, page 215) build into Daisy.

Plot 2D soil profile.

```
< profile (column column)
;; Shared parameters are described in section 44.1.
(extra)
(where screen)
(title title)
(device device)
(canvas canvas)
(size size)
(legend auto) >
```

- *column*: **column** component (see chapter 26)
Column whose soil profile to plot.

44.3 soil

A 'common' model (see 44.1, page 215) build into Daisy.

Generate a 2D gnuplot graph with soil content.


```

< soil (file file)
      (top 0 [cm])
      (max max)
      (min min)
      (dimension dimension)
      (missing missing ...) ; Has default value.
      (filter)
      (original original ...)
      (dim_line dim_line)
      (bottom bottom)
      (type block)
      (left 0 [cm])
      (right right)
      (when when) ; Has partial value.
      (sizeratio -1 [])
      (use_log false)
      (samples 25)
      ;; Shared parameters are described in section 44.1.
      (extra)
      (where screen)
      (title title)
      (device device)
      (canvas canvas)
      (size size)
      (legend auto) >

```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *top*: number [cm]
Parameter (default 0)
Higest z value in plot.
- *max*: number [<user>]
Optional parameter
Fixed highest value. By default determine this from the data.
- *min*: number [<user>]
Optional parameter
Fixed lowest value. By default determine this from the data.
- *dimension*: string (see section 4.1.5)
Optional parameter
Dimension for data. By default, use dimension from file.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

< *tag allowed...* >

- *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *bottom*: number [**cm**]
Optional parameter
Deepest z value in plot. By default, derive value from data file.
- *type*: string (see section 4.1.5)
State variable (default 'block')
Plot type. Valid options are 'block' and 'contour'.
- *left*: number [**cm**]
Parameter (default 0)
Minimum x value in plot.
- *right*: number [**cm**]
Optional parameter
Maximum x value in plot. By default, derive value from data file.
- *when*: **Time** fixed component (see section 107.27)
Submodel (has partially specified default value)
Use value closest to this time.
- *sizeratio*: number (dimensionless)
Parameter (default -1)
If positive, ratio of y-axes length to x-axis length. If negative, use same ratio for units on the two axes.
- *use_log*: boolean (see section 4.1.2)
Parameter (default false)
log10 of each value
- *samples*: integer
Parameter (default 25)
Number of sample lines for the 'smooth' and 'contour' types.

44.4 time

A ‘common’ model (see 44.1, page 215) build into Daisy.

Generate a gnuplot graph with times series.

```
< time (source source ...)
      (end end)
      (begin begin)
      (ymin ymin)
      (ymax ymax)
      (y2min y2min)
      (y2max y2max)
      ;; Shared parameters are described in section 44.1.
      (extra)
      (where screen)
      (title title)
      (device device)
      (canvas canvas)
      (size size)
      (legend auto) >
```

- *source*: **source** component (see chapter 83) sequence
Time series to plot.
- *end*: **Time** fixed component (see section 107.27)
Optional submodel
Last date at x-axis.
- *begin*: **Time** fixed component (see section 107.27)
Optional submodel
First date at x-axis.
- *ymin*: number [**<user>**]
Optional parameter
Fixed lowest value on left y-axis. By default determine this from the data.
- *ymax*: number [**<user>**]
Optional parameter
Fixed highest value on right y-axis. By default determine this from the data.
- *y2min*: number [**<user>**]
Optional parameter
Fixed lowest value on left y-axis. By default determine this from the data.
- *y2max*: number [**<user>**]
Optional parameter
Fixed highest value on right y-axis. By default determine this from the data.

44.5 vector

A ‘common’ model (see 44.1, page 215) build into Daisy.

Generate a 2D gnuplot vector field.

```

< vector (file file)
        (top 0 [cm])
        (dimension dimension)
        (factor factor)
        (end end)
        (begin begin)
        (missing missing ...) ; Has default value.
        (filter)
        (original original ...)
        (dim_line dim_line)
        (bottom bottom)
        (left 0 [cm])
        (right right)
        (when when)
        ;; Shared parameters are described in section 44.1.
        (extra)
        (where screen)
        (title title)
        (device device)
        (canvas canvas)
        (size size)
        (legend auto) >

```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *top*: number [cm]
Parameter (default 0)
Higest z value in plot.
- *dimension*: string (see section 4.1.5)
Optional parameter
Dimension for data. By default, use dimension from file.
- *factor*: number (dimension not specified)
Parameter
Multiply vector values with this factor.
- *end*: **Time** fixed component (see section 107.27)
Optional submodel
Ignore values after this time.
- *begin*: **Time** fixed component (see section 107.27)
Optional submodel
Ignore values before this time.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

`< tag allowed... >`

- *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
 - *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.
- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.
 - *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
 - *bottom*: number [**cm**]
Optional parameter
Deepest z value in plot. By default, derive value from data file.
 - *left*: number [**cm**]
Parameter (default 0)
Minimum x value in plot.
 - *right*: number [**cm**]
Optional parameter
Maximum x value in plot. By default, derive value from data file.
 - *when*: **Time** fixed component (see section 107.27)
Optional submodel
Use value closest to this time.

44.6 xy

A 'common' model (see 44.1, page 215) build into Daisy.
Generate a gnuplot graph with up to two x-axes.

```

< xy (source source ...)
      (ymin ymin)
      (ymax ymax)
      (y2min y2min)
      (y2max y2max)
      (xmin xmin)
      (xmax xmax)
      (x2min x2min)
      (x2max x2max)
      ;; Shared parameters are described in section 44.1.
      (extra)
      (where screen)
      (title title)
      (device device)
      (canvas canvas)
      (size size)
      (legend auto) >

```

- *source*: **xy**source component (see chapter 105) sequence
XY series to plot.
- *ymin*: number [**<user>**]
Optional parameter
Fixed lowest value on left y-axis. By default determine this from the data.
- *ymax*: number [**<user>**]
Optional parameter
Fixed highest value on right y-axis. By default determine this from the data.
- *y2min*: number [**<user>**]
Optional parameter
Fixed lowest value on left y-axis. By default determine this from the data.
- *y2max*: number [**<user>**]
Optional parameter
Fixed highest value on right y-axis. By default determine this from the data.
- *xmin*: number [**<user>**]
Optional parameter
Fixed lowest value on left x-axis. By default determine this from the data.
- *xmax*: number [**<user>**]
Optional parameter
Fixed highest value on right x-axis. By default determine this from the data.
- *x2min*: number [**<user>**]
Optional parameter
Fixed lowest value on left x-axis. By default determine this from the data.
- *x2max*: number [**<user>**]
Optional parameter
Fixed highest value on right x-axis. By default determine this from the data.

44.7 multi

Generate multiple graphs for the gnuplot command file.

```
< multi  (before)
          (after)
          (graph graph ...) >
```

- *before*: string (see section 4.1.5) sequence
Parameter (default: an empty sequence)
List of extra gnuplot commands. The commands will be inserted right before the first graph.
- *after*: string (see section 4.1.5) sequence
Parameter (default: an empty sequence)
List of extra gnuplot commands. The commands will be inserted right after the last graph.
- *graph*: **gnuplot** component (see chapter 44) sequence
Graphs to plot.

Chapter 45

groundwater

The 'groundwater' component is responsible for specifying the groundwater table at each timestep.

45.1 common

All groundwater models can log height.

```
< common >
```

Log Variables

- *height*: number [**cm**]
Groundwater level. Positive numbers indicate free drainage.

45.2 aquitard

A 'common' model (see 45.1, page 225) build into Daisy.

Aquitard groundwater, free drainage.

```
< aquitard (K_aquitard 0.001 [cm/h])  
          (Z_aquitard 200 [cm])  
          (h_aquifer h_aquifer)  
          (pressure_table pressure_table) >
```

- *K_aquitard*: number [**cm/h**]
Parameter (default 0.001)
Conductivity of the aquitard.
- *Z_aquitard*: number [**cm**]
Parameter (default 200)
Thickness of the aquitard. The aquitard begins below the bottommost soil horizon.
- *h_aquifer*: number [**cm**]
Optional state variable
Pressure potential in the aquifer below the aquitard. By default. this is Z_aquitard. You can alternatively specify the pressure as a virtual groundwater level. See 'pressure_table'.
- *pressure_table*: **depth** component (see chapter 34)
Optional component
Height of groundwater the corresponds to the pressure in the aquifer.

If you drilled a well down to the aquifer, this is number what the water level in the well would be as height above ground (a negative number). This is different from the actual groundwater table, because the aquitart block the water, and the pipes lead the water away. You can alternatively specify the pressure directly, with 'h_aquifer'.

45.3 deep

A 'common' model (see 45.1, page 225) build into Daisy.
Deep groundwater, free drainage.

45.4 extern

A 'common' model (see 45.1, page 225) build into Daisy.
Look up groundwater table in an scope.

```
< extern (table table)
      (initial_table initial_table) >
```

- *table*: **number** component (see chapter 59)
Expression that evaluates to groundwater table in.
- *initial_table*: number [**cm**]
Optional parameter
Groundwater level for initialization of soil water.

45.5 file

A 'common' model (see 45.1, page 225) build into Daisy.
Read groundwater table from a file.

```
< file file
      (offset 0 [cm]) >
```

- *file*: string (see section 4.1.5)
Parameter
Name of file to read data from. The format of each line in the file is 'YEAR MONTH DAY HEIGHT', where HEIGHT should in cm above ground (i.e. a negative number). Linear interpolation is used between the datapoints.
- *offset*: number [**cm**]
Parameter (default 0)
Add this to depth from file.

45.6 fixed

A 'common' model (see 45.1, page 225) build into Daisy.
Fixed high groundwater level.

```
< fixed table >
```

- *table*: number [**cm**]
Parameter
Groundwater level (negative number below surface).

45.7 flux

A ‘common’ model (see 45.1, page 225) build into Daisy.

Flux groundwater, free drainage.

```
< flux flux >
```

- *flux*: number [cm/h]
Parameter
Constant flux to groundwater.

45.8 lysimeter

A ‘common’ model (see 45.1, page 225) build into Daisy.

Lysimeter bottom.

45.9 source

A ‘common’ model (see 45.1, page 225) build into Daisy.

Read groundwater table from a source.

```
< source source
      (offset 0 [cm]) >
```

- *source*: **source** component (see chapter 83)
Groundwater table time series.
- *offset*: number [cm]
Parameter (default 0)
Add this to depth from source.

45.10 static

A ‘common’ model (see 45.1, page 225) build into Daisy.

Static groundwater level. Provided for backward compatibility, use ‘deep’ or ‘fixed’ instead.

```
< static (table 1 [cm])
      (obsolete obsolete) >
```

- *table*: number [cm]
Parameter (default 1)
Groundwater level. Positive numbers indicate free drainage.
- *obsolete*: string (see section 4.1.5)
Optional parameter
Use ‘deep’ or ‘fixed’ instead’.

Chapter 46

heatrect

Heat transport in rectangular grid.

46.1 Mollerup

Finite volume solution to heat transfer by Mikkel Mollerup.

Used by movement rectangle heat (see 56.2, page 307) .

```
< Mollerup (solver solver) ; Default cxsparse value.  
      (debug 0) >
```

- *solver*: **solver** component (see chapter 82)
Component (default ‘*cxsparse*’)
Model used for solving matrix equation system.
- *debug*: integer
Parameter (default 0)
Enable additional debug message. A value of 0 means no message, higher numbers means more messages.

46.2 linear

Linear temperature interpolation between top and bottom.

46.3 none

No heat transport.

46.4 top

Soil temperature is equal to upper boundary. This can be used for a poorly isolated soil column.

Chapter 47

horizon

A ‘horizon’ is a soil type with specific physical properties. It is the responsibility of the ‘horizon’ component to specify these properties.

Used by SoilLayer @ horizon (see 107.19, page 610) , and SoilRegion @ horizon (see 107.20, page 611) .

```
< component  (HorHeat HorHeat)                                ; Has default value.
              (hydraulic hydraulic)                            ; Default hypres value.
              (tortuosity tortuosity)                          ; Default linear value.
              (C_per_N C_per_N)
              (anisotropy 1 [])
              (dry_bulk_density dry_bulk_density)
              (quartz quartz)
              (SOM_C_per_N 11 11 11)
              (SOM_fractions SOM_fractions ...)
              (turnover_factor 1 [])
              (root_homogeneity 1 [])
              (root_retardation 1 [])
              (K_factor 1 [])
              (Nitrification Nitrification)                    ; Default soil value.
              (secondary_domain secondary_domain)              ; Default none value.
              (attributes)
              (r_pore_min 0.1 [um])
              (CEC CEC) >
```

- *HorHeat*: **HorHeat** fixed component (see section 107.18)
Submodel (has fully specified default value)
Heat capacity and conductivity.
- *hydraulic*: **hydraulic** component (see chapter 48)
Component (default ‘hypres’)
The hydraulic propeties of the soil.
- *tortuosity*: **tortuosity** component (see chapter 90)
Component (default ‘linear’)
The soil tortuosity.
- *C_per_N*: number [g C/g N]
Optional parameter
Total C/N ratio for this horizon. This is the combined initial C/N ratio for all organic matter pools in the horizon. The C/N ration of the AOM and SMB pools is assumed to be known, given that this number is used to find the

common C/N ration for the SOM pools. The C/N ration for the SOM pools will then gradually move towards the values specified by 'SOM_C_per_N'. By default, the values given by 'SOM_C_per_N' will be used for initialization.

- *anisotropy*: number (dimensionless)
Parameter (default 1)
Horizontal saturated water conductivity relative to vertical saturated water conductivity. The higher this value, the faster the water will move towards drain pipes.
- *dry_bulk_density*: number [**g/cm³**]
Optional parameter
The soils dry bulk density. By default, this is calculated from the soil constituents.
- *quartz*: number [**<fraction>**]
Optional parameter
Fraction of quartz in soil minerals. By default, this is calculated from the soil constituents. $\text{quartz} = \text{clay} * 0.15 + \text{silt} * 0.6 + \text{sand} * 0.7$ See also [Møberg et al., 1988]
- *SOM_C_per_N*: number [**g C/g N**] sequence
Parameter (has default value with length 3)

(SOM_C_per_N 11 11 11 [**g C/g N**])

Parameter description:

C/N ratio for each SOM pool in this soil. If 'C_per_N' is specified, this is used as a goal only. If 'C_per_N' is unspecified, the SOM pools will be initialized with this value.

- *SOM_fractions*: number (dimensionless) sequence
Optional parameter
Fraction of humus in each SOM pool, typically slow, fast and inert. Negative numbers mean unspecified, let Daisy find appropriate values.
- *turnover_factor*: number (dimensionless)
Parameter (default 1)
Factor multiplied to the turnover rate for all organic matter pools in this horizon.
- *root_homogeneity*: number (dimensionless)
Parameter (default 1)
Factor multiplied to the root density of all crops in this horizon. The idea is to emulate the effect of roots not being distributed evenly in the soil.
- *root_retardation*: number (dimensionless)
Parameter (default 1)
Factor multiplied to root penetration speed.
- *K_factor*: number (dimensionless)
Parameter (default 1)
Factor multiplied to the hydraulic conductivity.
- *Nitrification*: **nitrification** component (see chapter 58)
Component (default 'soil')
The soil nitrification process.

- *secondary_domain*: **secondary** component (see chapter 78)
Component (default 'none')
Secondary matrix domain for solute movement.
- *attributes*: submodel (see section 4.1.7) sequence
Optional submodel (default: an empty sequence)
List of additional attributes for this horizon. Intended for use with pedotransfer functions.

< key value >

– *key*: string (see section 4.1.5)
Parameter
Name of attribute.

– *value*: number [**<user>**]
Parameter
Value of attribute.

- *r_pore_min*: number [**um**]
Parameter (default 0.1)
Smallest pores in the soil.

- *CEC*: number [**cmolc/kg**]
Optional parameter
Cation-Exchange Capacity. By default this will be calculated from a pedotransfer function based on Danish cultivated soils. The pedotransfer function is known to be invalid for podzolic B-horizons. See also [Krogh et al., 2000]

47.1 BSI3

A horizon using BSI3 texture classification.

```

< BSI3  (clay clay)
        (silt silt)
        (sand sand)
        (humus humus)
        (chalk 0 [<fraction>])
        (normalize false)
        (other 0 [<fraction>])
        ;; Shared parameters are described in chapter 47.
        (HorHeat HorHeat)                ; Has default value.
        (hydraulic hydraulic)            ; Default hypres value.
        (tortuosity tortuosity)          ; Default linear value.
        (C_per_N C_per_N)
        (anisotropy 1 [])
        (dry_bulk_density dry_bulk_density)
        (quartz quartz)
        (SOM_C_per_N 11 11 11)
        (SOM_fractions SOM_fractions ...)
        (turnover_factor 1 [])
        (root_homogeneity 1 [])
        (root_retardation 1 [])
        (K_factor 1 [])
        (Nitrification Nitrification)    ; Default soil value.
        (secondary_domain secondary_domain) ; Default none value.
        (attributes)
        (r_pore_min 0.1 [um])
        (CEC CEC) >

```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 2 [um].
- *silt*: number [<fraction>]
Parameter
Mineral particles between 2 [um] and 60 [um].
- *sand*: number [<fraction>]
Parameter
Mineral particles between 60 [um] and 2000 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.
- *chalk*: number [<fraction>]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [<fraction>]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.2 DIN3

A 'BSI3' model (see 47.1, page 233) build into Daisy.
'DIN3' is another name for 'BSI3'

47.3 MIT3

A 'BSI3' model (see 47.1, page 233) build into Daisy.
'MIT3' is another name for 'BSI3'

47.4 BSI7

A horizon using BSI7 texture classification.

```
< BSI7 (clay clay)
    (fine_sand fine_sand)
    (medium_sand medium_sand)
    (coarse_sand coarse_sand)
    (fine_silt fine_silt)
    (medium_silt medium_silt)
    (coarse_silt coarse_silt)
    (humus humus)
    (chalk 0 [<fraction>])
    (normalize false)
    (other 0 [<fraction>])
    ;; Shared parameters are described in chapter 47.
    (HorHeat HorHeat) ; Has default value.
    (hydraulic hydraulic) ; Default hypres value.
    (tortuosity tortuosity) ; Default linear value.
    (C_per_N C_per_N)
    (anisotropy 1 [])
    (dry_bulk_density dry_bulk_density)
    (quartz quartz)
    (SOM.C_per_N 11 11 11)
    (SOM.fractions SOM.fractions ...)
    (turnover_factor 1 [])
    (root_homogeneity 1 [])
    (root_retardation 1 [])
    (K_factor 1 [])
    (Nitrification Nitrification) ; Default soil value.
    (secondary_domain secondary_domain) ; Default none value.
    (attributes)
    (r_pore_min 0.1 [um])
    (CEC CEC) >
```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 2 [um].
- *fine_sand*: number [<fraction>]
Parameter
Mineral particles between 60 [um] and 200 [um].

- *medium_sand*: number [**<fraction>**]
Parameter
Mineral particles between 200 [um] and 600 [um].
- *coarse_sand*: number [**<fraction>**]
Parameter
Mineral particles between 600 [um] and 2000 [um].
- *fine_silt*: number [**<fraction>**]
Parameter
Mineral particles between 2 [um] and 6 [um].
- *medium_silt*: number [**<fraction>**]
Parameter
Mineral particles between 6 [um] and 20 [um].
- *coarse_silt*: number [**<fraction>**]
Parameter
Mineral particles between 20 [um] and 60 [um].
- *humus*: number [**<fraction>**]
Parameter
Humus content of soil.
- *chalk*: number [**<fraction>**]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [**<fraction>**]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.5 MIT7

A 'BSI7' model (see 47.4, page 235) build into Daisy.

'MIT7' is another name for 'BSI7'

47.6 DIN5

A horizon using DIN5 texture classification.

```

< DIN5  (clay clay)
        (silt silt)
        (fine_sand fine_sand)
        (medium_sand medium_sand)
        (coarse_sand coarse_sand)
        (humus humus)
        (chalk 0 [<fraction>])
        (normalize false)
        (other 0 [<fraction>])
        ;; Shared parameters are described in chapter 47.
        (HorHeat HorHeat) ; Has default value.
        (hydraulic hydraulic) ; Default hypres value.
        (tortuosity tortuosity) ; Default linear value.
        (C_per_N C_per_N)
        (anisotropy 1 [])
        (dry_bulk_density dry_bulk_density)
        (quartz quartz)
        (SOM_C_per_N 11 11 11)
        (SOM_fractions SOM_fractions ...)
        (turnover_factor 1 [])
        (root_homogeneity 1 [])
        (root_retardation 1 [])
        (K_factor 1 [])
        (Nitrification Nitrification) ; Default soil value.
        (secondary_domain secondary_domain) ; Default none value.
        (attributes)
        (r_pore_min 0.1 [um])
        (CEC CEC) >

```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 2 [um].
- *silt*: number [<fraction>]
Parameter
Mineral particles between 2 [um] and 60 [um].
- *fine_sand*: number [<fraction>]
Parameter
Mineral particles between 60 [um] and 200 [um].
- *medium_sand*: number [<fraction>]
Parameter
Mineral particles between 200 [um] and 600 [um].
- *coarse_sand*: number [<fraction>]
Parameter
Mineral particles between 600 [um] and 2000 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.
- *chalk*: number [<fraction>]
Parameter (default 0)
Chalk content of soil.

- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [**<fraction>**]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.7 GEUS7

A horizon using GEUS7 texture classification.

```
< GEUS7 (clay clay)
    (fine_sand fine_sand)
    (medium_sand medium_sand)
    (coarse_sand coarse_sand)
    (fine_silt fine_silt)
    (medium_silt medium_silt)
    (coarse_silt coarse_silt)
    (humus humus)
    (chalk 0 [<fraction>])
    (normalize false)
    (other 0 [<fraction>])
    ;; Shared parameters are described in chapter 47.
    (HorHeat HorHeat) ; Has default value.
    (hydraulic hydraulic) ; Default hypres value.
    (tortuosity tortuosity) ; Default linear value.
    (C_per_N C_per_N)
    (anisotropy 1 [])
    (dry_bulk_density dry_bulk_density)
    (quartz quartz)
    (SOM_C_per_N 11 11 11)
    (SOM_fractions SOM_fractions ...)
    (turnover_factor 1 [])
    (root_homogeneity 1 [])
    (root_retardation 1 [])
    (K_factor 1 [])
    (Nitrification Nitrification) ; Default soil value.
    (secondary_domain secondary_domain) ; Default none value.
    (attributes)
    (r_pore_min 0.1 [um])
    (CEC CEC) >
```

- *clay*: number [**<fraction>**]
Parameter
Mineral particles up to 2 [um].
- *fine_sand*: number [**<fraction>**]
Parameter
Mineral particles between 63 [um] and 200 [um].
- *medium_sand*: number [**<fraction>**]
Parameter

Mineral particles between 200 [um] and 600 [um].

- *coarse_sand*: number [<fraction>]
Parameter
Mineral particles between 600 [um] and 2000 [um].
- *fine_silt*: number [<fraction>]
Parameter
Mineral particles between 2 [um] and 6 [um].
- *medium_silt*: number [<fraction>]
Parameter
Mineral particles between 6 [um] and 20 [um].
- *coarse_silt*: number [<fraction>]
Parameter
Mineral particles between 20 [um] and 63 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.
- *chalk*: number [<fraction>]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [<fraction>]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.8 ISSS3

A horizon using ISSS3 texture classification.

```

< ISSS3  (clay clay)
          (silt silt)
          (sand sand)
          (humus humus)
          (chalk 0 [<fraction>])
          (normalize false)
          (other 0 [<fraction>])
          ;; Shared parameters are described in chapter 47.
          (HorHeat HorHeat) ; Has default value.
          (hydraulic hydraulic) ; Default hypres value.
          (tortuosity tortuosity) ; Default linear value.
          (C_per_N C_per_N)
          (anisotropy 1 [])
          (dry_bulk_density dry_bulk_density)
          (quartz quartz)
          (SOM_C_per_N 11 11 11)
          (SOM_fractions SOM_fractions ...)
          (turnover_factor 1 [])
          (root_homogeneity 1 [])
          (root_retardation 1 [])
          (K_factor 1 [])
          (Nitrification Nitrification) ; Default soil value.
          (secondary_domain secondary_domain) ; Default none value.
          (attributes)
          (r_pore_min 0.1 [um])
          (CEC CEC) >

```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 2 [um].
- *silt*: number [<fraction>]
Parameter
Mineral particles between 2 [um] and 20 [um].
- *sand*: number [<fraction>]
Parameter
Mineral particles between 20 [um] and 2000 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.
- *chalk*: number [<fraction>]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [<fraction>]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.9 ISSS4

A horizon using ISSS4 texture classification.

```
< ISSS4  (clay clay)
          (silt silt)
          (fine_sand fine_sand)
          (coarse_sand coarse_sand)
          (humus humus)
          (chalk 0 [<fraction>])
          (normalize false)
          (other 0 [<fraction>])
          ;; Shared parameters are described in chapter 47.
          (HorHeat HorHeat) ; Has default value.
          (hydraulic hydraulic) ; Default hypres value.
          (tortuosity tortuosity) ; Default linear value.
          (C_per_N C_per_N)
          (anisotropy 1 [])
          (dry_bulk_density dry_bulk_density)
          (quartz quartz)
          (SOM_C_per_N 11 11 11)
          (SOM_fractions SOM_fractions ...)
          (turnover_factor 1 [])
          (root_homogeneity 1 [])
          (root_retardation 1 [])
          (K_factor 1 [])
          (Nitrification Nitrification) ; Default soil value.
          (secondary_domain secondary_domain) ; Default none value.
          (attributes)
          (r_pore_min 0.1 [um])
          (CEC CEC) >
```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 2 [um].
- *silt*: number [<fraction>]
Parameter
Mineral particles between 2 [um] and 20 [um].
- *fine_sand*: number [<fraction>]
Parameter
Mineral particles between 20 [um] and 200 [um].
- *coarse_sand*: number [<fraction>]
Parameter
Mineral particles between 200 [um] and 2000 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.
- *chalk*: number [<fraction>]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)

If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.

- *other*: number [**<fraction>**]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.10 USDA3

A horizon using USDA3 texture classification.

```
< USDA3  (clay clay)
          (silt silt)
          (sand sand)
          (humus humus)
          (chalk 0 [<fraction>])
          (normalize false)
          (other 0 [<fraction>])
          ;; Shared parameters are described in chapter 47.
          (HorHeat HorHeat) ; Has default value.
          (hydraulic hydraulic) ; Default hypres value.
          (tortuosity tortuosity) ; Default linear value.
          (C_per_N C_per_N)
          (anisotropy 1 [])
          (dry_bulk_density dry_bulk_density)
          (quartz quartz)
          (SOM_C_per_N 11 11 11)
          (SOM_fractions SOM_fractions ...)
          (turnover_factor 1 [])
          (root_homogeneity 1 [])
          (root_retardation 1 [])
          (K_factor 1 [])
          (Nitrification Nitrification) ; Default soil value.
          (secondary_domain secondary_domain) ; Default none value.
          (attributes)
          (r_pore_min 0.1 [um])
          (CEC CEC) >
```

- *clay*: number [**<fraction>**]
Parameter
Mineral particles up to 2 [um].
- *silt*: number [**<fraction>**]
Parameter
Mineral particles between 2 [um] and 50 [um].
- *sand*: number [**<fraction>**]
Parameter
Mineral particles between 50 [um] and 2000 [um].
- *humus*: number [**<fraction>**]
Parameter
Humus content of soil.

- *chalk*: number [**<fraction>**]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [**<fraction>**]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.11 FAO3

A 'USDA3' model (see 47.10, page 242) build into Daisy.
'FAO3' is another name for 'USDA3'

47.12 USDA7

A horizon using USDA7 texture classification.

```
< USDA7 (clay clay)
      (silt silt)
      (very_fine_sand very_fine_sand)
      (fine_sand fine_sand)
      (medium_sand medium_sand)
      (coarse_sand coarse_sand)
      (very_coarse_sand very_coarse_sand)
      (humus humus)
      (chalk 0 [<fraction>])
      (normalize false)
      (other 0 [<fraction>])
      ;; Shared parameters are described in chapter 47.
      (HorHeat HorHeat) ; Has default value.
      (hydraulic hydraulic) ; Default hypres value.
      (tortuosity tortuosity) ; Default linear value.
      (C_per_N C_per_N)
      (anisotropy 1 [])
      (dry_bulk_density dry_bulk_density)
      (quartz quartz)
      (SOM_C_per_N 11 11 11)
      (SOM_fractions SOM_fractions ...)
      (turnover_factor 1 [])
      (root_homogeneity 1 [])
      (root_retardation 1 [])
      (K_factor 1 [])
      (Nitrification Nitrification) ; Default soil value.
      (secondary_domain secondary_domain) ; Default none value.
      (attributes)
      (r_pore_min 0.1 [um])
      (CEC CEC) >
```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 2 [um].
- *silt*: number [<fraction>]
Parameter
Mineral particles between 2 [um] and 50 [um].
- *very_fine_sand*: number [<fraction>]
Parameter
Mineral particles between 50 [um] and 100 [um].
- *fine_sand*: number [<fraction>]
Parameter
Mineral particles between 100 [um] and 250 [um].
- *medium_sand*: number [<fraction>]
Parameter
Mineral particles between 250 [um] and 500 [um].
- *coarse_sand*: number [<fraction>]
Parameter
Mineral particles between 500 [um] and 1000 [um].
- *very_coarse_sand*: number [<fraction>]
Parameter
Mineral particles between 1000 [um] and 2000 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.
- *chalk*: number [<fraction>]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [<fraction>]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.13 FAO7

A ‘USDA7’ model (see 47.12, page 243) build into Daisy.

‘FAO7’ is another name for ‘USDA7’

47.14 USPRA3

A horizon using USPRA3 texture classification.

```
< USPRA3 (clay clay)
          (silt silt)
          (sand sand)
          (humus humus)
          (chalk 0 [<fraction>])
          (normalize false)
          (other 0 [<fraction>])
          ;; Shared parameters are described in chapter 47.
          (HorHeat HorHeat) ; Has default value.
          (hydraulic hydraulic) ; Default hypres value.
          (tortuosity tortuosity) ; Default linear value.
          (C_per_N C_per_N)
          (anisotropy 1 [])
          (dry_bulk_density dry_bulk_density)
          (quartz quartz)
          (SOM_C_per_N 11 11 11)
          (SOM_fractions SOM_fractions ...)
          (turnover_factor 1 [])
          (root_homogeneity 1 [])
          (root_retardation 1 [])
          (K_factor 1 [])
          (Nitrification Nitrification) ; Default soil value.
          (secondary_domain secondary_domain) ; Default none value.
          (attributes)
          (r_pore_min 0.1 [um])
          (CEC CEC) >
```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 5 [um].
- *silt*: number [<fraction>]
Parameter
Mineral particles between 5 [um] and 50 [um].
- *sand*: number [<fraction>]
Parameter
Mineral particles between 50 [um] and 2000 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.
- *chalk*: number [<fraction>]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [<fraction>]
Parameter (default 0)

Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.15 USPRA4

A horizon using USPRA4 texture classification.

```
< USPRA4 (clay clay)
          (silt silt)
          (fine_sand fine_sand)
          (coarse_sand coarse_sand)
          (humus humus)
          (chalk 0 [<fraction>])
          (normalize false)
          (other 0 [<fraction>])
          ;; Shared parameters are described in chapter 47.
          (HorHeat HorHeat) ; Has default value.
          (hydraulic hydraulic) ; Default hypres value.
          (tortuosity tortuosity) ; Default linear value.
          (C_per_N C_per_N)
          (anisotropy 1 [])
          (dry_bulk_density dry_bulk_density)
          (quartz quartz)
          (SOM_C_per_N 11 11 11)
          (SOM_fractions SOM_fractions ...)
          (turnover_factor 1 [])
          (root_homogeneity 1 [])
          (root_retardation 1 [])
          (K_factor 1 [])
          (Nitrification Nitrification) ; Default soil value.
          (secondary_domain secondary_domain) ; Default none value.
          (attributes)
          (r_pore_min 0.1 [um])
          (CEC CEC) >
```

- *clay*: number [<fraction>]
Parameter
Mineral particles up to 5 [um].
- *silt*: number [<fraction>]
Parameter
Mineral particles between 5 [um] and 50 [um].
- *fine_sand*: number [<fraction>]
Parameter
Mineral particles between 50 [um] and 250 [um].
- *coarse_sand*: number [<fraction>]
Parameter
Mineral particles between 250 [um] and 2000 [um].
- *humus*: number [<fraction>]
Parameter
Humus content of soil.

- *chalk*: number [**<fraction>**]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.
- *other*: number [**<fraction>**]
Parameter (default 0)
Other soil components. Typically larger mineral components (stones, pebbles). These will be ignored by Daisy, but can be included here to ensure the sum is 1.0.

47.16 numeric

A horizon using explicit texture classification.

```
< numeric  (humus humus)
            (fractions fractions ...)
            (limits limits ...)
            (chalk 0 [<fraction>])
            (normalize false)
            ;; Shared parameters are described in chapter 47.
            (HorHeat HorHeat)                ; Has default value.
            (hydraulic hydraulic)              ; Default hypres value.
            (tortuosity tortuosity)             ; Default linear value.
            (C_per_N C_per_N)
            (anisotropy 1 [])
            (dry_bulk_density dry_bulk_density)
            (quartz quartz)
            (SOM_C_per_N 11 11 11)
            (SOM_fractions SOM_fractions ...)
            (turnover_factor 1 [])
            (root_homogeneity 1 [])
            (root_retardation 1 [])
            (K_factor 1 [])
            (Nitrification Nitrification)       ; Default soil value.
            (secondary_domain secondary_domain) ; Default none value.
            (attributes)
            (r_pore_min 0.1 [um])
            (CEC CEC) >
```

- *humus*: number [**<fraction>**]
Parameter
Humus content of soil.
- *fractions*: number [**<fraction>**] sequence
Parameter
Fraction of particles between the corresponding numerical limits.
- *limits*: number [**um**] sequence
Parameter
Numerical limits for particle sizes.

- *chalk*: number [**<fraction>**]
Parameter (default 0)
Chalk content of soil.
- *normalize*: boolean (see section 4.1.2)
Parameter (default false)
If this is true, normalize the mineral fraction to 1.0. Otherwise, give an error if the sum is not 1.0.

Chapter 48

hydraulic

This component is responsible for specifying the soils hydraulic properties.

Used by horizon component hydraulic (see 47, page 231) , and Soil @ Hydraulic (see 107.21, page 611) .

48.1 B_BaC

Brooks and Corey retention curve model with Burdine theory.

```
< B_BaC  (Theta_sat Theta_sat)
          (lambda lambda)
          (K_sat K_sat)
          (K_at_h K_at_h)
          (Theta_res 0 [<fraction>])
          (h_b h_b)
          (1 2 [])
          (p p) >
```

- *Theta_sat*: number [<fraction>]
State variable
Saturation point.
- *lambda*: number (dimensionless)
Parameter
Pore size index.
- *K_sat*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
- *K_at_h*: submodel (see section 4.1.7)
Optional submodel
Water conductivity at specified pressure.

```
< h K >
```

- *h*: number [**cm**]
Parameter
Soil water pressure.
- *K*: number [**cm/h**]
Parameter
Water conductivity.

- *Theta_res*: number [**<fraction>**]
Parameter (default 0)
Soil residual water.
- *h_b*: number [**cm**]
Parameter
Bubbling pressure.
- *l*: number (dimensionless)
Parameter (default 2)
Burdine form parameter. Ignored if 'p' is set.
- *p*: number (dimensionless)
Optional parameter
Hydraulic conductivity form parameter. By default $p=1+1+2/\lambda$.

48.2 B_BaC_Bimodal

Brooks and Corey retention curve model with Burdine theory. Bimodal hydraulic conductivity curve.

```
< B_BaC_Bimodal  (Theta_sat Theta_sat)
                  (lambda lambda)
                  (K_sat K_sat)
                  (Theta_res 0 [<fraction>])
                  (h_b h_b)
                  (Theta_b Theta_b)
                  (K_b K_b) >
```

- *Theta_sat*: number [**<fraction>**]
State variable
Saturation point.
- *lambda*: number (dimensionless)
Parameter
Pore size index.
- *K_sat*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
- *Theta_res*: number [**<fraction>**]
Parameter (default 0)
Soil residual water.
- *h_b*: number [**cm**]
Parameter
Bubbling pressure.
- *Theta_b*: number (dimensionless)
Parameter
Water content at 'h_b'.
- *K_b*: number [**cm/h**]
Parameter
Water conductivity at 'h_b'.

48.3 B_C

Campbell retention curve model with Burdine theory. See also [Campbell, 1974, Burdine, 1953]

```
< B_C (b b)
      (Theta_sat Theta_sat)
      (K_sat K_sat)
      (K_at_h K_at_h)
      (h_b h_b)
      (l 2 [])
      (p p) >
```

- *b*: number (dimensionless)
Parameter
Campbell parameter.
- *Theta_sat*: number [**<fraction>**]
State variable
Saturation point.
- *K_sat*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
- *K_at_h*: submodel (see section 4.1.7)
Optional submodel
Water conductivity at specified pressure.

```
< h K >
```

- *h*: number [**cm**]
Parameter
Soil water pressure.
- *K*: number [**cm/h**]
Parameter
Water conductivity.
- *h_b*: number [**cm**]
Parameter
Bubbling pressure.
- *l*: number (dimensionless)
Parameter (default 2)
Burdine form parameter. Ignored if 'p' is set.
- *p*: number (dimensionless)
Optional parameter
Hydraulic conductivity form parameter. By default $p=1+1+2*b$.

48.4 B_C_inverse

Campbell retention curve model with Burdine theory.

This implementation is based on inverse modelling, you specify water at wilting point (pF 4.2) and field capacity (pF 2.0), from which the retention curve (the *b* and *h_b* parameters) is derived. Based on this, the conductivity curve is fully specified by a single point. See also [Campbell, 1974, Burdine, 1953]

```

< B_C_inverse (Theta_sat Theta_sat)
              (K_sat K_sat)
              (K_at_h K_at_h)
              (Theta_fc Theta_fc)
              (Theta_wp Theta_wp) >

```

- *Theta_sat*: number [**<fraction>**]
State variable
Saturation point.
- *K_sat*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
- *K_at_h*: submodel (see section 4.1.7)
Optional submodel
Water conductivity at specified pressure.

```

< h K >

```

 - *h*: number [**cm**]
Parameter
Soil water pressure.
 - *K*: number [**cm/h**]
Parameter
Water conductivity.
- *Theta_fc*: number [**<fraction>**]
Parameter
Field capacity.
- *Theta_wp*: number [**<fraction>**]
Optional parameter
Wilting point. By default, this value will be estimated from texture. See also [Madsen and Platou, 1983]

48.5 B_vG

van Genuchten retention curve model with Burdine theory.

```

< B_vG (alpha alpha)
        (Theta_sat Theta_sat)
        (K_sat K_sat)
        (K_at_h K_at_h)
        (Theta_res 0 [<fraction>])
        (1 2 [])
        (n n) >

```

- *alpha*: number [**cm⁻¹**]
Parameter
van Genuchten alpha.
- *Theta_sat*: number [**<fraction>**]
State variable
Saturation point.
- *K_sat*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.

- *K_at_h*: submodel (see section 4.1.7)
Optional submodel
Water conductivity at specified pressure.

```
< h K >
```

 - *h*: number [**cm**]
Parameter
Soil water pressure.
 - *K*: number [**cm/h**]
Parameter
Water conductivity.
- *Theta_res*: number [**<fraction>**]
Parameter (default 0)
Soil residual water.
- *l*: number (dimensionless)
Parameter (default 2)
tortuosity parameter.
- *n*: number (dimensionless)
Parameter
van Genuchten n.

48.6 Cosby_et_al

Modified Campbell retention curve model with Burdine theory. Parameters estimated from soil texture as specified by Cosby et al.

48.7 MACRO

van Genuchten retention curve model with Mualem theory. The near saturated retention and hydraulic properties have been adjusted to take macropores into account. See also [Larsbo and Jarvis, 2003]

```
< MACRO (alpha alpha)
      (Theta_sat Theta_sat)
      (K_sat K_sat)
      (Theta_res 0 [<fraction>])
      (h_b h_b)
      (l 0.5 [])
      (Theta_b Theta_b)
      (K_b K_b)
      (n n)
      (n_ma n_ma)
      (enable_K_macro true)
      (enable_Theta_macro true) >
```

- *alpha*: number [**cm⁻¹**]
Parameter
van Genuchten alpha.
- *Theta_sat*: number [**<fraction>**]
State variable
Saturation point.

- *K_sat*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
- *Theta_res*: number [<**fraction**>]
Parameter (default 0)
Soil residual water.
- *h_b*: number [**cm**]
Parameter
Pressure at boundary point of change between matrix and macropores domains.
- *l*: number (dimensionless)
Parameter (default 0.5)
tortuosity parameter.
- *Theta_b*: number [<**fraction**>]
Parameter
Water content at boundary point.
- *K_b*: number [**cm/h**]
Optional parameter
Water conductivity at boundary point.
- *n*: number (dimensionless)
Parameter
van Genuchten n.
- *n_ma*: number (dimensionless)
Parameter
Macropore size distribution factor.
- *enable_K_macro*: boolean (see section 4.1.2)
Parameter (default true)
Include contribution from macropores in conductivity curve.
- *enable_Theta_macro*: boolean (see section 4.1.2)
Parameter (default true)
Include contribution from macropores in retention curve.

48.8 M_BaC

Brooks and Corey retention curve model with Mualem theory.

```
< M_BaC (Theta_sat Theta_sat)
        (lambda lambda)
        (K_sat K_sat)
        (K_at_h K_at_h)
        (Theta_res 0 [<fraction>])
        (h_b h_b)
        (l 0.5 [])
        (p p) >
```

- *Theta_sat*: number [<**fraction**>]
State variable
Saturation point.

- *lambda*: number (dimensionless)
Parameter
Pore size index.
- *K_sat*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
- *K_at_h*: submodel (see section 4.1.7)
Optional submodel
Water conductivity at specified pressure.

```

      <  h  K  >
      - h: number [cm]
        Parameter
        Soil water pressure.
      - K: number [cm/h]
        Parameter
        Water conductivity.
```
- *Theta_res*: number [<**fraction**>]
Parameter (default 0)
Soil residual water.
- *h_b*: number [**cm**]
Parameter
Bubbling pressure.
- *l*: number (dimensionless)
Parameter (default 0.5)
Burdine form parameter. Ignored if 'p' is set.
- *p*: number (dimensionless)
Optional parameter
Hydraulic conductivity form parameter. By default $p=1+2+2/\lambda$.

48.9 M_BaC_Bimodal

Brooks and Corey retention curve model with Mualem theory. Bimodal hydraulic conductivity curve.

```

< M_BaC_Bimodal  (Theta_sat Theta_sat)
                  (lambda lambda)
                  (K_sat K_sat)
                  (Theta_res 0 [<fraction>])
                  (h_b h_b)
                  (Theta_b Theta_b)
                  (K_b K_b) >
```

- *Theta_sat*: number [<**fraction**>]
State variable
Saturation point.
- *lambda*: number (dimensionless)
Parameter
Pore size index.

- *K_{sat}*: number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
- *Theta_{res}*: number [<fraction>]
Parameter (default 0)
Soil residual water.
- *h_b*: number [**cm**]
Parameter
Bubbling pressure.
- *Theta_b*: number (dimensionless)
Parameter
Water content at 'h_b'.
- *K_b*: number [**cm/h**]
Parameter
Water conductivity at 'h_b'.

48.10 M_BivG

Bimodal van Genuchten retention curve model with Mualem theory (Durner, 1994).

- ```
< M_BivG (Theta_sat Theta_sat)
 (K_sat K_sat)
 (K_at_h K_at_h)
 (Theta_res 0 [<fraction>])
 (1 0.5 [])
 (alpha1 alpha1)
 (n1 n1)
 (alpha2 alpha2)
 (n2 n2)
 (w2 w2) >
```
- *Theta<sub>sat</sub>*: number [<fraction>]  
State variable  
Saturation point.
  - *K<sub>sat</sub>*: number [**cm/h**]  
Optional parameter  
Water conductivity of saturated soil.
  - *K<sub>at\_h</sub>*: submodel (see section 4.1.7)  
Optional submodel  
Water conductivity at specified pressure.  

```
< h K >
```

    - *h*: number [**cm**]  
Parameter  
Soil water pressure.
    - *K*: number [**cm/h**]  
Parameter  
Water conductivity.
  - *Theta<sub>res</sub>*: number [<fraction>]  
Parameter (default 0)  
Soil residual water.



- *l*: number (dimensionless)  
Parameter (default 0.5)  
tortuosity parameter.
- *alpha1*: number [**cm**<sup>-1</sup>]  
Parameter  
van Genuchten alpha. for mode 1
- *n1*: number (dimensionless)  
Parameter  
van Genuchten n. for mode 1
- *alpha2*: number [**cm**<sup>-1</sup>]  
Parameter  
van Genuchten alpha. for mode 2
- *n2*: number (dimensionless)  
Parameter  
van Genuchten n. for mode 2
- *w2*: number (dimensionless)  
Parameter  
weight of mode 2

## 48.11 M\_C

Campbell retention curve model with Mualem theory. See also [Campbell, 1974, Mualem, 1976]

```
< M_C (b b)
 (Theta_sat Theta_sat)
 (K_sat K_sat)
 (K_at_h K_at_h)
 (h_b h_b)
 (1 0.5 [])
 (p p) >
```

- *b*: number (dimensionless)  
Parameter  
Campbell parameter.
- *Theta\_sat*: number [**<fraction>**]  
State variable  
Saturation point.
- *K\_sat*: number [**cm/h**]  
Optional parameter  
Water conductivity of saturated soil.
- *K\_at\_h*: submodel (see section 4.1.7)  
Optional submodel  
Water conductivity at specified pressure.

```
< h K >
```

- *h*: number [**cm**]  
Parameter  
Soil water pressure.

- $K$ : number [**cm/h**]  
Parameter  
Water conductivity.
- $h_b$ : number [**cm**]  
Parameter  
Bubbling pressure.
- $l$ : number (dimensionless)  
Parameter (default 0.5)  
Mualem form parameter. Ignored if 'p' is set.
- $p$ : number (dimensionless)  
Optional parameter  
Hydraulic conductivity form parameter. By default  $p=1+2+2*b$ .

## 48.12 M\_vG

van Genuchten retention curve model with Mualem theory.

- ```
< M_vG (alpha alpha)
      (Theta_sat Theta_sat)
      (K_sat K_sat)
      (K_at_h K_at_h)
      (Theta_res 0 [<fraction>])
      (1 0.5 [])
      (n n) >
```
- α : number [**cm⁻¹**]
Parameter
van Genuchten alpha.
 - Θ_{sat} : number [<**fraction**>]
State variable
Saturation point.
 - K_{sat} : number [**cm/h**]
Optional parameter
Water conductivity of saturated soil.
 - K_{at_h} : submodel (see section 4.1.7)
Optional submodel
Water conductivity at specified pressure.
- ```
< h K >
```
- $h$ : number [**cm**]  
Parameter  
Soil water pressure.
  - $K$ : number [**cm/h**]  
Parameter  
Water conductivity.
  - $\Theta_{res}$ : number [<**fraction**>]  
Parameter (default 0)  
Soil residual water.

- *l*: number (dimensionless)  
Parameter (default 0.5)  
tortuosity parameter.
- *n*: number (dimensionless)  
Parameter  
van Genuchten *n*.

### 48.13 *M\_vG\_compact*

van Genuchten retention curve model with Mualem theory and compaction.

```
< M_vG_compact (Theta_sat Theta_sat)
 (Theta_res 0 [<fraction>])
 (ref_alpha ref_alpha)
 (ref_n ref_n)
 (ref_K_sat ref_K_sat)
 (mod_alpha mod_alpha)
 (mod_n mod_n)
 (mod_K_sat mod_K_sat) >
```

- *Theta\_sat*: number [<fraction>]  
State variable  
Saturation point.
- *Theta\_res*: number [<fraction>]  
Parameter (default 0)  
Soil residual water.
- *ref\_alpha*: number [ $\text{cm}^{-1}$ ]  
Parameter  
Reference van Genuchten alpha.
- *ref\_n*: number (dimensionless)  
Parameter  
Reference van Genuchten *n*.
- *ref\_K\_sat*: number [ $\text{cm/h}$ ]  
Parameter  
Reference water conductivity of saturated soil.
- *mod\_alpha*: plf [<fraction> → <none>]  
Parameter  
Porosity modifier for van Genuchten alpha.
- *mod\_n*: plf [<fraction> → <none>]  
Parameter  
Porosity modifier for van Genuchten *n*.
- *mod\_K\_sat*: plf [<fraction> → <none>]  
Parameter  
Porosity modifier for water conductivity of saturated soil.

### 48.14 *M\_vGip*

Modified van Genuchten retention curve model with Mualem theory.

```

< M_vGip (alpha alpha)
 (Theta_sat Theta_sat)
 (K_sat K_sat)
 (K_at_h K_at_h)
 (Theta_res 0 [<fraction>])
 (l 0.5 [])
 (n n)
 (he -2 [cm]) >

```

- *alpha*: number [**cm**<sup>-1</sup>]  
Parameter  
van Genuchten alpha.
- *Theta\_sat*: number [<**fraction**>]  
State variable  
Saturation point.
- *K\_sat*: number [**cm/h**]  
Optional parameter  
Water conductivity of saturated soil.
- *K\_at\_h*: submodel (see section 4.1.7)  
Optional submodel  
Water conductivity at specified pressure.

```
< h K >
```

- *h*: number [**cm**]  
Parameter  
Soil water pressure.
- *K*: number [**cm/h**]  
Parameter  
Water conductivity.
- *Theta\_res*: number [<**fraction**>]  
Parameter (default 0)  
Soil residual water.
- *l*: number (dimensionless)  
Parameter (default 0.5)  
tortuosity parameter.
- *n*: number (dimensionless)  
Parameter  
van Genuchten n.
- *he*: number [**cm**]  
Parameter (default -2)  
Bubbling pressure.

### 48.15 M\_vGp

van Genuchten retention curve model with Mualem theory. A  $p_m(h)$  function is multiplied to the conductivity to simulate the change near macropores.

$p_m = (1/(-h \cdot X + 1))^f$ ;  $h > h_m$   $p_m = (1/(-h_m \cdot X + 1))^f$ ;  $h \leq h_m$   $X = 1 \text{ cm}^{-1}$  See also [Børgesen et al., 2006]

```

< M_vGp (alpha alpha)
 (Theta_sat Theta_sat)
 (K_sat K_sat)
 (Theta_res 0 [<fraction>])
 (l 0.5 [])
 (n n)
 (h_m h_m)
 (f f) >

```

- *alpha*: number [**cm**<sup>-1</sup>]  
Parameter  
van Genuchten alpha.
- *Theta\_sat*: number [<**fraction**>]  
State variable  
Saturation point.
- *K\_sat*: number [**cm/h**]  
Optional parameter  
Water conductivity of saturated soil.
- *Theta\_res*: number [<**fraction**>]  
Parameter (default 0)  
Soil residual water.
- *l*: number (dimensionless)  
Parameter (default 0.5)  
tortuosity parameter.
- *n*: number (dimensionless)  
Parameter  
van Genuchten n.
- *h\_m*: number [**cm**]  
Parameter  
Pressure point of chance between matrix and macropores.
- *f*: number (dimensionless)  
Parameter  
Macropores conductivity curve shape parameter.

## 48.16 hypres

van Genuchten retention curve model with Mualem theory. Parameters specified by the HYPRES transfer function. See also [Wösten et al., 1999]

Used by horizon component hydraulic (see 47, page 231) .

```

< hypres (K_sat K_sat)
 (K_at_h K_at_h)
 (Theta_res 0.01 [<fraction>])
 (topsoil topsoil) >

```

- *K\_sat*: number [**cm/h**]  
Optional parameter  
Water conductivity of saturated soil.

- *K\_at\_h*: submodel (see section 4.1.7)  
Optional submodel  
Water conductivity at specified pressure.  

```
< h K >
```

  - *h*: number [**cm**]  
Parameter  
Soil water pressure.
  - *K*: number [**cm/h**]  
Parameter  
Water conductivity.
- *Theta\_res*: number [<fraction>]  
Parameter (default 0.01)  
Soil residual water.
- *topsoil*: boolean (see section 4.1.2)  
Optional parameter  
If set true this horizon will be initialized as a topsoil (i.e. the plowing layer), if set false it will be initialized as a subsoil. By default, the horizon will be initialized as a topsoil if and only if it is the topmost horizon in the soil profile.

## 48.17 hyprop

HYPROP-FIT parameters for bimodal van Genuchten retention curve. It includes adsorptive water retention and scaling (code 1211). See also [Iden and Durner, 2014, Peters, 2013, Pertassek and Durner, 2017]

```
< hyprop (alpha alpha ...)
(m m ...)
(ha ha)
(a -1.5 [])
(Theta_sat Theta_sat)
(K_sat K_sat)
(K_at_h K_at_h)
(Theta_res 0 [<fraction>])
(M_intervals 500)
(n n ...)
(h0 -6.30957e+06 [cm])
(w w ...)
(omega omega)
(vapor_conductivity true)
(tau tau)
(M_start M_start) >
```

- *alpha*: number [**cm**<sup>-1</sup>] sequence  
Parameter  
van Genuchten alpha parameters for each mode.
- *m*: number (dimensionless) sequence  
Optional parameter  
van Genuchten m parameters for each mode. By default (1 - (1 / n)).
- *ha*: number [**cm**]  
Optional parameter  
Pressure at air entry for the adsorptive retention. By default, 1/max (alpha).

- *a*: number (dimensionless)  
Parameter (default -1.5)  
Value description: Fixed value. See also [Peters, 2013]  
Slope on the log-log scale.
- *Theta\_sat*: number [<fraction>]  
State variable  
Saturation point.
- *K\_sat*: number [cm/h]  
Optional parameter  
Water conductivity of saturated soil.
- *K\_at\_h*: submodel (see section 4.1.7)  
Optional submodel  
Water conductivity at specified pressure.
 

$$< \quad h \quad K \quad >$$
  - *h*: number [cm]  
Parameter  
Soil water pressure.
  - *K*: number [cm/h]  
Parameter  
Water conductivity.
- *Theta\_res*: number [<fraction>]  
Parameter (default 0)  
Soil residual water.
- *M\_intervals*: integer  
Parameter (default 500)  
Number of intervals for numeric integration of K.
- *n*: number (dimensionless) sequence  
Parameter  
van Genuchten n parameters for each mode.
- *h0*: number [cm]  
Parameter (default -6.30957e+06)  
Value description: The pressure at oven dryness for 105 dg C See also [Schneider and Goss, 2012]  
Pressure where the water content is zero.
- *w*: number [<fraction>] sequence  
Parameter  
Weight of each mode. Sum must be 1.
- *omega*: number [<fraction>]  
Parameter  
Weight of film conductivity.
- *vapor\_conductivity*: string (see section 4.1.5)  
Parameter (default 'true')  
Include vapor conductivity. Possible values, 'true', 'false', 'only'. 'false': include capillary and film conductivity, but not vapor. 'true': include capillary, film, and vapor conductivity. 'only': include vapor conductivity but not capillary or film.

- *tau*: number (dimensionless)  
Parameter  
Capillary conductivity parameter.
- *M\_start*: number [**cm**]  
Optional parameter  
Start point of numeric intergration of K.  
Must be below wilting point, and below h\_surf for steady state exfiltration.  
By default same as h0.

## 48.18 linear

Hysteresis using linear transformation.

$$\text{Theta}^*(h) = a \text{ Theta}(h) + b$$

$$K^*(h) = c K(h) + d$$

where Theta and K are defined by either 'wet' or 'dry'.

'a', 'b', 'c', and 'd' are recalculated whenever the node switches between wetting and drying.

```
< linear (d d)
 (a a)
 (b b)
 (debug false)
 (wet wet)
 (dry dry)
 (is_wetting is_wetting)
 (c c) >
```

- *d*: number [**cm/h**]  
State variable  
Offset added to hydraulic conductivity (K).
- *a*: number (dimensionless)  
State variable  
Factor multiplied with water content (Theta).
- *b*: number [**cm<sup>3</sup>/cm<sup>3</sup>**]  
State variable  
Offset added to water content (Theta).
- *debug*: boolean (see section 4.1.2)  
Parameter (default false)  
Let us know when we switch between drying and wetting.
- *wet*: **hydraulic** component (see chapter 48)  
The hydraulic propeties of wetting soil.
- *dry*: **hydraulic** component (see chapter 48)  
The hydraulic propeties of drying soil.
- *is\_wetting*: boolean (see section 4.1.2)  
State variable  
True iff 'a', 'b', 'c', and 'd' are applied to the 'wet' model.
- *c*: number (dimensionless)  
State variable  
Factor multiplied with hydraulic conductivity (K).



## 48.19 mod\_C

Modified Campbell retention curve model with Burdine theory.

```
< mod_C (b b)
 (Theta_sat Theta_sat)
 (K_sat K_sat)
 (K_at_h K_at_h)
 (h_b h_b) >
```

- *b*: number (dimensionless)  
Parameter  
Campbell parameter.
- *Theta\_sat*: number [**<fraction>**]  
State variable  
Saturation point.
- *K\_sat*: number [**cm/h**]  
Optional parameter  
Water conductivity of saturated soil.
- *K\_at\_h*: submodel (see section 4.1.7)  
Optional submodel  
Water conductivity at specified pressure.

```
< h K >
```

- *h*: number [**cm**]  
Parameter  
Soil water pressure.
- *K*: number [**cm/h**]  
Parameter  
Water conductivity.
- *h\_b*: number [**cm**]  
Parameter  
Bubbling pressure.

## 48.20 old2

Reads a file of lines in the format < pF Theta Cw2 K >, where pF is the water pressure, Theta is the water content at that pressure, cw2 is dTheta/dh at that pressure [ $\text{cm}^{-1}$ ], and K is the water conductivity at that pressure [ $\text{cm/h}$ ].

There must be exactly 501 lines, with pF starting at 0 and ending at 'inc' \* 500, increasing with 'inc' on each line.

```
< old2 file
 (M_intervals 500)
 (inc 0.01 [pF]) >
```

- *file*: string (see section 4.1.5)  
Parameter  
The file to read.
- *M\_intervals*: integer  
Parameter (default 500)  
Number of intervals for numeric integration of K.

- *inc*: number [**pF**]  
Parameter (default 0.01)  
Increment in pF between each line in the file.

## 48.21 table

Reads a ddf file with column pF Theta Cw2 K.

```
< table (file file)
 (Theta_sat Theta_sat)
 (missing missing ...) ; Has default value.
 (filter)
 (original original ...)
 (dim_line dim_line)
 (K_sat K_sat)
 (Theta_res 0 [<fraction>])
 (M_intervals 500) >
```

- *file*: string (see section 4.1.5)  
Parameter  
Name of Daisy file where data is found.
- *Theta\_sat*: number [<**fraction**>]  
Optional state variable  
Saturation point.
- *missing*: string (see section 4.1.5) sequence  
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:

List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence  
Submodel (default: an empty sequence)  
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

- *tag*: string (see section 4.1.5)  
Parameter  
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence  
Parameter  
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence  
Optional parameter  
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim\_line*: boolean (see section 4.1.2)  
Optional parameter  
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *K\_sat*: number [cm/h]  
Optional parameter  
Water conductivity of saturated soil.
- *Theta\_res*: number [<fraction>]  
Parameter (default 0)  
Soil residual water.
- *M\_intervals*: integer  
Parameter (default 500)  
Number of intervals for numeric integration of K.

## 48.22 wepp

van Genuchten retention curve model with Mualem theory. Parameters specified by the HYPRES transfer function. Tillage and weather dynamics from WEPP. CECr from Krogh et al. See also [Wösten et al., 1999, Flanagan and Nearing, 1995]

```
< wepp (Theta_res 0.01 [<fraction>])
 (consolidate_factor 0.667 [])
 (consolidate_factor_water 0 [])
 (delta_pmx fixed delta_pmx_fixed)
 (time_consolidation 0.005 [d-1])
 (average_depth average_depth)
 (CECr CECr)
 (consolidated_bulk_density consolidated_bulk_density)
 (p_t p_t)
 (delta_pmx 0 [kg/m3])
 (delta_pc 0 [kg/m3])
 (RRi 0 [kg/m3])
 (br 0.3 [kg DM/m2])
 (allow_negative_delta_pc false)
 (crust_thickness 0.005 [m])
 (day_count 0 [d])
 (R_c 0 [m])
 (E_a 0 [J/m2])
 (RRt 0.013 [m])
 (freeze_effect 0 [kg/m3])
 (freeze_on 1 [<fraction>])
 (freeze_off 0 [<fraction>])
 (frozen false) >
```

- *Theta\_res*: number [<fraction>]  
Parameter (default 0.01)  
Soil residual water.
- *consolidate\_factor*: number (dimensionless)  
Parameter (default 0.667)  
The ratio between consolidated and loose bulk density.
- *consolidate\_factor\_water*: number (dimensionless)  
Parameter (default 0)

An additional water dependent consolidation factor. This is multiplied with the relative water content [0-1] at tillage, and added to 'consolidate\_factor'.

- *delta\_pmx\_fixed*: number [**kg/m<sup>3</sup>**]  
Optional parameter  
Fixed value for 'delta\_pmx'. By default 'delta\_pmx' will be calculated from clay content and dry bulk density.
- *time\_consolidation*: number [**d<sup>-1</sup>**]  
Parameter (default 0.005)  
Controls consolidation as a function of passing time.  $\text{consolidation} = \max * (1 - \exp(-\text{time\_consolidation} * \text{day\_count}))$  Larger values, faster consolidation.
- *average\_depth*: number [**cm**]  
Optional parameter  
Average depth of horizon (positive). Used for estimating 'CECr' iff unspecified. If unspecified, the value from 'Soil' is used.
- *CECr*: number [**cmolc/kg clay**]  
Optional parameter  
Ratio of the cation exchange content of the clay (CECc) to the clay content. Used for calculating 'consolidated\_bulk\_density', iff unspecified. By default calculated from 'average\_depth', CEC, and texture.  
  
If you have not specified 'CEC' in the horizon, setting 'CECr' to 50.0 will make most sense due to the way CEC is estimated from texture.
- *consolidated\_bulk\_density*: number [**kg/m<sup>3</sup>**]  
Optional parameter  
Consolidated dry bulk density after rain and time. By default, this will be calculated from texture, CEC, and depth.
- *p\_t*: number [**kg/m<sup>3</sup>**]  
Optional state variable  
Dry bulk density after last tillage. By default, this will be consolidated dry bulk density per wepp.
- *delta\_pmx*: number [**kg/m<sup>3</sup>**]  
State variable (default 0)  
Potential dry bulk density change due to rain.
- *delta\_pc*: number [**kg/m<sup>3</sup>**]  
State variable (default 0)  
Potential dry bulk density change due to time.
- *RRi*: number [**kg/m<sup>3</sup>**]  
State variable (default 0)  
Random roughness after last tillage.
- *br*: number [**kg DM/m<sup>2</sup>**]  
State variable (default 0.3)  
Buried residue 0-15 cm at last tillage.
- *allow\_negative\_delta\_pc*: boolean (see section 4.1.2)  
Parameter (default false)  
Allow 'delta\_pc' to be negative after tillage.

- *crust\_thickness*: number [**m**]  
Parameter (default 0.005)  
Thickness of soil layer that form crust after rain.
- *day\_count*: number [**d**]  
State variable (default 0)  
Number of days since last tillage.
- *R\_c*: number [**m**]  
State variable (default 0)  
Rain since last tillage.
- *E\_a*: number [**J/m<sup>2</sup>**]  
State variable (default 0)  
Energy in rain since last tillage.
- *R<sub>Rt</sub>*: number [**m**]  
State variable (default 0.013)  
Random roughness.
- *freeze\_effect*: number [**kg/m<sup>3</sup>**]  
Parameter (default 0)  
Maximum decrease of bulk density from frost. The freeze effect works by decreasing 'R\_c'. 'R\_c' will never be decreased below zero.
- *freeze\_on*: number [**<fraction>**]  
Parameter (default 1)  
Volumetric ice content above this will activate the frozen state. The freeze effect will be activated if the soil is not already frozen.
- *freeze\_off*: number [**<fraction>**]  
Parameter (default 0)  
Volumetric ice content below this will deactivate frozen state.
- *frozen*: boolean (see section 4.1.2)  
State variable (default false)  
True if the soil is considered frozen. The soil will be considered frozen after the volumetric ice content reaches above 'freeze\_on', and will stay frozen until the volumetric ice content fall below 'freeze\_off'. When the soil enter the frozen state the 'freeze\_effect' trigger.

## Log Variables

- *alpha*: number [**cm<sup>-1</sup>**]  
van Genuchten alpha.
- *rho\_b*: number [**g/cm<sup>3</sup>**]  
Current dry bulk density.
- *Theta\_sat*: number [**<fraction>**]  
Saturated water content.
- *K\_sat*: number [**cm/h**]  
Water conductivity of saturated soil.
- *l*: number (dimensionless)  
tortuosity parameter.

- *Theta\_fc*: number [**<fraction>**]  
Water content at field capacity (pF 2.0).
- *Theta\_wp*: number [**<fraction>**]  
Water content at wilting point (pF 4.2).
- *n*: number (dimensionless)  
van Genuchten n.
- *K\_bare*: number [**mm/h**]  
Bare soil conductivity.
- *K\_15*: number [**cm/h**]  
Water conductivity for h = -1.5 cm.
- *K\_30*: number [**cm/h**]  
Water conductivity for h = -3.0 cm.
- *K\_60*: number [**cm/h**]  
Water conductivity for h = -6.0 cm.
- *K\_120*: number [**cm/h**]  
Water conductivity for h = -12.0 cm.
- *K\_fc*: number [**cm/h**]  
Water conductivity at field capacity (pF 2).
- *K\_wp*: number [**cm/h**]  
Water conductivity at wilting point (pF 4.2).

### 48.23 Styczen

A ‘wepp’ model (see 48.22, page 267) build into Daisy.

Parameterization of for Danish soils. See also [Wösten et al., 1999, Flanagan and Nearing, 1995]

```
< (delta_pmx_fixed 80 [kg/m3])
 (CECr 50 [cmolc/kg clay])
 (allow_negative_delta_pc true)
 (freeze_effect 100 [kg/m3])
 (freeze_on 0.19 [<fraction>])
 (freeze_off 0.01 [<fraction>]) >
```

### 48.24 yolo

Yolo soil. Haverkamp et.al., 1977.

```
< yolo (M_intervals 500) >
```

- *M\_intervals*: integer  
Parameter (default 500)  
Number of intervals for numeric integration of K.

# Chapter 49

## integer

Generic representation of integers.

### 49.1 \*

Use the product of its operands.

< "\*" *operands* ... >

- *operands*: **integer** component (see chapter 49) sequence  
The operands for this function.

### 49.2 +

Use the sum of its operands.

< "+" *operands* ... >

- *operands*: **integer** component (see chapter 49) sequence  
The operands for this function.

### 49.3 -

Negate integer or subtract integers. With one operand, negates it. With more than one operand, subtracts all but the first from the first.

< "-" *operands* ... >

- *operands*: **integer** component (see chapter 49) sequence  
The operands for this function.

### 49.4 cond

Return the value of the first clause whose condition is true.

< cond *clauses* ... >

- *clauses*: **IntegerCondClause** fixed component (see section 107.28) sequence  
List of clauses to match for.

## 49.5 **const**

Always give the specified value.

`< const value >`

- *value*: integer  
Parameter  
Fixed value for this integer.

## 49.6 **div**

Divide the first operand by the rest.

`< div operands ... >`

- *operands*: **integer** component (see chapter 49) array of length 2  
The operands for this function.

## 49.7 **max**

Use the largest value of its operands.

`< max operands ... >`

- *operands*: **integer** component (see chapter 49) sequence  
The operands for this function.

## 49.8 **min**

Use the smallest value of its operands.

`< min operands ... >`

- *operands*: **integer** component (see chapter 49) sequence  
The operands for this function.

## 49.9 **mod**

Modulo the first operand by the rest.

`< mod operands ... >`

- *operands*: **integer** component (see chapter 49) array of length 2  
The operands for this function.

## 49.10 **sqr**

Take the square of its argument.

`< sqr operand >`

- *operand*: **integer** component (see chapter 49)  
Operand for this function.



# Chapter 50

## listsum

Condense a list of numbers to a single number.

### 50.1 arithmetic

Arithmetic average of all members of a list.

### 50.2 count

Count all members of a list.

### 50.3 max

The maximum of all members of a list.

### 50.4 min

The minimum of all members of a list.

### 50.5 sum

The sum of all members of a list.



# Chapter 51

## litter

Litter, surface residuals, or mulch below canopy.

< component >

### Log Variables

- *cover*: number [<fraction>]  
Fraction of surface area covered by litter.

### 51.1 none

The effect of surface residuals is ignored by the model.

Used by column default Litter (see 26.1, page 153) .

### 51.2 permanent

A permanent litter layer cover the ground, as for example in a forest.

```
< permanent (albedo albedo)
 (vapor_flux_factor 1 [<fraction>])
 (interception_capacity interception_capacity) >
```

- *albedo*: number (dimensionless)  
Optional parameter  
Reflection factor. By default, the surface albedo will be used.
- *vapor\_flux\_factor*: number [<fraction>]  
Parameter (default 1)  
Reduction factor for potential evaporation below litter.
- *interception\_capacity*: number [**mm**]  
Parameter  
Storage capacity of litter.

### 51.3 residue

A dynamic litter layer based on applied fertilizer and crop residuals.

A 'mulch area index' is calculated from the surface organic materials, and from that a mulch cover is calculated based on Beer's law similarly to how the crop cover is calculated from the leaf area index. See also [Scopel et al., 2004]

```
< residue (albedo albedo)
 (vapor_flux_factor 0 [<fraction>])
 (water_capacity water_capacity)
 (specific_AI specific_AI)
 (extinction_coefficient extinction_coefficient) >
```

- *albedo*: number (dimensionless)  
Optional parameter  
Reflection factor. By default, the surface albedo will be used.
- *vapor\_flux\_factor*: number [<fraction>]  
Parameter (default 0)  
Reduction factor for potential evaporation below litter. Only area covered by residue is affected.
- *water\_capacity*: number [L/kg]  
Parameter  
Water holding capacity of surface residulas.
- *specific\_AI*: number [m<sup>2</sup>/kg DM]  
Parameter  
Area covered per litter mass.
- *extinction\_coefficient*: number (dimensionless)  
Parameter  
Beer's law extinction coefficient for litter.

## Log Variables

- *mass*: number [kg DM/m<sup>2</sup>]  
Total mass of mulch layer.

## 51.4 Maize

A 'residue' model (see 51.3, page 275) build into Daisy.

Maize crop residues in La Tinaja. See also [Scopel et al., 1998]

```
< (water_capacity 3.8 [L/kg])
 (specific_AI 3.7 [m2/kg DM])
 (extinction_coefficient 0.8 []) >
```

## 51.5 Millet

A 'residue' model (see 51.3, page 275) build into Daisy.

Millet crop residues in Planaltina. See also [Macena et al., 2003]

```
< (water_capacity 3.2 [L/kg])
 (specific_AI 3.9 [m2/kg DM])
 (extinction_coefficient 0.45 []) >
```

## 51.6 mulch

A 'residue' model (see 51.3, page 275) build into Daisy.

A decomposing mulch layer.

The bottom of the mulch layer may decompose based on the conditions in the top of soil. See also [Findeling et al., 2007]

```

< mulch (alpha 0 [])
 (retention retention) ; Default PASTIS value.
 (T_ref 10 [dg C])
 (decompose_heat_factor decompose_heat_factor) ; Has default value.
 (decompose_water_factor decompose_water_factor) ; Has default value.
 (decompose_SMB_pool 1)
 (decompose_SMB_KM 0 [g C/cm3])
 (SMB_ref SMB_ref)
 (DOC_name DOC)
 (DON_name DON)
 (density density)
 (Theta_res Theta_res)
 (particle_density particle_density)
 (decompose_height decompose_height)
 (evaporate_depth 1000 [cm])
 (soil_height soil_height)
 (h_min -3.16228e+06 [cm])
 (factor_exch factor_exch)
 (Si Si)
 (use_soil_decompose true)
 ;; Shared parameters are described in section 51.3.
 (albedo albedo)
 (vapor_flux_factor 0 [<fraction>])
 (water_capacity water_capacity)
 (specific_AI specific_AI)
 (extinction_coefficient extinction_coefficient) >

```

- *alpha*: number (dimensionless)  
Parameter (default 0)  
Interception parameter. The fraction of water hitting the litter will be determined by:  
$$\exp(-\alpha (\Theta_{\text{sat}} - \Theta_{\text{res}}) / (\Theta_{\text{sat}} - \Theta))$$
  
If alpha is 0 (default), all water hitting the canopy will be intercepted.
- *retention*: **retention** component (see chapter 71)  
Component (default 'PASTIS')  
The retention curve to use.
- *T\_ref*: number [dg C]  
Parameter (default 10)  
Reference temperature for decomposition. The heat factor on decomposition will be scaled so it is 1 at this temperature..
- *decompose\_heat\_factor*: plf [dg C → <none>]  
Parameter (has default value with 0 points)  
Heat factor on decomposition.
- *decompose\_water\_factor*: plf [cm → <none>]  
Parameter (has default value with 0 points)  
Water potential factor on decomposition.
- *decompose\_SMB\_pool*: integer  
Parameter (default 1)  
SMB pool for Michaelis-Menten kinetics. Use 0 for SMB1, 1 for SMB2, or -1 for all SMB pools.

- *decompose\_SMB\_KM*: number [**g C/cm<sup>3</sup>**]  
Parameter (default 0)  
Michaelis-Menten kinetics parameter. Decompose rate is modified by  $C / (KM + C)$ , where C is the carbon content in the pool specified by 'decompose\_SMB\_pool'.
- *SMB\_ref*: number [**g C/cm<sup>3</sup>**]  
Optional parameter  
Reference SMB carbon for decomposition of mulch. The SMB factor for decomposition will be scaled so it is 1 at this amount of SMB carbon. By default, it will not be scaled.
- *DOC\_name*: string (see section 4.1.5)  
Parameter (default 'DOC')  
Name of compound representing dissolved organic carbon.
- *DON\_name*: string (see section 4.1.5)  
Parameter (default 'DON')  
Name of compound representing dissolved organic nitrogen.
- *density*: number [**kg DM/m<sup>3</sup>**]  
Parameter  
Bulk density of mulch layer.
- *Theta\_res*: number (dimensionless)  
Parameter  
Water content where biological activity stops
- *particle\_density*: number [**kg DM/m<sup>3</sup>**]  
Optional parameter  
Particle density of mulch layer. If unspecified, use bulk density instead.
- *decompose\_height*: number [**cm**]  
Parameter  
Height of mulch layer considered in contact with the soil. Only this part of the mulch layer will decompose
- *evaporate\_depth*: number [**cm**]  
Parameter (default 1000)  
Depth into mulch layer affected by evaporation. Only water in this part of the mulch layer will evaporate.  
  
The mulch layer is assumed to be filled up from the bottom.  
  
If W is the amount of water in the mulch, R is the residual water ( $\Theta_{res} * C$ ), C is the water capacity, D is this parameter, and H is the height of the mulch layer, the maximum evaporation (Ea\_max) can be found as:  
  
$$Ea_{max} = \max(0, (W - R) - (C - R) * (H - \min(D, H)) / H)$$
- *soil\_height*: number [**cm**]  
Parameter  
Height of soil layer (a negative number) contributing to decay.
- *h\_min*: number [**cm**]  
Parameter (default -3.16228e+06)  
Water pressure where biological activity stops.

- *factor\_exch*: number (dimensionless)  
Parameter  
Limiting factor when calculating Darcy exchange between mulch and soil. It is intended to emulate poor contact between the two media.
- *Si*: number [**<fraction>**]  
Parameter  
Water content where diffusion to wash off begins relative to Theta\_sat. Theta\_sat = 100 %
- *use\_soil\_decompose*: boolean (see section 4.1.2)  
Parameter (default true)  
Use temperature and moisture of top soil for turnover and decomposition. The depth of the top soil is determined by 'soil\_height'.

## Log Variables

- *h*: number [**cm**]  
Mulch water potential at stat of timestep.
- *water*: number [**mm**]  
Total water in mulch.
- *Theta*: number (dimensionless)  
Relative water content.
- *factor*: number (dimensionless)  
The combined effect of SMB, T, h, and contact. This affects both turnover and chemical decomposition in litter.
- *T*: number [**dg C**]  
Temperature of water in mulch.
- *h\_factor*: number (dimensionless)  
Water potential effect on turnover.
- *T\_factor*: number (dimensionless)  
Temperature effect on turnover.
- *SMB.C*: number [**g C/cm<sup>3</sup>**]  
Microbial C in top soil.
- *height*: number [**cm**]  
Total height of mulch layer.
- *contact*: number [**<fraction>**]  
Fraction of mulch layer in contact with soil. DOM in this part of the mulch is degraded.
- *protected\_water*: number [**mm**]  
Amount of water potentially protected from evaporation. Only when the water content exceeds this amount it will evaporate.
- *h1*: number [**cm**]  
Estimated mulch water potential at end of timestep.
- *h\_soil*: number [**cm**]  
Soil water potential.

- *K\_soil*: number [**cm/h**]  
Soil water conductivity.
- *E\_darcy*: number [**cm/h**]  
Potential mulch-soil water exchange as calculated by modified Darcy. Positive down.
- *T\_soil*: number [**dg C**]  
Temperature of top soil.
- *SMB\_factor*: number (dimensionless)  
Microbical effect on turnover.
- *DOC\_gen*: number [**g/cm<sup>2</sup>/h**]  
Dissolved organic carbon generated from turnover.
- *DON\_gen*: number [**g/cm<sup>2</sup>/h**]  
Dissolved organic nitrogen generated from turnover.
- *SOC\_gen*: number [**g/cm<sup>2</sup>/h**]  
Stationary organic carbon generated from turnover.
- *SON\_gen*: number [**g/cm<sup>2</sup>/h**]  
Stationary organic nitrogen generated from turnover.



# Chapter 52

## log

Running a simulation is uninteresting, unless you can get access to the results in one way or another. The purpose of the 'log' component is to provide this access. Most 'log' models does this by writing a summary of the state to a log file.

### 52.1 checkpoint

Create a checkpoint of the entire simulation state, suitable for later hot start.

```
< checkpoint (where checkpoint)
 (when when) ; Default finished value. >
```

- *where*: string (see section 4.1.5)  
Parameter (default 'checkpoint')  
File name prefix for the generated checkpoint. The time will be appended, together with the '.dai' suffix.
- *when*: **condition** component (see chapter 28)  
Component (default 'finished')  
Make a checkpoint every time this condition is true.

### 52.2 harvest

Create a log of all harvests.

```
< harvest (where "harvest.dlf")
 (print_header true)
 (print_tags true)
 (print_dimension true)
 (print_N true)
 (print_C false) >
```

- *where*: string (see section 4.1.5)  
Parameter (default 'harvest.dlf')  
Name of the log file to create.
- *print\_header*: string (see section 4.1.5)  
Parameter (default 'true')  
If this is set to 'false', no header is printed. If this is set to 'true', a full header is printed. If this is set to 'fixed', a small fixed size header is printed.
- *print\_tags*: boolean (see section 4.1.2)  
Parameter (default true)  
Print a tag line in the file.

- *print\_dimension*: boolean (see section 4.1.2)  
Parameter (default true)  
Print a line with units after the tag line.
- *print\_N*: boolean (see section 4.1.2)  
Parameter (default true)  
Print nitrogen content of harvest.
- *print\_C*: boolean (see section 4.1.2)  
Parameter (default false)  
Print carbon content of harvest.

**Table columns (common):**

- *stem\_DM*: [Mg DM/ha]  
Stem dry matter removed by harvest.
- *dead\_DM*: [Mg DM/ha]  
Yellow leaves dry matter removed by harvest.
- *leaf\_DM*: [Mg DM/ha]  
Green leaves dry matter removed by harvest.
- *sorg\_DM*: [Mg DM/ha]  
Storage organ (grains or tuber) dry matter removed by harvest. For some crops, only the economically important part of the storage organ is counted.

**Table columns (if *print\_N* is set):**

- *stem\_N*: [kg N/ha]  
Stem nitrogen removed by harvest.
- *dead\_N*: [kg N/ha]  
Yellow leaves nitrogen removed by harvest.
- *leaf\_N*: [kg N/ha]  
Green leaves nitrogen removed by harvest.
- *sorg\_N*: [kg N/ha]  
Storage organ (grains or tuber) nitrogen removed by harvest.

**Table columns (if *print\_C* is set):**

- *stem\_C*: [kg C/ha]  
Stem carbon removed by harvest.
- *dead\_C*: [kg C/ha]  
Yellow leaves carbon removed by harvest.
- *leaf\_C*: [kg C/ha]  
Green leaves carbon removed by harvest.
- *sorg\_C*: [kg C/ha]  
Storage organ (grains or tuber) carbon removed by harvest.

**Table columns (common):**

- *WStress*: [d]  
Number of days worth of production lost due to water stress.  
This is just a rough measure. It is calculated by weighting the water stress variation over the day with the amount of global radiation received in the same period. So water stress at noon with high radiation counts a lot more than water stress near sunset with low radiation. All days are counted the same, so water stress a day with low radiation is counted the same as water stress a day with high radiation.
- *NStress*: [d]  
Number of days worth of production lost due to nitrogen stress.  
This is just a rough measure. It is calculated by weighting the nitrogen stress variation over the day with the amount of global radiation received in the same period. So nitrogen stress at noon with high radiation counts a lot more than nitrogen stress near sunset with low radiation. All days are counted the same, so nitrogen stress a day with low radiation is counted the same as nitrogen stress a day with high radiation.
- *WP\_ET*: [kg/m<sup>3</sup>]  
A measure for water usage efficiency. Specifically the economic yield (storage organ) divided by the total evapotranspiration in the period from emergence to maturity or harvest.

## 52.3 select

Select variables to log.

```
< select (volume volume) ; Default box value.
 (summary)
 (entries entries ...)
 (from from)
 (to to)
 (when when)
 (time_columns time_columns)
 (parameter_names)
 (active active) ; Default true value.
 (print_initial print_initial) >
```

- *volume*: **volume** component (see chapter 102)  
Component (default 'box')  
Soil volume to log.
- *summary*: **summary** component (see chapter 86) sequence  
Component (default: an empty sequence)  
Summaries for this log file.
- *entries*: **select** component (see chapter 80) sequence  
What to log in each column.
- *from*: number [cm]  
Optional parameter  
Default 'from' value for all entries. By default, use the top of the soil. OBSOLETE: Use (volume box (top FROM)) instead.
- *to*: number [cm]  
Optional parameter

Default 'to' value for all entries. By default, use the bottom of the soil.  
 OBSOLETE: Use (volume box (bottom TO)) instead.

- *when*: **condition** component (see chapter 28)  
 Add entries to the log file when this condition is true.
- *time\_columns*: boolean (see section 4.1.2)  
 Optional parameter  
 Iff true, add columns for year, month, mday and hour in the beginning of the lines. By default, this will be true if you have not specified any time entries yourself.
- *parameter\_names*: string (see section 4.1.5) sequence  
 Parameter (default: an empty sequence)  
 List of string parameters to print to the table header.  
 For example, if you have defined 'column' and 'crop' parameters for this table log parameterization, you can print them to the log file header by specifying '(parameter\_names column crop)'.
- *active*: **condition** component (see chapter 28)  
 Component (default 'true')  
 Add data when this condition is true. E.g. count percolation only when there is no crop.
- *print\_initial*: boolean (see section 4.1.2)  
 Optional parameter  
 Print a line with initial values when logging starts. By default, an initial line will be printed if any entry has 'handle' set to 'current'.

## 52.4 extern

A 'select' model (see 52.3, page 283) build into Daisy.

Log simulation state for extern use.

```
< extern (numbers numbers ...)
 (where where)
 (print_initial false)
 ;; Shared parameters are described in section 52.3.
 (volume volume) ; Default box value.
 (summary)
 (entries entries ...)
 (from from)
 (to to)
 (when when)
 (time_columns time_columns)
 (parameter_names)
 (active active) ; Default true value. >
```

- *numbers*: submodel (see section 4.1.7) sequence  
 Optional submodel  
 Initial numeric values. By default, none.  

```
< (value value)
 (name name) >
```

  - *value*: number (dimension not specified)  
 State variable  
 Numeric value.

- *name*: string (see section 4.1.5)  
State variable  
Name to refer to number with.
- *where*: string (see section 4.1.5)  
Optional parameter  
Name of the extern log to use. By default, use the model name.

## 52.5 table

A ‘select’ model (see 52.3, page 283) build into Daisy.

Write results in a tabular Daisy log file.

```
< table (where where)
 (missing_value "00.00")
 (print_header true)
 (print_tags true)
 (print_dimension true)
 (flush false)
 (record_separator " ")
 (field_separator " ")
 (array_separator " ")
 ;; Shared parameters are described in section 52.3.
 (volume volume) ; Default box value.
 (summary)
 (entries entries ...)
 (from from)
 (to to)
 (when when)
 (time_columns time_columns)
 (parameter_names)
 (active active) ; Default true value.
 (print_initial print_initial) >
```

- *where*: string (see section 4.1.5)  
Parameter  
Name of the log file to create.
- *missing\_value*: string (see section 4.1.5)  
Parameter (default ‘00.00’)  
String to print when the path doesn’t match anything. This can be relevant for example if you are logging a crop, and there are no crops on the field.
- *print\_header*: string (see section 4.1.5)  
Parameter (default ‘true’)  
If this is set to ‘false’, no header is printed. If this is set to ‘true’, a full header is printed. If this is set to ‘fixed’, a small fixed size header is printed.
- *print\_tags*: boolean (see section 4.1.2)  
Parameter (default true)  
Print a tag line in the file.
- *print\_dimension*: boolean (see section 4.1.2)  
Parameter (default true)  
Print a line with units after the tag line.

- *flush*: boolean (see section 4.1.2)  
Parameter (default false)  
Flush to disk after each entry (for debugging).
- *record\_separator*: string (see section 4.1.5)  
Parameter (default ' ')  
String to print between records (time steps).
- *field\_separator*: string (see section 4.1.5)  
Parameter (default ' ')  
String to print between fields.
- *array\_separator*: string (see section 4.1.5)  
Parameter (default ' ')  
String to print between array entries.

## 52.6 column

A 'table' model (see 52.5, page 285) defined in 'log-std.dai'.

A log table for a specific column.

```
< column (column "*")
 (parameter_names parameter_names ...) ; Has default value.
 (colfid colfid) ; Default cond value. >
```

- *column*: string (see section 4.1.5)  
Parameter (default '\*')  
Name of column to log. Use "\*" to log all columns.
- *colfid*: **string** component (see chapter 85)  
Component (default 'cond')

```
(colfid cond ("string-equal" "${column}" "*")
 "")
 ((true)
 "${column}_"))
```

Parameter description:

File component name indicating column logged.

## 52.7 Field nitrogen

A 'column' model (see 52.6, page 286) defined in 'log-std.dai'.

Nitrogen input, output, transformation and content for the system.

The intended use of this log is large scale nitrogen balance, for example reservoir management. It provide information about how much nitrogen is in the field (down to a specified depth), where it is located in the field (surface, soil matrix or in biopores), what form is has (crop, soil organic matter, or mineral) as well as the sources, sinks and amounts of nitrogen entering or leaving the system, and transformation between the four forms. It does not provide information about internal translocation of nitrogen between surface, soil matrix and biopores, use see the 'Soil nitrogen' log instead for that.

For the balances of this log to work, you must include the entire root zone

```

< "Field nitrogen" (summary summary ...) ; Has default value.
 (unit "kg N/ha")
 (where where) ; Has default value.
 (entries entries ...) ; Has default value.
 (when when) ; Default hourly value. >

```

- *unit*: string (see section 4.1.5)  
Parameter (default 'kg N/ha')  
Base unit.

**Table columns:**

- *Min-Surface-Fertilizer*: [kg N/ha]  
Mineral fertilizer added above the soil surface.
- *Min-Soil-Fertilizer*: [kg N/ha]  
Mineral fertilizer incorporated into the soil.
- *Deposition*: [kg N/ha]  
Nitrogen added to the soil surface as atmospheric deposition.
- *Matrix-Leaching*: [kg N/ha]  
Loss from leaching through the soil matrix below the specified depth. This can be negative in case of capillary rise.
- *Biopore-Leaching*: [kg N/ha]  
Loss from leaching through biopores below the specified depth.
- *Soil-Drain*: [kg N/ha]  
Loss of nitrogen from the soil matrix to drain pipes.
- *Biopore-Drain*: [kg N/ha]  
Loss of nitrogen from the biopores to drain pipes.
- *Surface-Loss*: [kg N/ha]  
Loss of mineral nitrogen stored above surface.  
  
This includes volatilization, surface runoff, mineral nitrogen on leaves removed at harvest, and chemical transformations.
- *Min-Surface*: [kg N/ha]  
Mineral nitrogen stored above the soil surface. This includes leaves, ponded water, etc.
- *Min-Soil*: [kg N/ha]  
Mineral nitrogen stored in the soil matrix.
- *Min-Biopores*: [kg N/ha]  
Mineral nitrogen stored in the biopores.
- *Error*: [kg N/ha]  
Mineral nitrogen 'borrowed from the future'.  
  
This can be non-zero if the processes that remove nitrogen from some place in the soil are fast compared to the timestep. If this happens, you should rerun the simulation with a shorter timestep. Or readjust the system to slow down those processes that remove mineral nitrogen.
- *Mineralization*: [kg N/ha]  
NH<sub>4</sub> produced in the soil matrix by mineralization of organic matter.

- *Immobilization:* [kg N/ha]  
NO<sub>3</sub> immobilized as a result of organic matter turnover.
- *Crop-Uptake:* [kg N/ha]  
Nitrogen uptake by the roots.
- *Volatilization:* [kg N/ha]  
NH<sub>4</sub>-fertilizer lost during the application.
- *N<sub>2</sub>O-Nitrification:* [kg N/ha]  
N<sub>2</sub>O production in the soil matrix due to nitrification.  
  
The N<sub>2</sub>O produced is assumed to escape to the atmosphere, and will not be traced by the model.
- *NO<sub>3</sub>-Denitrification:* [kg N/ha]  
Loss of NO<sub>3</sub>-N due to denitrification.
- *N<sub>2</sub>O-Denitrification:* [kg N/ha]  
N<sub>2</sub>O-N released by denitrification.
- *Fixated:* [kg N/ha]  
Atmospheric nitrogen fixated by the crop.
- *Org-Fertilizer:* [kg N/ha]  
Organically bound nitrogen supplied by fertilizers.
- *Seed:* [kg N/ha]  
Nitrogen supplied in seeds when sowing.
- *Harvest:* [kg N/ha]  
Nitrogen removed as part of the harvest.
- *Residuals-Surface:* [kg N/ha]  
Above ground plant residuals.  
  
This includes both exfoliation during crop growth, and any above ground residuals left after harvest.
- *Residuals-Soil:* [kg N/ha]  
Below ground plant residuals.  
  
This includes root death and rhizodeposition during plant growth, as well as the roots left after harvest.
- *Org-Surface:* [kg N/ha]  
Nitrogen content of organic material on the soil surface.  
  
This does not include nitrogen in the crop.
- *Org-Soil:* [kg N/ha]  
Nitrogen content of organic material below the soil surface.  
  
This does not include nitrogen in living roots.
- *Crop:* [kg N/ha]  
Nitrogen content of the crop, excluding dead leaves.



## 52.8 Field water

A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’.

Information about water input, output and content for the system.

The intended use of this log is large scale water balance, for example reservoir management. It provide information about how much water is in the field (down to a specified depth), where it is located in the field (surface, soil matrix or in biopores), as well as the sources, sinks and amounts of water entering or leaving the system. It does not provide information about internal translocation of water between surface, soil matrix and biopores, use see the ‘Soil water’ log instead for that.

For the balances of this log to work, you must include the entire root zone, as well as the biopore zone.

### Table columns:

- *Precipitation:* [mm]  
Total amount of water entering the system as rain and snow.
- *Irrigation:* [mm]  
Total amount of water added to the system from irrigation.
- *Potential evapotranspiration:* [mm]  
There is energy enough to evaporate this amount of water.
- *Actual evapotranspiration:* [mm]  
Amount of water removed by evaporation and transpiration.  
  
The evaporation part covers water evaporated from the soil surface, the snow pack, or intercepted water on the canopy. The transpiration covers water removed from the soil through the root system and the stomata on the leaves.  
  
The actual evapotranspiration will be lower than the potential evapotranspiration when the available water is insufficient.
- *Matrix percolation:* [mm]  
This is the amount of water leaving the system through the soil bottom. It can be negative if there is capillary rise.
- *Biopore percolation:* [mm]  
This is the amount of water leaving the system through biopores.
- *Matrix drain flow:* [mm]  
Amount of water leaving the system through the soil matrix to drain pipes.  
  
Note that the total amount of water in the drain pipes may be higher than this if you have specified the ‘to’ parameter, as the whole soil profile will contribute to the drain flow, and only the contributions from the soil interval being logged is counted here.
- *Biopore drain flow:* [mm]  
Amount of water leaving the system through biopores to drain pipes.  
  
Note that the total amount of water in the drain pipes may be higher than this if you have specified the ‘to’ parameter, as the whole soil profile will contribute to the drain flow, and only the contributions from the soil interval being logged is counted here.
- *Runoff:* [mm]  
Amount of water running of the surface.

This is intended to simulate water runoff from a surface with a slop. However, since Daisy is a one dimensional model, the water have nowhere go to. For typical setups, this value will be zero.

- *Biopore water*: [mm]  
The total amount of biopore water in the soil interval
- *Soil matrix water*: [mm]  
Total water content of the soil.
- *Surface water*: [mm]  
Total water stored on the surface. This includes the snow pack, intercepted water, ponding, and water in the litter layer.

## 52.9 Soil nitrogen

A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’.

Nitrogen input, output, transformation and content for the soil.

```
< "Soil nitrogen" (summary summary ...) ; Has default value.
 (unit "kg N/ha")
 (where where) ; Has default value.
 (entries entries ...) ; Has default value.
 (when when) ; Default hourly value. >
```

- *unit*: string (see section 4.1.5)  
Parameter (default ‘kg N/ha’)  
Base unit.

### Table columns:

- *NO3-Matrix-In*: [kg N/ha]  
NO3-N infiltration.
- *NO3-Biopore-In*: [kg N/ha]  
NO3-N infiltration.
- *NO3-Leak-Matrix*: [kg N/ha]  
NO3-N leaching.
- *NO3-Leak-Biopore*: [kg N/ha]  
NO3-N leaching.
- *NO3-to-Biopores*: [kg N/ha]  
Movement of NO3-N from soil matrix to biopores.
- *NO3-from-Biopores*: [kg N/ha]  
Movement of NO3-N from soil matrix to biopores.
- *NO3-Soil-Drain*: [kg N/ha]  
NO3-N lost to drains directly from the soil matrix.
- *NO3-Indirect-Drain*: [kg N/ha]  
NO3-N lost to drains from the soil matrix, through biopores.
- *NO3-Biopore-Drain*: [kg N/ha]  
NO3-N lost to drains from biopores.
- *NO3-Incorp*: [kg N/ha]  
NO3-N incorporated directly into the soil.

- *NO<sub>3</sub>-Tillage*: [kg N/ha]  
NO<sub>3</sub>-N added by tillage operations.
- *NO<sub>3</sub>-Uptake*: [kg N/ha]  
NO<sub>3</sub>-N removed by plant roots.
- *NO<sub>3</sub>-Soil*: [kg N/ha]  
NO<sub>3</sub>-N content of the soil matrix (excluding biopores).
- *NO<sub>3</sub>-Biopores*: [kg N/ha]  
NO<sub>3</sub>-N content of the biopores.
- *NO<sub>3</sub>-Error*: [kg N/ha]  
NO<sub>3</sub>-N borrowed from the future.  
  
This may be temporarily non-zero if some process remove NO<sub>3</sub> too fast compared to the time step.
- *NH<sub>4</sub>-Matrix-In*: [kg N/ha]  
NH<sub>4</sub>-N infiltration.
- *NH<sub>4</sub>-Biopore-In*: [kg N/ha]  
NH<sub>4</sub>-N infiltration.
- *NH<sub>4</sub>-Leak-Matrix*: [kg N/ha]  
NH<sub>4</sub>-N leaching.
- *NH<sub>4</sub>-Leak-Biopore*: [kg N/ha]  
NH<sub>4</sub>-N leaching.
- *NH<sub>4</sub>-to-Biopores*: [kg N/ha]  
Movement of NH<sub>4</sub>-N from soil matrix to biopores.
- *NH<sub>4</sub>-from-Biopores*: [kg N/ha]  
Movement of NH<sub>4</sub>-N from soil matrix to biopores.
- *NH<sub>4</sub>-Soil-Drain*: [kg N/ha]  
NH<sub>4</sub>-N lost to drains directly from the soil matrix.
- *NH<sub>4</sub>-Indirect-Drain*: [kg N/ha]  
NH<sub>4</sub>-N lost to drains from the soil matrix, through biopores.
- *NH<sub>4</sub>-Biopore-Drain*: [kg N/ha]  
NH<sub>4</sub>-N lost to drains from biopores.
- *NH<sub>4</sub>-Incorp*: [kg N/ha]  
NH<sub>4</sub>-N incorporated directly into the soil.
- *NH<sub>4</sub>-Tillage*: [kg N/ha]  
NH<sub>4</sub>-N added by tillage operations.
- *NH<sub>4</sub>-Uptake*: [kg N/ha]  
NH<sub>4</sub>-N removed by plant roots.
- *NH<sub>4</sub>-Soil*: [kg N/ha]  
NH<sub>4</sub>-N content of the soil matrix (excluding biopores).
- *NH<sub>4</sub>-Biopores*: [kg N/ha]  
NH<sub>4</sub>-N content of the biopores.

- *NH<sub>4</sub>-Error*: [kg N/ha]  
NH<sub>4</sub>-N borrowed from the future.  
  
This may be temporarily non-zero if some process remove NH<sub>4</sub> too fast compared to the time step.
- *NO<sub>3</sub>-Denitrification*: [kg N/ha]  
NO<sub>3</sub>-N removed by denitrification.
- *N<sub>2</sub>O-Denitrification*: [kg N/ha]  
N<sub>2</sub>O-N released by denitrification.
- *NH<sub>4</sub>-Nitrification*: [kg N/ha]  
NH<sub>4</sub>-N removed by nitrification.
- *NO<sub>3</sub>-Nitrification*: [kg N/ha]  
NO<sub>3</sub>-N added by nitrification.
- *N<sub>2</sub>O-Nitrification*: [kg N/ha]  
NH<sub>4</sub>-N lost as N<sub>2</sub>O as a byproduct of the nitrification process.  
  
NH<sub>4</sub>-Nitrification = NO<sub>3</sub>-Nitrification + N<sub>2</sub>O-Nitrification
- *NH<sub>4</sub>-Mineralization*: [kg N/ha]  
NH<sub>4</sub>-N added from mineralization of organic matter in the soil.
- *NO<sub>3</sub>-Immobilization*: [kg N/ha]  
NO<sub>3</sub>-N removed by the organic matter turnover processes.
- *Residuals-N*: [kg N/ha]  
Nitrogen in organic matter added to the soil from roots.
- *Tillage-Org\_N*: [kg N/ha]  
Nitrogen in organic matter added to the soil by tillage operations.
- *Bioincorporation*: [kg N/ha]  
Nitrogen in organic matter added to the soil by bioincorporation.  
  
This includes dead leaves removed from the soil surface by earthworms.
- *AOM*: [kg N/ha]  
Total amount of nitrogen in added organic matter in the soil.  
  
AOM consists of fertilizer and plant residuals that has not yet been decomposed.
- *SOM*: [kg N/ha]  
Total amount of nitrogen in soil humus.  
  
This consist of dead organic matter that can no longer be traced to its origins.
- *SMB*: [kg N/ha]  
Nitrogen in soil microbiological organisms.  
  
The living part of the soil (excluding plant roots).
- *Buffer*: [kg N/ha]  
Nitrogen in humus added by fertilizer but not yet available for turnover.  
  
This is usually zero.

## 52.10 Soil water

A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’.

Information about water input, output and content for the soil.

The intended use of this log is small scale water management, for example irrigation optimization. Above ground water is considered external to the system. The balances provided by the log will work on any subset of the soil that matches the discretization.

### Table columns:

- *Matrix infiltration:* [mm]  
Amount of water entering the soil through the matrix system.  
This is the normal infiltration channel.
- *Matrix percolation:* [mm]  
Amount of water leaving the soil through the matrix system.  
This is the normal percolation channel.
- *Biopore infiltration:* [mm]  
Amount of water entering the soil through the biopores.
- *Biopore percolation:* [mm]  
Amount of water leaving the soil through the biopore system.
- *Subsoil irrigation:* [mm]  
mount of water incorporated directly into the soil.  
his include both subsoil irrigation and any water in fertilizer irectly incorpo-  
rated into the soil.
- *Matrix direct drain flow:* [mm]  
Flow from matrix directly to the drain pipes.  
This does not count water that flow to the drain pipes through biopores.  
Note that water flow to the drain pipes from all soil layers, both above and  
below the pipes, and even from the automatically inserted aquitard horizon  
that lies below the normal soil horizons. So if you have specified less than  
the full soil profile, that is, if you have set the ‘from’ or ‘to’ parameters, the  
amount you see logged here will be less than the total amount of water in the  
drain pipes.  
If the soil has no drain pipes this amount will be zero.
- *Matrix indirect drain flow:* [mm]  
Flow from matrix indirectly to the drain pipes.  
This only counts water that flow to the drain pipes through biopores.
- *Biopore drain flow:* [mm]  
Water in the interval that flows from the biopores to the drain pipes.  
Note that water flow to the drain pipes from all soil layers, both above and  
below the pipes, and even from the automatically inserted aquitard horizon  
that lies below the normal soil horizons. So if you have specified less than  
the full soil profile, that is, if you have set the ‘from’ or ‘to’ parameters, the  
amount you see logged here will be less than the total amount of water in the  
drain pipes.  
If the soil has no biopores connected directly to the drain, this amount will  
be zero.

- *Root extraction:* [mm]  
Water in the soil interval extracted by the roots.  
  
This number will be equal to the transpiration if the soil interval being logged includes the entire root zone.
- *Freezing:* [mm]  
Water in the soil interval turned into ice.  
  
This number will be zero unless the 'enable\_ice' SoilHeat parameter has been set.
- *Tillage:* [mm]  
Net amount of water being added to the soil interval by tillage operations.  
  
This number will be zero if the soil interval includes the all the soil affected by the tillage operation, which is the normal case. The number will be negative if more water have been added than removed by the tillage operation.
- *Matrix to biopores:* [mm]  
Amount of water leaving the soil matrix to the biopores. This does not include water that flows to drain connected biopores.
- *Biopores to matrix:* [mm]  
Amount of water leaving the biopores to the matrix.
- *Matrix water:* [mm]  
The total amount of soil matrix water in the soil interval
- *Biopore water:* [mm]  
The total amount of biopore water in the soil interval

## 52.11 biopore

A 'column' model (see 52.6, page 286) defined in 'log-std.dai'.

A log table for a specific biopore class.

```
< biopore (biopore "*")
 (parameter_names parameter_names ...) ; Has default value.
 (bioporefid bioporefid) ; Default cond value. >
```

- *biopore:* string (see section 4.1.5)  
Parameter (default '\*')  
Name of biopore to log. Use "\*" to log all biopore classes.
- *bioporefid:* **string** component (see chapter 85)  
Component (default 'cond')

```
(bioporefid cond (("string-equal" "${biopore}" "*")
 biopores)
 ((true)
 "${biopore}"))
```

Parameter description:

File component name indicating biopore logged.

## 52.12 chemical

A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’.

A log table for a specific chemical.

```
< chemical (chemical "*")
 (parameter_names parameter_names ...) ; Has default value.
 (chemfid chemfid) ; Default cond value. >
```

- *chemical*: string (see section 4.1.5)  
Parameter (default ‘\*’)  
Name of chemical to log.
- *chemfid*: **string** component (see chapter 85)  
Component (default ‘cond’)

```
(chemfid cond ("string-equal" "${chemical}" "*")
 chemicals)
((true)
 "${chemical}"))
```

Parameter description:

File name component indicating chemical logged.

## 52.13 Field chemical

A ‘chemical’ model (see 52.12, page 295) defined in ‘log-std.dai’.

Content, transport and transformation of chemicals in the field.

```
< "Field chemical" (summary summary ...) ; Has default value.
 (unit "g/ha")
 (where where) ; Has default value.
 (entries entries ...) ; Has default value.
 (when when) ; Default hourly value. >
```

- *unit*: string (see section 4.1.5)  
Parameter (default ‘g/ha’)  
Base unit.

### Table columns:

- *Spray*: [g/ha]  
Applied to the surface.  
  
Chemicals can be added to the surface through the spray, fertilize and irrigate management operations.
- *Deposit*: [g/ha]  
Atmospheric deposition.  
  
This is usually zero when tracking pesticides, but can be relevant for atmospheric pollutants, and is also significant when for nitrogen.
- *Harvest*: [g/ha]  
Removed by harvest.  
  
Some amount of the chemical may be left on the canopy, and removed during harvest. It is usually not significant for the overall balance, but might be important to the consumers, depending on the chemical.

- *Dissipate*: [g/ha]  
Dissipating from the canopy.  
Some chemicals evaporate or decompose when stored on the canopy.
- *Litter Decompose*: [g/ha]  
Decomposed while stored in the litter pack.
- *Litter Transform*: [g/ha]  
Added from chemical transformation in the litter pack.
- *Surface Decompose*: [g/ha]  
Decomposed while stored on the soil surface.
- *Surface Transform*: [g/ha]  
Added from chemical transformation on the soil surface.
- *Runoff*: [g/ha]  
Lost from soil surface due to lateral water flow.
- *Leak matrix*: [g/ha]  
Lost due to leaching.
- *Leak biopores*: [g/ha]  
Lost due to leaching.
- *Biopore drain*: [g/ha]  
Lost from biopores to drain pipes.
- *Soil drain*: [g/ha]  
Lost from the soil matrix to drain pipes.
- *Surface drain*: [g/ha]  
Amount moving directly from surface to drain, through biopores.  
This never come in contact with the soil matrix.
- *External*: [g/ha]  
Added to the soil through some external mechanism, such as subsoil irrigation.
- *Uptake*: [g/ha]  
Removed from the soil matrix through plant roots.
- *Soil Decompose*: [g/ha]  
Decomposed while in the soil matrix.
- *Soil Transform*: [g/ha]  
Added through chemical transformation in the soil matrix.
- *Snow*: [g/ha]  
Amount stored in the snow pack.
- *Canopy*: [g/ha]  
Amount stored in on the canopy.
- *Litter*: [g/ha]  
Amount stored in the surface litter (mulch, residuals, fertilizer).
- *Surface*: [g/ha]  
Amount stored on the soil surface.



- *Soil*: [g/ha]  
Amount stored in the soil matrix.
- *Biopores*: [g/ha]  
Amount stored in the biopores.
- *Error*: [g/ha]  
Amount borrowed from the future.  
  
This can be temporarily non-zero if some process remove the chemical fast compared to the timestep of the simulation.

## 52.14 Soil chemical

A ‘chemical’ model (see 52.12, page 295) defined in ‘log-std.dai’.

Content, transport and transformation of chemicals in the soil.

```
< "Soil chemical" (summary summary ...) ; Has default value.
 (unit "g/ha")
 (where where) ; Has default value.
 (entries entries ...) ; Has default value.
 (when when) ; Default hourly value. >
```

- *unit*: string (see section 4.1.5)  
Parameter (default ‘g/ha’)  
Base unit.

### Table columns:

- *In-Matrix*: [g/ha]  
Infiltration through soil matrix.
- *In-Biopores*: [g/ha]  
Infiltration through biopores.
- *Leak-Matrix*: [g/ha]  
Leaching through the soil matrix.
- *Leak-Biopores*: [g/ha]  
Leaching.
- *Biopores to matrix*: [g/ha]  
Enter the soil matrix from the biopores.
- *Matrix to biopores*: [g/ha]  
Enter the biopores from the matrix.
- *Tillage*: [g/ha]  
Added through tillage operations.
- *Drain-Soil*: [g/ha]  
Lost to drains directly from the soil matrix. This does not include what enters the drains through the biopores.
- *Drain-Biopores*: [g/ha]  
Lost to drains from the biopores.
- *Drain-Biopores-Indirect*: [g/ha]  
Lost to drains from matrix through biopores. This does not include what enters the drain directly from the soil matrix.

- *External*: [g/ha]  
Added externally, for example through subsoil irrigation.
- *Uptake*: [g/ha]  
Lost with water uptake by plant roots.
- *Decompose*: [g/ha]  
Decomposed.
- *Transform*: [g/ha]  
Added by chemical transformation.
- *Content*: [g/ha]  
Total amount in soil matrix in the specified soil interval.
- *Biopores*: [g/ha]  
Total amount in biopores in the specified soil interval.
- *Error*: [g/ha]  
Borrowed from the future.  
  
This can be non-zero if some process remove the chemical too fast compared to the simulation timestep.

## 52.15 crop

A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’.

A log table for a specific crop.

```
< crop (crop "*")
 (parameter_names parameter_names ...) ; Has default value.
 (cropfid cropfid) ; Default cond value.
 (cropfid2 cropfid2) ; Default cond value. >
```

- *crop*: string (see section 4.1.5)  
Parameter (default ‘\*’)  
Name of crop to log. Use “\*” to log all crops.
- *cropfid*: **string** component (see chapter 85)  
Component (default ‘cond’)

```
(cropfid cond ("string-equal" "${crop}" "*")
 crop)
((true)
 "${crop}"))
```

Parameter description:

File component name indicating crop logged.

- *cropfid2*: **string** component (see chapter 85)  
Component (default ‘cond’)

```
(cropfid2 cond ("string-equal" "${crop}" "*")
 "")
((true)
 "${crop}_"))
```

Parameter description:

File component name indicating crop logged. Empty if not specified.

## 52.16 Crop

A ‘crop’ model (see 52.15, page 298) defined in ‘log-std.dai’.

Crop development and production.

**Table columns:**

- *DS*: [DS]  
Development stage of the crop. 0 is emergence, 1 is flowering, 2 is ripe.
- *Height*: [cm]  
Height of crop.
- *LAI*: [ $\text{m}^2/\text{m}^2$ ]  
Leaf area index.
- *Depth*: [cm]  
Effective depths of root zone.
- *WLeaf*: [Mg DM/ha]  
Dry matter content of leaves.
- *WDead*: [Mg DM/ha]  
Dry matter content of dead (yellow) leaves still on crop.
- *WStem*: [Mg DM/ha]  
Dry matter content of stem.
- *WSOrg*: [Mg DM/ha]  
Dry matter content of storage organ (grain, tuber).
- *WRoot*: [Mg DM/ha]  
Dry matter content of roots.
- *Fixated*: [kg N/ha]  
Atmospheric nitrogen fixated by the crop (root biome).
- *NLeaf*: [kg N/ha]  
Nitrogen content of the leaves.
- *NDead*: [kg N/ha]  
Nitrogen content of dead (yellow) leaves still on the crop.
- *NStem*: [kg N/ha]  
Nitrogen content of the stem.
- *NSOrg*: [kg N/ha]  
Nitrogen content of the storage organ (grain or tuber).
- *NRoot*: [kg N/ha]  
Nitrogen content of the roots.
- *water\_stress*: [<none>]  
Highest water stress experienced in the period.  
  
When the root system cannot fulfil the demand of water from the canopy, the crop experience water stress, and production suffers.  
  
The value is between 0 (no water stress) and 1 (no production).

- *nitrogen\_stress*: [**<none>**]  
Highest nitrogen stress experienced in the period.  
When the crop nitrogen content is below a critical level, crop experience nitrogen stress, and production suffers.  
The value is between 0 (no nitrogen stress) and 1 (no production).
- *Phenology*: [**<unknown>**]  
Current phenological stage.  
For most crops, this is -1 (not yet emerged), 0 (vegetative), 1 (reproductive), or 2 (ripe). Some crops may show other scaled, like BBCH.

# Chapter 53

## macro

Preferential flow in soil macro pores.

### 53.1 default

The area between 'height\_start' and 'height\_end' contains macropores, which are initiated when the water potential reach 'pressure\_initiate', and then immediately emptied down to 'pressure\_end'. The water entering the macropore is distributed in soil below as a source term, according to the 'distribution' parameter.

```
< default (distribution distribution)
 (height_start height_start)
 (height_end height_end)
 (pressure_initiate -3 [cm])
 (pressure_end -30 [cm])
 (pond_max 0.5 [mm]) >
```

- *distribution*: plf [**cm** → <**fraction**>]  
Parameter  
Distribution of macropore end points as a function of height. The function should start with '1' at 'height\_end', and then decrease to '0' at 'height\_start'. It can be constant, but may never increase. The value indicates the fraction of macropores which ends at the given where all macropores is assumed to start at the top.
- *height\_start*: number [**cm**]  
Optional parameter  
Macropores starts at this depth (a negative number). If not specified, use the last point in 'distribution'.
- *height\_end*: number [**cm**]  
Optional parameter  
Macropores ends at this depth (a negative number). If not specified, use the first point in 'distribution'.
- *pressure\_initiate*: number [**cm**]  
Parameter (default -3)  
Pressure needed to init pref.flow
- *pressure\_end*: number [**cm**]  
Parameter (default -30)  
Pressure after pref.flow has been init

- *pond\_max*: number [**mm**]  
Parameter (default 0.5)  
Maximum height of ponding before spilling into macropores. After macropores are activated pond will have this height.

### Log Variables

- *S<sub>p</sub>*: number [**h**<sup>-1</sup>]  
Macropore sink term.

## 53.2 none

No macropores.

## Chapter 54

# mactrans

Macropore transportation of solutes.

### 54.1 default

Solute follows water.

Used by tertiary old mactrans (see 89.3, page 481) .





## Chapter 55

# max\_exfiltration

Soil limited maximum exfiltration.

### 55.1 Hansen

Darcy formulated for water content with minimum water at surface.

Used by surface default max\_exfiltration (see 87.1, page 465) .

### 55.2 Theta

Darcy formulated for water content with minimum water at surface.

< Theta (h\_crit *h\_crit*) >

- *h\_crit*: number [cm]  
Optional parameter  
Surface potential for soil limited evaporation. If positive, use atmospheric potential (Kelvin's equation).

### 55.3 gradient

Darcy formulated for water content with minimum water at surface.

< gradient *max\_exfiltration\_gradient* >

- *max\_exfiltration\_gradient*: number (dimensionless)  
Parameter  
Maximal pressure gradient for calculating exfiltration. The gradient is assumed from center of top node to surface of top node.

### 55.4 steady-state

Steady state flow from top cell to surface.

< "steady-state" (h\_crit *h\_crit*) >

- *h\_crit*: number [cm]  
Optional parameter  
Surface potential for soil limited evaporation. If positive, use atmospheric potential (Kelvin's equation).



# Chapter 56

## movement

This component handles the movement in the soil.

```
< component (Tertiary Tertiary) >
```

- *Tertiary*: **tertiary** component (see chapter 89)  
Tertiary (that is, non-matrix) transport method.

### Log Variables

- *water\_failure\_level*: integer  
The number of the last water transport model to fail. It is -1 if the first model succeeded, and 0 if the first model failed but the second succeeded.
- *solute\_failure\_level*: integer  
The number of the last solute transport model to fail. It is -1 if the first model succeeded, and 0 if the first model failed but the second succeeded.

## 56.1 solute

Shared paramaters for handling solutes.

```
< solute (matrix_solute matrix_solute ...)
 (matrix_solid matrix_solid) ; Default none value.
 (sink_sorbed true)
 ;; Shared parameters are described in chapter 56.
(Tertiary Tertiary) >
```

- *matrix\_solute*: **transport** component (see chapter 92) sequence  
Matrix solute transport models. Each model will be tried in turn, until one succeeds. If none succeeds, the simulation ends.
- *matrix\_solid*: **transport** component (see chapter 92)  
Component (default 'none')  
Matrix solute transport model used for fully sorbed constituents.
- *sink\_sorbed*: boolean (see section 4.1.2)  
Parameter (default true)  
Substract sink term from sorbed matter.

## 56.2 rectangle

A 'solute' model (see 56.1, page 307) build into Daisy.

Two dimensional movement in a rectangular grid.

```
< rectangle (Geometry Geometry) ; Has partial value.
 (pipe_outlet pipe_outlet) ; Default deep value.
 (Tertiary Tertiary) ; Default none value.
 (matrix_solute matrix_solute ...) ; Has default value.
 (matrix_water matrix_water ...) ; Has default value.
 (drainpoints)
 (heat heat) ; Default Mollerup value.
 ;; Shared parameters are described in section 56.1.
 (matrix_solid matrix_solid) ; Default none value.
 (sink_sorbed true) >
```

- *Geometry*: **GeometryRect** fixed component (see section 107.17)  
Submodel (has partially specified default value)  
Discretization of the soil.

- *pipe\_outlet*: **depth** component (see chapter 34)  
Component (default ‘deep’)  
Water table in drain pipe outlet.

By default this will be considered deep enough to ensure free flow of water out of drains. If higher than the drain pipes, water may flow either way depending on the pressure in the soil.

- *matrix\_water*: **uzrect** component (see chapter 99) sequence  
Component (has default value with length 3)

```
(matrix_water Mollerup
 "v+h"
 const)
```

Parameter description:

Matrix water transport models. Each model will be tried in turn, until one succeeds. If none succeeds, the simulation ends.

- *drainpoints*: submodel (see section 4.1.7) sequence  
Submodel (default: an empty sequence)  
Location of cells with drain pipes.

```
< z x >
```

- *z*: number [**cm**]  
Parameter  
Vertical position.
- *x*: number [**cm**]  
Parameter  
Horizontal position.

- *heat*: **heatrect** component (see chapter 46)  
Component (default ‘Mollerup’)  
Heat transport model.

### 56.3 vertical

A ‘solute’ model (see 56.1, page 307) build into Daisy.  
One dimensional movement.

Used by column default Movement (see 26.1, page 153) .

```
< vertical (Geometry Geometry) ; Has default value.
 (Tertiary Tertiary) ; Default old value.
 (matrix_solute matrix_solute ...) ; Has default value.
 (matrix_water matrix_water ...) ; Has default value.
 ;; Shared parameters are described in section 56.1.
 (matrix_solid matrix_solid) ; Default none value.
 (sink_sorbed true) >
```

- *Geometry*: **Geometry1D** fixed component (see section 107.16)  
Submodel (has fully specified default value)  
Discretization of the soil.
- *matrix\_water*: **uzmodel** component (see chapter 98) sequence  
Component (has default value with length 2)

```
(matrix_water richards
 lr)
```

Parameter description:

Vertical matrix water transport models. Each model will be tried in turn, until one succeeds. If none succeeds, the simulation ends.



## Chapter 57

# net\_radiation

Calculate net radiation from meteorological data.

### 57.1 brunt

Brunt, 1932. Default parameterization by Jensen et.al., 1990. FAO recommendation. See also [Kjaersgaard et al., 2007b]

Used by bioclimate default net\_radiation (see 19.2, page 115) .

```
< brunt (A 0.64 [])
 (B 0.044 [1/sqrt(hPa)]) >
```

- *A*: number (dimensionless)  
Parameter (default 0.64)  
Brunt 'A' parameter (offset).
- *B*: number [1/sqrt(hPa)]  
Parameter (default 0.044)  
Brunt 'B' parameter (vapor pressure factor).

### 57.2 brutsaert

Brutsaert, 1975

### 57.3 idso\_jackson

Idso and Jackson, 1969

### 57.4 prata

Prata, 1996

### 57.5 satterlund

Satterlund, 1979

### 57.6 swinbank

Swinbank, 1963





## Chapter 58

# nitrification

The nitrification process, transforming ammonium into nitrate and nitrous oxide.

Used by horizon component Nitrification (see 47, page 231) .

```
< component (N2O_fraction 0.02 [<fraction>]) >
```

- *N2O\_fraction*: number [**<fraction>**]  
Parameter (default 0.02)  
Fraction of ammonium lost as N2O.

### 58.1 soil

$k_{10} * M / (k + M)$ . Michaelis-Menten kinetics, with nitrification based on total ammonium content.

Used by horizon component Nitrification (see 47, page 231) .

```
< soil (heat_factor heat_factor) ; Has default value.
(water_factor water_factor) ; Has default value.
(k 5e-05 [g N/cm3])
(k_10 2.08333e-07 [g N/cm3/h])
;; Shared parameters are described in chapter 58.
(N2O_fraction 0.02 [<fraction>]) >
```

- *heat\_factor*: plf [**dg C** → **<none>**]  
Parameter (has default value with 0 points)  
Heat factor.
- *water\_factor*: plf [**cm** → **<none>**]  
Parameter (has default value with 0 points)  
Water potential factor.
- *k*: number [**g N/cm<sup>3</sup>**]  
Parameter (default 5e-05)  
Half saturation constant.
- *k\_10*: number [**g N/cm<sup>3</sup>/h**]  
Parameter (default 2.08333e-07)  
Max rate.

### 58.2 solute

$k_{10} * C / (k + C)$ . Michaelis-Menten kinetics, with nitrification based on ammonium solute.

```

< solute (heat_factor heat_factor) ; Has default value.
 (water_factor water_factor) ; Has default value.
 (k k)
 (k_10 k_10)
 ;; Shared parameters are described in chapter 58.
 (N20_fraction 0.02 [<fraction>]) >

```

- *heat\_factor*: plf [**dg C** → **<none>**]  
 Parameter (has default value with 0 points)  
 Heat factor.
- *water\_factor*: plf [**cm** → **<none>**]  
 Parameter (has default value with 0 points)  
 Water potential factor.
- *k*: number [**g/cm<sup>3</sup>**]  
 Parameter  
 Half saturation constant.
- *k\_10*: number [**h<sup>-1</sup>**]  
 Parameter  
 Max rate.

# Chapter 59

## number

Generic representation of numbers.

Used by select component `expr` (see 80, page 427) , and biopore component `density` (see 20, page 117) .

### 59.1 \*

Use the product of its operands.

< "\*" *operands* ... >

- *operands*: **number** component (see chapter 59) sequence  
The operands for this function.

### 59.2 +

Use the sum of its operands.

< "+" *operands* ... >

- *operands*: **number** component (see chapter 59) sequence  
The operands for this function.

### 59.3 -

Negate number or subtract numbers. With one operand, negates it. With more than one operand, subtracts all but the first from the first.

< "-" *operands* ... >

- *operands*: **number** component (see chapter 59) sequence  
The operands for this function.

### 59.4 /

Divide the first operand by the rest.

< "/" *operands* ... >

- *operands*: **number** component (see chapter 59) sequence  
The operands for this function.

## 59.5 apply

Apply function to operand.

```
< apply function operand >
```

- *function*: function [**<user>** → **<user>**] (see chapter 42)  
Parameter  
Function to apply.
- *operand*: number [**<user>**]  
Parameter  
Operand for this function.

## 59.6 child

Numbers based on another number.

```
< child (value value) >
```

- *value*: **number** component (see chapter 59)  
Operand for this function.

## 59.7 convert

A ‘child’ model (see 59.6, page 316) build into Daisy.

Convert to specified dimension.

```
< convert value dimension >
```

- *value*: **number** component (see chapter 59)  
Operand for this function.
- *dimension*: string (see section 4.1.5)  
Parameter  
Dimension to convert to.

## 59.8 dim

A ‘child’ model (see 59.6, page 316) build into Daisy.

Specify dimension for number.

```
< dim value dimension
 (warn_known true) >
```

- *value*: **number** component (see chapter 59)  
Operand for this function.
- *dimension*: string (see section 4.1.5)  
Parameter  
Dimension to use.
- *warn\_known*: boolean (see section 4.1.2)  
Parameter (default true)  
Issue a warning if the dimensions is already known.

## 59.9 identity

A ‘child’ model (see 59.6, page 316) build into Daisy.

Pass value unchanged.

```
< identity (dimension dimension)
 ;; Shared parameters are described in section 59.6.
 (value value) >
```

- *dimension*: string (see section 4.1.5)  
Optional parameter  
Dimension of this value.

## 59.10 const

Always give the specified value.

```
< const value >
```

- *value*: number [**<user>**]  
Parameter  
Fixed value for this number.

## 59.11 initial\_zero

A ‘const’ model (see 59.10, page 317) build into Daisy.

Initial zero content.

Used by chemical base initial (see 23.1, page 125) .

```
< value >
```

## 59.12 zero\_gradient

A ‘const’ model (see 59.10, page 317) build into Daisy.

Assume same concentration in groundwater as in the bottom of the soil profile.

Used by chemical base C\_below (see 23.1, page 125) .

```
< value >
```

## 59.13 daisy/soil/soil.h

Find pressure (h) for a given water content (Theta).

```
< "daisy/soil/soil.h" (horizon horizon)
 (Theta Theta)
 (top_soil top_soil) >
```

- *horizon*: **horizon** component (see chapter 47)  
The soil horizon whose properties we want to examine.
- *Theta*: **number** component (see chapter 59)  
The water content we want to compare with.
- *top\_soil*: boolean (see section 4.1.2)  
Parameter  
Set this to true for the A horizon.

### 59.14 depth

Find soil value at specific depth.

```
< depth (h h)
 (column column)
 (z z) >
```

- *h*: **number** component (see chapter 59)  
The tension we want to compare with.
- *column*: **column** component (see chapter 26)  
The soil column whose properties we want to examine.
- *z*: **number** component (see chapter 59)  
The height we want to compare with.

### 59.15 depth\_K

A ‘depth’ model (see 59.14, page 318) build into Daisy.  
Find water conductivity (K) for a given pressure (h).

### 59.16 depth\_Theta

A ‘depth’ model (see 59.14, page 318) build into Daisy.  
Find water content (Theta) for a given pressure (h).

### 59.17 exp

Take the exponential of its argument.

```
< exp operand >
```

- *operand*: **number** component (see chapter 59)  
Operand for this function.

### 59.18 fetch

Fetch the value and dimension in the current scope.

```
< fetch name >
```

- *name*: string (see section 4.1.5)  
Parameter  
Name of a the symbol.

### 59.19 get

Get the value of symbol in the current scope.

```
< get name dimension >
```

- *name*: string (see section 4.1.5)  
Parameter  
Name of a the symbol.
- *dimension*: string (see section 4.1.5)  
Parameter  
Expected dimension for the symbol.

## 59.20 horizon

Find soil value at specific horizon.

```
< horizon (h h)
 (horizon horizon)
 (top_soil top_soil) >
```

- *h*: **number** component (see chapter 59)  
The tension we want to compare with.
- *horizon*: **horizon** component (see chapter 47)  
The soil horizon whose properties we want to examine.
- *top\_soil*: **boolean** (see section 4.1.2)  
Parameter  
Set this to true for the A horizon.

## 59.21 soil\_K

A ‘horizon’ model (see 59.20, page 319) build into Daisy.

Find hydraulic conductivity (K) for a given pressure (h).

## 59.22 soil\_Theta

A ‘horizon’ model (see 59.20, page 319) build into Daisy.

Find water content (Theta) for a given pressure (h).

## 59.23 soil\_heat\_capacity

A ‘horizon’ model (see 59.20, page 319) build into Daisy.

Find heat capacity for a given pressure (h).

## 59.24 soil\_heat\_conductivity

A ‘horizon’ model (see 59.20, page 319) build into Daisy.

Find heat conductivity for a given pressure (h).

## 59.25 if

Select between two numbers depending on a boolean expression.

```
< if if then else >
```

- *if*: **boolean** component (see chapter 21)  
Select which number to use.
- *then*: **number** component (see chapter 59)  
Use this if true.
- *else*: **number** component (see chapter 59)  
Use this if false.

## 59.26 initial\_C

Find initial content from concentration.

< **initial\_C** *C* >

- *C*: number [g/cm<sup>3</sup>]  
Parameter  
Initial concentration in soil water with sorbtion ignored. That is  $M = C * \text{Theta}$ .

## 59.27 initial\_NH4

A ‘initial\_C’ model (see 59.26, page 320) build into Daisy.

Initial NH4 concentration in soil water.

Used by chemical NH4 initial (see 23.37, page 143) .

< *C* >

## 59.28 initial\_NO3

A ‘initial\_C’ model (see 59.26, page 320) build into Daisy.

Initial NO3 concentration in soil water.

Used by chemical NO3 initial (see 23.38, page 144) .

< *C* >

## 59.29 let

Bind symbols in ‘clauses’ in a new scope, and evaluate ‘expr’ in that scope.

< **let** *clauses* ... *expr* >

- *clauses*: submodel (see section 4.1.7) sequence  
List of identifiers and values to bind in this scope.

< *identifier* *expr* >

– *identifier*: string (see section 4.1.5)

Parameter

Identifier to bind.

– *expr*: **number** component (see chapter 59)

Value to give it.

- *expr*: **number** component (see chapter 59)  
Expression to evaluate.

## 59.30 ln

Take the natural logarithm of its argument.

< **ln** *operand* >

- *operand*: **number** component (see chapter 59)  
Operand for this function.



## 59.31 log10

Take the base 10 logarithm of its argument.

```
< log10 operand >
```

- *operand*: **number** component (see chapter 59)  
Operand for this function.

## 59.32 max

Use the largest value of its operands.

```
< max operands ... >
```

- *operands*: **number** component (see chapter 59) sequence  
The operands for this function.

## 59.33 min

Use the smallest value of its operands.

```
< min operands ... >
```

- *operands*: **number** component (see chapter 59) sequence  
The operands for this function.

## 59.34 plf

Look up argumen in a piecewise linear function.

```
< plf operand
 (domain "<unknown>")
 (range "<unknown>")
 (points points ...) >
```

- *operand*: **number** component (see chapter 59)  
Operand for this function.
- *domain*: string (see section 4.1.5)  
Parameter (default '<unknown>')  
Unit for the operand of the function.
- *range*: string (see section 4.1.5)  
Parameter (default '<unknown>')  
Unit for the operand of the function.
- *points*: submodel (see section 4.1.7) sequence  
List of points (x y) defining the piecewise linear function. The x values must be ordered lowest first.

```
< x y >
```

- *x*: number [<user>]  
Parameter  
Operand.
- *y*: number [<user>]  
Parameter  
Value.

### 59.35 pow

Raise 'base' to the power of 'exponent'.

```
< pow base exponent >
```

- *base*: **number** component (see chapter 59)  
The base operand for this function.
- *exponent*: **number** component (see chapter 59)  
The exponent operand for this function.

### 59.36 solute\_uptake

Find root nitrogen uptake.

```
< solute_uptake (L L)
 (value value)
 (volume volume)
 (Theta Theta)
 (Theta_sat Theta_sat)
 (S_w S_w)
 (C_l C_l)
 (PotNUpt PotNUpt)
 (I_max I_max)
 (C_root_min C_root_min)
 (diffusion_coef diffusion_coef)
 (Rad Rad) >
```

- *L*: **number** component (see chapter 59)  
See root system.
- *value*: **number** component (see chapter 59)  
Value to return One of 'uptake', 'C\_root', 'D', 'alpha', 'beta'.
- *volume*: **number** component (see chapter 59)  
See root system.
- *Theta*: **number** component (see chapter 59)  
See root system.
- *Theta\_sat*: **number** component (see chapter 59)  
See root system.
- *S\_w*: **number** component (see chapter 59)  
See root system.
- *C\_l*: **number** component (see chapter 59)  
See root system.
- *PotNUpt*: **number** component (see chapter 59)  
See root system.
- *I\_max*: **number** component (see chapter 59)  
See root system.
- *C\_root\_min*: **number** component (see chapter 59)  
See root system.

- *diffusion\_coef*: **number** component (see chapter 59)  
See root system.
- *Rad*: **number** component (see chapter 59)  
See root system.

## 59.37 solute\_uptake2

Find root nitrogen uptake.

```
< solute_uptake2 (L L)
 (value value)
 (volume volume)
 (Theta Theta)
 (Theta_sat Theta_sat)
 (S_w S_w)
 (C_l C_l)
 (diffusion_coef diffusion_coef)
 (Rad Rad)
 (F1 F1)
 (K1 K1)
 (F2 F2)
 (K2 K2) >
```

- *L*: **number** component (see chapter 59)  
See root system.
- *value*: **number** component (see chapter 59)  
Value to return One of 'uptake', 'C\_root', 'C1', 'C2', 'D', 'alpha', 'beta', 'A', 'B'.
- *volume*: **number** component (see chapter 59)  
See root system.
- *Theta*: **number** component (see chapter 59)  
See root system.
- *Theta\_sat*: **number** component (see chapter 59)  
See root system.
- *S\_w*: **number** component (see chapter 59)  
See root system.
- *C\_l*: **number** component (see chapter 59)  
See root system.
- *diffusion\_coef*: **number** component (see chapter 59)  
See root system.
- *Rad*: **number** component (see chapter 59)  
See root system.
- *F1*: **number** component (see chapter 59)  
See root system.
- *K1*: **number** component (see chapter 59)  
See root system.

- *F2*: **number** component (see chapter 59)  
See root system.
- *K2*: **number** component (see chapter 59)  
See root system.

### 59.38 source

Extract information from a time series.

```
< source (source source)
 (end end)
 (begin begin) >
```

- *source*: **source** component (see chapter 83)  
The time series we want to extract a number from.
- *end*: **Time** fixed component (see section 107.27)  
Optional submodel  
Ignore values after this date.
- *begin*: **Time** fixed component (see section 107.27)  
Optional submodel  
Ignore values before or at this date.

### 59.39 source\_average

A ‘source’ model (see 59.38, page 324) build into Daisy.  
Find average number in time series.

### 59.40 source\_increase

A ‘source’ model (see 59.38, page 324) build into Daisy.  
Find increase in value during time series.

### 59.41 source\_sum

A ‘source’ model (see 59.38, page 324) build into Daisy.  
Calculate the sum of the values in a time series.

### 59.42 source\_unique

A ‘source’ model (see 59.38, page 324) build into Daisy.  
Find unique number in time series.

### 59.43 sqr

Take the square of its argument.

```
< sqr operand >
```

- *operand*: **number** component (see chapter 59)  
Operand for this function.

## 59.44 **sqrt**

Take the square root of its argument.

< **sqrt** *operand* >

- *operand*: **number** component (see chapter 59)  
Operand for this function.

## 59.45 **x**

The value of the symbol 'x' in the current scope.

Used by xysource loop x (see 105.6, page 576) .



# Chapter 60

## organic

Turnover of organic matter in the soil.

### 60.1 default

Mineralization and immobilization in soil. See also [Hansen et al., 1991, Bruun et al., 2003]

```
< default (am am ...) ; Has default value.
 (Bioincorporation Bioincorporation) ; Has default value.
 (ClayOM ClayOM) ; Default old value.
 (domsorp)
 (heat_factor heat_factor) ; Has default value.
 (water_factor water_factor) ; Has default value.
 (active_underground false)
 (K_NH4 0.020833 [h-1])
 (K_NO3 0.020833 [h-1])
 (CO2_threshold 0.0001 [h-1])
 (buffer buffer) ; Has default value.
 (smb smb ...) ; Has default value.
 (som som ...) ; Has default value.
 (initial_SOM initial_SOM ...)
 (stored_SOM_C)
 (stored_SOM_N)
 (dom)
 (pH_factor pH_factor) ; Has default value.
 (smb_tillage_factor)
 (som_tillage_factor)
 (min_AM_C 0.5 [g C/m2])
 (min_AM_N 0.05 [g N/m2])
 (init init) ; Has default value. >
```

- *am*: **am** component (see chapter 17) sequence  
Component (has default value with length 1)

(am root)

Parameter description:  
Added organic matter pools.

- *Bioincorporation*: **Bioincorporation** fixed component (see section 107.11)  
Submodel (has fully specified default value)  
Biological incorporation of litter.

- *ClayOM*: **ClayOM** component (see chapter 10)  
Component (default 'old')  
Clay effect model.
- *domsorp*: **domsorp** component (see chapter 36) sequence  
Component (default: an empty sequence)  
Interchange between DOM and SOM pools.
- *heat\_factor*: plf [**dg C** → **<none>**]  
Parameter (has default value with 0 points)  
Value description: Equation 6-13 See also [Hansen et al., 1990]  
Default heat factor, used if not specified by OM pool. If the PLF is empty, a built-in function will be used instead.
- *water\_factor*: plf [**cm** → **<none>**]  
Parameter (has default value with 0 points)  
Value description: Equation 6-16 See also [Hansen et al., 1990]  
Default water potential factor, used if not specified by OM pool. If the PLF is empty, a built-in function will be used instead.
- *active\_underground*: boolean (see section 4.1.2)  
Parameter (default false)  
Set this flag to turn on mineralization below the root zone.
- *K\_NH4*: number [**h<sup>-1</sup>**]  
Parameter (default 0.020833)  
Maximal immobilization rate for ammonium.
- *K\_NO3*: number [**h<sup>-1</sup>**]  
Parameter (default 0.020833)  
Maximal immobilization rate for nitrate.
- *CO2\_threshold*: number [**h<sup>-1</sup>**]  
Parameter (default 0.0001)  
Turnover rate above which pools will contribute to 'CO2\_fast'.
- *buffer*: submodel (see section 4.1.7)  
Submodel (has fully specified default value)

(**buffer** )

Parameter description:

Buffer between AOM pools and SOM.

```
< (N)
 (C)
 (where 1)
 (turnover_rate 1 [h-1])
 (turnover_halftime turnover_halftime) >
```

- *N*: number [**g N/cm<sup>3</sup>**] soil cells  
State variable (default: an empty sequence)  
Buffer nitrogen content.
- *C*: number [**g C/cm<sup>3</sup>**] soil cells  
State variable (default: an empty sequence)  
Buffer carbon content.



- *where*: integer  
Parameter (default 1)  
The SOM pool to move the buffer content into. The first and slow SOM pool is numbered '0', the second and faster is numbered '1'.
- *turnover\_rate*: number [ $\text{h}^{-1}$ ]  
Parameter (default 1)  
Turnover rate from buffer into SOM. Ignored if you specify 'turnover\_halftime'.
- *turnover\_halftime*: number [ $\text{h}$ ]  
Optional parameter  
Turnover halftime from buffer into SOM. Overrides 'turnover\_rate' if specified.

- *smb*: **SMB** component (see chapter 13) sequence  
Soil MicroBiomass pools. Initial value will be estimated based on equilibrium with AM and SOM pools.
- *som*: **SOM** component (see chapter 14) sequence  
Component (has default value with length 3)

```
(som "SOM-SLOW"
 "SOM-FAST"
 "SOM-INERT")
```

Parameter description:  
Soil Organic Matter pools.

- *initial\_SOM*: submodel (see section 4.1.7) sequence  
Optional submodel  
Layered initialization of soil SOM content.  

```
< end weight >
```

  - *end*: number [ $\text{cm}$ ]  
Parameter  
End point of this layer (a negative number).
  - *weight*: number [ $\text{kg C/m}^2$ ]  
Parameter  
Organic carbon content of this layer.
- *stored\_SOM\_C*: number [ $\text{g C/cm}^3$ ] sequence  
State variable (default: an empty sequence)  
Stored SOM C content for all pools and all cells. By default, this is equivalent to the initial C content of the SOM pools.
- *stored\_SOM\_N*: number [ $\text{g N/cm}^3$ ] sequence  
State variable (default: an empty sequence)  
Stored SOM N content for all pools and all cells. By default, this is equivalent to the initial N content of the SOM pools.
- *dom*: **DOM** fixed component (see section 107.1) sequence  
Submodel (default: an empty sequence)  
Dissolved Organic Matter pools.
- *pH\_factor*: plf [ $\text{pH} \rightarrow \text{<none>}$ ]  
Parameter (has default value with 0 points)  
Soil pH influence on organic matter turnover. By default, pH is ignored.

- *smb\_tillage\_factor*: plf [**d** → <**none**>] sequence  
Parameter (default: an empty sequence)  
Tillage influence on turnover rates for each SMB pool. If no value is given, tillage will have no influence.
- *som\_tillage\_factor*: plf [**d** → <**none**>] sequence  
Parameter (default: an empty sequence)  
Tillage influence on SOM turnover rates for each SOM pool. If no value is given, tillage will have no influence.
- *min\_AM\_C*: number [**g C/m<sup>2</sup>**]  
Parameter (default 0.5)  
Minimal amount of carbon in AOM ensuring it is not removed.
- *min\_AM\_N*: number [**g N/m<sup>2</sup>**]  
Parameter (default 0.05)  
Minimal amount of nitrogen in AOM ensuring it is not removed.
- *init*: submodel (see section 4.1.7)  
Submodel (has fully specified default value)

(init )

Parameter description:

Parameters for initialization of the SOM and SMB pools.

If the C content of all the pools have been specified explicitly, use those values. Otherwise, get the total C content from either the 'initial\_SOM' parameter if specified, or else from the humus content specified in the soil horizons.

If 'SOM\_fractions' has been specified, the pools will be initialized assuming the SMB pools are in equilibrium. Otherwise, also SOM pools expect the first will be assumed to be in equilibrium as well.

```
< (h -100 [cm])
 (root 800 [kg C/ha/y])
 (T T)
 (end end)
 (efficiency 0.5 0.5)
 (fractions 0 1 0)
 (dist 7 [cm])
 (input input)
 (bioinc 0 [kg C/ha/y])
 (variable_pool variable_pool)
 (variable_pool.2 variable_pool.2)
 (background_mineralization background_mineralization)
 (SOM_limit_where 0)
 (SOM_limit_lower 0.3 0.7 0)
 (SOM_limit_upper 0.7 0.3 0)
 (debug_equations)
 (debug_rows true)
 (debug_to_screen false)
 (top_summary top_summary) >
```

- *h*: number [**cm**]  
Parameter (default -100)  
Pressure used for equilibrium.

- *root*: number [**kg C/ha/y**]  
 Parameter (default 800)  
 Amount of carbon added to the organic matter system from dead roots.  
 This is part of the total amount specified by the 'input' parameter.
- *T*: number [**dg C**]  
 Optional parameter  
 Temperature used for equilibrium.  
 By default, the yearly average from the weather component will be used.
- *end*: number [**cm**]  
 Optional parameter  
 Depth of non-root input.  
 The input will distribute uniformly down to this size, after subtracting the part of the input allocated to the 'root' parameter.  
 By default, the end of the first horizon will be used.
- *efficiency*: number [**<fraction>**] sequence  
 Parameter (has default value with length 2)  
  
 (**efficiency 0.5 0.5 []**)  
 Parameter description:  
 The efficiency this pool can be digested by each of the SMB pools. This is only used if you specify the input parameter.
- *fractions*: number [**<fraction>**] sequence  
 Parameter (has default value with length 3)  
  
 (**fractions 0 1 0 []**)  
 Parameter description:  
 Destinations for AOM input. The first numbers corresponds to each SMB pool, while the last number correspond to the SOM buffer. This is only used if you specify the input parameter.
- *dist*: number [**cm**]  
 Parameter (default 7)  
 Distance to go down in order to decrease the root density to half the original.
- *input*: number [**kg C/ha/y**]  
 Optional parameter  
 Amount of carbon added to the organic matter system.  
 If this is unspecified, the input rate from the initial added matter pools will be used instead.
- *bioinc*: number [**kg C/ha/y**]  
 Parameter (default 0)  
 Amount of carbon added to the organic matter system from bioincorporation.  
 This is part of the total amount specified by the 'input' parameter.
- *variable-pool*: integer  
 Optional parameter  
 If neither the C content nor 'SOM\_fractions' are specified, equilibrium is assumed for all SOM pools except the one specified by this parameter. If you set this to -1 (or any number not corresponding to a SOM pool), equilibrium will be assumed for all pools, and the humus content specified by the horizon will be ignored. Note, the numbering is zero-based, so '0' specifies SOM1. By default, the slowest active pool will be used.

- *variable\_pool\_2*: integer  
Optional parameter  
If 'background\_mineralization' is specified, this pool is no longer assumed to be in equilibrium. Note, the numbering is zero-based, so '0' specifies SOM1. By default, the second slowest active pool will be used.
- *background\_mineralization*: number [**kg N/ha/y**]  
Optional parameter  
The background mineralization is the mineralization from all SMB and SOM pools, but not from the AOM pools.  
If neither the C content of individual pools nor 'SOM\_fractions' are specified, the SOM and SMB pools will be initialized so all pools in the top soil (above 'end', usually the first horizon) are in equilibrium except those specified by 'variable\_pool' and 'variable\_pool\_2', usually SOM1 and SOM2. These two will be initialized so the background mineralization will be the specified number. The subsoil is not affected by this parameter.  
If the background mineralization is unspecified, 'variable\_pool\_2' will be assumed to be in equilibrium instead.
- *SOM\_limit\_where*: integer  
Parameter (default 0)  
This is the SOM pool that must be within the limits specified by 'SOM\_limit\_lower' and 'SOM\_limit\_upper'. Use negative number to disable. Note, the numbering is zero-based, so '0' specifies SOM1.
- *SOM\_limit\_lower*: number [**<fraction>**] sequence  
Parameter (has default value with length 3)  
  
(SOM\_limit\_lower 0.3 0.7 0 [])  
  
Parameter description:  
Lower limit for for automatic SOM partitioning.  
The SOM pool specified by 'SOM\_limit\_where' must contain at least the fraction of the total SOM content given in this list, where the first number correspond to the SOM1 fraction, the second number to SOM2, etc. If the fraction is below the one given in this list, the SOM partitioning in this list will be used instead.  
If the SOM partitioning have been specified directly, either by the 'SOM\_fractions' horizon parameter or by specifying the C content of each pool, this parameter will be ignored. The limit is also ignore for soil layers below 'end'.
- *SOM\_limit\_upper*: number [**<fraction>**] sequence  
Parameter (has default value with length 3)  
  
(SOM\_limit\_upper 0.7 0.3 0 [])  
  
Parameter description:  
Upper limit for for automatic SOM partitioning. Works like 'SOM\_limit\_lower'.
- *debug\_equations*: integer sequence  
Parameter (default: an empty sequence)  
Print equations used for initialization for the specified intervals.
- *debug\_rows*: boolean (see section 4.1.2)  
Parameter (default true)  
Print summari information for each row.

- *debug\_to\_screen*: boolean (see section 4.1.2)  
Parameter (default false)  
If true, print debug information to screen, else to the 'daisy.log' file.
- *top\_summary*: string (see section 4.1.5)  
Optional parameter  
Name of file to print a summary of the organic carbon and nitrogen content in the zone down to the 'end' parameter. If unspecified, no such file will be generated, but the summary will still be found in 'daisy.log'.

## Log Variables

- *CO2*: number [g CO<sub>2</sub>-C/cm<sup>3</sup>/h] soil cells  
CO<sub>2</sub> evolution in soil from all pools.
- *humus*: number [g/cm<sup>3</sup>] soil cells  
Total organic matter in the soil layer.
- *NO<sub>3</sub>\_source*: number [g N/cm<sup>3</sup>/h] soil cells  
Mineralization this time step (negative numbers mean immobilization).
- *NH<sub>4</sub>\_source*: number [g N/cm<sup>3</sup>/h] soil cells  
Mineralization this time step (negative numbers mean immobilization).
- *fertilized\_N*: number [g N/cm<sup>2</sup>/h]  
Amount of organic bound nitrogen applied. This includes nitrogen incorporated directly in the soil.
- *fertilized\_C*: number [g C/cm<sup>2</sup>/h]  
Amount of organic bound carbon applied. This includes carbon incorporated directly in the soil.
- *tillage\_N\_top*: number [g N/m<sup>2</sup>/h]  
Amount of nitrogen added to surface during tillage. This is a negative number.
- *tillage\_C\_top*: number [g C/m<sup>2</sup>/h]  
Amount of carbon added to surface during tillage. This is a negative number.
- *tillage\_N\_soil*: number [g N/cm<sup>3</sup>/h] soil cells  
Amount of nitrogen added to soil during tillage.
- *tillage\_C\_soil*: number [g C/cm<sup>3</sup>/h] soil cells  
Amount of carbon added to surface during tillage.
- *total\_C*: number [g C/cm<sup>3</sup>] soil cells  
Total organic C in the soil layer.
- *total\_N*: number [g N/cm<sup>3</sup>] soil cells  
Total organic N in the soil layer.
- *CO<sub>2</sub>\_fast*: number [g CO<sub>2</sub>-C/cm<sup>3</sup>/h] soil cells  
CO<sub>2</sub> evolution in soil from pools faster than 'CO<sub>2</sub>\_threshold'.
- *top\_CO<sub>2</sub>*: number [g CO<sub>2</sub>-C/cm<sup>2</sup>/h]  
CO<sub>2</sub> evolution at surface.
- *top\_DM*: number [kg DM/m<sup>2</sup>]  
Added organic dry matter on top of surface.
- *abiotic\_factor*: number (dimensionless) soil cells  
Product of current heat and water factors.

## 60.2 SOM2000

A ‘default’ model (see 60.1, page 327) build into Daisy.

Using pre Daisy v7 parameterization of SMB-FAST. Also, disable respiration for bioincorporation. See also [Hansen et al., 1991, Bruun et al., 2003]

```
< (Bioincorporation Bioincorporation) ; Has default value.
 (smb smb ...) ; Has default value. >
```

## 60.3 SOM2025

A ‘default’ model (see 60.1, page 327) build into Daisy.

Using Daisy v7 parameterization of SMB-FAST. See also [Hansen et al., 1991, Bruun et al., 2003]

Used by column default OrganicMatter (see 26.1, page 153) .

```
< (smb smb ...) ; Has default value. >
```

## 60.4 none

Ignore all soil organic matter dynamics.

# Chapter 61

## parser

To start the simulation, many parameters must be specified and state variables must be given an initial value. It is the responsibility of the 'parser' component to read these data from an external source (typically a setup file), and convert them into the internal format.

Used by Toplevel @ input (see 107.30, page 624) .

### 61.1 file

Read a setup file containing lots of parentheses.

< file *where* >

- *where*: string (see section 4.1.5)  
Parameter  
File to read from.





# Chapter 62

## pet

The 'pet' component should calculate the potential evapotranspiration from meteorological data, as well as the crop and soil state.

```
< component >
```

### Log Variables

- *wet*: number [mm/h]  
Potential evapotranspiration for a wet system.
- *dry*: number [mm/h]  
Potential evapotranspiration for a dry system.
- *reference\_evapotranspiration*: number [mm/h]  
Reference evapotranspiration for a dry system.

### 62.1 FAO\_PM

Potential evapotranspiration using Penman-Monteith. See also [Allen et al., 1998]  
Used by bioclimate FAO56\_daily pet (see 19.3, page 115) .

```
< FAO_PM (use_wet false)
 (rb 20 [s/m]) >
```

- *use\_wet*: boolean (see section 4.1.2)  
Parameter (default false)  
Use wet PM for wet surface.  
  
This will give significantly higher potential evapotranspiration especially when feed with daily weather data.
- *rb*: number [s/m]  
Parameter (default 20)  
Boundary layer resistance for wet surface.

### Log Variables

- *reference\_evapotranspiration\_wet*: number [mm/h]  
Reference evapotranspiration for a wet system.
- *potential\_evapotranspiration\_wet*: number [mm/h]  
Potential evapotranspiration for a wet system.
- *potential\_evapotranspiration\_dry*: number [mm/h]  
Potential evapotranspiration for a dry system.

## 62.2 FAO\_PM\_hourly

Potential evapotranspiration using Penman-Monteith. Modified to work with hourly data. See also [Allen et al., 1998, Allen et al., 2006]

Used by bioclimate FAO56\_hourly pet (see 19.4, page 116) .

```
< FAO_PM_hourly (use_wet false)
 (rb 20 [s/m]) >
```

- *use\_wet*: boolean (see section 4.1.2)  
Parameter (default false)  
Use wet PM for wet surface.  
  
This will give significantly higher potential evapotranspiration especially when feed with daily weather data.
- *rb*: number [s/m]  
Parameter (default 20)  
Boundary layer resistance for wet surface.

### Log Variables

- *reference\_evapotranspiration\_wet*: number [mm/h]  
Reference evapotranspiration for a wet system.
- *potential\_evapotranspiration\_wet*: number [mm/h]  
Potential evapotranspiration for a wet system.
- *potential\_evapotranspiration\_dry*: number [mm/h]  
Potential evapotranspiration for a dry system.

## 62.3 Hargreaves

Potential evapotranspiration based on temperature. See also [Hargreaves and Samani, 1985]

## 62.4 Makkink

Potential evapotranspiration using Makkink's Equation.  $E_M = \beta_M M_0 + \beta_M M_1 * (s * Si) / (\lambda (s + \gamma))$  .

$E_M$  is the reference evapotranspiration [mm/d].  $\beta_M M_0$  [mm] and  $\beta_M M_1$  [] are empirical parameters.  $Si$  is global radiation [MJ/m<sup>2</sup>/d].  $s$  is the slope of the vapour pressure curve [Pa/K].  $\lambda$  is latent heat of vapourization [MJ/kg].  $\gamma$  is the psychrometric constant [Pa/K]. See also [Makkink, 1957]

```
< Makkink (use_daily_temperature true)
 (beta_M0 beta_M0)
 (beta_M1 beta_M1) >
```

- *use\_daily\_temperature*: boolean (see section 4.1.2)  
Parameter (default true)  
Use daily temperature values even if hourly are available.
- *beta\_M0*: number [mm/d]  
Parameter  
Makkink parameter
- *beta\_M1*: number (dimensionless)  
Parameter  
Makkink parameter

## 62.5 AslyngHansen82

A ‘Makkink’ model (see 62.4, page 338) build into Daisy.

Danish calibration. See also [Aslyng et al., 1982, Hansen, 1984]

```
< (beta_M0 0 [mm/d])
 (beta_M1 0.7 []) >
```

## 62.6 Makkink57

A ‘Makkink’ model (see 62.4, page 338) build into Daisy.

Original parameterization. See also [Makkink, 1957]

```
< (beta_M0 -0.12 [mm/d])
 (beta_M1 0.61 []) >
```

## 62.7 deBruin87

A ‘Makkink’ model (see 62.4, page 338) build into Daisy.

Dutch calibration. See also [de Bruin, 1987]

```
< (beta_M0 0 [mm/d])
 (beta_M1 0.65 []) >
```

## 62.8 PM

Potential evapotranspiration using Penman-Monteith.

```
< PM (rb 20 [s/m]) >
```

- *rb*: number [s/m]

Parameter (default 20)

Boundary layer resistance for wet surface. used for bare soil only.

## 62.9 makkink

Potential evapotranspiration using Makkink’s Equation (old version).

## 62.10 weather

Potential evapotranspiration using weather data.



## Chapter 63

# phenology

The development process.

```
< component (DS -1 [DS])
 (DAP 0 [d])
 (partial_day_length 0 [h])
 (day_length 0 [h]) >
```

- *DS*: number [DS]  
State variable (default -1)  
Development stage [-1:2], 0 is emergence, 1 is flowering.
- *DAP*: number [d]  
State variable (default 0)  
Days after planting.
- *partial\_day\_length*: number [h]  
State variable (default 0)  
Number of light hours this day, so far.
- *day\_length*: number [h]  
State variable (default 0)  
Number of light hours yesterday.

### 63.1 TSum

Crop phenology model purely based on temperature sums. The length of emergence, and the vegetative and reproductive phase are all based on the specified temperature sums. Temperatures below the specified thresholds do not contribute to the temperature sum. Cut stress and leaf respiration does not affect this phenology model.

```
< TSum (EmrTSum EmrTSum)
 (EmrThrs 0 [dg C])
 (VegTSum VegTSum)
 (VegThrs 0 [dg C])
 (RepTSum RepTSum)
 (RepThrs 0 [dg C])
 ;; Shared parameters are described in chapter 63.
 (DS -1 [DS])
 (DAP 0 [d])
 (partial_day_length 0 [h])
 (day_length 0 [h]) >
```

- *EmrTSum*: number [dg C d]  
Parameter  
Soil temperature sum at emergence.
- *EmrThrs*: number [dg C]  
Parameter (default 0)  
Minimum soil temperature for emergence. Temperature below this will not count in the sum.
- *VegTSum*: number [dg C d]  
Parameter  
Air temperature sum for vegetative fase.
- *VegThrs*: number [dg C]  
Parameter (default 0)  
Minimum air temperature for development in vegetative fase. Temperature below this will not count in the sum.
- *RepTSum*: number [dg C d]  
Parameter  
Air temperature sum for vegetative fase.
- *RepThrs*: number [dg C]  
Parameter (default 0)  
Minimum air temperature for development in vegetative fase. Temperature below this will not count in the sum.

## 63.2 default

Default crop phenology model.

Used by crop Maize Devel (see 29.12, page 174) , crop Pioneer Maize Devel (see 29.19, page 175) , crop Ikuwala Maize Devel (see 29.13, page 174) , crop Silage Maize Devel (see 29.14, page 174) , crop Pea Devel (see 29.17, page 174) , crop Potato; Koege Devel (see 29.20, page 175) , crop Potato; SCRI Devel (see 29.21, page 175) , crop Potato; FertOrgaNic Devel (see 29.22, page 175) , crop Potato; Agria Devel (see 29.23, page 175) , crop Potato; Folva Devel (see 29.24, page 175) , crop Potato; Triada Devel (see 29.26, page 175) , crop Rye Devel (see 29.29, page 176) , crop Spring Barley Devel (see 29.31, page 176) , crop Spring Barley; Foulum Devel (see 29.32, page 176) , crop Sugar Beet Devel (see 29.35, page 176) , crop Fodder Beet Devel (see 29.9, page 174) , crop Spring Rape Devel (see 29.33, page 176) , crop Spring Wheat Devel (see 29.34, page 176) , crop Winter Barley Devel (see 29.40, page 177) , crop Winter Barley; Koge Devel (see 29.42, page 177) , crop Winter Rape Devel (see 29.43, page 177) , crop Winter Wheat Devel (see 29.44, page 177) , crop Winter Wheat; Foulum Devel (see 29.46, page 177) , crop Winter Wheat JG Devel (see 29.45, page 177) , crop Grass to grain Devel (see 29.10, page 174) , crop Grass Devel (see 29.11, page 174) , crop Ryegrass Devel (see 29.30, page 176) , crop Wclover Devel (see 29.36, page 176) , crop Beetroot Devel (see 29.2, page 173) , crop Broccoli Devel (see 29.3, page 173) , crop Broccoli - transplanted Devel (see 29.4, page 173) , crop Brussels sprouts Devel (see 29.5, page 173) , crop Brussels sprouts - transplanted Devel (see 29.6, page 173) , crop Celeriac Devel (see 29.7, page 174) , crop Celeriac - transplanted Devel (see 29.8, page 174) , crop Potato; SCRI - AArhus Devel (see 29.27, page 176) , crop Early potato Devel (see 29.28, page 176) , crop White cabbage Devel (see 29.37, page 177) , crop White cabbage - transplanted Devel (see 29.38, page 177) , crop Early white cabbage -

transplanted Devel (see 29.39, page 177) , crop Onion Devel (see 29.15, page 174) ,  
and crop Onion - planting of sets Devel (see 29.16, page 174) .

```
< default (modify_DS modify_DS)
 (EmrTSum EmrTSum)
 (EmrSMF EmrSMF) ; Has default value.
 (DS_Emr 0.01 [DS])
 (DSRate1 DSRate1)
 (DSRate2 DSRate2)
 (TempEff1 TempEff1)
 (TempEff2 TempEff2)
 (PhotEff1 PhotEff1)
 (DSMature 2 [DS])
 (DSRepeat 4 [DS])
 (DSSetBack 1.7 [DS])
 (defined_until_ds 2 [DS])
 ;; Shared parameters are described in chapter 63.
 (DS -1 [DS])
 (DAP 0 [d])
 (partial_day_length 0 [h])
 (day_length 0 [h]) >
```

- *modify\_DS*: **number** component (see chapter 59)  
Optional component  
Expression for modifying DS. The original value is available as a free variable named 'x'. The expression will be used as a new value for DS.
- *EmrTSum*: number [**dg C d**]  
Parameter  
Soil temperature sum at emergence.
- *EmrSMF*: plf [**cm** → **<none>**]  
Parameter (has default const value 1 [])  
Soil moisture (h-function) effect on emergence.
- *DS\_Emr*: number [**DS**]  
Parameter (default 0.01)  
Development stage at emergence.
- *DSRate1*: number [**DS/d**]  
Parameter  
Development rate in the vegetative stage.
- *DSRate2*: number [**DS/d**]  
Parameter  
Development rate in the reproductive stage.
- *TempEff1*: plf [**dg C** → **<none>**]  
Parameter  
Temperature effect, vegetative stage.
- *TempEff2*: plf [**dg C** → **<none>**]  
Parameter  
Temperature effect, reproductive stage.
- *PhotEff1*: plf [**h** → **<none>**]  
Parameter  
Photoperiode effect, vegetative stage.

- *DSMature*: number [DS]  
Parameter (default 2)  
Development stage at maturation.
- *DSRepeat*: number [DS]  
Parameter (default 4)  
Development stage when DS set back is activated.
- *DSSetBack*: number [DS]  
Parameter (default 1.7)  
Development stage set back at DSRepeat.
- *defined\_until\_ds*: number [DS]  
Parameter (default 2)  
This parameterization is only valid until the specified development state.



## Chapter 64

# photosynthesis

Leaf photosynthesis.

```
< component (min_PAR 0.1 [W/m2]) >
```

- *min\_PAR*: number [W/m<sup>2</sup>]  
Parameter (default 0.1)  
Minimum PAR at top of canopy for photosynthesis. If radiation is below this amount, photosynthesis will be disabled.

### 64.1 Farquhar

Base parameterization for Farquhar derived photosynthesis models.

Farquhar et al. (1980) photosynthesis and Ball et al. (1987) stomataconductance model coupled as described by Collatz et al., 1991.

```
< Farquhar (Xn 0.00116 [mol/mol/s])
(Gamma25 3.69 [Pa])
(Ea_Gamma 29000 [J/mol])
("N-dist" N-dist) ; Default exp value.
(Stomatacon Stomatacon) ; Default Leuning value.
;; Shared parameters are described in chapter 64.
(min_PAR 0.1 [W/m2]) >
```

- *Xn*: number [mol/mol/s]  
Parameter (default 0.00116)  
Value description: page 550 See also [de Pury and Farquhar, 1997]  
Slope of relationship between leaf rubisco N and Vmax.  $Xn = 1.16E-3$  mol/mol/s for wheat (de Pury & Farquhar, 1997)
- *Gamma25*: number [Pa]  
Parameter (default 3.69)  
CO2 compensation point of photosynthesis.  $\Gamma_{25} = 3.69$  Pa for wheat (Collatz et al., 1991)
- *Ea\_Gamma*: number [J/mol]  
Parameter (default 29000)  
Activation energy for Gamma.  $Ea_{\Gamma} = 29000$  (Jordan & Ogren, 1984)
- *N-dist*: **rubiscoNdist** component (see chapter 75)  
Component (default 'exp')  
Rubisco N-distribution in the canopy layer.

- *Stomatacon*: **stomatacon** component (see chapter 84)  
Component (default 'Leuning')  
Stomata conductance of water vapor.

## Log Variables

- *LAI*: number []  
Leaf area index for the canopy used in photosynthesis.
- *ABA\_effect*: number (dimensionless)  
Water stress effect induced by ABA and crown water potential
- *ci\_vector*: number [**Pa**] canopy intervals  
CO<sub>2</sub> pressure in Stomatal in each layer.
- *Vm\_vector*: number [**mmol/m<sup>2</sup>/s**] canopy intervals  
Photosynthetic capacity in each layer.
- *Jm\_vector*: number [**mmol/m<sup>2</sup>/s**] canopy intervals  
Potential rate of electron transport in each layer.
- *pn\_vector*: number [**mol/m<sup>2</sup> leaf/s**] canopy intervals  
Net photosynthesis.
- *cs\_vector*: number [**Pa**] canopy intervals  
CO<sub>2</sub> pressure at leaf surface.
- *hs\_vector*: number [<**fraction**>] canopy intervals  
Relative humidity at leaf surface.
- *gs\_vector*: number [**mol/m<sup>2</sup> leaf/s**] canopy intervals  
Stomata conductance in each layer.
- *Nleaf\_vector*: number [**mol N/m<sup>2</sup>**] canopy intervals  
Distribution of photosynthetic N-leaf.
- *Ass\_vector*: number [**mol CH<sub>2</sub>O/m<sup>2</sup>/h**] canopy intervals  
Brutto assimilate.
- *LAI\_vector*: number [**m<sup>2</sup> leaf/m<sup>2</sup> field**] canopy intervals  
LAI.
- *ci\_middel*: number [**Pa**]  
Stomata average CO<sub>2</sub> pressure.
- *Gamma*: number [**Pa**]  
CO<sub>2</sub> compensation point of photosynthesis.
- *gbw*: number [**mol/m<sup>2</sup> leaf/s**]  
Boundary lauer conductance.
- *gs*: number [**mol/m<sup>2</sup> field/s**]  
Stomata conductance.
- *gs\_ms*: number [**m/s**]  
Stomata conductance.
- *Ass*: number [**g CH<sub>2</sub>O/m<sup>2</sup>/h**]  
'Net' leaf assimilate of CO<sub>2</sub> (brutto photosynthesis).

- *Res*: number [**g CH<sub>2</sub>O**/m<sup>2</sup>/h]  
Farquhar leaf respiration.
- *PAR\_*: number [**mol**/m<sup>2</sup>/h]  
PAR.
- *Vmax*: number [[**mmol**/m<sup>2</sup>/s]]  
Photosynthetic Rubisco capacity.
- *jm*: number [[**mmol**/m<sup>2</sup>/s]]  
Potential rate of electron transport.
- *leafPhotN*: number [[**mol N**/m<sup>2</sup>]]  
Content of photosynthetic active leaf N.
- *fraction\_total*: number []  
Fraction of leaf contributing to the photosynthesis.

## 64.2 FC\_C3

A ‘Farquhar’ model (see 64.1, page 345) build into Daisy.

Photosynthesis for C3 crops described by Faquhar et al. (1980). See also [de Pury and Farquhar, 1997]

```
< FC_C3 (S 710 [J/mol/K])
 (beta 0.95 [?])
 (TempEff TempEff)
 (Kc25 40.4 [Pa])
 (Ko25 24800 [Pa])
 (H 220000 [J/mol])
 (c_Vm 26.35 [?])
 (Ea_Vm 65330 [J/mol])
 (Eda_Vm 202900 [J/mol])
 (Ea_Jm 37000 [J/mol])
 (Ea_ko 36000 [J/mol])
 (Ea_kc 59400 [J/mol])
 (Ea_rd 66400 [J/mol])
 (Sv 650 [J/mol/K])
 (theta 0.7 [?])
 (alfa 0.08 [mol/mol])
 ;; Shared parameters are described in section 64.1.
 (min_PAR 0.1 [W/m2])
 (Xn 0.00116 [mol/mol/s])
 (Gamma25 3.69 [Pa])
 (Ea_Gamma 29000 [J/mol])
 ("N-dist" N-dist) ; Default exp value.
 (Stomatacon Stomatacon) ; Default Leuning value. >
```

- *S*: number [**J/mol/K**]  
Parameter (default 710)  
Electron transport temperature response parameter,(De Pury & Farquhar, 1997)
- *beta*: number (dimension not specified)  
Parameter (default 0.95)  
Curvature, Collatz et al., 1991

- *TempEff*: plf [**dg C** → **<none>**]  
Parameter  
Temperature factor for assimilate production.
- *Kc25*: number [**Pa**]  
Parameter (default 40.4)  
Micahaelis-Menten constant of Rubisco for CO<sub>2</sub>.  $Kc25 = 40.4$  Pa for wheat (Collatz et al., 1991)
- *Ko25*: number [**Pa**]  
Parameter (default 24800)  
Micahaelis-Menten constant of Rubisco for O<sub>2</sub> at 25 degrees.  $Ko25 = 24800$  Pa for wheat (Collatz et al., 1991)
- *H*: number [**J/mol**]  
Parameter (default 220000)  
Curvature parameter of  $J_m$ , (De Pury & Farquhar, 1997)
- *c\_Vm*: number (dimension not specified)  
Parameter (default 26.35)  
Temperature scaling constant for  $V_{max}$ .  $c\_Vm = 26.35$  (Bernacchi et al., 2001)
- *Ea\_Vm*: number [**J/mol**]  
Parameter (default 65330)  
Activation energy for  $V_{max}$ .  $Ea\_Vm = 65330$  J/mol (Ball, 1988)
- *Eda\_Vm*: number [**J/mol**]  
Parameter (default 202900)  
Deactivation energy for  $V_{max}$ .  $Eda\_Vm = 202900$  J/mol
- *Ea\_Jm*: number [**J/mol**]  
Parameter (default 37000)  
Actimation energy for  $J_m$ .  $Ea\_Jm = 37000$  J/mol (Farquhar et al., 1980).
- *Ea\_ko*: number [**J/mol**]  
Parameter (default 36000)  
Actimation energy for  $k_o$ .  $Ea\_ko = 36000$  J/mol (Badger & Collatz, 1977).
- *Ea\_kc*: number [**J/mol**]  
Parameter (default 59400)  
Actimation energy for  $k_c$ .  $Ea\_kc = 59400$  J/mol (Badger & Collatz, 1977)
- *Ea\_rd*: number [**J/mol**]  
Parameter (default 66400)  
Actimation energy for  $r_d$ .  $Ea\_rd = 66400$  J/mol (Farquhar et al., 1980)
- *Sv*: number [**J/mol/K**]  
Parameter (default 650)  
Entropy term.  $Sv = 650$  J/mol/K
- *theta*: number (dimension not specified)  
Parameter (default 0.7)  
Curvature of leaf response of electron transport to irradiance, (De Pury & Farquhar, 1997)
- *alfa*: number [**mol/mol**]  
Parameter (default 0.08)  
Fraction of PAR effectively absorbed by PSII,

## 64.3 FC\_C4

A 'Farquhar' model (see 64.1, page 345) build into Daisy.

C4 photosynthesis and stomatal conductance model by Collatz et al., 1992.

```
< FC_C4 (alpha 0.04 [mol/mol])
 (beta 0.93 [?])
 (theta 0.83 [?])
 (Q10k 1.8 [?])
 (Q10vm 2.4 [?])
 (Q10rd 2 [?])
 (kj 0.6 [?])
 (paab 0.86 [?])
 ;; Shared parameters are described in section 64.1.
 (min_PAR 0.1 [W/m2])
 (Xn 0.00116 [mol/mol/s])
 (Gamma25 3.69 [Pa])
 (Ea_Gamma 29000 [J/mol])
 ("N-dist" N-dist) ; Default exp value.
 (Stomatacon Stomatacon) ; Default Leuning value. >
```

- *alpha*: number [mol/mol]  
Parameter (default 0.04)  
Initial slope of photosynthetic light response.  $\alpha = 0.04$  (Collatz et al., 1992)
- *beta*: number (dimension not specified)  
Parameter (default 0.93)  
Curvature parameter
- *theta*: number (dimension not specified)  
Parameter (default 0.83)  
Curvature parameter
- *Q10k*: number (dimension not specified)  
Parameter (default 1.8)  
 $Q_{10k} = 1.8$  (Collatz et al., 1992)
- *Q10vm*: number (dimension not specified)  
Parameter (default 2.4)  
 $Q_{10vm} = 2.4$  (Collatz et al., 1992)
- *Q10rd*: number (dimension not specified)  
Parameter (default 2)  
 $Q_{10rd} = 2.0$  (Collatz et al., 1992)
- *kj*: number (dimension not specified)  
Parameter (default 0.6)  
Initial slope of photosynthetic CO<sub>2</sub> response,  $k_j = 0.6 \text{ mol/m}^2/\text{s}$  (Collatz et al., 1992)
- *paab*: number (dimension not specified)  
Parameter (default 0.86)  
Leaf absorbtivity to PAR.  $paab = 0.86$  (Collatz et al., 1992)

## 64.4 GL

Goudriaan and Laar, 1978.

Used by crop default LeafPhot (see 29.1, page 171) , crop Maize LeafPhot (see 29.12, page 174) , crop Pioneer Maize LeafPhot (see 29.19, page 175) , crop Pea LeafPhot (see 29.17, page 174) , crop Potato; Koege LeafPhot (see 29.20, page 175) , crop Potato; SCRI LeafPhot (see 29.21, page 175) , crop Potato; FertOrgaNic LeafPhot (see 29.22, page 175) , crop Potato; Folva LeafPhot (see 29.24, page 175) , crop Potato; Triada LeafPhot (see 29.26, page 175) , crop Rye LeafPhot (see 29.29, page 176) , crop Spring Barley LeafPhot (see 29.31, page 176) , crop Sugar Beet LeafPhot (see 29.35, page 176) , crop Fodder Beet LeafPhot (see 29.9, page 174) , crop Spring Rape LeafPhot (see 29.33, page 176) , crop Spring Wheat LeafPhot (see 29.34, page 176) , crop Winter Barley LeafPhot (see 29.40, page 177) , crop Winter Barley; Koge LeafPhot (see 29.42, page 177) , crop Winter Rape LeafPhot (see 29.43, page 177) , crop Winter Wheat LeafPhot (see 29.44, page 177) , crop Winter Wheat JG LeafPhot (see 29.45, page 177) , crop Grass to grain LeafPhot (see 29.10, page 174) , crop Ryegrass LeafPhot (see 29.30, page 176) , crop Wclove LeafPhot (see 29.36, page 176) , crop Beetroot LeafPhot (see 29.2, page 173) , crop Broccoli LeafPhot (see 29.3, page 173) , crop Brussels sprouts LeafPhot (see 29.5, page 173) , crop Celeriac LeafPhot (see 29.7, page 174) , crop Potato; SCRI - AArhus LeafPhot (see 29.27, page 176) , crop Early potato LeafPhot (see 29.28, page 176) , crop White cabbage LeafPhot (see 29.37, page 177) , crop Early white cabbage - transplanted LeafPhot (see 29.39, page 177) , crop Onion LeafPhot (see 29.15, page 174) , and crop Onion - planting of sets LeafPhot (see 29.16, page 174)

```

< GL (Qeff Qeff)
 (Fm Fm)
 (TempEff TempEff)
 (DSEff DSEff) ; Has default value.
 (DAPEff DAPEff) ; Has default value.
 ;; Shared parameters are described in chapter 64.
 (min_PAR 0.1 [W/m²]) >

```

- *Qeff*: number [(g CO<sub>2</sub>/m<sup>2</sup>/h)/(W/m<sup>2</sup>)]  
Parameter  
Quantum efficiency at low light.
- *Fm*: number [g CO<sub>2</sub>/m<sup>2</sup>/h]  
Parameter  
Maximum assimilation rate.
- *TempEff*: plf [dg C → <none>]  
Parameter  
Temperature factor for assimilate production.
- *DSEff*: plf [DS → <none>]  
Parameter (has default const value 1 [])  
Development stage factor for assimilate production.
- *DAPEff*: plf [d → <none>]  
Parameter (has default const value 1 [])  
Age factor for assimilate production. Age is given as day after planting.

## Chapter 65

# ponddamp

Dampening affect of ponding on soil erosion from rain.

### 65.1 EUROSEM

$KH = \exp(-b \cdot h)$  See also [Morgan et al., 1998]  
< EUROSEM (b 2 [mm<sup>-1</sup>]) >

- *b*: number [mm<sup>-1</sup>]  
Parameter (default 2)  
Exponential degradation coefficient. The range of 'b' is from 0.9 to 3.1, a default value of 2 is proposed by the EUROSEM project.

### 65.2 Hairsine91

$KH = (h / dds)^{-0.8}$  See also [Hairsine and Rose, 1991]

### 65.3 Park82

$KH = 2.7183 * \exp(-h / dds)$  See also [Park et al., 1982]

### 65.4 none

$KH = 1.0$

Used by reaction colgen\_Jarvis99 ponddamp (see 70.6, page 387) .





# Chapter 66

## program

Run a program.

Used by Toplevel @ run (see 107.30, page 624) .

### 66.1 AM\_table

Generate a table of fertilizers.

### 66.2 Daisy

The Daisy crop/soil/atmosphere model. See also [Hansen et al., 1990, Hansen, 2002, Hansen et al., 1991]

```
< Daisy (harvest)
 (stop stop)
 (weather weather)
 (time time) ; Has partial value.
 (column column ...)
 (timestep timestep)
 (exchange)
 (scope scope) ; Default null value.
 (output output ...)
 (activate_output activate_output) ; Default true value.
 (log_time_columns log_time_columns ...) ; Has default value.
 (log_prefix "")
 (log_suffix "")
 (print_time print_time) ; Default periodic value.
 (manager manager)
 (previous previous)
 (next_large next_large)
 (message_timestep true)
 (minimal_timestep minimal_timestep)
 (stop_when stop_when) ; Default false value. >
```

- *harvest*: **Harvest** fixed component (see section 107.15) sequence  
Submodel (default: an empty sequence)  
Total list of all crop yields.
- *stop*: **Time** fixed component (see section 107.27)  
Optional submodel

Latest time where the simulation stops. By default, the simulation will run until the manager request it to stop.

- *weather*: **weather** component (see chapter 103)  
Optional component  
Weather model for providing climate information during the simulation. Can be overwritten by column specific weather.
- *time*: **Time** fixed component (see section 107.27)  
Submodel (has partially specified default value)  
Current time in the simulation.
- *column*: **column** component (see chapter 26) sequence  
List of columns to use in this simulation.
- *timestep*: **Timestep** fixed component (see section 107.25)  
Optional submodel  
Length of large timestep in simulation. The default value is 1 hour, anything else is unlikely to work.
- *exchange*: **scope** component (see chapter 76) sequence  
Component (default: an empty sequence)  
List of exchange items for communicating with external models.
- *scope*: **scopesel** component (see chapter 77)  
Component (default 'null')  
Scope to evaluate expressions in.
- *output*: **log** component (see chapter 52) sequence  
List of logs for output during the simulation.
- *activate\_output*: **condition** component (see chapter 28)  
Component (default 'true')  
Activate output logs when this condition is true. You can use the 'after' condition to avoid logging during an initialization period.
- *log\_time\_columns*: string (see section 4.1.5) sequence  
Parameter (has default value with length 4)

(log\_time\_columns year month mday hour)

Parameter description:

List of default time components to include in log files.

Choose between: 'year': Year 'month': Month 'week': Week number (first Thursday is in week 1) 'yday': Julian day 'mday': Day in month 'wday': Weekday (Sunday = 7) 'hour': Hour 'minute': Minute 'second': Second 'microsecond': Microsecond

- *log\_prefix*: string (see section 4.1.5)  
Parameter (default '')  
Prefix for log file names. Set it to 'log/' to put all files in a subdir.
- *log\_suffix*: string (see section 4.1.5)  
Parameter (default '')  
Suffix for log file names. Set it to '.csv' to make Excel happy.

- *print\_time*: **condition** component (see chapter 28)  
Component (default 'periodic')  
Print simulation time whenever this condition is true. The simulation time will also be printed whenever there are any news to report, like emergence of crop or various management operations. Good values for this parameter would be hourly, daily or monthly.
- *manager*: **action** component (see chapter 15)  
Specify the management operations to perform during the simulation.
- *previous*: **Time** fixed component (see section 107.27)  
Optional submodel  
Previous time in the simulation.
- *next\_large*: **Time** fixed component (see section 107.27)  
Optional submodel  
End of next large timestep.
- *message\_timestep*: boolean (see section 4.1.2)  
Parameter (default true)  
Show messages about timestep modifications.
- *minimal\_timestep*: **Timestep** fixed component (see section 107.25)  
Optional submodel  
Minimum length of timestep in simulation. By default, this is the same as 'timestep'.
- *stop\_when*: **condition** component (see chapter 28)  
Component (default 'false')  
Stop when this is true.

## Log Variables

- *dt*: number [**h**]  
Current timestep used by simulation.

## 66.3 GP2D

Write root density table using 2D extension to Gerwitz and Page

```
< GP2D (DS 2 [DS])
 (row_position 0 [cm])
 (WRoot WRoot)
 (row_width row_width)
 (Geometry Geometry) ; Has partial value.
 (soil_depth soil_depth)
 (crop_depth crop_depth)
 (crop_width crop_width) >
```

- *DS*: number [**DS**]  
Parameter (default 2)  
Development stage [0-2]. Not currently used.
- *row\_position*: number [**cm**]  
Parameter (default 0)  
Position of row on x-axis.

- *WRoot*: number [g DM/m<sup>2</sup>]  
Parameter  
Total root dry matter.
- *row\_width*: number [cm]  
Parameter  
Distance between rows.
- *Geometry*: **GeometryRect** fixed component (see section 107.17)  
Submodel (has partially specified default value)  
Discretization of the soil.
- *soil\_depth*: number [cm]  
Parameter  
Limit on root depth by soil (no crops have roots below this).
- *crop\_depth*: number [cm]  
Parameter  
Limit of root depth by crop (no soil have roots below this).
- *crop\_width*: number [cm]  
Parameter  
Maximum horizontal distance of roots from plant.

## 66.4 KM2

Read KM2 precipitation data.

```
< KM2 (file file) >
```

- *file*: string (see section 4.1.5)  
Parameter  
Name of KM2 file where data is found.

## 66.5 Osvaldo

Find the modelling error. This is done between one set of measured data and multiple sets of simulated data.

```
< Osvaldo (pars pars)
 (par_file_prefix par_file_prefix)
 (par_file_suffix par_file_suffix)
 (par_width par_width)
 (measured_file measured_file) >
```

- *pars*: integer  
Parameter  
Number of simulated dataset to compare.
- *par\_file\_prefix*: string (see section 4.1.5)  
Parameter  
Beginning of file name before simulation number.
- *par\_file\_suffix*: string (see section 4.1.5)  
Parameter  
End of file name after simulation number.

- *par\_width*: integer  
Parameter  
Number of digits in simulated file name.
- *measured\_file*: string (see section 4.1.5)  
Parameter  
Name of file containing measurements.

## 66.6 RS2WG

Prepare data from RainSim to the weather generator.

```
< RS2WG (rshourly_origin rshourly_origin)
 (wgcycle_begin wgcycle_begin)
 (wgcycle_length wgcycle_length)
 (rshourly_file rshourly_file)
 (rsdaily_file rsdaily_file)
 (dwfhourly_file dwfhourly_file)
 (hint_file hint_file)
 (hlim hlim ...) >
```

- *rshourly\_origin*: integer  
Parameter  
Start of hourly data from RainSim.
- *wgcycle\_begin*: integer  
Parameter  
Start of weather cycle for Weather Generator.
- *wgcycle\_length*: integer  
Parameter  
Length (in years) of weather cycle for Weather Generator.
- *rshourly\_file*: string (see section 4.1.5)  
Parameter  
Name of file with hourly data from RainSim.
- *rsdaily\_file*: string (see section 4.1.5)  
Parameter  
Name of file with daily precipitation for the Weather Generator.
- *dwfhourly\_file*: string (see section 4.1.5)  
Parameter  
Daisy weather file with hourly data from RainSim.
- *hint\_file*: string (see section 4.1.5)  
Parameter  
Hourly precipitation interval file.
- *hlim*: number [mm] sequence  
Parameter  
Upper limits for each precipitation interval.

## 66.7 WG2DWF

Prepare data from the weather generator for Daisy.

```

< WG2DWF (rshourly_origin rshourly_origin)
 (wgcycle_begin wgcycle_begin)
 (wgcycle_length wgcycle_length)
 (wg_file wg_file)
 (dwfdaily_file dwfdaily_file)
 (monthly_file monthly_file)
 (yearly_file yearly_file)
 (dint_file dint_file)
 (dlim dlim ...)
 (header header) >

```

- *rshourly\_origin*: integer  
Parameter  
Start of hourly data from RainSim.
- *wgcycle\_begin*: integer  
Parameter  
Start of weather cycle for Weather Generator.
- *wgcycle\_length*: integer  
Parameter  
Length (in years) of weather cycle for Weather Generator.
- *wg\_file*: string (see section 4.1.5)  
Parameter  
Name of file generated by the Weather Generator.
- *dwfdaily\_file*: string (see section 4.1.5)  
Parameter  
Daisy weather file with daily data from Weather Generator.
- *monthly\_file*: string (see section 4.1.5)  
Parameter  
Monthly index file.
- *yearly\_file*: string (see section 4.1.5)  
Parameter  
Yearly index file..
- *dint\_file*: string (see section 4.1.5)  
Parameter  
Daily precipitation interval file.
- *dlim*: number [mm] sequence  
Parameter  
Upper limits for each precipitation interval.
- *header*: string (see section 4.1.5)  
Parameter  
Stuff to put in the Daisy weather file header.

## 66.8 astdump

Generate a dump of the abstract syntax tree.

```

< astdump (where "astdump.txt") >

```

- *where*: string (see section 4.1.5)  
Parameter (default 'astdump.txt')  
Name of file to store results in.

## 66.9 batch

Run a sequence of programs.

```
< batch (directory ".")
 (run run ...) >
```

- *directory*: string (see section 4.1.5)  
Parameter (default '.')  
Directory in which to initialize, check and run the programs.
- *run*: **program** component (see chapter 66) sequence  
List of programs to run. The programs will be run in the sequence listed.

## 66.10 cd

Change working directory.

```
< cd directory >
```

- *directory*: string (see section 4.1.5)  
Parameter  
Name of directory to change into.

## 66.11 cpedata

Manipulate data from Agrovand.

```
< cpedata (file file)
 (value value)
 (factor 1 [])
 (begin begin)
 (debug 0)
 (day day)
 (missing missing ...) ; Has default value.
 (filter)
 (original original ...)
 (dim_line dim_line)
 (handle handle)
 (weight weight)
 (origin origin) ; Has partial value.
 (every_hour false) >
```

- *file*: string (see section 4.1.5)  
Parameter  
Name of Daisy file where data is found.
- *value*: string (see section 4.1.5)  
Parameter  
Tag used for value.
- *factor*: number (dimensionless)  
Parameter (default 1)  
Multiply printed value with this number.
- *begin*: **Time** fixed component (see section 107.27)  
Optional submodel  
If specified, only print entries after this date.

- *debug*: integer  
Parameter (default 0)  
Debug level, 0 means no debug information.

- *day*: string (see section 4.1.5)  
Parameter  
Tag used for day.

- *missing*: string (see section 4.1.5) sequence  
Parameter (has default value with length 2)

(missing "" "00.00")

Parameter description:  
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence  
Submodel (default: an empty sequence)  
Only include data from rows that passes all these filters.

< tag allowed... >

- *tag*: string (see section 4.1.5)  
Parameter  
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence  
Parameter  
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence  
Optional parameter  
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim\_line*: boolean (see section 4.1.2)  
Optional parameter  
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.

- *handle*: string (see section 4.1.5)  
Parameter  
This option determine how the specified variable should be handled.  
average: print mean of values within timestep.  
difference: print difference between last value in timestep, and the last value in the previous timestep.  
sum: print sum of all values within current timestep.

- *weight*: string (see section 4.1.5)  
Optional parameter  
Tag used for weight. If you specify this, 'value' will be given this weight.

- *origin*: **Time** fixed component (see section 107.27)  
Submodel (has partially specified default value)  
Day 1.



- *every\_hour*: boolean (see section 4.1.2)  
Parameter (default false)  
If true, print zeroes for hours with no data.

## 66.12 docmodel

Document specific models.

```
< docmodel (component component)
 (format format) ; Default LaTeX value.
 (where where)
 (models models ...) >
```

- *component*: string (see section 4.1.5)  
Parameter  
Component to find the models in.
- *format*: **format** component (see chapter 41)  
Component (default 'LaTeX')  
Text format used for the document.
- *where*: string (see section 4.1.5)  
Parameter  
Name of file to store results in.
- *models*: string (see section 4.1.5) sequence  
Parameter  
Models to document.

## 66.13 document

Generate the components part of the reference manual.

```
< document (format format) ; Default LaTeX value.
 (where "components.tex")
 (print_parameterizations false) >
```

- *format*: **format** component (see chapter 41)  
Component (default 'LaTeX')  
Text format used for the document.
- *where*: string (see section 4.1.5)  
Parameter (default 'components.tex')  
Name of file to store results in.
- *print\_parameterizations*: boolean (see section 4.1.2)  
Parameter (default false)  
Include a copy of all loaded parameterizations in document.

## 66.14 extract

Extract data from a data file.

```

< extract (expr expr)
 (file file)
 (missing missing ...) ; Has default value.
 (filter)
 (original original ...)
 (dim_line dim_line)
 (handle handle)
 (valid valid) ; Default true value.
 (columns columns ...)
 (rows rows ...)
 (column_separator "+")
 (row_separator "+") >

```

- *expr*: **number** component (see chapter 59)  
Expression to extract. The expression can refer to the value in a specific column by the tag for that column.

- *file*: string (see section 4.1.5)  
Parameter  
Name of Daisy file where data is found.

- *missing*: string (see section 4.1.5) sequence  
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:  
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence  
Submodel (default: an empty sequence)  
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

- *tag*: string (see section 4.1.5)  
Parameter  
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence  
Parameter  
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence  
Optional parameter  
List of dimensions of the data in the data file.  
If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim\_line*: boolean (see section 4.1.2)  
Optional parameter  
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *handle*: string (see section 4.1.5)  
Parameter  
How to handle extracted data.

- *valid*: **boolean** component (see chapter 21)  
Component (default 'true')  
Ignore entries if this boolean expression is false.
- *columns*: string (see section 4.1.5) sequence  
Parameter  
List of entries to show as columns in the table. Each combination of values will be one column.
- *rows*: string (see section 4.1.5) sequence  
Parameter  
List of entries to show as columns in the table. Each combination of values will be one row.
- *column\_separator*: string (see section 4.1.5)  
Parameter (default '+')  
String to separate values within a column descriptor.
- *row\_separator*: string (see section 4.1.5)  
Parameter (default '+')  
String to separate values within a row descriptor.

## 66.15 gnuplot

Generate a gnuplot command file.

```
< gnuplot (cd false)
 (extra)
 (command_file "daisy.gnuplot")
 (graph graph ...) >
```

- *cd*: boolean (see section 4.1.2)  
Parameter (default false)  
Set this flag to add a 'cd' command to the current working directory. This is useful under MS Windows when dragging the file to a gnuplot icon.
- *extra*: string (see section 4.1.5) sequence  
Parameter (default: an empty sequence)  
List of extra gnuplot commands. The commands will be inserted right before the list of graphs.
- *command\_file*: string (see section 4.1.5)  
Parameter (default 'daisy.gnuplot')  
File name for gnuplot commands.
- *graph*: **gnuplot** component (see chapter 44) sequence  
Graphs to plot.

## 66.16 hmovie

Manipulate data from Agrovand.

```

< hmovie (file file)
 (dim dim)
 (cd false)
 (missing missing ...) ; Has default value.
 (filter)
 (original original ...)
 (dim_line dim_line)
 (command_file "daisy.gnuplot")
 (x_min x_min)
 (x_max x_max)
 (y_min y_min)
 (y_max y_max)
 (z_min z_min)
 (z_max z_max)
 (x_ticks x_ticks ...)
 (x_tic_max x_tic_max)
 (output_base output_base)
 (output_width output_width)
 (tags tags ...)
 (missing_value missing_value) >

```

- *file*: string (see section 4.1.5)  
Parameter  
Name of Daisy file where data is found.
- *dim*: string (see section 4.1.5)  
Parameter  
Dimension of all axes.
- *cd*: boolean (see section 4.1.2)  
Parameter (default false)  
Set this flag to add a 'cd' command to the current working directory. This is useful under MS Windows when dragging the file to a gnuplot icon.
- *missing*: string (see section 4.1.5) sequence  
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:  
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence  
Submodel (default: an empty sequence)  
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

- *tag*: string (see section 4.1.5)  
Parameter  
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence  
Parameter  
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence  
Optional parameter  
List of dimensions of the data in the data file.  
  
If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.  
  
By default Daisy will use the names specified in data file.
- *dim\_line*: boolean (see section 4.1.2)  
Optional parameter  
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *command\_file*: string (see section 4.1.5)  
Parameter (default 'daisy.gnuplot')  
File name for gnuplot commands.
- *x\_min*: number [**<user>**]  
Parameter  
Min value of x-axes.
- *x\_max*: number [**<user>**]  
Parameter  
Max value of x-axes.
- *y\_min*: number [**<user>**]  
Parameter  
Min value of y-axes.
- *y\_max*: number [**<user>**]  
Parameter  
Max value of y-axes.
- *z\_min*: number [**<user>**]  
Parameter  
Min value of z-axes.
- *z\_max*: number [**<user>**]  
Parameter  
Max value of z-axes.
- *x\_tics*: submodel (see section 4.1.7) sequence  
List of tic markers for the x-axes.  
  

**< value name >**

  - *value*: number (dimension not specified)  
Parameter  
Tic position.
  - *name*: string (see section 4.1.5)  
Parameter  
Tic name.
- *x\_tic\_max*: number [**<user>**]  
Parameter  
Highest synthetic z value for at x tics.

- *output\_base*: string (see section 4.1.5)  
Parameter  
Prefix used for all output files.
- *output\_width*: integer  
Parameter  
Number of digits in output file name.
- *tags*: submodel (see section 4.1.7) sequence  
List of column tags for the data file.  

```
< x y tag >
```

  - *x*: number (dimension not specified)  
Parameter  
X position.
  - *y*: number (dimension not specified)  
Parameter  
Y position.
  - *tag*: string (see section 4.1.5)  
Parameter  
Column name.
- *missing\_value*: number [**<user>**]  
Parameter  
Replace missing values with this.

## 66.17 hydraulic

Generate a table of the retention curve and hydraulic conductivity.

```
< hydraulic horizon
 (T 20 [dg C])
 (top_soil top_soil)
 (intervals 50)
 (print_Cw2 false)
 (print_M false)
 (min_pF 0 [pF])
 (max_pF 5 [pF]) >
```

- *horizon*: **horizon** component (see chapter 47)  
The hydraulic model to show in the table.
- *T*: number [**dg C**]  
Parameter (default 20)  
Temperature for conductivity.
- *top\_soil*: boolean (see section 4.1.2)  
Parameter  
Set to true for the plowing layer.
- *intervals*: integer  
Parameter (default 50)  
Number of intervals in the table.
- *print\_Cw2*: boolean (see section 4.1.2)  
Parameter (default false)  
Set to true to include Cw2 to the table.

- *print\_M*: boolean (see section 4.1.2)  
Parameter (default false)  
Set to true to include M to the table.
- *min\_pF*: number [pF]  
Parameter (default 0)  
Minimal pF in table.
- *max\_pF*: number [pF]  
Parameter (default 5)  
Maximal pF in table.

## 66.18 minimize

Find local minimum for program.

The optimization will stop if the worst guess fails to improve more than 'epsilon' within 'min\_iter' iterations, or if the total number of iterations exceeds 'max\_iter'. See also [Nelder and Mead, 1965]

```
< minimize (expr expr)
 (scope scope)
 (run run)
 (parameters parameters ...)
 (parameter parameter ...)
 (limit limit)
 (simplex simplex ...)
 (epsilon epsilon)
 (min_iter 10)
 (max_iter max_iter) >
```

- *expr*: **number** component (see chapter 59)  
Expression to minimize.
- *scope*: **scopesel** component (see chapter 77)  
Scope to evaluate expressions in.
- *run*: **program** component (see chapter 66)  
Program to optimize.
- *parameters*: submodel (see section 4.1.7) sequence  
Optional submodel  
List of (NAME VAL1 VAL2). NAME is the name of a parameter to optimize, VAL1 and VAL2 are two different legal values for the parameter.

```
< name value... >
```

- *name*: string (see section 4.1.5)  
Parameter  
Name of parameter.
- *value*: number [<user>] array of length 2  
Parameter  
Two values.

- *parameter*: string (see section 4.1.5) sequence  
Optional parameter  
List of parameters to optimize.

- *limit*: **boolean** component (see chapter 21)  
Limit parameter values so this expression is true.
- *simplex*: submodel (see section 4.1.7) sequence  
Optional submodel  
List of points defining the initial simplex. You must define one more point than you have parameters.  

```
< value... >
```

  - *value*: number (dimension not specified) sequence  
Parameter  
Value of each parameter.
- *epsilon*: number (dimension not specified)  
Parameter  
Minimal improvement of worst point to be considered for 'min\_iter'.
- *min\_iter*: integer  
Parameter (default 10)  
Maximal number of iterations with no improvement of worst point.
- *max\_iter*: integer  
Parameter  
Stop after this number of iterations.

## 66.19 nwaps

Nwaps collects results from multiple directories.

```
< nwaps (file)
 (fractiles 0 0.1 0.5 0.9 1)
 (missing missing ...) ; Has default value.
 (directory)
 (parent_directory ".")
 (scenario scenario ...) ; Has default value.
 (scn_sep "-")
 (input_suffix ".dlf")
 (output_prefix out_)
 (output_suffix ".csv")
 (summary_prefix sum_)
 (success_file SUCCESS)
 (failure_file FAILURE) >
```

- *file*: string (see section 4.1.5) sequence  
Parameter (default: an empty sequence)  
Files to look for data in. If empty, use all applicable.
- *fractiles*: number [**<fraction>**] sequence  
Parameter (has default value with length 5)

```
(fractiles 0 0.1 0.5 0.9 1 [])
```

Parameter description:  
Fractiles to include in summary.

- *missing*: string (see section 4.1.5) sequence  
Parameter (has default value with length 2)



```
(missing "" "00.00")
```

Parameter description:

List of strings indicating missing values.

- *directory*: string (see section 4.1.5) sequence  
Parameter (default: an empty sequence)  
Directories to look for data in. If empty, use all applicable.
- *parent\_directory*: string (see section 4.1.5)  
Parameter (default ‘.’)  
Directory to look for subfolders in.
- *scenario*: string (see section 4.1.5) sequence  
Parameter (has default value with length 1)

```
(scenario Scenario)
```

Parameter description:

Names of scenarios.

- *scn\_sep*: string (see section 4.1.5)  
Parameter (default ‘-’)  
String separating scenarios in directory names.
- *input\_suffix*: string (see section 4.1.5)  
Parameter (default ‘.dlf’)  
Look for files with this suffix.
- *output\_prefix*: string (see section 4.1.5)  
Parameter (default ‘out\_’)  
Prepend this to output file names.
- *output\_suffix*: string (see section 4.1.5)  
Parameter (default ‘.csv’)  
Append this to output file names.
- *summary\_prefix*: string (see section 4.1.5)  
Parameter (default ‘sum\_’)  
Prepend this to summary file names.
- *success\_file*: string (see section 4.1.5)  
Parameter (default ‘SUCCESS’)  
Only combine directories containing this file.
- *failure\_file*: string (see section 4.1.5)  
Parameter (default ‘FAILURE’)  
Don’t combine directories containing this file.

## 66.20 post-process

Extract a subset of a soil profile log file.

```

< "post-process" (file file)
 (top top)
 (before before)
 (after after)
 (dimension dimension)
 (where where)
 (missing missing ...) ; Has default value.
 (filter)
 (original original ...)
 (dim_line dim_line)
 (bottom bottom)
 (left left)
 (right right)
 (handle_z all)
 (handle_x all)
 (time_columns time_columns ...)
 (print_header true) >

```

- *file*: string (see section 4.1.5)  
Parameter  
Name of Daisy file where data is found.
- *top*: number [**cm**]  
Optional parameter  
Only include values below this height.
- *before*: **Time** fixed component (see section 107.27)  
Optional submodel  
Only include values before this time.
- *after*: **Time** fixed component (see section 107.27)  
Optional submodel  
Only include values after this time.
- *dimension*: string (see section 4.1.5)  
Optional parameter  
Dimension for data output. By default, use dimension from file.
- *where*: string (see section 4.1.5)  
Parameter  
Name of output file.
- *missing*: string (see section 4.1.5) sequence  
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:  
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence  
Submodel (default: an empty sequence)  
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

- *tag*: string (see section 4.1.5)  
Parameter  
Name of column in Daisy log file to filter for.

- *allowed*: string (see section 4.1.5) sequence  
Parameter  
List of allowable values in filter.
- *original*: string (see section 4.1.5) sequence  
Optional parameter  
List of dimensions of the data in the data file.  
  
If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.  
  
By default Daisy will use the names specified in data file.
- *dim\_line*: boolean (see section 4.1.2)  
Optional parameter  
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *bottom*: number [**cm**]  
Optional parameter  
Only include values above this height.
- *left*: number [**cm**]  
Optional parameter  
Only include values to the right of this position.
- *right*: number [**cm**]  
Optional parameter  
Only include values to the left of this position.
- *handle\_z*: string (see section 4.1.5)  
Parameter (default 'all')  
How to handle multiple values for specified dimension. Legal values are: all: output all values. sum: output the sum of the values, multiplied by the length of the cell average: output the lenbth weighted average content of the cells.
- *handle\_x*: string (see section 4.1.5)  
Parameter (default 'all')  
How to handle multiple values for specified dimension. Legal values are: all: output all values. sum: output the sum of the values, multiplied by the length of the cell average: output the lenbth weighted average content of the cells.
- *time\_columns*: string (see section 4.1.5) sequence  
Optional parameter  
List of time components to include in output. By default, use the same as the source.  
  
Choose between: 'year': Year 'month': Month 'week': Week number (first Thursday is in week 1) 'yday': Julian day 'mday': Day in month 'yday': Weekday (Sunday = 7) 'hour': Hour 'minute': Minute 'second': Second 'microsecond': Microsecond
- *print\_header*: string (see section 4.1.5)  
Parameter (default 'true')  
If this is set to 'false', no header is printed. If this is set to 'true', a full header is printed. If this is set to 'fixed', a small fixed size header is printed.

## 66.21 rootmatch

Match root data with GP2D model.

```
< rootmatch (file file)
 (gnuplot gnuplot) ; Has default value.
 (SpRtLength 100 [m/g])
 (DensRtTip 0.1 [cm/cm3])
 (debug 0)
 (row.position 0 [cm])
 (row.distance row_distance)
 (missing missing ...) ; Has default value.
 (filter)
 (original original ...)
 (dimline dim_line)
 (pos_dim pos_dim)
 (dens_dim dens_dim)
 (tag_x tag_x)
 (tag_z_min tag_z_min)
 (tag_z_max tag_z_max)
 (tag_density tag_density)
 (show_data true)
 (show_match false)
 (x_offset 0 [cm])
 (tabular false)
 (min_dist -1 [cm])
 (SoilDepth SoilDepth) >
```

- *file*: string (see section 4.1.5)  
Parameter  
Name of Daisy file where data is found.
- *gnuplot*: submodel (see section 4.1.7)  
Submodel (has fully specified default value)

(gnuplot )

Parameter description:

Irrigation model for first season. If missing, don't irrigate.

```
< (x_offset 0 [cm])
 (x_start 0 [cm])
 (x_end x_end)
 (z_end z_end)
 (dx 1 [cm])
 (dz dz) >
```

- *x\_offset*: number [**cm**]  
Optional parameter (default 0)  
Subtract this from printed x values.
- *x\_start*: number [**cm**]  
Parameter (default 0)  
Start table here.
- *x\_end*: number [**cm**]  
Optional parameter  
End table here. By default, this is 'x\_start' plus 'row.distance'.

- *z\_end*: number [**cm**]  
Optional parameter  
End table here at this depth.
- *dx*: number [**cm**]  
Parameter (default 1)  
Horizontal interval size.
- *dz*: number [**cm**]  
Optional parameter  
Vertical interval size. By default identical to ‘dx’.
- *SpRtLength*: number [**m/g**]  
Parameter (default 100)  
Specific root length
- *DensRtTip*: number [**cm/cm<sup>3</sup>**]  
Parameter (default 0.1)  
Root density at (potential) penetration depth.
- *debug*: integer  
Parameter (default 0)  
Show debug messages if larger than zero.
- *row\_position*: number [**cm**]  
State variable (default 0)  
Horizontal position of row crops.
- *row\_distance*: number [**cm**]  
State variable  
Distance between rows of crops.
- *missing*: string (see section 4.1.5) sequence  
Parameter (has default value with length 2)  
  
(missing "" "00.00")  
  
Parameter description:  
List of strings indicating missing values.
- *filter*: submodel (see section 4.1.7) sequence  
Submodel (default: an empty sequence)  
Only include data from rows that passes all these filters.  
  - < *tag allowed...* >
  - *tag*: string (see section 4.1.5)  
Parameter  
Name of column in Daisy log file to filter for.
  - *allowed*: string (see section 4.1.5) sequence  
Parameter  
List of allowable values in filter.
- *original*: string (see section 4.1.5) sequence  
Optional parameter  
List of dimensions of the data in the data file.  
  
If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.  
  
By default Daisy will use the names specified in data file.

- *dim\_line*: boolean (see section 4.1.2)  
Optional parameter  
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *pos\_dim*: string (see section 4.1.5)  
Parameter  
Position dimension
- *dens\_dim*: string (see section 4.1.5)  
Parameter  
Root density dimension.
- *tag\_x*: string (see section 4.1.5)  
Parameter  
Name of column indicating horizontal distance from row.
- *tag\_z\_min*: string (see section 4.1.5)  
Parameter  
Name of column indicating minimal vertical distance from row.
- *tag\_z\_max*: string (see section 4.1.5)  
Parameter  
Name of column indicating maximal vertical distance from row.
- *tag\_density*: string (see section 4.1.5)  
Parameter  
Name of column containing measured root density.
- *show\_data*: boolean (see section 4.1.2)  
Parameter (default true)  
Show comparison of observed and modelled values.
- *show\_match*: boolean (see section 4.1.2)  
Parameter (default false)  
Show matching observed and modelled values with errorbars.
- *x\_offset*: number [cm]  
State variable (default 0)  
Subtract this from the x values shown with 'show\_match'.
- *tabular*: boolean (see section 4.1.2)  
Parameter (default false)  
Show parameters in tabular form for easy import to spreadsheet.
- *min\_dist*: number [cm]  
State variable (default -1)  
Ignore root data closer than this to the row in both dimensions.
- *SoilDepth*: number [cm]  
Optional parameter  
Specifies how to handle soil imposed maximum root depth. If positive, always use this depth. If negative, disable maximum root depth (making the root zone infinite). If not specified, use the value that gives the best fit.  
  
Note that unless you specify a negative value, the root zone will be limited by DensRtTip, that is, depth where the corresponding 1D GP would give densities less than DensRtTip, will instead have a zero density.

## 66.22 spawn

Spawn a number of programs in parallel.

```
< spawn (file)
 (program)
 (directory)
 (input_directory "..")
 (exe exe)
 (parallel parallel) >
```

- *file*: string (see section 4.1.5) sequence  
Parameter (default: an empty sequence)  
Setup files containing programs to run. By default, use the present setup file.
- *program*: string (see section 4.1.5) sequence  
Parameter (default: an empty sequence)  
Names of programs to run.
- *directory*: string (see section 4.1.5) sequence  
Parameter (default: an empty sequence)  
Directories to run the programs in. By default, this will be the names of the programs.
- *input\_directory*: string (see section 4.1.5)  
Parameter (default ‘..’)  
When trying to open files from the current directory look here instead.
- *exe*: string (see section 4.1.5)  
Optional parameter  
Name of executable to spawn. By default, the currently running executable.
- *parallel*: integer  
Optional parameter  
Maximum number of programs to run in parallel. By default this is determined by the hardware. Select 0 to spawn all in parallel.

## 66.23 weather

Weather data from a data file.

```
< weather (file file)
 (fractiles)
 (weather weather)
 (end end) ; Has partial value.
 (begin begin) ; Has partial value. >
```

- *file*: string (see section 4.1.5)  
Optional parameter  
Base name for files containing daily average data. By default, create no such files
- *fractiles*: number [**<fraction>**] sequence  
Parameter (default: an empty sequence)  
List of fractiles print in ‘file’.
- *weather*: **weather** component (see chapter 103)  
Weather source to analyse.

- *end*: **Time** fixed component (see section 107.27)  
Submodel (has partially specified default value)  
Last date to analyse.
- *begin*: **Time** fixed component (see section 107.27)  
Submodel (has partially specified default value)  
First date to analyze.

## 66.24 write

Write string to file.

```
< write (where screen)
 (what what) >
```

- *where*: string (see section 4.1.5)  
Parameter (default 'screen')  
File to write it in. If the value is 'screen', write the string to the screen.
- *what*: string (see section 4.1.5)  
Parameter  
String to write.

## 66.25 wse

Generate a table of the water stress effect.

```
< wse wse
 (intervals 10) >
```

- *wse*: **wse** component (see chapter 104)  
The water stress effect to show in the table.
- *intervals*: integer  
Parameter (default 10)  
Number of intervals in the table.



# Chapter 67

## raddist

The 'raddist' component calculates the radiation distribution in the canopy.

### 67.1 default

Default model of radiation distribution in the canopy.

Used by bioclimate default raddist (see 19.2, page 115) .

### 67.2 sun-shade

Sun-shade model of radiation distribution in the canopy.

Used by bioclimate SSOC raddist (see 19.5, page 116) .

```
< "sun-shade" (sigma_PAR 0.15 [])
 (sigma_NIR 0.83 [])
 (Ps_PAR 0.1 [])
 (Ps_NIR 0.18 [])
 (Ps_PAR_SWE Ps_PAR_SWE) ; Has default value.
 (Ps_NIR_SWE Ps_NIR_SWE) ; Has default value. >
```

- *sigma\_PAR*: number (dimensionless)  
Parameter (default 0.15)  
Leaf scattering coefficient of PAR. sigma\_PAR=0,15 (Houborg, 2006)
- *sigma\_NIR*: number (dimensionless)  
Parameter (default 0.83)  
Leaf scattering coefficient of NIR. sigma\_NIR=0,83 (Houborg, 2006)
- *Ps\_PAR*: number (dimensionless)  
Parameter (default 0.1)  
Soil reflection coefficient of PAR, Ps\_PAR = 0.1 (Houborg, 2006)
- *Ps\_NIR*: number (dimensionless)  
Parameter (default 0.18)  
Soil reflection coefficient of NIR, Ps\_NIR = 0.18 (Houborg, 2006)
- *Ps\_PAR\_SWE*: plf [**pF** → <none>]  
Parameter (has default const value 1 [])  
Effect of soil water on Ps\_PAR.
- *Ps\_NIR\_SWE*: plf [**pF** → <none>]  
Parameter (has default const value 1 [])  
Effect of soil water on Ps\_BIR.

## Log Variables

- *IRb0*: number [ $\mathbf{W\ m^{-2}}$ ]  
Beam radiation above the canopy
- *IRd0*: number [ $\mathbf{W\ m^{-2}}$ ]  
Diffuse radiation above the canopy
- *Ph\_PAR*: number []  
Canopy reflection coefficeint of beam PAR for horizontal leaves
- *Pcb\_PAR*: number []  
Canopy reflection coefficeint of beam PAR for uniform leaf-angle distribution
- *Pscb\_PAR*: number []  
Canopy-soil reflection coefficeint of beam PAR for uniform leaf-angle distribution
- *Pscd\_PAR*: number []  
Canopy-soil reflection coefficeint of diffuse PAR for uniform leaf-angle distribution
- *Ph\_NIR*: number []  
Canopy reflection coefficeint of beam NIR for horizontal leaves
- *Pcb\_NIR*: number []  
Canopy reflection coefficeint of beam NIR for uniform leaf-angle distribution
- *Pscb\_NIR*: number []  
Canopy-soil reflection coefficeint of beam NIR for uniform leaf-angle distribution
- *Pscd\_NIR*: number []  
Canopy-soil reflection coefficeint of diffuse NIR for uniform leaf-angle distribution

## Chapter 68

# rainergy

Energy in rain.

### 68.1 Brown87

Energy as a semi-empirical function of rain intensity. The energy content in the fraction that hits the canopy is ignored. See also [Brown and Foster, 1987]

Used by reaction colgen\_Jarvis99 rainergy (see 70.6, page 387) .

### 68.2 EUROSEM

Kinetic energy model taking vegetation into account. See also [Morgan et al., 1998]

Used by reaction colgen\_Morgan98 rainergy (see 70.8, page 389) .



# Chapter 69

## rate

Specify a rate or a halftime.

Used by AOM component turnover (see 8, page 53) , SMB component turnover (see 13, page 65) , and SOM component turnover (see 14, page 67) .

### 69.1 halftime

A rate specified through the equivalent halftime ( $\text{rate} = \ln 2 / \text{halftime}$ ).

Used by chemical FOCUS canopy\_dissipation (see 23.2, page 135) , chemical Bentazon decompose (see 23.9, page 138) , chemical IPU decompose (see 23.10, page 138) , chemical MCPP decompose (see 23.12, page 138) , chemical Ioxynil decompose (see 23.11, page 138) , chemical 2,4-D decompose (see 23.7, page 138) , chemical Heptachlor decompose (see 23.16, page 139) , and chemical DDT decompose (see 23.15, page 138) .

```
< halftime halftime >
```

- *halftime*: number [h]  
Parameter  
Halftime of the rate to use.

### 69.2 rate

Specify rate directly.

Used by chemical base litter\_diffusion (see 23.1, page 125) , chemical nutrient canopy\_dissipation (see 23.31, page 143) , chemical DOC litter\_diffusion (see 23.33, page 143) , chemical classic canopy\_dissipation (see 23.28, page 142) , chemical colloid canopy\_dissipation (see 23.29, page 142) , chemical solid canopy\_dissipation (see 23.30, page 143) , chemical Tracer canopy\_dissipation (see 23.27, page 142) , chemical Atrazine decompose (see 23.8, page 138) , chemical Pendimethalin decompose (see 23.13, page 138) , chemical PFAS canopy\_dissipation (see 23.18, page 141) , reaction DOM\_turnover turnover (see 70.1, page 383) , AOM AOM-SLOW turnover (see 8.4, page 55) , AOM AOM-FAST turnover (see 8.2, page 54) , AOM AOM-DIRECT turnover (see 8.1, page 54) , AOM CROP-SLOW turnover (see 8.7, page 55) , AOM CROP-FAST turnover (see 8.3, page 54) , SMB SMB-SLOW turnover (see 13.3, page 66) , SMB SMB-FAST turnover (see 13.1, page 66) , SOM SOM-SLOW turnover (see 14.4, page 68) , SOM SOM-SLOW-OLD turnover (see 14.5, page 68) , SOM SOM-FAST turnover (see 14.1, page 68) , and SOM SOM-INERT turnover (see 14.3, page 68) .

```
< rate rate >
```

- *rate*: number [ $\mathbf{h}^{-1}$ ]  
Parameter  
Rate to use.

### 69.3 zero

A ‘rate’ model (see 69.2, page 381) build into Daisy.

A rate of zero.

< *rate* >

# Chapter 70

## reaction

Generic transformations between soil chemicals.

### 70.1 DOM\_turnover

Turnover of dissolved organic matter. See also [Garnier et al., 2003]

Used by chemistry DOM reaction (see 24.2, page 145) .

```
< DOM_turnover (DOC_name DOC)
 (DON_name DON)
 (turnover turnover) ; Default rate value.
 (max_N_depletion max_N_depletion) ; Default rate value.
 (where 1)
 (efficiency 0.62 [<fraction>])
 (C_per_N_goal 10 [g C/g N]) >
```

- *DOC\_name*: string (see section 4.1.5)  
Parameter (default 'DOC')  
Name of compound representing dissolved organic carbon.
- *DON\_name*: string (see section 4.1.5)  
Parameter (default 'DON')  
Name of compound representing dissolved organic nitrogen.
- *turnover*: **rate** component (see chapter 69)  
Component (default 'rate')

```
(turnover rate 0.04125 [h-1])
```

Parameter description:

Turnover of DOM. Garnier et al. (2003) table 1 (T=15 degC)  $1.49 \cdot 10 / 15 = 0.99 \text{ [d}^{-1}\text{]}$

- *max\_N\_depletion*: **rate** component (see chapter 69)  
Component (default 'rate')

```
(max_N_depletion rate 0.1 [h-1])
```

Parameter description:

Max depletion of N.

- *where*: integer  
Parameter (default 1)  
Target for DOM turnover. Should be a number between 0 and the number of SMB pools. 0 means the SMB1, 1 means SMB2. If there are 2 SMB pools, 2 will mean soil buffer.
- *efficiency*: number [**<fraction>**]  
Parameter (default 0.62)  
Value description: Table 1 See also [Garnier et al., 2003]  
Fraction of carbon not lost as CO<sub>2</sub> during turnover.
- *C\_per\_N\_goal*: number [**g C/g N**]  
Parameter (default 10)  
Value description: Table 2 See also [Garnier et al., 2003]  
Target C/N for DOM turnover. If non-negative, mineral N will be added or removed to achieve the goal.

## Log Variables

- *NO<sub>3</sub>*: number [**g N/cm<sup>3</sup>/h**] soil cells  
Rate of nitrate generated.
- *NH<sub>4</sub>*: number [**g N/cm<sup>3</sup>/h**] soil cells  
Rate of ammonium generated.
- *DON*: number [**g N/cm<sup>3</sup>/h**] soil cells  
Rate of dissolved organic N consumed.
- *DOC*: number [**g C/cm<sup>3</sup>/h**] soil cells  
Rate of dissolved organic C consumed.
- *CO<sub>2</sub>*: number [**g C/cm<sup>3</sup>/h**] soil cells  
Rate of carbon dioxide generated.
- *SOC*: number [**g C/cm<sup>3</sup>/h**] soil cells  
Rate of dissolved organic C immobilized.
- *SON*: number [**g N/cm<sup>3</sup>/h**] soil cells  
Rate of dissolved organic N immobilized.
- *h\_factor*: number (dimensionless) soil cells  
Soil water potential effect on turnover rate.
- *T\_factor*: number (dimensionless) soil cells  
Soil temperature effect on turnover rate.
- *N\_factor*: number (dimensionless) soil cells  
Soil nitrogen effect on turnover rate.

## 70.2 Python

Python interface for chemical reactions above and below ground.

```
< Python (soil soil)
 (top top)
 (module module)
 (extra extra ...)
 (in in ...)
 (texture texture ...)
 (out out ...) >
```



- *soil*: string (see section 4.1.5)  
Optional parameter  
Name of Python function for soil reactions.
- *top*: string (see section 4.1.5)  
Optional parameter  
Name of Python function for above ground reactions.
- *module*: string (see section 4.1.5)  
Parameter  
Python module to find the functions.
- *extra*: string (see section 4.1.5) sequence  
Parameter  
Extra input parameters to Python soil function.  
Options include:  
 z [cm]: Height above soil surface (always negative). rho\_b [g S/cm<sup>3</sup>]: Soil dry bulk density. Theta\_sat [cm<sup>3</sup> W/cm<sup>3</sup> V]: Saturated water per system volume. Theta [cm<sup>3</sup> W/cm<sup>3</sup> V]: Actual water per system volume. Theta\_primary [cm<sup>3</sup> W/cm<sup>3</sup> V]: Water in primary domain per system volume. Theta\_secondary [cm<sup>3</sup> W/cm<sup>3</sup> V]: Water in secondary domain per sys. volume. h [cm]: Soil water potential. T [dg C]: Temperature. CO2\_C [g C/cm<sup>3</sup> V/h]: CO2 production rate per system volume. CO2\_C\_fast [g C/cm<sup>3</sup> V/]: Fast CO2 production rate per system volume. SMB\_C [g C/cm<sup>3</sup> V]: Fast pool SMB carbon per system volume. ORG\_C [g C/cm<sup>3</sup> V]: Stationary carbon per system volume.
- *in*: submodel (see section 4.1.7) sequence  
List of solutes to pass to Python as input.  
 < *chemical handle tag* >  
 – *chemical*: string (see section 4.1.5)  
Parameter  
Name of chemical.  
 – *handle*: string (see section 4.1.5)  
Optional parameter  
How to access it. One of 'C\_primary', 'C\_secondary', 'C\_average', 'M\_primary', 'M\_secondary', or 'M\_total'.  
 – *tag*: string (see section 4.1.5)  
Optional parameter  
Name to use as a Python parameter.
- *texture*: submodel (see section 4.1.7) sequence  
List of texture classes to pass to Python.  
 < *tag size* >  
 – *tag*: string (see section 4.1.5)  
Parameter  
Name of Python parameter.  
 – *size*: number [um]  
Parameter  
Include particles below this size.
- *out*: submodel (see section 4.1.7) sequence  
List of Python as output.  
 < *chemical handle tag* >

- *chemical*: string (see section 4.1.5)  
Parameter  
Name of chemical.
- *handle*: string (see section 4.1.5)  
Optional parameter  
Where to add it. One of 'primary', 'secondary'.
- *tag*: string (see section 4.1.5)  
Optional parameter  
Name to use both for logging and as Python output.

### 70.3 adsorption

Maintain equilibrium between solute and sorbed form.

```
< adsorption (equilibrium equilibrium)
 (solute solute)
 (sorbed sorbed)
 (adsorption_rate adsorption_rate)
 (desorption_rate desorption_rate) >
```

- *equilibrium*: **adsorption** component (see chapter 16)  
Function for calculating equilibrium between solute and sorbed form.
- *solute*: string (see section 4.1.5)  
Parameter  
Name of solute form of chemical.
- *sorbed*: string (see section 4.1.5)  
Parameter  
Name of sorbed form of chemical.
- *adsorption\_rate*: **number** component (see chapter 59)  
Transformation rate from solute to sorbed form.
- *desorption\_rate*: **number** component (see chapter 59)  
Optional component  
Transformation rate from sorbed to solute form. By default, this is identical to 'adsorption\_rate'.

### Log Variables

- *adsorption\_source*: number [g/cm<sup>3</sup>/h] soil cells  
Converted from solute to sorbed form this timestep (may be negative).

### 70.4 bound\_release

Release of chemicals bound to colloids from surface soil.

This follows the generation of colloids on the surface. The colloid generation model should already have calculated the amount of released colloids as either a fraction of the readily available colloids (Jarvis99), or as a fraction of the total amount of soil in the mixing layer (Styczen88, Morgan98). The same fraction of the immobile chemical on the surface is released in the colloid bound form.

This reaction must be listed after the colloid generation reaction in the setup file.

```
< bound_release (bound bound)
 (colloid colloid)
 (immobile immobile) >
```

- *bound*: string (see section 4.1.5)  
Optional parameter  
Chemical bound to colloids. If unspecified, the colloid bound form will not be traced.
- *colloid*: string (see section 4.1.5)  
Optional parameter (default 'colloid')  
Name of colloid whose release we mimic.
- *immobile*: string (see section 4.1.5)  
Parameter  
Immobile (or mixed form) chemical in the soil surface.

### Log Variables

- *release*: number [g/cm<sup>2</sup>/h]  
Release rate of immobile chemical as colloids.

## 70.5 colgen

Shared parameter and log variable for colloid generation models.

```
< colgen (ponddamp ponddamp)
 (colloid colloid) >
```

- *ponddamp*: **ponddamp** component (see chapter 65)  
Model for calculating 'KH'.
- *colloid*: string (see section 4.1.5)  
Parameter  
Colloid to generate.

### Log Variables

- *surface\_release*: number [<**fraction**>]  
Fraction of available soil particles released as colloids this timestep.
- *dds*: number [mm]  
Median raindrop size.
- *KH*: number [<**fraction**>]  
Ponding factor.
- *D*: number [g/cm<sup>2</sup>/h]  
Depletion of detachable particles from top soil.

## 70.6 colgen\_Jarvis99

A 'colgen' model (see 70.5, page 387) build into Daisy.

Colloid generation emulating the MACRO model. See also [Jarvis et al., 1999, Brubaker et al., 1992]

```

< colgen_Jarvis99 (ponddamp ponddamp) ; Default none value.
 (rainergy rainergy) ; Default Brown87 value.
 (Ms Ms)
 (tillage_replenish_all false)
 (Mmax Mmax)
 (Mmax_tillage_factor Mmax_tillage_factor) ; Has default value.
 (kd kd)
 (kr kr)
 (zi zi)
 ;; Shared parameters are described in section 70.5.
 (colloid colloid) >

```

- *rainergy*: **rainergy** component (see chapter 68)  
Component (default 'Brown87')  
Model for calculating energy in rain.
- *Ms*: number [g/g]  
Optional state variable  
Current concentration of detachable particles in top soil. By default, 10% of Mmax.
- *tillage\_replenish\_all*: boolean (see section 4.1.2)  
Parameter (default false)  
Set Ms = Mmax after tillage.
- *Mmax*: number [g/g]  
Optional parameter  
Maximum amount of detachable particles. By default, method 1 of Brubaker et al, 1992, will be used.
- *Mmax\_tillage\_factor*: plf [**d** → <none>]  
Parameter (has default const value 1 [])  
Factor to modify Mmax with as a fuction of days after tillage.
- *kd*: number [g/J]  
Parameter  
Detachment rate coefficient.
- *kr*: number [g/cm<sup>2</sup>/h]  
Parameter  
Replenishment rate coefficient.
- *zi*: number [cm]  
Optional parameter  
Thickness of surface soil layer. By default, the value of 'z\_mixing' from 'Surface' is used.

## Log Variables

- *As*: number [g/cm<sup>2</sup>]  
Current amount of detachable particles in top soil.
- *P*: number [g/cm<sup>2</sup>/h]  
Replenishment of detachable particles to top soil.
- *KE*: number [J/cm<sup>2</sup>/h]  
Kinertic energy avalable for colloid generation.

- $E$ : number [ $\text{J}/\text{cm}^2/\text{mm}$ ]  
Kinetic energy in rain.

## 70.7 colloid-generation

A ‘colgen\_Jarvis99’ model (see 70.6, page 387) defined in ‘chemistry-base.dai’.

Release of colloids in soil surface from heavy rain. Simulated with Jarvis99. Does not take vegetation into account See also [Jarvis et al., 1999, Brubaker et al., 1992]

Used by chemistry colloid reaction (see 24.6, page 146) .

## 70.8 colgen\_Morgan98

A ‘colgen’ model (see 70.5, page 387) build into Daisy.

Colloid generation using kinetic energy, emulating EUROSEM. See also [Morgan et al., 1998]

```
< colgen_Morgan98 (rainergy rainergy) ; Default EUROSEM value.
 (kd kd)
 ;; Shared parameters are described in section 70.5.
 (ponddamp ponddamp)
 (colloid colloid) >
```

- *rainergy*: **rainergy** component (see chapter 68)  
Component (default ‘EUROSEM’)  
Model for calculating energy in rain.
- *kd*: number [ $\text{g}/\text{J}$ ]  
Parameter  
Detachment rate coefficient. The EUROSEM user manual list values between 0.8 and 6.0 [ $\text{g}/\text{J}$ ] for various soils in Table A9.1.

### Log Variables

- $KE$ : number [ $\text{J}/\text{cm}^2/\text{h}$ ]  
Kinertic energy available for colloid generation.
- $E$ : number [ $\text{J}/\text{cm}^2/\text{mm}$ ]  
Kinetic energy in rain.

## 70.9 colgen\_Styczen88

A ‘colgen’ model (see 70.5, page 387) build into Daisy.

Colloid generation using rainfall momentum. See also [Styczen and Høeg-Schmidt, 1988]

```
< colgen_Styczen88 (Ae Ae)
 (MA MA)
 (droplet_diameter droplet_diameter)
 ;; Shared parameters are described in section 70.5.
 (ponddamp ponddamp)
 (colloid colloid) >
```

- $Ae$ : number [ $\text{h}^2/\text{g}/\text{cm}^2$ ]  
Parameter  
Soil resistance factor.

- *MA*: number [**<fraction>**]  
Optional parameter  
Protective cover (mulch factor). By default, use the cover predicted by the litter model.
- *droplet\_diameter*: number [**mm**]  
Parameter  
Size of droplets from vegetation.

### Log Variables

- *DH*: number [**kg<sup>2</sup>/m/s<sup>2</sup>**]  
Squared vegetation droplet momentum.
- *CM*: number [**<fraction>**]  
Vegetation factor.
- *MR*: number [**(N s)<sup>2</sup>/m<sup>2</sup>/s**]  
Squared direct rainfall momentum.

## 70.10 default

Transformation between two soil chemicals.

```
< default (transform transform)
 (A A)
 (B B) >
```

- *transform*: **transform** component (see chapter 91)  
Transformation process between 'A' to 'B'.
- *A*: string (see section 4.1.5)  
Parameter  
Name of first soil component in equilibrium.
- *B*: string (see section 4.1.5)  
Parameter  
Name of second soil component in equilibrium.

### Log Variables

- *S<sub>AB</sub>*: number [**g/cm<sup>3</sup>/h**] soil cells  
Converted from A to B this timestep (may be negative).

## 70.11 denitrification

Denitrification in soil, (conversion of nitrate to atmospheric nitrogen).

In this model, it is made proportional to the CO<sub>2</sub> development, as specified by the parameter *alpha*, with a maximum rate specified by the parameter 'K'. The denitrification is also affected by temperature and water pressure. Additional denitrification from CO<sub>2</sub> produced from fast OM pools can be triggered by setting *alpha\_fast* or *water\_factor\_fast* different. This additional denitrification is limited by *K\_fast*.

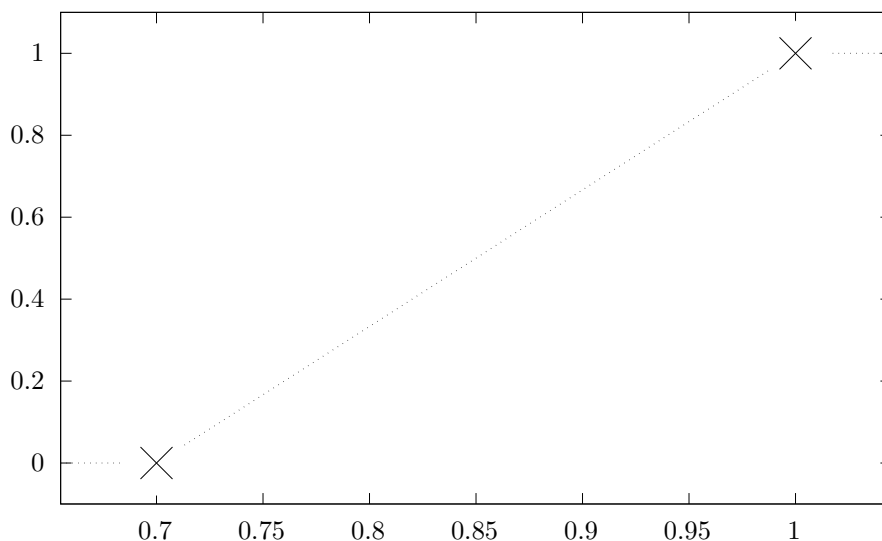
Used by chemistry N reaction (see 24.3, page 145) .

```

< denitrification (denprod denprod) ; Default Parton1996 value.
 (alpha 0.1 [(g NO3-N/h)/(g CO2-C/h)])
 (K 0.020833 [h-1])
 (K_fast K_fast)
 (alpha_fast alpha_fast)
 (heat_factor heat_factor)
 (water_factor water_factor) ; Has default value.
 (water_factor_fast water_factor_fast)
 (redox_height redox_height) >

```

- *denprod*: **denprod** component (see chapter 32)  
Component (default 'Parton1996')  
Denitrification products.
- *alpha*: number [(g NO3-N/h)/(g CO2-C/h)]  
Parameter (default 0.1)  
Anaerobic denitrification constant for slow pools.
- *K*: number [h<sup>-1</sup>]  
Parameter (default 0.020833)  
Maximum fraction of nitrate converted at each time step from slow pools.
- *K\_fast*: number [h<sup>-1</sup>]  
Optional parameter  
Maximum fraction of nitrate converted at each time step from fast pools. By default this is identical to 'K'.
- *alpha\_fast*: number [(g NO3-N/h)/(g CO2-C/h)]  
Optional parameter  
Anaerobic denitrification constant for fast pools. This applies to the CO2 produced from turnover of fast OM pools. By default, this is identical to *alpha*.
- *heat\_factor*: plf [dg C → <none>]  
Optional parameter  
Heat factor. By default, use a build in function valid for temperate climates.
- *water\_factor*: plf [<fraction> → <none>]  
Optional parameter (has default value with 2 points)



```
(water_factor (0.7 [] 0 []) (1 [] 1 []))
```

Parameter description:

Water potential factor for slow pools. This is a function of the current water content as a fraction of the maximal water content.

- *water\_factor\_fast*: plf [**<fraction>** → **<none>**]  
Optional parameter  
Water potential factor for fast pools By default, this is identical to the 'water\_factor' parameter.
- *redox\_height*: number [**cm**]  
Optional parameter  
Height (a negative number) below which redox processes start. All NO<sub>3</sub> below this height will be denitrified immediately. By default no redox denitrification occurs.

## Log Variables

- *converted*: number [**g/cm<sup>3</sup>/h**] sequence  
Amount of denitrification.
- *converted\_fast*: number [**g/cm<sup>3</sup>/h**] soil cells  
Additional denitrification due to turnover in fast pools.
- *converted\_redox*: number [**g/cm<sup>3</sup>/h**] soil cells  
Additional denitrification due to chemical redox processes.
- *potential*: number [**g/cm<sup>3</sup>/h**] soil cells  
Potential amount of denitrification at anarobic conditions.
- *potential\_fast*: number [**g/cm<sup>3</sup>/h**] soil cells  
Additional potential due to turnover in fast pools.

## 70.12 equilibrium

Equilibrium between two soil chemicals.

```
< equilibrium (equilibrium equilibrium)
 (secondary false)
 (surface false)
 (A A)
 (B B)
 (k_AB k_AB)
 (k_BA k_BA)
 (colloid colloid)
 (primary primary) >
```

- *equilibrium*: **equilibrium** component (see chapter 39)  
Function for calculating equilibrium between A and B.
- *secondary*: boolean (see section 4.1.2)  
Parameter (default false)  
Equilibrium should happen in the secondary domain. There will only be a reaction when there is water in the secondary domain (inter-aggregate pores), at both the beginning and end of the timestep. By default, only the content of the primary domain (soil-bound and intra-aggregate pores), will be included



in the reaction. There is no way to use this model to specify an equilibrium reaction in the tertiary domain (biopores).

- *surface*: boolean (see section 4.1.2)  
Parameter (default false)  
Equilibrium should happen in the surface.
- *A*: string (see section 4.1.5)  
Parameter  
Name of first soil component in equilibrium.
- *B*: string (see section 4.1.5)  
Parameter  
Name of second soil component in equilibrium.
- *k<sub>AB</sub>*: **number** component (see chapter 59)  
Transformation rate from soil component 'A' to 'B'.
- *k<sub>BA</sub>*: **number** component (see chapter 59)  
Optional component  
Transformation rate from soil component 'B' to 'A'. By default, this is identical to 'k<sub>AB</sub>'.
- *colloid*: string (see section 4.1.5)  
Optional parameter  
Let 'rho.b' denote content of specified chemical. This might affect the evaluation of the 'k<sub>AB</sub>' and 'k<sub>BA</sub>' parameter expressions, as well as the 'equilibrium' model. By default, 'rho.b' will be the soil dry bulk density.
- *primary*: boolean (see section 4.1.2)  
Optional parameter  
Equilibrium should happen in the primary domain. If true, the content of the primary soil domain (soil-bound and intra-aggregate pores), will be included in the reaction. By default, this will be true if 'secondary' is false, and be false if 'secondary' is true.

## Log Variables

- *S<sub>AB</sub>*: number [g/cm<sup>3</sup>/h] soil cells  
Converted from A to B in soil this timestep (may be negative).
- *surface<sub>AB</sub>*: number [g/cm<sup>2</sup>/h]  
Converted from A to B on surface this timestep (may be negative).

## 70.13 filter\_velocity

Filtration of soil colloids.

```
< filter_velocity (immobile immobile)
 (mobile mobile)
 (fc_primary fc_primary)
 (fc_secondary fc_secondary) >
```

- *immobile*: string (see section 4.1.5)  
Optional parameter  
Immobile colloids in the soil. By default, filtered colloids are not tracked.

- *mobile*: string (see section 4.1.5)  
Parameter  
Mobile colloids dissolved in soil water.
- *fc\_primary*: number [ $\text{cm}^{-1}$ ]  
Parameter  
Filter coefficient in the primary domain
- *fc\_secondary*: number [ $\text{cm}^{-1}$ ]  
Parameter  
Filter coefficient in secondary domain

### Log Variables

- *F\_primary*: number [ $\text{g}/\text{cm}^3/\text{h}$ ] soil cells  
Filtration in the primary domain (intra-aggregate pores).
- *F\_secondary*: number [ $\text{g}/\text{cm}^3/\text{h}$ ] soil cells  
Filtration in secondary domain (inter-aggregate pores).

## 70.14 colloid-filter

A ‘filter\_velocity’ model (see 70.13, page 393) defined in ‘chemistry-base.dai’.  
Filtration of colloids in the soil matrix.  
Used by chemistry colloid reaction (see 24.6, page 146) .

## 70.15 nitrification

Nitrification. The actual nitrification specification is part of the horizon models, this reaction just applies the models and logs the result.

Used by chemistry N reaction (see 24.3, page 145) .

```
< nitrification >
```

### Log Variables

- *NO3*: number [ $\text{g}/\text{cm}^3/\text{h}$ ] soil cells  
Amount of nitrate generated this hour.
- *NH4*: number [ $\text{g}/\text{cm}^3/\text{h}$ ] soil cells  
Amount of ammonium consumed this hour.
- *N2O*: number [ $\text{g}/\text{cm}^3/\text{h}$ ] soil cells  
Amount of nitrous oxide generated this hour.

## 70.16 shoot

Pool based on shoot DM. Generate chemical (M) on canopy up to maximum (M\_max). M\_max depends on the shoot DM and DS.  $\text{dM}/\text{dt} = \text{fill\_rate} * (\text{M\_max} - \text{M})$ .

```
< shoot (chemical chemical)
 (crop all)
 (fill fill)
 (M_max M_max)
 (DS_factor DS_factor) ; Has default value. >
```

- *chemical*: string (see section 4.1.5)  
Parameter  
Chemical to generate.
- *crop*: string (see section 4.1.5)  
Parameter (default 'all')  
Crop whose shoot is used as basis. Special value 'all' means all crops on the field.
- *fill*: **rate** component (see chapter 69)  
Pool fill rate.
- *M\_max*: plf [ $\text{g}/\text{cm}^2 \rightarrow \text{g}/\text{cm}^2$ ]  
Parameter  
Maximum amount of chemical per shoot DM.
- *DS\_factor*: plf [**DS**  $\rightarrow$  <none>]  
Parameter (has default const value 1 [])  
Influence of DS on M\_max.

## 70.17 shoot2

Pool based on shoot DM. Generate chemical (M) on canopy up to maximum (M\_max). M\_max depends on the shoot DM and DS.

$\text{dM}/\text{dt} = \text{growth\_rate} + \text{fill\_rate} * (\text{M\_max} - \text{M})$ .

$\text{growth\_rate} = \max(0.0, (\text{M\_max}(t_1) - \text{M\_max}(t_0)) / (t_1 - t_0))$  See also [Jorgensen et al., 2020, García-Jorgensen et al., 2024]

```
< shoot2 (chemical chemical)
 (crop all)
 (fill fill)
 (M_max M_max)
 (DS_factor DS_factor) ; Has default value.
 (M_pot 0 [g/cm^2]) >
```

- *chemical*: string (see section 4.1.5)  
Parameter  
Chemical to generate.
- *crop*: string (see section 4.1.5)  
Parameter (default 'all')  
Crop whose shoot is used as basis. Special value 'all' means all crops on the field.
- *fill*: **rate** component (see chapter 69)  
Pool fill rate.
- *M\_max*: plf [ $\text{g}/\text{cm}^2 \rightarrow \text{g}/\text{cm}^2$ ]  
Parameter  
Maximum amount of chemical per shoot DM.
- *DS\_factor*: plf [**DS**  $\rightarrow$  <none>]  
Parameter (has default const value 1 [])  
Influence of DS on M\_max.
- *M\_pot*: number [ $\text{g}/\text{cm}^2$ ]  
State variable (default 0)  
Potential chemical content.

## 70.18 sorption

Kinetic linear sorption equilibrium. Faster than the 'equilibrium' reaction model, more flexible than the 'adsorption' reaction model.

```
< sorption (solute solute)
 (K_clay K_clay)
 (K_OC K_OC)
 (K_d K_d)
 (sorbed sorbed)
 (colloid colloid)
 (k_sorption k_sorption)
 (k_desorption k_desorption)
 (soil_enrichment_factor 1 []) >
```

- *solute*: string (see section 4.1.5)  
Parameter  
Name of solute form of chemical. If this chemical has equilibrium sorption, only the colute part is used.
- *K\_clay*: number [ $\text{cm}^3/\text{g}$ ]  
Optional parameter  
Clay dependent distribution parameter. It is multiplied with the soil clay fraction to get the clay part of the 'K\_d' factor. If 'K\_OC' is specified, 'K\_clay' defaults to 0.
- *K\_OC*: number [ $\text{cm}^3/\text{g}$ ]  
Optional parameter  
Humus dependent distribution parameter. It is multiplied with the soil organic carbon fraction to get the carbon part of the 'K\_d' factor. By default, 'K\_OC' is equal to 'K\_clay'.
- *K\_d*: number [ $\text{cm}^3/\text{g}$ ]  
Optional parameter  
Equillibrium parameter:  $M = C (K_d \rho_b + \Theta)$  Here M is the total amount, C is solute concentration, K\_d is this parameter,  $\rho_b$  is the dry bulk density, and  $\Theta$  is the volumetric water content. By default, K\_d is calculated from K\_clay and K\_OC.
- *sorbed*: string (see section 4.1.5)  
Parameter  
Name of sorbed form of chemical.
- *colloid*: string (see section 4.1.5)  
Optional parameter  
Sorp to this chemical instead of to the soil matrix.
- *k\_sorption*: number [ $\text{h}^{-1}$ ]  
Parameter  
Sorption rate.
- *k\_desorption*: number [ $\text{h}^{-1}$ ]  
Optional parameter  
Desorption rate. By default, this is identical to 'k.sorption'.
- *soil\_enrichment\_factor*: number (dimensionless)  
Parameter (default 1)  
Multiply K\_d with this number if 'colloid' is set. This represents how much more accesible colloids is compared to the soil matrix.

**Log Variables**

- *surface\_sorption*: number  $[\mathbf{g}/\mathbf{cm}^2/\mathbf{h}]$   
Sorption on surface this timestep (may be negative).
- *S\_sorption*: number  $[\mathbf{g}/\mathbf{cm}^3/\mathbf{h}]$  soil cells  
Sorption in soil this timestep (may be negative).
- *S\_sorption\_primary*: number  $[\mathbf{g}/\mathbf{cm}^3/\mathbf{h}]$  soil cells  
Sorption in primary domain this timestep (may be negative).
- *S\_sorption\_secondary*: number  $[\mathbf{g}/\mathbf{cm}^3/\mathbf{h}]$  soil cells  
Sorption in secondary domain this timestep (may be negative).



# Chapter 71

## retention

Specify a retention or a halftime.

### 71.1 PASTIS

Retention curve of Mulch used in PASTIS.  $h(\text{Theta}) = -(-h_{\text{min}})^{(1 - ((\text{Theta} - \text{Theta}_{\text{res}}) / (2/3 \text{Theta}_{\text{sat}} - \text{Theta}_{\text{res}})))}$

Used by litter mulch retention (see 51.6, page 276) .

### 71.2 exp

Retention curve of Mulch used in Exponential.  $\text{Theta}(h) = \exp(k h) (\text{Theta}_{\text{sat}} - \text{Theta}_{\text{res}}) + \text{Theta}_{\text{res}}$   $h(\text{Theta}) = \ln((\text{Theta} - \text{Theta}_{\text{res}}) / (\text{Theta}_{\text{sat}} - \text{Theta}_{\text{res}})) / k$

```
< exp (Theta_sat Theta_sat)
 (k k)
 (Theta_res Theta_res)
 (h_min h_min) >
```

- *Theta\_sat*: number [**<fraction>**]  
Optional parameter  
Residual water.
- *k*: number [**cm<sup>-1</sup>**]  
Parameter  
 $\text{Theta}(h) = \exp(k h) (\text{Theta}_{\text{sat}} - \text{Theta}_{\text{res}}) + \text{Theta}_{\text{res}}$
- *Theta\_res*: number [**<fraction>**]  
Optional parameter  
Residual water.
- *h\_min*: number [**cm**]  
Optional parameter  
Pressure below residual water.





# Chapter 72

## root

Root processes.

```
< component (rootdens rootdens)
(Rad 0.005 [cm])
(ABAprod ABAprod) ; Default none value.
(NH4_uptake NH4_uptake)
(NO3_uptake NO3_uptake) ; Default fixed_sink value.
(DptEmr 10 [cm])
(PenPar1 0.25 [cm/dg C/d])
(PenPar2 4 [dg C])
(PenpFFac PenpFFac) ; Has default value.
(PenClayFac PenClayFac) ; Has default value.
(PenWaterFac PenWaterFac) ; Has default value.
(PenDSFac PenDSFac) ; Has default value.
(DensityDSFac DensityDSFac) ; Has default value.
(MaxPen 100 [cm])
(MaxWidth MaxWidth)
(h_wp -15000 [cm])
(MxNH4Up 2.5e-07 [g/cm/h])
(MxNO3Up 2.5e-07 [g/cm/h])
(Rxylem 10 [])
(PotRtDpt PotRtDpt)
(Depth Depth)
(Density Density ...)
(ABAConc 0 [g/cm3])
(h_x 0 [cm])
(partial_soil_temperature 0 [dg C h])
(partial_day 0 [h])
(soil_temperature 0 [dg C])
(water_stress_days 0 [d]) >
```

- *rootdens*: **rootdens** component (see chapter 73)  
Optional component  
Root density model.
- *Rad*: number [**cm**]  
Parameter (default 0.005)  
Root radius.
- *ABAprod*: **ABAprediction** component (see chapter 7)  
Component (default ‘none’)

ABA production model.

- *NH<sub>4</sub>\_uptake*: **solute\_uptake** component (see chapter 81)  
Optional component  
NH<sub>4</sub> uptake model. By default, this will be identical to the NO<sub>3</sub> uptake model.
- *NO<sub>3</sub>\_uptake*: **solute\_uptake** component (see chapter 81)  
Component (default 'fixed\_sink')  
NO<sub>3</sub> uptake model.
- *DptEmr*: number [**cm**]  
Parameter (default 10)  
Penetration at emergence.
- *PenPar1*: number [**cm/dg C/d**]  
Parameter (default 0.25)  
Penetration rate parameter, coefficient.
- *PenPar2*: number [**dg C**]  
Parameter (default 4)  
Penetration rate parameter, threshold.
- *PenpFFac*: plf [**pF** → <none>]  
Parameter (has default const value 1 [])  
Moisture dependent factor to multiply 'PenPar1' with. If pressure is less than -1 cm, pF will be assumed to be 0.
- *PenClayFac*: plf [<**fraction**> → <none>]  
Parameter (has default const value 1 [])  
Clay dependent factor to multiply 'PenPar1' with.
- *PenWaterFac*: plf [<**fraction**> → <none>]  
Parameter (has default const value 1 [])  
Water dependent factor to multiply 'PenPar1' with. The factor is a function of relative water content (Theta/Theta\_sat).
- *PenDSFac*: plf [**DS** → <none>]  
Parameter (has default const value 1 [])  
Development stage dependent factor to multiply 'PenPar1' with.
- *DensityDSFac*: plf [**DS** → <none>]  
Parameter (has default const value 1 [])  
DS based factor for effective root density.
- *MaxPen*: number [**cm**]  
Parameter (default 100)  
Maximum penetration depth.
- *MaxWidth*: number [**cm**]  
Optional parameter  
Maximum horizontal distance of roots from plant.
- *h\_wp*: number [**cm**]  
Parameter (default -15000)  
Matrix potential at wilting point.

- *MxNH4Up*: number [g/cm/h]  
Parameter (default 2.5e-07)  
Maximum NH4 uptake per unit root length.
- *MxNO3Up*: number [g/cm/h]  
Parameter (default 2.5e-07)  
Maximum NO3 uptake per unit root length.
- *Rxylem*: number (dimensionless)  
Parameter (default 10)  
Transport resistance in xyleme.
- *PotRtDpt*: number [cm]  
Optional state variable  
Potential root penetration depth.
- *Depth*: number [cm]  
Optional state variable  
Rooting Depth.
- *Density*: number [cm/cm<sup>3</sup>] soil cells  
Optional state variable  
Root density in soil layers.
- *ABAConc*: number [g/cm<sup>3</sup>]  
State variable (default 0)  
ABA concentration in water uptake.
- *h\_x*: number [cm]  
State variable (default 0)  
Root extraction at surface.
- *partial\_soil\_temperature*: number [dg C h]  
State variable (default 0)  
Soil temperature hours this day, so far.
- *partial\_day*: number [h]  
State variable (default 0)  
Hours we have accumulated soil temperature this day.
- *soil\_temperature*: number [dg C]  
State variable (default 0)  
Average soil temperature yesterday.
- *water\_stress\_days*: number [d]  
State variable (default 0)  
Number of days production has halted due to water stress. This is the sum of water stress for each hour, multiplied with the fraction of the radiation of that day that was received that hour.

## Log Variables

- *EffectiveDensity*: number [cm/cm<sup>3</sup>] soil cells  
Effective root density in soil layers. This takes heterogeneous distribution of roots in soil into account.
- *H2OExtraction*: number [cm<sup>3</sup>/cm<sup>3</sup>/h] soil cells  
Extraction of H2O in soil layers.

- *NH<sub>4</sub>Extraction*: number [**g N/cm<sup>3</sup>/h**] soil cells  
Extraction of NH<sub>4</sub>-N in soil layers.
- *NO<sub>3</sub>Extraction*: number [**g N/cm<sup>3</sup>/h**] soil cells  
Extraction of NO<sub>3</sub>-N in soil layers.
- *ABAExtraction*: number [**g/cm<sup>3</sup>/h**] soil cells  
Extraction of ABA in soil layers.
- *water\_stress*: number (dimensionless)  
Fraction of requested water we didn't get.
- *production\_stress*: number (dimensionless)  
SVAT induced stress, or -1 if not applicable.
- *Ept*: number [**mm/h**]  
Potential transpiration.
- *H<sub>2</sub>O Upt*: number [**mm/h**]  
H<sub>2</sub>O uptake.
- *NH<sub>4</sub>Upt*: number [**g N/m<sup>2</sup>/h**]  
NH<sub>4</sub>-N uptake.
- *NO<sub>3</sub>Upt*: number [**g N/m<sup>2</sup>/h**]  
NO<sub>3</sub>-N uptake.

## 72.1 classic

Classic root system model.

Used by crop default Root (see 29.1, page 171) , crop simple Root (see 29.47, page 178) , crop Maize Root (see 29.12, page 174) , crop Pioneer Maize Root (see 29.19, page 175) , crop Pea Root (see 29.17, page 174) , crop Potato; Koege Root (see 29.20, page 175) , crop Potato; SCRI Root (see 29.21, page 175) , crop Potato; FertOrgaNic Root (see 29.22, page 175) , crop Rye Root (see 29.29, page 176) , crop Spring Barley Root (see 29.31, page 176) , crop Spring Barley; Foulum Root (see 29.32, page 176) , crop Sugar Beet Root (see 29.35, page 176) , crop Fodder Beet Root (see 29.9, page 174) , crop Spring Rape Root (see 29.33, page 176) , crop Spring Wheat Root (see 29.34, page 176) , crop Winter Barley Root (see 29.40, page 177) , crop Winter Barley; Koge Root (see 29.42, page 177) , crop Winter Rape Root (see 29.43, page 177) , crop Winter Wheat Root (see 29.44, page 177) , crop Winter Wheat JG Root (see 29.45, page 177) , crop Grass to grain Root (see 29.10, page 174) , crop Ryegrass Root (see 29.30, page 176) , crop Wclover Root (see 29.36, page 176) , crop Beetroot Root (see 29.2, page 173) , crop Broccoli Root (see 29.3, page 173) , crop Broccoli - transplanted Root (see 29.4, page 173) , crop Brussels sprouts Root (see 29.5, page 173) , crop Brussels sprouts - transplanted Root (see 29.6, page 173) , crop Celeriac Root (see 29.7, page 174) , crop Celeriac - transplanted Root (see 29.8, page 174) , crop Potato; SCRI - AArhus Root (see 29.27, page 176) , crop White cabbage Root (see 29.37, page 177) , crop White cabbage - transplanted Root (see 29.38, page 177) , crop Onion Root (see 29.15, page 174) , vegetation permanent Root (see 100.3, page 545) , vegetation afforestation Root (see 100.1, page 542) , and vegetation Grass Root (see 100.6, page 547) .

# Chapter 73

## rootdens

Root density calculations.

Used by root component rootdens (see 72, page 401) .

```
< component (SpRtLength 100 [m/g]) >
```

- *SpRtLength*: number [m/g]  
Parameter (default 100)  
Specific root length

### 73.1 Anders Pedersen

Use exponential function for root density. In this variant of Gerwitz and Page, 'a' is specified as a function of development stage. See also [Gerwitz and Page, 1974]

```
< "Anders Pedersen" (a_DS a_DS)
 (q q)
 ;; Shared parameters are described in chapter 73.
 (SpRtLength 100 [m/g]) >
```

- *a\_DS*: plf [DS  $\rightarrow$  cm<sup>-1</sup>]  
Parameter  
Form parameter as a function of development stage.
- *q*: number [cm]  
Parameter  
Extra root length below max rooting depth. Root density will decrease linearly from the GP calculated amount at max rooting depth to zero 'q' further down.

### Log Variables

- *a*: number [cm<sup>-1</sup>]  
Form parameter. Calculated from 'a\_DS'.
- *L0*: number [cm/cm<sup>3</sup>]  
Root density at soil surface.

### 73.2 DS\_Depth

Specify root density as a function of development stage.

```
< DS_Depth (entries entries ...)
 ;; Shared parameters are described in chapter 73.
 (SpRtLength 100 [m/g]) >
```

- *entries*: submodel (see section 4.1.7) sequence  
A list of pairs, where the first element of each pair is a development stage (usually a number between 0 (emergence) and 2 (ripe), and the second element is a PLF specifying the relative root density as a function of soil depth in cm (a positive number).

To find the absolute root density, Daisy will interpolate the relative root density distribution specified for the entries before and after the current development stage, and scale them to match the current total root mass.

```
< index density >
```

- *index*: number (dimension not specified)  
Parameter  
Index for specifying root density.
- *density*: plf [<unknown> → <none>]  
Parameter  
Relative root density as a function of root depth .

### 73.3 DS\_Rel

Specify root density as a function of development stage.

```
< DS_Rel (entries entries ...)
;; Shared parameters are described in chapter 73.
(SpRtLength 100 [m/g]) >
```

- *entries*: submodel (see section 4.1.7) sequence  
A list of pairs, where the first element of each pair is a development stage (usually a number between 0 (emergence) and 2 (ripe), and the second element is a PLF specifying the relative root density as a function of soil depth relative to the total root depth.

To find the absolute root density, Daisy will interpolate the relative root density distribution specified for the entries before and after the current development stage, and scale them to match the current total root mass.

```
< index density >
```

- *index*: number (dimension not specified)  
Parameter  
Index for specifying root density.
- *density*: plf [<unknown> → <none>]  
Parameter  
Relative root density as a function of root depth .

### 73.4 Depth\_Depth

Specify root density as a function of development stage.

```
< Depth_Depth (entries entries ...)
;; Shared parameters are described in chapter 73.
(SpRtLength 100 [m/g]) >
```

- *entries*: submodel (see section 4.1.7) sequence  
A list of pairs, where the first element of each pair is the root depth, (a positive number), and the second element is a PLF specifying the relative root density as a function of soil depth in cm (a positive number).

To find the absolute root density, Daisy will interpolate the relative root density distribution specified for the entries before and after the current development stage, and scale them to match the current total root mass.

- ```
< index density >
```
- *index*: number (dimension not specified)
Parameter
Index for specifying root density.
 - *density*: plf [<unknown> → <none>]
Parameter
Relative root density as a function of root depth .

73.5 GP1D

Use exponential function for root density. See also [Gerwitz and Page, 1974]

```
< GP1D (DensRtTip 0.1 [cm/cm3])
      (DensIgnore DensIgnore)
      ;; Shared parameters are described in chapter 73.
      (SpRtLength 100 [m/g]) >
```

- *DensRtTip*: number [cm/cm³]
Parameter (default 0.1)
Root density at (potential) penetration depth.
- *DensIgnore*: number [cm/cm³]
Optional parameter
Ignore cells with less than this root density. By default, this is the same as DensRtTip.

Log Variables

- *a*: number [cm⁻¹]
Form parameter. Calculated from 'DensRtTip'.
- *k*: number (dimensionless)
Scale factor due to soil limit.

Some roots might be below the soil imposed maximum root depth, or in areas with a density lower than the limit specified by DensIgnore. These roots will be re distributed within the root zone by multiplying the density with this scale factor.

- *L0*: number [cm/cm³]
Root density at soil surface.

73.6 GP2D

Use exponential function for root density in row crops.

This is a two dimension model (z, x), where the z-axis is vertical, and the x-axis is horizontal and ortogonal to the row. The row is assumed to be uniform (dense), allowing us to ignore that dimension.

We assume the root density decrease with horizontal distance to row, as well as depth below row. See also [Gerwitz and Page, 1974]

```

< GP2D (DensRtTip 0.1 [cm/cm3])
      (DensIgnore DensIgnore)
      (debug 0)
      (row_position 0 [cm])
      (row_distance row_distance)
      ;; Shared parameters are described in chapter 73.
      (SpRtLength 100 [m/g]) >

```

- *DensRtTip*: number [**cm/cm³**]
Parameter (default 0.1)
Root density at (potential) penetration depth.
- *DensIgnore*: number [**cm/cm³**]
Optional parameter
Ignore cells with less than this root density. By default, this is the same as *DensRtTip*.
- *debug*: integer
Parameter (default 0)
Add debug messages if larger than 0.
- *row_position*: number [**cm**]
State variable (default 0)
Horizontal position of row crops.
- *row_distance*: number [**cm**]
State variable
Distance between rows of crops.

Log Variables

- *k*: number (dimensionless)
Scale factor due to soil limit.

Some roots might be below the soil imposed maximum root depth, or in areas with a density lower than the limit specified by 'DensIgnore'. These roots will be re distributed within the root zone by multiplying the density with this scale factor.

- *a_z*: number [**cm⁻¹**]
Form parameter. Calculated from 'DensRtTip'.
- *a_x*: number [**cm⁻¹**]
Form parameter. Calculated from 'DensRtTip'.
- *L00*: number [**cm/cm³**]
Root density at row crop at soil surface.

73.7 Gerwitz+Page74

Use exponential function for root density. See also [Gerwitz and Page, 1974]

```

< "Gerwitz+Page74" (DensRtTip 0.1 [cm/cm3])
                  (MinDens 0 [cm/cm3])
                  ;; Shared parameters are described in chapter 73.
                  (SpRtLength 100 [m/g]) >

```


- *DensRtTip*: number [**cm/cm³**]
Parameter (default 0.1)
Root density at (potential) penetration depth.
- *MinDens*: number [**cm/cm³**]
Parameter (default 0)
Minimal root density Root density will never be below this, as long as there is enough root mass. Extra root mass will be distributed according to Gerwitz and Page. If there are too little root mass, the root will have the same density all the way down.

Log Variables

- *a*: number [**cm⁻¹**]
Form parameter. Calculated from 'DensRtTip'.
- *L0*: number [**cm/cm³**]
Root density at soil surface.

73.8 growth

Dynamic root growth model.

```
< growth (Depth Depth)
  (DensRtTip 0.1 [cm/cm3])
  (row_position 0 [cm])
  (row_distance row_distance)
  (depth_factor depth_factor) ; Has default value.
  (Width Width)
  (WRoot WRoot)
  ;; Shared parameters are described in chapter 73.
  (SpRtLength 100 [m/g]) >
```

- *Depth*: number [**cm**]
Optional state variable
Expected depth of root zone (positive).
- *DensRtTip*: number [**cm/cm³**]
Parameter (default 0.1)
Root density at (potential) penetration depth.
- *row_position*: number [**cm**]
State variable (default 0)
Horizontal position of row crops.
- *row_distance*: number [**cm**]
Optional state variable
Distance between rows of crops.
- *depth_factor*: plf [**cm** → <none>]
Parameter (has default const value 1 [])
Depth (negative) affect on root growth. Specify a value less than one to decrease root growth, and larger than one to increase root growth at the specified depth.
- *Width*: number [**cm**]
Optional state variable
Expected width of root zone.

- *WRoot*: number [g DM/m²]
Optional state variable
Total root mass.

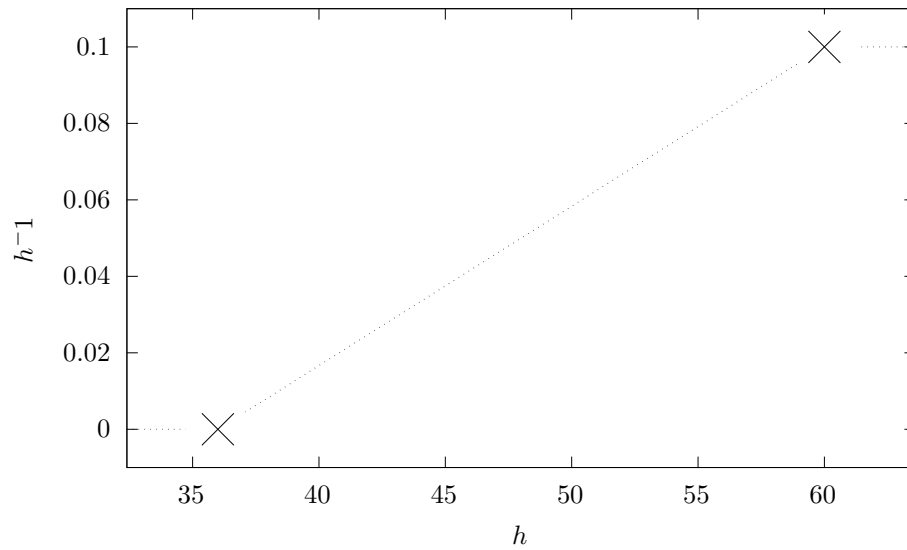
73.9 local

Dynamic root growth model.

```
< local (Depth Depth)
  (DensRtTip 0.1 [cm/cm3])
  (row_position 0 [cm])
  (row_distance row_distance)
  (depth_factor depth_factor) ; Has default value.
  (Width Width)
  (neighbor_effect 0 [cm/d])
  (max_internal_growth_rate 0.1 [d-1])
  (h_threshold 0 [cm])
  (death_rate death_rate) ; Has default value.
  (death_T_factor death_T_factor) ; Default T_min_15 value.
  (emerging true)
  (flooded flooded ...)
  ;; Shared parameters are described in chapter 73.
  (SpRtLength 100 [m/g]) >
```

- *Depth*: number [cm]
Optional state variable
Expected depth of root zone (positive).
- *DensRtTip*: number [cm/cm³]
Parameter (default 0.1)
Root density at (potential) penetration depth.
- *row_position*: number [cm]
Parameter (default 0)
Horizontal position of row crops.
- *row_distance*: number [cm]
Optional parameter
Distance between rows of crops.
- *depth_factor*: plf [cm → <none>]
Parameter (has default const value 1 [])
Depth (negative) affect on root growth. Specify a value less than one to decrease root growth, and larger than one to increase root growth at the specified depth.
- *Width*: number [cm]
Optional state variable
Expected width of root zone.
- *neighbor_effect*: number [cm/d]
Parameter (default 0)
How fast root density in neighbor cells affect growth. After expansion, excess roots are distributed proportional to existing root density in each cell, plus a part of the neighbor cells.

- *max_internal_growth_rate*: number [$\mathbf{d}^{-1}$]
Parameter (default 0.1)
Root density within root zone cannot grow faster than this.
- *h_threshold*: number [$\mathbf{cm}$]
Parameter (default 0)
Root starts dying if pressure is above this threshold.
- *death_rate*: plf [$\mathbf{h} \rightarrow \mathbf{h}^{-1}$]
Parameter (has default value with 2 points)



```
(death_rate (36 [h] 0 [h-1]) (60 [h] 0.1 [h-1]))
```

Parameter description:

Death rate as function of temperature modified time flooded.

- *death_T_factor*: function [$\mathbf{dg} \mathbf{C} \rightarrow \langle \mathbf{none} \rangle$] (see chapter 42)
Parameter (default 'T_min_15')
Temperature effect on flooding.
- *emerging*: boolean (see section 4.1.2)
State variable (default true)
True until root mass is large enough to sustain root length in RZ. Until then, L within RZ is set to DensRtTip.
- *flooded*: number [$\mathbf{h}$] soil cells
Optional state variable
Time flooded, adjusted for temperature. The temperature adjustment is the same used for mineralization.

Log Variables

- *D*: number [$\mathbf{cm/cm^3/d}$] soil cells
Death.
- *E*: number [$\mathbf{cm/cm^3/d}$] soil cells
Expansion

- *expansion_volume*: number [**cm**³]
Expansion volume.
- *E_tot*: number [**cm/d**]
Total expansion root length.
- *A_tot*: number [**cm**²]
Total expansion area.
- *I*: number [**cm/cm**³/**d**] soil cells
Internal adjustment.
- *I_tot*: number [**cm/d**]
Total internal adjustment root length.
- *D_tot*: number [**cm/d**]
Total death root length.

Chapter 74

rubiscoN

Find total amount of Rubisco-N in leaves.

74.1 default

Standard estimation of RuBisCo N.

Used by crop default RubiscoN (see 29.1, page 171) .

```
< default (fraction 0.75 [<fraction>])  
          (offset 0 [g N/m2])  
          (subtract_Nf false)  
          (subtract_Pt false) >
```

- *fraction*: number [<**fraction**>]
Parameter (default 0.75)
Value description: Equation 6 See also [Boegh et al., 2002]
Fraction of remaining N in leaves that RuBisCo-N. First structural and luxury N may be subtracted, as per the other parameters.
- *offset*: number [**g N/m²**]
Parameter (default 0)
Subtract this amount of N per LAI.
- *subtract_Nf*: boolean (see section 4.1.2)
Parameter (default false)
Subtract N corresponding to the non functional concentration in leaves.
- *subtract_Pt*: boolean (see section 4.1.2)
Parameter (default false)
Subtract N above the critical concentration in leaves.

Chapter 75

rubiscoNdist

The 'rubiscoNdist' component calculates the rubisco N distribution for photosynthesis in the canopy.

75.1 exp

Boegh et al.(2002) rubisco N-distribution model in the canopy for photosynthesis.

Used by photosynthesis Farquhar N-dist (see 64.1, page 345) .

```
< exp (kn 0.713 []) >
```

- *kn*: number (dimensionless)
Parameter (default 0.713)
Extinction coefficient of nitrogen in the canopy, $kn = 0.713$ (De Pury & Farquhar, 1997)

75.2 expr

expr rubisco N-distribution model in the canopy.

```
< expr value  
      (obsolete obsolete)  
      (f_photo 1 []) >
```

- *value*: **number** component (see chapter 59)
Expression that evaluates to the relative rubisco N intensity where 1 is the value in top of the canopy.
- *obsolete*: string (see section 4.1.5)
Optional parameter
Experimental, avoid
- *f_photo*: number (dimensionless)
Parameter (default 1)
Fraction of photosynthetically active N in canopy. According to (Boegh et al., 2002) $f_photo = 0.75$. However, non-functional N is already subtracted from leaf-N in the cropN_std module, therefore $f_photo = 1.0$ as default.

75.3 forced

Forced rubisco capacity distribution model in the canopy.

```
< forced value  
      (obsolete obsolete) >
```

- *value*: **number** component (see chapter 59)
Expression that evaluates to the relative rubisco capacity where 1 is the value in top of the canopy.
- *obsolete*: string (see section 4.1.5)
Optional parameter
Experimental, avoid

75.4 uniform

Uniform rubisco N-distribution model in the canopy for photosynthesis.

< **uniform** (**f_photo** 1 []) >

- *f_photo*: number (dimensionless)
Parameter (default 1)
Fraction of photosynthetically active N in canopy, $f_{\text{photo}} = 0.75$ (Boegh et al., 2002). However, non-functional N is already subtracted from leaf-N in the `cropN_std` module, therefore $f_{\text{photo}} = 1.0$ as default.

Chapter 76

scope

A scope maps names to values.

76.1 exchange

Exchange values with an external model.

< **exchange** (entries *entries* ...) >

- *entries*: **exchange** component (see chapter 40) sequence
List of items to exchange.

Chapter 77

scopesel

A method to choose a scope in a Daisy simulation.

77.1 multi

Combine multiple named scopes.

The entries in the combined scope will have the form `<scope>.<key>`, where `<scope>` is the name of the scope containing the entry, and `<key>` is the name of the entry in `<scope>`.

```
< multi  name... >
```

- *name*: string (see section 4.1.5) sequence
Parameter
Names of scope to select.

77.2 name

Select named scope.

```
< name  name >
```

- *name*: string (see section 4.1.5)
Parameter
Name of scope to select.

77.3 null

Select the empty scope.

Used by action `extern_fertigation` scope (see 15.9, page 78) , program Daisy scope (see 66.2, page 353) , and column default scope (see 26.1, page 153) .

Chapter 78

secondary

Specify secondary domain.

The secondary domain consist typically of soil fractures or other inter-aggregate pores small enough to be dominated by capillarity, yet so large that water moves fast enough that the solute equilibrium with the primary domain (typically intra-aggregate pores) can not be maintained.

This allows a pulse of water to be move through saturated or near saturated soil without solutes in the new water being mixed with solutes in the old water. The effects are twofold: It allows solutes applied to the surface to reach deeper soil layers much faster than it would otherwise, and it protects solutes in the soil matrix from being washed out with fast moving new water.

Used by horizon component `secondary_domain` (see 47, page 231) .

78.1 `alpha`

Shared base class for non-empty secondary domains.

```
< alpha (alpha alpha) >
```

- *alpha*: number [h^{-1}]
Parameter
Exchange rate between primary and secondary water.

78.2 `cracks`

A ‘alpha’ model (see 78.1, page 421) build into Daisy.

Secondary domain specified by aperture and density of soil cracks.

```
< cracks (density density)  
      (use_secondary true)  
      (aperture aperture)  
      ;; Shared parameters are described in section 78.1.  
      (alpha alpha) >
```

- *density*: number [m^{-1}]
Parameter
Density of cracks.
- *use_secondary*: boolean (see section 4.1.2)
Parameter (default `true`)
Divide soil matrix into two domains for solute transport. Set this to `false` to make cracks affect only the conductivity curve.

- *aperture*: number [m]
Parameter
Average distance between walls in cracks.

78.3 pressure

A ‘alpha’ model (see 78.1, page 421) build into Daisy.

Horizon has secondary domain specified by pressure threshold.

The secondary domain consist of water in matrix pores larger than what corresponds to the specified pressure.

```
< pressure    (K 0 [cm/h])
              (h_lim h_lim)
              (use_secondary true)
              ;; Shared parameters are described in section 78.1.
              (alpha alpha) >
```

- *K*: number [cm/h]
Parameter (default 0)
Water conductivity when secondary domain is active. If the secondary domain is already included in the normal conductivity curve, specify 0.0 to use that value instead.
- *h_lim*: number [cm]
Parameter
Minimal pressure needed for activating secondary domain.
- *use_secondary*: boolean (see section 4.1.2)
Parameter (default true)
Divide soil matrix into two domains for solute transport. Set this to false to make h_lim affect only the conductivity curve.

78.4 none

No secondary domain.

There is always full equilibrium between solute in different size matrix pores.

Used by horizon component secondary_domain (see 47, page 231) .

Chapter 79

seed

Initial growth after emergence.

The initial growth process governs the growth of the crop until the point where there is enough leaf area for photosynthesis to take over.

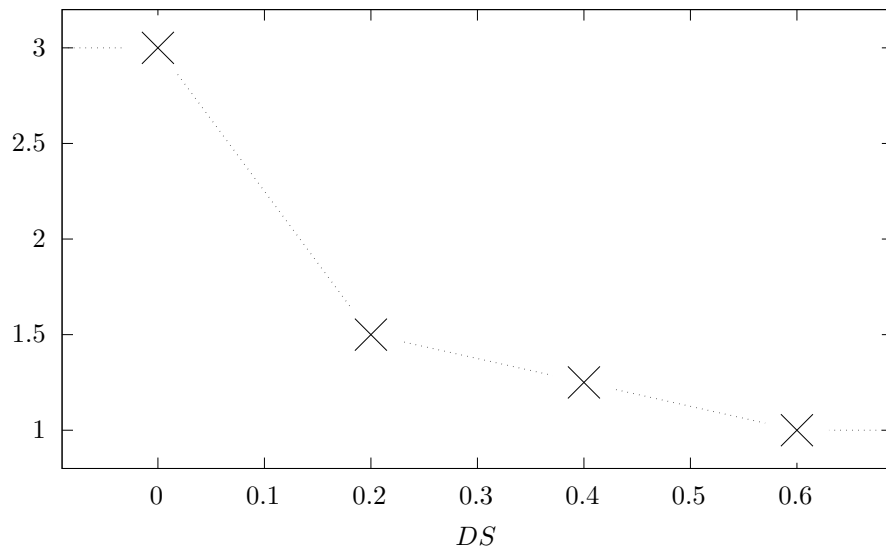
79.1 LAI

Initial crop growth is governed by a forced LAI function.

Used by crop default Seed (see 29.1, page 171) , crop Maize Seed (see 29.12, page 174) , crop Pioneer Maize Seed (see 29.19, page 175) , crop Ikuwala Maize Seed (see 29.13, page 174) , crop Potato; Koege Seed (see 29.20, page 175) , crop Potato; FertOrgaNic Seed (see 29.22, page 175) , crop Potato; Agria Seed (see 29.23, page 175) , crop Potato; Folva Seed (see 29.24, page 175) , crop Potato; Triada Seed (see 29.26, page 175) , crop Sugar Beet Seed (see 29.35, page 176) , crop Fodder Beet Seed (see 29.9, page 174) , crop Winter Rape Seed (see 29.43, page 177) , crop Grass to grain Seed (see 29.10, page 174) , crop Ryegrass Seed (see 29.30, page 176) , crop Wclover Seed (see 29.36, page 176) , crop Beetroot Seed (see 29.2, page 173) , crop Broccoli Seed (see 29.3, page 173) , crop Brussels sprouts Seed (see 29.5, page 173) , crop Celeriac Seed (see 29.7, page 174) , crop White cabbage Seed (see 29.37, page 177) , crop Early white cabbage - transplanted Seed (see 29.39, page 177) , and crop Onion Seed (see 29.15, page 174) .

```
< LAI    (DSLAI05 0.15 [DS])  
          (SpLAIfac SpLAIfac)    ; Has default value.  
          (InitCAI true) >
```

- *DSLAI05*: number [DS]
Parameter (default 0.15)
DS at CAI=0.5; initial phase.
- *SpLAIfac*: plf [DS → <none>]
Parameter (has default value with 4 points)



```
(SpLAIfac (0 [DS] 3 []) (0.2 [DS] 1.5 []) (0.4 [DS] 1.25 [])
           (0.6 [DS] 1 []))
```

Parameter description:

Factor defining maximum specific leaf weight. Only used during the initial phase.

- *InitCAI*: boolean (see section 4.1.2)
State variable (default true)
Initial CAI development phase.

79.2 release

Initial crop growth is governed by carbon released from seeds.

Used by crop Winter Wheat JG Seed (see 29.45, page 177) .

```
< release (C C)
          (rate rate)
          (T_factor T_factor)
          (initial_weight initial_weight)
          (DM_fraction DM_fraction)
          (C_fraction C_fraction)
          (N_fraction N_fraction) >
```

- *C*: number [**g C/m²**]
Optional state variable
Unreleased carbon left in seeds.
- *rate*: number [**h⁻¹**]
Parameter
Release rate of seed carbon to assimilate pool.
- *T_factor*: plf [**dg C → <fraction>**]
Optional parameter
Soil temperature effect on release rate. By default use the same as for maintenance respiration: $0.4281 (\exp(0.57 - 0.024 T + 0.0020 T^2) - \exp(0.57 - 0.042$

$T - 0.0051 T^2$)) This will give a rate of 1 at 20 dg C, slightly above 2 at 30 dg C, and a bit below 0.5 at 10 dg C.

- *initial_weight*: number [g w.w./m²]
Optional parameter
Initial seed weight to use when not specified by the sow operation. If not specified here, specifying seed amount when sowing is mandatory.
- *DM_fraction*: number [<fraction>]
Parameter
Dry matter content in seeds.
- *C_fraction*: number [<fraction>]
Parameter
Carbon content in seeds.
- *N_fraction*: number [<fraction>]
Parameter
Nitrogen content in seeds.

Chapter 80

select

Select part of state.

```
< component (multi sum)
              (expr expr)
              (flux flux)
              (dimension dimension)
              (factor 1 [?])
              (offset 0 [?])
              (accumulate false)
              (handle handle)
              (tag tag)
              (when when)
              (documentation documentation)
              (path path ...)
              (spec spec)
              (interesting_content interesting_content)
              (negate false) >
```

- *multi*: string (see section 4.1.5)
Optional parameter (default 'sum')
This option determine how to handle mutiple matches within a timestep. This could be two crops on the same column, or one crop on two columns.

min: Use smallest value

max: Use largest value

sum: Use the sum of all matches, weighted by relative column area if the matches are from different columns.
- *expr*: **number** component (see chapter 59)
Optional component
Expression for findig the value for the log file, given the internal value 'x'. For example '(expr (ln x))' will give you the natural logarithm of the value.
- *flux*: boolean (see section 4.1.2)
Optional parameter
OBSOLETE. This value will be used if 'handle' is not specified. A value of true then means 'sum', and false means 'current'.
- *dimension*: string (see section 4.1.5)
Optional parameter
The unit for numbers in this column. These will be printed in the second line

of the log file. The character '&' will be replaced with the log timestep. If you do not specify the dimension explicitly, a value will be interfered from 'spec' if available.

- *factor*: number (dimension not specified)
Parameter (default 1)
Factor to multiply the calculated value with, before logging. OBSOLETE: Use 'expr' instead.
- *offset*: number (dimension not specified)
Parameter (default 0)
Offset to add to the calculated value, before logging. OBSOLETE: Use 'expr' instead.
- *accumulate*: boolean (see section 4.1.2)
Parameter (default false)
Log accumulated values.
- *handle*: string (see section 4.1.5)
Optional parameter
This option determine how the specified variable should be logged.

min: Log the smallest value seen since last time the variable was logged. If 'accumulate' is true, use the smallest value ever.

max: Log the largest value seen since last time the variable was logged. If 'accumulate' is true, use the largest value ever.

average: Log the arithmetic average value seen since last time the variable was logged. If 'accumulate' is true, use the average of all values.

sum: Accumulate flux value since last time the variable was logged. If 'accumulate' is true, accumulate since the start of the log.

sum: Accumulate content value since last time the variable was logged. If 'accumulate' is true, accumulate since the start of the log.

current: Log the current value for the variable. If 'accumulate' is true, the printed values will be accumulated.
- *tag*: string (see section 4.1.5)
Optional parameter
Tag to identify the column. These will be printed in the first line of the log file. The default tag is the last element in the path.
- *when*: **condition** component (see chapter 28)
Optional component
OBSOLETE. If you set this variable, 'flux' will be set to true. This overwrites any direct setting of 'flux'.
- *documentation*: string (see section 4.1.5)
Optional parameter
Documentation for this entry.
- *path*: string (see section 4.1.5) sequence
Parameter
Sequence of attribute names leading to the variable you want to log in this column. The first name should be one of the attributes of the daisy component itself. What to specify as the next name depends on the type of the attribute you selected before.

If the value of that attribute itself is a fixed component, you should specify the name of an attribute in that component as the second name.

If the value is a library component, you should specify the name of the model or parameterization you are interested in, and then the name of the attribute inside the model you want to log.

The last attribute in the patch should be a number, a number sequence, a string, or an integer. These are the only values which can be logged by this model.

You can use the special value "*" to match everything at a given level, for example all crops. This way the path can specify multiple values, they will be added before they are printed in the log file. All values that start with a "\$" will work like "*". They are intended to be mapped with the 'set' attribute in the 'table' log model.

- *spec*: submodel (see section 4.1.7)

Optional submodel

Specification for the attribute to be logged of the form

library model submodel* attribute

Unlike path, the attribute may occur several different places in the simulation, if the model is used at several places. Also, there is no wildcards, so only a single model can be matches. The spec is used for helping Daisy establish a unique dimension and description for the attribute.

```
< library model submodels_and_attribute... >
```

- *library*: string (see section 4.1.5)

Parameter

Name of library where the attribute belong. Use 'fixed' to denote a fixed component.

- *model*: string (see section 4.1.5)

Parameter

Name of model or fixed component where the attribute belongs.

- *submodels_and_attribute*: string (see section 4.1.5) sequence

Parameter

Name of submodels and attribute.

- *interesting_content*: boolean (see section 4.1.2)

Optional parameter

True if the content of this column is interesting enough to warrant an initial line in the log file. By default, this is true iff 'handle' is 'current'.

- *negate*: boolean (see section 4.1.2)

Parameter (default false)

Switch sign of value. I.e. upward fluxes become downward fluxes.

80.1 array

Log all members of an array.

80.2 quiver

Convert edge flow to node based flow vectors.

80.3 value

Log a single numeric value.

80.4 content

A ‘value’ model (see 80.3, page 430) build into Daisy.

Extract content at specified location. The ”location” may be a line, plane or volume if one or more dimension parameters are left out. In that case, the weighted average is used.

```
< content (x x)
          (z z)
          (height height)
          (y y)
          ;; Shared parameters are described in section 80.3.
          (multi sum)
          (expr expr)
          (flux flux)
          (dimension dimension)
          (factor 1 [?])
          (offset 0 [?])
          (accumulate false)
          (handle handle)
          (tag tag)
          (when when)
          (documentation documentation)
          (path path ...)
          (spec spec)
          (interesting_content interesting_content)
          (negate false) >
```

- *x*: number [**cm**]
Optional parameter
Specify width (distance from left side) to measure content. The value will be a weighted average of all cells containing width. By default, cell in all widths will be included.
- *z*: number [**cm**]
Optional parameter
Specify height (negative below surface) to measure content. The value will be a weighted average of all cells containing height. By default, cell in all heights will be included.
- *height*: number [**cm**]
Optional parameter
OBSOLETE: Use ‘z’ instead.
- *y*: number [**cm**]
Optional parameter
Specify length (distance from front) to measure content. The value will be a weighted average of all cells containing length. By default, cell in all lengths will be included.

80.5 flow

A ‘value’ base model (see 80.3, page 430) build into Daisy.

Common base for logging flow through a specific plane.

```
< flow (volume volume) ; Default box value.
      (density false)
      ;; Shared parameters are described in section 80.3.
      (multi sum)
      (expr expr)
      (flux flux)
      (dimension dimension)
      (factor 1 [?])
      (offset 0 [?])
      (accumulate false)
      (handle handle)
      (tag tag)
      (when when)
      (documentation documentation)
      (path path ...)
      (spec spec)
      (interesting_content interesting_content)
      (negate false) >
```

- *volume*: **volume** component (see chapter 102)
Component (default ‘box’)
Soil volume to log flow into.
- *density*: boolean (see section 4.1.2)
Parameter (default false)
If true, divide value with volume height.

80.6 flow_back

A ‘flow’ model (see 80.5, page 431) build into Daisy.

Extract flow from back of specified volume.

80.7 flow_bottom

A ‘flow’ model (see 80.5, page 431) build into Daisy.

Extract flow from bottom of specified volume.

```

< flow_bottom (to to)
                ;; Shared parameters are described in section 80.5.
                (multi sum)
                (expr expr)
                (flux flux)
                (volume volume)                                ; Default box value.
                (dimension dimension)
                (factor 1 [?])
                (offset 0 [?])
                (accumulate false)
                (handle handle)
                (tag tag)
                (when when)
                (documentation documentation)
                (path path ...)
                (spec spec)
                (interesting_content interesting_content)
                (negate false)
                (density false) >

```

- *to*: number [cm]
Optional parameter
Specify height (negative) to measure interval. By default, measure to the bottom. OBSOLETE: Use (volume box (bottom TO)) instead.

80.8 flux_bottom

A ‘flow_bottom’ model (see 80.7, page 431) build into Daisy.

Flux entering bottom of specified volume. OBSOLETE: Use ‘(flow_bottom (density true))’ instead.

Used by log Field water entries (see 52.8, page 289) , log Soil water entries (see 52.10, page 293) , log Field nitrogen entries (see 52.7, page 286) , log Soil nitrogen entries (see 52.9, page 290) , log Field chemical entries (see 52.13, page 295) , and log Soil chemical entries (see 52.14, page 297) .

```
< (density true) >
```

80.9 flow_front

A ‘flow’ model (see 80.5, page 431) build into Daisy.

Extract flow from front of specified volume.

80.10 flow_left

A ‘flow’ model (see 80.5, page 431) build into Daisy.

Extract flow from left of specified volume.

80.11 flow_right

A ‘flow’ model (see 80.5, page 431) build into Daisy.

Extract flow from right of specified volume.

80.12 flow_top

A ‘flow’ model (see 80.5, page 431) build into Daisy.

Extract flow from top of specified volume.

```
< flow_top  (from from)
            ;; Shared parameters are described in section 80.5.
            (multi sum)
            (expr expr)
            (flux flux)
            (volume volume)                                ; Default box value.
            (dimension dimension)
            (factor 1 [?])
            (offset 0 [?])
            (accumulate false)
            (handle handle)
            (tag tag)
            (when when)
            (documentation documentation)
            (path path ...)
            (spec spec)
            (interesting_content interesting_content)
            (negate false)
            (density false) >
```

- *from*: number [**cm**]
Optional parameter
Specify height (negative) to measure from. By default, measure from the top.
OBSOLETE: Use (volume box (top FROM)) instead.

80.13 flux_top

A ‘flow_top’ model (see 80.12, page 433) build into Daisy.

Flux leaving top of specified volume. OBSOLETE: Use ‘(flow_top (negate true) (density true))’ instead.

Used by log Soil water entries (see 52.10, page 293) , log Soil nitrogen entries (see 52.9, page 290) , and log Soil chemical entries (see 52.14, page 297) .

```
<  (density true) >
```

80.14 index

A ‘value’ model (see 80.3, page 430) build into Daisy.

Extract content at specified array index.

```

< index (index index)
      ;; Shared parameters are described in section 80.3.
      (multi sum)
      (expr expr)
      (flux flux)
      (dimension dimension)
      (factor 1 [?])
      (offset 0 [?])
      (accumulate false)
      (handle handle)
      (tag tag)
      (when when)
      (documentation documentation)
      (path path ...)
      (spec spec)
      (interesting_content interesting_content)
      (negate false) >

```

- *index*: integer
Parameter
Specify array index to select.

80.15 number

A ‘value’ model (see 80.3, page 430) build into Daisy.

Extract specified number. If used on an array, it will treat them as individual numbers as specified by the ‘handle’ parameter.

Used by log Crop entries (see 52.16, page 299) , log Field water entries (see 52.8, page 289) , log Field nitrogen entries (see 52.7, page 286) , and log Field chemical entries (see 52.13, page 295) .

80.16 volume_base

A ‘value’ model (see 80.3, page 430) build into Daisy.

Shared parameters for volume based logs.

```

< volume_base (volume volume) ; Default box value.
  (density false)
  (space sum)
  (density_z density_z)
  (density_x density_x)
  (density_y density_y)
  (min_root_density -1 [cm/cm3])
  (min_root_crop "*")
  ;; Shared parameters are described in section 80.3.
  (multi sum)
  (expr expr)
  (flux flux)
  (dimension dimension)
  (factor 1 [?])
  (offset 0 [?])
  (accumulate false)
  (handle handle)
  (tag tag)
  (when when)
  (documentation documentation)
  (path path ...)
  (spec spec)
  (interesting_content interesting_content)
  (negate false) >

```

- *volume*: **volume** component (see chapter 102)
Component (default 'box')
Soil volume to log.
- *density*: boolean (see section 4.1.2)
Parameter (default false)
If true, divide total content with volume. Otherwise, obey 'density_z', 'density_x', and 'density_y'. Only used if 'space' is 'sum'.
- *space*: string (see section 4.1.5)
Optional parameter (default 'sum')
This option determine how to handle mutiple cells within the volume.
min: Use smallest value
max: Use largest value
sum: Use the weighted sum of all cells. If this is set, the 'density_z', 'density_x', and 'density_y' parameters take effect.
- *density_z*: boolean (see section 4.1.2)
Parameter
If true, divide total content with volume height. This parameter is ignored if 'density' is true. Only used if 'space' is 'sum'.
- *density_x*: boolean (see section 4.1.2)
Parameter
If true, divide total content with volume width. This parameter is ignored if 'density' is true. Only used if 'space' is 'sum'.
- *density_y*: boolean (see section 4.1.2)
Parameter
If true, divide total content with volume depth. This parameter is ignored if 'density' is true. Only used if 'space' is 'sum'.

- *min_root_density*: number [**cm/cm³**]
Parameter (default -1)
Minimum root density in cells.

Set this parameter to a positive amount in order to log only cells within the (dynamic) root zone. If the root density in the cell is above this amount, the full amount of the data being logged will be included. If the root density is below, the amount included will be scaled down accordingly. That is, if there are no roots, the data for the cell will be scaled to zero, while if there is only half the specified minimum root density, the data for the cell will be scaled to 0.5.
- *min_root_crop*: string (see section 4.1.5)
Parameter (default '*')
Name of crop whose roots should be used for the root density requirements. Set this to "*" to use all roots.

80.17 interval

A 'volume_base' model (see 80.16, page 434) build into Daisy.

Summarize specified interval. This is similar to 'volume', except for the default values of 'density_x' and 'density_y', and the unique 'from' and 'to' parameters.

Used by log Field water entries (see 52.8, page 289) , log Soil water entries (see 52.10, page 293) , log Field nitrogen entries (see 52.7, page 286) , log Soil nitrogen entries (see 52.9, page 290) , log Field chemical entries (see 52.13, page 295) , and log Soil chemical entries (see 52.14, page 297) .

```
< interval (from from)
           (to to)
           (density_z false)
           (density_x true)
           (density_y true)
           ;; Shared parameters are described in section 80.16.
           (multi sum)
           (expr expr)
           (flux flux)
           (volume volume) ; Default box value.
           (dimension dimension)
           (factor 1 [?])
           (offset 0 [?])
           (accumulate false)
           (handle handle)
           (tag tag)
           (when when)
           (documentation documentation)
           (path path ...)
           (spec spec)
           (interesting_content interesting_content)
           (negate false)
           (density false)
           (space sum)
           (min_root_density -1 [cm/cm3])
           (min_root_crop "*") >
```

- *from*: number [**cm**]
Optional parameter

Specify height (negative) to measure from. By default, measure from the top.
OBSOLETE: Use (volume box (top FROM)) instead.

- *to*: number [**cm**]
Optional parameter
Specify height (negative) to measure interval. By default, measure to the bottom. OBSOLETE: Use (volume box (bottom TO)) instead.

80.18 volume

A ‘volume_base’ model (see 80.16, page 434) build into Daisy.

Summarize specified volume.

```
< (density_z false)
  (density_x false)
  (density_y false) >
```

80.19 water

A ‘volume_base’ model (see 80.16, page 434) build into Daisy.

Shared parameters for water limited volumn logging.

```
< water (h h)
  (h_ice 0 [cm])
  ;; Shared parameters are described in section 80.16.
  (multi sum)
  (expr expr)
  (flux flux)
  (volume volume) ; Default box value.
  (dimension dimension)
  (factor 1 [?])
  (offset 0 [?])
  (accumulate false)
  (handle handle)
  (tag tag)
  (when when)
  (documentation documentation)
  (path path ...)
  (spec spec)
  (interesting_content interesting_content)
  (negate false)
  (density false)
  (space sum)
  (density_z density_z)
  (density_x density_x)
  (density_y density_y)
  (min_root_density -1 [cm/cm3])
  (min_root_crop "*") >
```

- *h*: number [**cm**]
Parameter
Pressure to log water content for.
- *h_ice*: number [**cm**]
Parameter (default 0)
Pressure at which all air is out of the matrix. When there are no ice, this is

0.0. When there are ice, the ice is presumed to occupy the large pores, so it is h ($\text{Theta_sat} - X_{\text{ice}}$).

80.20 water_interval

A 'water' model (see 80.19, page 437) build into Daisy.

Summarize water content in the specified interval. This is similar to 'water_volume', except for the default values of 'density_x' and 'density_y', and the unique 'from' and 'to' parameters.

```
< water_interval (from from)
                  (to to)
                  (density_z false)
                  (density_x true)
                  (density_y true)
                  ;; Shared parameters are described in section 80.19.
                  (multi sum)
                  (h h)
                  (expr expr)
                  (flux flux)
                  (volume volume) ; Default box value.
                  (dimension dimension)
                  (factor 1 [?])
                  (offset 0 [?])
                  (accumulate false)
                  (handle handle)
                  (tag tag)
                  (when when)
                  (documentation documentation)
                  (path path ...)
                  (spec spec)
                  (interesting_content interesting_content)
                  (negate false)
                  (density false)
                  (space sum)
                  (min_root_density -1 [cm/cm³])
                  (min_root_crop "*")
                  (h_ice 0 [cm]) >
```

- *from*: number [cm]
Optional parameter
Specify height (negative) to measure from. By default, measure from the top.
OBSOLETE: Use (volume box (top FROM)) instead.
- *to*: number [cm]
Optional parameter
Specify height (negative) to measure interval. By default, measure to the bottom. OBSOLETE: Use (volume box (bottom TO)) instead.

80.21 water_volume

A 'water' model (see 80.19, page 437) build into Daisy.

Summarize water content in the specified volume.

```
< (density_z false)
  (density_x false)
  (density_y false) >
```


Chapter 81

solute_uptake

The 'solute_uptake' component calculates uptake of nitrogen through roots.

Used by root component NH4_uptake (see 72, page 401) .

81.1 diffusion

N is transported to the root surface based on diffusion and convection.

The convection is based on concentration in the soil, while the diffusion is based on concentration in both soil and at root surfaces.

This base model find variables 'A' and 'B', so the transport of nitrogen to the root surface is calculated as $A - B * C_{Root}$, where C_{Root} is the concentration at the root surface. See also [Hansen and Abrahamsen, 2009]

< diffusion >

Log Variables

- *alpha*: number (dimensionless) soil cells
Internal variable.
- *A*: number (dimensionless) soil cells
 C_{Root} independent N uptake.
- *B*: number (dimensionless) soil cells
 C_{Root} dependent N uptake factor.
- *D*: number [cm^2/h] soil cells
Diffusion variable.
- *beta*: number (dimensionless) soil cells
Internal variable.

81.2 fixed_sink

A 'diffusion' model (see 81.1, page 441) build into Daisy.

Find the highest value of 'CRoot' that meets the N demand. See also [Hansen and Abrahamsen, 2009]

Used by root component NO3_uptake (see 72, page 401) .

< fixed_sink >

Log Variables

- *CRoot*: number [g N/cm³]
Nitrogen concentration at root surface.

81.3 variable_sink

A ‘diffusion’ model (see 81.1, page 441) build into Daisy.

Nitrogen uptake limited by the root ability to assimilate N.

The transport of N to the root surface is given as

$$A - B C$$

The assimilation of N from root surface is given as

$$F1 C / (K1 + C) + F2 C / (K2 + C)$$

We find the C for which

$$A - B C = F1 C / (K1 + C) + F2 C / (K2 + C)$$

If this is higher than the demand, uptake is scaled down. See also [Tsay et al., 2007, Miller and Cramer, 2005]

```
< variable_sink (K1 7e-07 [g N/cm3])
                (K2 7e-05 [g N/cm3])
                (F_relative 0.1 [<fraction>]) >
```

- *K1*: number [g N/cm³]
Parameter (default 7e-07)
Value description: 50 uM See also [Tsay et al., 2007]
K parameter for high affinity uptake.
- *K2*: number [g N/cm³]
Parameter (default 7e-05)
Value description: 5 mM See also [Tsay et al., 2007]
K parameter for low affinity uptake.
- *F_relative*: number [<fraction>]
Parameter (default 0.1)
Fraction of max uptake accounted for by high affinity uptake.

Log Variables

- *CRoot*: number [g N/cm³] soil cells
Nitrogen concentration at root surface.
- *C1*: number [g N/cm³] soil cells
CRoot calculated assuming $C \gg K1$. Solution to $A - B C = F1 + F2 C / (K2 + C)$.
- *C2*: number [g N/cm³] soil cells
CRoot calculated assuming $C \ll K2$. Solution to $A - B C = F1 C / (K1 + C) + F2 C / K2$.
- *CEst*: number [g N/cm³] soil cells
Best of C1 or C2..

81.4 none

No uptake.

Chapter 82

solver

A way to solve the matrix equation ' $A x = b$ '.

82.1 cxsparse

Solve equation using CXSparse library described in:

Direct Methods for Sparse Linear Systems, T. A. Davis, SIAM, Philadelphia, Sept. 2006. Part of the SIAM Book Series on the Fundamentals of Algorithms.

The uBLAS interface was provided by Gunter Winkler <guwi17@gmx.de>.

Used by heatrect Mollerup solver (see 46.1, page 229) , transport Mollerup solver (see 92.2, page 487) , uzrect Mollerup solver (see 99.1, page 537) , and svat SSOC solver (see 88.2, page 474) .

82.2 none

Don't solve the equation.

82.3 ublas

Solve equation using UBLAS lu functions.

Chapter 83

source

Time series, with possible error bars and formatting information.

83.1 arithmetic

Read a daisy log, weather or data file. Calculate a single value for each time step, based on the value in the various columns.

```
< arithmetic (expr expr)
              (file file)
              (with with)
              (timestep h)
              (missing missing ...) ; Has default value.
              (filter)
              (original original ...)
              (dim_line dim_line)
              (style style)
              (title title)
              (accumulate false)
              (handle normal)
              (default_hour 8)
              (time_offset time_offset) ; Has default value.
              (valid valid) ; Default true value. >
```

- **expr: number** component (see chapter 59)
Expression for calculating the value for this source for each row. The expression can refer to the value in a specific column by the tag for that column.
- **file:** string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- **with:** string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.

By default, data from dwf and dlf files will be drawn with lines, and data from ddf files will be drawn with points.
- **timestep:** string (see section 4.1.5)
Parameter (default 'h')
Multiple with this dimension when accumulating.

- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

- *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.

- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.

- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.

By default the name of the 'expr' object.

- *accumulate*: boolean (see section 4.1.2)
Parameter (default false)
Accumulate values.

- *handle*: string (see section 4.1.5)
Parameter (default 'normal')
Determine how to handle multiple simultaneously. Possible values are:

all: show all values.

sum: use the sum of the values.

normal: use the arithmetic average of the values, and calculate the standard deviation.
- *default_hour*: integer
Parameter (default 8)
Hour to assume when nothing else is specified;
- *time_offset*: **Timestep** fixed component (see section 107.25)
Submodel (has fully specified default value)
Add this to time from sources. By default, use unmodified times.
- *valid*: **boolean** component (see chapter 21)
Component (default 'true')
Ignore entries if this boolean expression is false.

83.2 column

Read a single column from a Daisy log, weather or data file.

```
< column (file file)
      (with with)
      (timestep h)
      (dimension dimension)
      (factor factor)
      (offset 0 [?])
      (missing missing ...) ; Has default value.
      (filter)
      (original original ...)
      (dim_line dim_line)
      (style style)
      (title title)
      (accumulate false)
      (handle normal)
      (default_hour 8)
      (time_offset time_offset) ; Has default value.
      (tag tag)
      (reset_offset false) >
```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.

By default, data from dwf and dlf files will be drawn with lines, and data from ddf files will be drawn with points.

- *timestep*: string (see section 4.1.5)
Parameter (default 'h')
Multiple with this dimension when accumulating.
- *dimension*: string (see section 4.1.5)
Optional parameter
Dimension of data to plot. By default this is the same as 'original'. If 'factor' is not specified, Daisy will attempt to convert the data.
- *factor*: number (dimension not specified)
Optional parameter
Multiply all data by this number. By default Daisy will convert from 'original' to 'dimension'.
- *offset*: number (dimension not specified)
Parameter (default 0)
Add this number to all data.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

(missing "" "00.00")

Parameter description:
List of strings indicating missing values.
- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.
 - < tag allowed... >
 - *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
 - *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.
- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.
- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over.

Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.

- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.
By default the same as 'tag'.
- *accumulate*: boolean (see section 4.1.2)
Parameter (default false)
Accumulate values.
- *handle*: string (see section 4.1.5)
Parameter (default 'normal')
Determine how to handle multiple simultaneously. Possible values are:
all: show all values.
sum: use the sum of the values.
normal: use the arithmetic average of the values, and calculate the standard deviation.
- *default_hour*: integer
Parameter (default 8)
Hour to assume when nothing else is specified;
- *time_offset*: **Timestep** fixed component (see section 107.25)
Submodel (has fully specified default value)
Add this to time from sources. By default, use unmodified times.
- *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file where data is found.
- *reset_offset*: boolean (see section 4.1.2)
Parameter (default false)
Set offset to first value read. Useful for plotting already accumulated data from a later date.

83.3 combine

Combine data from multiple sources with a single expression.

```
< combine (expr expr)
          (source source ...)
          (with with)
          (style style)
          (title title)
          (accumulate false) >
```

- *expr*: **number** component (see chapter 59)
Expression for calculating the value for this source for each row. A row is any date found in any of the member of 'source'. The expression may refer to the value of each source by its title.

- *source*: **source** component (see chapter 83) sequence
List of sources for data. The style information for the sources is ignored, but the dates, title and value is used as specified by 'expr' to calculate the combines date and value pairs.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
By default, let the first source decide.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.
By default the name of the 'expr' object.
- *accumulate*: boolean (see section 4.1.2)
Parameter (default false)
Accumulate values.

83.4 merge

Merge multiple timeseries into one. Any errorbars on the original timeseries are ignored, but the merged timeseries may have errorbars if there are multiple values for the same time.

```
< merge  (source source ...)
          (with with)
          (dimension dimension)
          (style style)
          (title title)
          (accumulate false) >
```

- *source*: **source** component (see chapter 83) sequence
List of timeseries to merge.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
By default, let the first source decide.

- *dimension*: string (see section 4.1.5)
Optional parameter
Dimension of data to plot. By default use the first source with a known dimension.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Parameter
Name of data series for the legend on the graph.
- *accumulate*: boolean (see section 4.1.2)
Parameter (default false)
Accumulate values.

Chapter 84

stomatacon

The 'Stomatacon' component calculates the stomata conductance of water vapour.

84.1 WSF

Common water stress effect parameters.

```
< WSF  (beta 0 [cm3/g])  
        (ABA_min 0 [g/cm3])  
        (delta 0 [MPa-1]) >
```

- *beta*: number [cm³/g]
Parameter (default 0)
Effect of ABA concentration. The effect is $\exp(-\text{beta} (\text{—ABA—} - \text{ABA_min}))$, where —ABA— is the ABA concentration in the xylem.
- *ABA_min*: number [g/cm³]
Parameter (default 0)
Level of ABA with unstressed production.
- *delta*: number [MPa⁻¹]
Parameter (default 0)
Effect of crown water potential. The effect is $\exp(-\text{delta } \text{—psi}_c\text{—})$, where psi_c is the crown potential.

84.2 BB_base

A 'WSF' base model (see 84.1, page 453) build into Daisy.

Common parameters for Ball&Berry derived models.

```
< BB_base  (m m)  
            (b b)  
            ;; Shared parameters are described in section 84.1.  
            (beta 0 [cm3/g])  
            (ABA_min 0 [g/cm3])  
            (delta 0 [MPa-1]) >
```

- *m*: number (dimensionless)
Parameter
Stomatal slope factor. Ball and Berry (1982): $m = 9$ for soyabean. Wang and Leuning(1998): $m = 11$ for wheat

- *b*: number [mol/m²/s]
Parameter
Stomatal intercept. Ball and Berry (1982) & Wang and Leuning(1998): (0.01 mol/m²/s)

84.3 BB

A ‘BB.base’ model (see 84.2, page 453) build into Daisy.

Stomata conductance calculated by the Ball & Berry model. See also [Ball et al., 1987]

84.4 Leuning

A ‘BB.base’ model (see 84.2, page 453) build into Daisy.

Stomata conductance calculated by the Leuning model. See also [Leuning, 1995]

Used by photosynthesis Farquhar Stomatacon (see 64.1, page 345) .

```
< Leuning (Do 1500 [[Pa]])
    ;; Shared parameters are described in section 84.2.
    (m m)
    (b b)
    (beta 0 [cm3/g])
    (ABA_min 0 [g/cm3])
    (delta 0 [MPa-1]) >
```

- *Do*: number [[Pa]]
Parameter (default 1500)
Empirical coefficient.

84.5 MNA

A ‘WSF’ model (see 84.1, page 453) build into Daisy.

Stomata conductance calculated by the model given by Eq. 14.

```
< MNA (alpha alpha)
    (m m)
    (max max)
    (b b)
    (obsolete obsolete)
    (lambda lambda)
    ;; Shared parameters are described in section 84.1.
    (beta 0 [cm3/g])
    (ABA_min 0 [g/cm3])
    (delta 0 [MPa-1]) >
```

- *alpha*: number (dimensionless)
Parameter
Humidity effect
- *m*: number [mol H₂O/m² leaf/s]
Parameter
Conductivity factor.
- *max*: number [mol H₂O/m² leaf/s]
Optional parameter
Maximal conductivity. By default, there is no maximum.

- *b*: number [**mol/m²/s**]
Parameter
Stomatal intercept. Ball and Berry (1982) & Wang and Leuning(1998): (0.01 mol/m²/s)
- *obsolete*: string (see section 4.1.5)
Optional parameter
We can't find a source to this equation.
- *lambda*: number [**umol CO₂/m² leaf/s**]
Parameter
Net photosynthesis effect

84.6 SHA12

A 'WSF' model (see 84.1, page 453) build into Daisy.

Stomata conductance calculated by the model given by Eq. 12. See also [Ahmadi et al., 2009]

```
< SHA12 (alpha 1 [])
      (m m)
      (min min)
      (lambda 1 [])
      ;; Shared parameters are described in section 84.1.
      (beta 0 [cm3/g])
      (ABA_min 0 [g/cm3])
      (delta 0 [MPa-1]) >
```

- *alpha*: number (dimensionless)
Parameter (default 1)
Humidity effect
- *m*: number (dimension not specified)
Parameter
Slope parameter, dimension depends on alpha and lambda.
- *min*: number [**mol H₂O/m² leaf/s**]
Optional parameter
Minimal conductivity.
- *lambda*: number (dimensionless)
Parameter (default 1)
Net photosynthesis effect

84.7 SHA14

A 'WSF' model (see 84.1, page 453) build into Daisy.

Stomata conductance calculated by the model given by Eq. 14. See also [Ahmadi et al., 2009]

```
< SHA14 (alpha alpha)
      (m m)
      (max max)
      (lambda lambda)
      ;; Shared parameters are described in section 84.1.
      (beta 0 [cm3/g])
      (ABA_min 0 [g/cm3])
      (delta 0 [MPa-1]) >
```

- *alpha*: number (dimensionless)
Parameter
Humidity effect
- *m*: number [**mol H₂O/m² leaf/s**]
Parameter
Conductivity factor.
- *max*: number [**mol H₂O/m² leaf/s**]
Optional parameter
Maximal conductivity. By default, there is no maximum.
- *lambda*: number [**m² leaf s/umol CO₂**]
Parameter
Net photosynthesis effect

Chapter 85

string

Generic representation of strings.

85.1 cond

Return the value of the first clause whose condition is true.

Used by log chemical chemfid (see 52.12, page 295) , log crop cropfid (see 52.15, page 298) , log biopore bioporefid (see 52.11, page 294) , and log column colfid (see 52.6, page 286) .

`< cond clauses ... >`

- *clauses*: **StringerCondClause** fixed component (see section 107.29) sequence
List of clauses to match for.

85.2 identity

Return the specified value.

`< identity (value value) >`

- *value*: string (see section 4.1.5)
Parameter
Constant value.

85.3 number

Extract the value of a number.

`< number (number number) >`

- *number*: **number** component (see chapter 59)
Number to manipulate.

85.4 dimension

A ‘number’ model (see 85.3, page 457) build into Daisy.

Extract the dimension of a number as a string.

85.5 value

A ‘number’ model (see 85.3, page 457) build into Daisy.

Extract the value of a number as a string.

```
< value (precision precision)  
      ;; Shared parameters are described in section 85.3.  
      (number number) >
```

- *precision*: integer
Optional parameter
Number of decimals after point. By default, use a floating format.

Chapter 86

summary

Summary reports for log parameterizations.

```
< component (title title) >
```

- *title*: string (see section 4.1.5)
Optional parameter
Title of this summary. By default, use the name of the parameterization.

86.1 Rsqr

Calculate coefficient of determination.

```
< Rsqr (measure measure ...)
      (print_data false)
      ;; Shared parameters are described in chapter 86.
      (title title) >
```

- *measure*: submodel (see section 4.1.7) sequence
Measured data.

```
< (time time)
  (data data ...) >
```

 - *time*: **Time** fixed component (see section 107.27)
Optional submodel
Time of measurement. Measured data will be compared to the first simulated value at or after this time. By default, compare with the first available simulated data.
 - *data*: submodel (see section 4.1.7) sequence
Data measured at this time.

```
< tag value... >
```

 - * *tag*: string (see section 4.1.5)
Parameter
Name of simulated data to compare with.
 - * *value*: number (dimension not specified) sequence
Parameter
Measured data.
- *print_data*: boolean (see section 4.1.2)
Parameter (default false)
Print a table with all data in the summary.

86.2 RsqrW

Calculate coefficient of determination.

```
< RsqrW (measure measure ...)
      (print_data false)
      ;; Shared parameters are described in chapter 86.
      (title title) >
```

- *measure*: submodel (see section 4.1.7) sequence
Measured data.

```
< (time time)
  (data data ...) >
```

 - *time*: **Time** fixed component (see section 107.27)
Optional submodel
Time of measurement. Measured data will be compared to the first simulated value at or after this time. By default, compare with the first available simulated data.
 - *data*: submodel (see section 4.1.7) sequence
Data measured at this time.

```
< weight tag value... >
```

 - * *weight*: number (dimensionless)
Parameter
Weight given this measurement.
 - * *tag*: string (see section 4.1.5)
Parameter
Name of simulated data to compare with.
 - * *value*: number (dimension not specified) sequence
Parameter
Measured data.
- *print_data*: boolean (see section 4.1.2)
Parameter (default false)
Print a table with all data in the summary.

86.3 balance

A summary model providing a balance for a log parameterization.

Used by log Field chemical summary (see 52.13, page 295) .

```
< balance (content content ...)
      (where where)
      (precision 2)
      (input input ...)
      (output output ...)
      (require_top false)
      ;; Shared parameters are described in chapter 86.
      (title title) >
```

- *content*: string (see section 4.1.5) sequence
Parameter
Tags of columns in log file representing content.
- *where*: string (see section 4.1.5)
Optional parameter

File name to store the summary. By default, the summary will be stored in daisy.log and the screen.

- *precision*: integer
Parameter (default 2)
Number of digits to print after decimal point.
- *input*: string (see section 4.1.5) sequence
Parameter
Tags of columns in log file representing inputs.
- *output*: string (see section 4.1.5) sequence
Parameter
Tags of columns in log file representing outputs.
- *require_top*: boolean (see section 4.1.2)
Parameter (default false)
If the balance only hold true when logging the top of the soil, i.e. the ‘from’ parameter of the log model is 0, this flag should be set.

86.4 Biopore chemical balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Used by log Soil chemical summary (see 52.14, page 297) .

86.5 Biopore water balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Water balance for the biopore domain alone.
Used by log Soil water summary (see 52.10, page 293) .

86.6 Field crop nitrogen balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Crop nitrogen balance for the complete system.
Used by log Field nitrogen summary (see 52.7, page 286) .

86.7 Field mineral nitrogen balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Mineral balance for the complete system.
Used by log Field nitrogen summary (see 52.7, page 286) .

86.8 Field nitrogen balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Nitrogen balance for the complete system.
Used by log Field nitrogen summary (see 52.7, page 286) .

86.9 Field organic nitrogen balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Organic nitrogen balance for the complete system.
Used by log Field nitrogen summary (see 52.7, page 286) .

86.10 Field water balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Water balance for the complete system.
Used by log Field water summary (see 52.8, page 289) .

86.11 Matrix chemical balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Used by log Soil chemical summary (see 52.14, page 297) .

86.12 Matrix water balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Balance for the matrix domain in the soil.
Used by log Soil water summary (see 52.10, page 293) .

86.13 N_balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.
Unit is kg, precion is g.

86.14 Soil N balance

A ‘N_balance’ model (see 86.13, page 462) defined in ‘log-std.dai’.
Nitrogen balance for a soil interval.
Used by log Soil nitrogen summary (see 52.9, page 290) .

86.15 Soil NH4+ balance

A ‘N_balance’ model (see 86.13, page 462) defined in ‘log-std.dai’.
NH4+ balance for a soil interval.
Used by log Soil nitrogen summary (see 52.9, page 290) .

86.16 Soil NO3- balance

A ‘N_balance’ model (see 86.13, page 462) defined in ‘log-std.dai’.
NO3- balance for a soil interval.
Used by log Soil nitrogen summary (see 52.9, page 290) .

86.17 Soil mineral-N balance

A ‘N_balance’ model (see 86.13, page 462) defined in ‘log-std.dai’.

Mineral N balance for a soil interval.

Used by log Soil nitrogen summary (see 52.9, page 290) .

86.18 Soil organic-N balance

A ‘N_balance’ model (see 86.13, page 462) defined in ‘log-std.dai’.

Organic nitrogen balance for a soil interval.

Used by log Soil nitrogen summary (see 52.9, page 290) .

86.19 Soil chemical balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.

Used by log Soil chemical summary (see 52.14, page 297) .

86.20 Soil water balance

A ‘balance’ model (see 86.3, page 460) defined in ‘log-std.dai’.

Balance for all water in the soil.

Used by log Soil water summary (see 52.10, page 293) .

86.21 fractiles

Show fractiles for specified tags.

```
< fractiles (fractiles fractiles ...)
           (tags tags ...)
           (first first)
           ;; Shared parameters are described in chapter 86.
           (title title) >
```

- *fractiles*: number [**<fraction>**] sequence
Parameter
List of fractiles to summarize.
- *tags*: string (see section 4.1.5) sequence
Parameter
List of tags to summarize.
- *first*: string (see section 4.1.5)
Optional parameter
If set, put this in the first column of the summary.

86.22 simple

A simple log file summary model.

```

< simple  (fetch fetch ...)
          (where where)
          (precision 2)
          (period period)
          (print_sum true)
          (sum_name Sum)
          ;; Shared parameters are described in chapter 86.
          (title title) >

```

- *fetch*: **FetchPretty** fixed component (see section 107.14) sequence
List of columns to fetch for the summary.
- *where*: string (see section 4.1.5)
Optional parameter
File name to store the summary. By default, the summary will be stored in daisy.log and the screen.
- *precision*: integer
Parameter (default 2)
Number of digits to print after decimal point.
- *period*: string (see section 4.1.5)
Optional parameter
Set this to 'y', 'm', 'w', 'd' or 'h' to get fluxes per time period instead of total amount.
- *print_sum*: boolean (see section 4.1.2)
Parameter (default true)
Print sum of all the summary lines.
- *sum_name*: string (see section 4.1.5)
Parameter (default 'Sum')
Name of the sum of all the entries.

Chapter 87

surface

Surface processes.

87.1 default

Keep track of soil surface.

Used by column default Surface (see 26.1, page 153) .

```
< default (max_exfiltration max_exfiltration) ; Default Hansen value.
          (temperature_change_rate 0.5 [h-1])
          (EpFactor 0.6 [])
          (EpFactor_SWE EpFactor_SWE) ; Has default value.
          (albedo_dry 0.15 [])
          (albedo_wet 0.08 [])
          (pond_section pond_section ...)
          (DetentionCapacity 1000 [mm])
          (ReservoirConstant 1 [h-1])
          (LocalDetentionCapacity 10 [mm])
          (z_mixing 0.1 [cm])
          (R_mixing 1e+09 [h/mm]) >
```

- *max_exfiltration*: **max_exfiltration** component (see chapter 55)
Component (default 'Hansen')
Soil limited exfiltration.
- *temperature_change_rate*: number [h⁻¹]
Parameter (default 0.5)
Relative change of surface temperature. If you set this to a negative value, surface temperature will reflect temperature of incoming water or ponded water, if any, snow, if any, or air temperature. Set this parameter to a positive value to dampen changes. Especially useful if you feed the model with daily weather data and average temperature.
- *EpFactor*: number (dimensionless)
Parameter (default 0.6)
Value description: See figure 4 in the cited paper.

The autumn value can be lower, due to mulching. With a crop factor of 1.2 a combined Kc of 1.15 is reached at LAI=5. See also [Kjaersgaard et al., 2008]
Conversion of reference evapotranspiration to potential evaporation for bare soil.

- *EpFactor_SWE*: plf [**pF** $\rightarrow$ **<none>**]
Parameter (has default const value 1 [])
Effect of soil water on EpFactor.
- *albedo_dry*: number (dimensionless)
Parameter (default 0.15)
Albedo of dry soil (pF ≥ 3)
- *albedo_wet*: number (dimensionless)
Parameter (default 0.08)
Albedo of wet soil (pF ≤ 1.7)
- *pond_section*: number [**mm**] sequence
Optional state variable
Amount of ponding on each section of the surface. By default, there will be no ponding. In an 1D simulation, there will only be one section. In general, there will be a section for each numeric cell in the soil matrix with an edge towards the surface.
- *DetentionCapacity*: number [**mm**]
State variable (default 1000)
Amount of ponding the surface can retain. If ponding in any part of the surface is above this, exceed will runoff.
- *ReservoirConstant*: number [**h**⁻¹]
Parameter (default 1)
Fraction of ponding above DetentionCapacity that runoffs each hour.
- *LocalDetentionCapacity*: number [**mm**]
State variable (default 10)
Amount of ponding the surface can retain locally. If ponding in any part of the surface is above this, exceed will be distributed to the rest of the surface.
- *z_mixing*: number [**cm**]
Parameter (default 0.1)
Depth of mixing layer in the top of the soil. The mixing layer affect exchange between soil colloids, soil water and the surface, especially in connection with intense rainfall.
- *R_mixing*: number [**h/mm**]
Parameter (default 1e+09)
Resistance to mixing inorganic compounds between soil and ponding.

Log Variables

- *T*: number [**dg C**]
Temperature of water or air directly above the surface.
- *pond*: number [**mm**]
Amount of ponding on the surface. Negative numbers indicate soil exfiltration.
- *EvapSoilSurface*: number [**mm/h**]
Water evaporated from the surface, including the pond and exfiltration.
- *Eps*: number [**mm/h**]
Potential evaporation from the surface.
- *runoff*: number [**mm/h**]
Amount of water runoff from ponding this hour.

- *ME*: number [mm/h]
Soil limit on exfiltration rate.

87.2 simple

Don't keep track of surface stuff.

87.3 const_flux

A 'simple' model (see 87.2, page 467) build into Daisy.

Constant flux upper boundary for soil.

```
< const_flux flux >
```

- *flux*: number [cm/h]
Parameter
Flux up from soil.

87.4 none

A 'const_flux' model (see 87.3, page 467) build into Daisy.

Zero flux upper boundary for soil.

```
< flux >
```

87.5 const_pressure

A 'simple' model (see 87.2, page 467) build into Daisy.

Constant pressure upper boundary for soil.

```
< const_pressure pressure >
```

- *pressure*: number [cm]
Parameter
Soil upper boundary pressure.

87.6 source_flux

A 'simple' model (see 87.2, page 467) build into Daisy.

Dynamic flux upper boundary for soil.

```
< source_flux (source source)  
              (interpolate false) >
```

- *source*: **source** component (see chapter 83)
Flux time series.
- *interpolate*: boolean (see section 4.1.2)
Parameter (default false)
If true, interpolate values. If false, use last read.

Chapter 88

svat

The task of the 'svat' component is to calculate the production stress, given the potential evapotranspiration, the actual evaporation from the surface, meteorological data, and the vegetation and soil state.

88.1 PMSW

Peter van der Keur's SVAT model. See also [van der Keur et al., 2001]

```
< PMSW (w 0.0025 [m])
        (albedo 0.2 [])
        (b1 0.53 [])
        (b2 0.0065 [])
        (b3 0.1 [])
        (b4 0.9 [])
        (ndif 2.5 [])
        (c_d 0.05 [])
        (z_0s 0.01 [m])
        (z0_def 0.005 [m])
        (alpha_u 3 [])
        (arac 0.00662 [])
        (alpha_k 2 [])
        (alpha_r 0.5 [])
        (theta_w 0.05 [cm3/cm3])
        (theta_c 0.25 [cm3/cm3])
        (rcmin_const 30 [s/m])
        (rcmax 1000 [])
        (tref 298 [])
        (zeta 0.0002 [])
        (f3const 0.0016 [])
        (spar 100 [])
        (tmin 0 [dg C])
        (tmax 55 [dg C])
        (nu_1 26.5 [])
        (nu_2 0.57 [])
        (nu_3 0.008 [])
        (dt1 -5 [dg C])
        (dt2 5 [dg C])
        (acc 0.01 []) >
```

- *w*: number [m]

Parameter (default 0.0025)
average leaf width

- *albedo*: number (dimensionless)
Parameter (default 0.2)
Bulk albedo
- *b1*: number (dimensionless)
Parameter (default 0.53)
Brunt coefficient 1
- *b2*: number (dimensionless)
Parameter (default 0.0065)
Brunt coefficient 2
- *b3*: number (dimensionless)
Parameter (default 0.1)
Brunt coefficient 3
- *b4*: number (dimensionless)
Parameter (default 0.9)
Brunt coefficient 4
- *ndif*: number (dimensionless)
Parameter (default 2.5)
Eddy diffusivity decay constant in crop
- *c_d*: number (dimensionless)
Parameter (default 0.05)
Mean drag coefficient for a leaf
- *z_0s*: number [**m**]
Parameter (default 0.01)
Roughness length for soil surface, SG (1990)
- *z0_def*: number [**m**]
Parameter (default 0.005)
Roughness length for soil surface, Oke
- *alpha_u*: number (dimensionless)
Parameter (default 3)
attenuation coefficient for wind speed
- *arac*: number (dimensionless)
Parameter (default 0.00662)
leaf boundary layer resistance coefficient
- *alpha_k*: number (dimensionless)
Parameter (default 2)
Att. coefficient of eddy diffusivity through sparse canopy
- *alpha_r*: number (dimensionless)
Parameter (default 0.5)
Att. coefficient for vegetation in ACOEFF()
- *theta_w*: number [**cm³/cm³**]
Parameter (default 0.05)
Soil water content at 'wilting point'

- *theta_c*: number [cm^3/cm^3]
Parameter (default 0.25)
Soil water content at 'field capacity'
- *rcmin_const*: number [s/m]
Parameter (default 30)
Constant minimum canopy resistance
- *rcmax*: number (dimensionless)
Parameter (default 1000)
Maximum canopy resistance
- *tref*: number (dimensionless)
Parameter (default 298)
Reference/optimum temperature in temperature dependent constraint function
- *zeta*: number (dimensionless)
Parameter (default 0.0002)
Coefficient in vapor pressure dependent constraint function
- *f3const*: number (dimensionless)
Parameter (default 0.0016)
Coefficient in temperature dependent constraint function
- *spar*: number (dimensionless)
Parameter (default 100)
Reference value of photosynthetically active part of Si
- *tmin*: number [$^{\circ}\text{C}$]
Parameter (default 0)
Minimum temperature for canopy conductance
- *tmax*: number [$^{\circ}\text{C}$]
Parameter (default 55)
Maximum temperature for canopy conductance
- *nu_1*: number (dimensionless)
Parameter (default 26.5)
coefficient in Jarvis (1976) constraint function f_temp
- *nu_2*: number (dimensionless)
Parameter (default 0.57)
coefficient in Lohammar (1980) constraint function f_def
- *nu_3*: number (dimensionless)
Parameter (default 0.008)
coefficient in Steward (1988) constraint function f_theta
- *dt1*: number [$^{\circ}\text{C}$]
Parameter (default -5)
lower solution limit in Newton-Raphson method
- *dt2*: number [$^{\circ}\text{C}$]
Parameter (default 5)
upper solution limit in Newton-Raphson method
- *acc*: number (dimensionless)
Parameter (default 0.01)
iteration accuracy in Newton-Raphson method

Log Variables

- *ha*: number [W/m^2]
Sensible heat flux from source- to screen height
- *netrad_brunt*: number [W/m^2]
Net radiation by Brunt
- *netlong_brunt*: number [W/m^2]
Net long radiation by Brunt
- *r_a*: number [s/m]
bulk aerodynamic resistance, neutral conditions
- *r_astab*: number [s/m]
bulk aerodynamic resistance, stability corrected
- *r_aa*: number [s/m]
aerodynamic resistance mean source-ref, uncorrected
- *r_aastab1*: number [s/m]
aerodynamic resistance mean source-ref, corrected 'method 1'
- *r_aastab2*: number [s/m]
aerodynamic resistance mean source-ref, corrected, 'method 2'
- *r_as*: number [s/m]
aerodynamic resistance from soil to mean source
- *r_ac*: number [s/m]
aerodynamic resistance from leaf to mean source
- *r_sc_1*: number [s/m]
Bulk canopy resistance (Noilhan et al., 1991)
- *r_sc_2*: number [s/m]
Bulk canopy resistance (Verma et al., 1993)
- *tskin*: number [$\text{dg } ^\circ\text{C}$]
soil/skin temperature
- *tcan*: number [$\text{dg } ^\circ\text{C}$]
canopy temperature at mean source
- *tleaf*: number [$\text{dg } ^\circ\text{C}$]
Leaf temperature
- *e_c_abs*: number [Pa]
vapor pressure at mean source height
- *e_slabs*: number [Pa]
saturated vapor pressure at leaf surface
- *hl*: number [W/m^2]
Sensible heat flux from leaf to mean source
- *hs*: number [W/m^2]
Sensible heat flux from soil to mean source
- *lea*: number [W/m^2]
Latent heat flux from source- to screen height

- *lel*: number [$\mathbf{W/m^2}$]
Latent heat flux from leaf to mean source
- *gflux*: number [$\mathbf{W/m^2}$]
Ground heat flux
- *dtcta*: number [$\mathbf{dg\ C}$]
Temperature gradient between mean source og screen height
- *dtltc*: number [$\mathbf{dg\ C}$]
Temperature gradient between leaf and mean source
- *dtstc*: number [$\mathbf{dg\ C}$]
Temperature gradient between soil and mean source
- *dtcta_star*: number [$\mathbf{dg\ C}$]
corrected temp gradient between mean source og screen height
- *dtltc_star*: number [$\mathbf{dg\ C}$]
corrected temperature gradient between leaf and mean source
- *dtstc_star*: number [$\mathbf{dg\ C}$]
corrected temperature gradient between soil and mean source
- *theta_0_20*: number [$\mathbf{cm^3/cm^3}$]
Averaged soil water content in upper 20 cm
- *f_1*: number (dimensionless)
Constraint function (Noilhan) related to solar radiation
- *f1_dolman*: number (dimensionless)
???
- *f_2*: number (dimensionless)
Constraint function (Noilhan) related to vapor pressure
- *f_3*: number (dimensionless)
Constraint function (Noilhan) related to air temperature
- *f_4*: number (dimensionless)
Constraint function (Noilhan) related to soil water content
- *f_temp*: number (dimensionless)
Constraint function (Verma) related to air temperature
- *f_def*: number (dimensionless)
Constraint function (Verma) related to vapor pressure
- *f_theta*: number (dimensionless)
Constraint function (Steward) related to soil water content
- *f_etep*: number (dimensionless)
Constraint function defined by crop_ea/crop_ep
- *r_sc_js*: number [$\mathbf{s/m}$]
Bulk canopy resistance: $f1_dolman*f_def*f3*f4$
- *r_sc*: number [$\mathbf{s/m}$]
Bulk canopy resistance: $f1_dolman*f_def*f3*f_etep$

- *rcmin_star*: number [**s/m**]
minimum canopy resistance
- *pstress*: number (dimensionless)
crop production stress
- *ustar_raa*: number (dimensionless)
friction velocity from RAA()
- *ustar_raastab1*: number (dimensionless)
friction velocity from RAASTAB1()
- *ustar_raastab2*: number (dimensionless)
friction velocity from RAASTAB2()
- *env_lai_factor*: number (dimensionless)
 $LAI * F_i$
- *e_pa*: number [**Pa**]
vapor pressure at 2 m
- *e_abs*: number [**kg/m³**]
absolute vapor pressure
- *tair*: number [**degr.C**]
air temperature
- *srad*: number [**W/m²**]
global radiation
- *u_ref*: number [**m/s**]
friction velocity from ??
- *prec*: number [**mm**]
precipitation

88.2 SSOC

Sun-Shade Open Canopy. See also [Plauborg et al., 2010]

```
< SSOC (solver solver) ; Default cxsparse value.
    (hypostomatous true)
    (max_iteration 1500)
    (z_0b 0.0006 [m])
    (epsilon_leaf 0.98 [<fraction>])
    (epsilon_soil 0.95 [<fraction>])
    (epsilon_soil_SWE epsilon_soil_SWE) ; Has default value.
    (b_SSOC 0.01 [mol/m2/s]) >
```

- *solver*: **solver** component (see chapter 82)
Component (default ‘cxsparse’)
Model used for solving the energy balance equation system.
- *hypostomatous*: boolean (see section 4.1.2)
Parameter (default true)
True for hypostomatous leaves. False for amphistomatous leaves (possessing stomata on both surfaces).

- *max_iteration*: integer
Parameter (default 1500)
Largest number of iterations before giving up on convergence.
- *z_0b*: number [**m**]
Parameter (default 0.0006)
Bare soil roughness height for momentum.
- *epsilon_leaf*: number [**<fraction>**]
Parameter (default 0.98)
Leaf emmisivity for long wave radiation.
- *epsilon_soil*: number [**<fraction>**]
Parameter (default 0.95)
Soil emmisivity for long wave radiation.
- *epsilon_soil_SWE*: plf [**pF** → **<none>**]
Parameter (has default const value 1 [])
Effect of soil water on epsilon_soil.
- *b_SSOC*: number [**mol/m²/s**]
Parameter (default 0.01)
Stomatal intercept, also defined in Stomatacon component.

Log Variables

- *LAI*: number [**m²/m²**]
Leaf area index.
- *s*: number [**Pa/K**]
Slope of water vapour pressure curve.
- *lambda*: number [**J/kg**]
Latent heat of vaporization in atmosphere.
- *rho_a*: number [**kg/m³**]
Air density.
- *gamma*: number [**Pa/K**]
Psychrometric constant.
- *T_s*: number [**K**]
Soil surface temperature.
- *T_0*: number [**K**]
Surface temperature (large scale).
- *T_c*: number [**K**]
Canopy-point temperature.
- *T_sun*: number [**K**]
Temperature of sunlit leaves.
- *T_shadow*: number [**K**]
Temperature of shadow leaves.
- *g_a*: number [**m/s**]
Heat conductance in the atmosphere - from canopy point to reference height (screen height).

- *g_H_s_c*: number [**m/s**]
Heat conductance from soil surface to canopy point.
- *g_H_sun_c*: number [**m/s**]
Heat conductance from sunlit leaves to canopy point.
- *gb_W_sun*: number [**m/s**]
Water conductance for sunlit leaves boundary layer.
- *g_W_sun_c*: number [**m/s**]
Water conductance from sunlit leaves to canopy point.
- *G_W_sun_c*: number [**W/m²/K**]
Scaled water conductance from sunlit leaves to canopy point.
- *g_H_shadow_c*: number [**m/s**]
Heat conductance from shadow leaves to canopy point.
- *gb_W_shadow*: number [**m/s**]
Water conductance for shadow leaves boundary layer.
- *g_W_shadow_c*: number [**m/s**]
Water conductance from shadow leaves to canopy point.
- *e_a*: number [**Pa**]
Vapour pressure of water in the atmosphere.
- *e_sat_air*: number [**Pa**]
Saturated vapour pressure of water in the air.
- *e_c*: number [**Pa**]
Vapour pressure of water in the canopy.
- *R_abs_soil*: number [**W/m²**]
Absorbed radiation in soil.
- *R_eq_abs_soil*: number [**W/m²**]
Absorbed radiation in soil at equilibrium.
- *R_abs_sun*: number [**W/m²**]
Absorbed radiation in sunlit leaves.
- *R_eq_abs_sun*: number [**W/m²**]
Absorbed radiation in sunlit leaves at equilibrium.
- *R_abs_shadow*: number [**W/m²**]
Absorbed radiation in shadow leaves.
- *R_eq_abs_shadow*: number [**W/m²**]
Absorbed radiation in shadow leaves at equilibrium.
- *sun_LAI_fraction_total*: number []
Sunlit fraction of leaf area in the canopy.
- *cover*: number []
Vegetation cover.
- *H_soil*: number [**W/m²**]
Sensible heat flux from the soil.

- *H_sun*: number [W/m^2]
Sensible heat flux from the sunlit leaves to the canopy point.
- *H_shadow*: number [W/m^2]
Sensible heat flux from the shadow leaves to canopy point.
- *H_c.a*: number [W/m^2]
Sensible heat flux from the canopy point to free atmosphere.
- *LE_sun*: number [W/m^2]
Latent heat flux from the sunlit leaves to the canopy point.
- *LE_shadow*: number [W/m^2]
Latent heat flux from the shadow leaves to the canopy point.
- *LE_atm*: number [W/m^2]
Latent heat flux from the canopy point to the free atmosphere.
- *E_trans*: number [mm/h]
Leaf transpiration.
- *G_R_soil*: number [$\text{W}/\text{m}^2/\text{K}$]
Radiation 'conductivity' from the soil.
- *G_R_sun*: number [$\text{W}/\text{m}^2/\text{K}$]
Radiation 'conductivity' from sunlit leaves.
- *G_R_shadow*: number [$\text{W}/\text{m}^2/\text{K}$]
Radiation 'conductivity' from shadow leaves.
- *G_R_leaf*: number [$\text{W}/\text{m}^2/\text{K}$]
Radiation 'conductivity' from the leaves at night.
- *G_H.a*: number [$\text{W}/\text{m}^2/\text{K}$]
Heat 'conductivity' from soil to free atmosphere.
- *k.h*: number [$\text{W}/\text{m}/\text{K}$]
Heat conductivity in soil.
- *T_z0*: number [K]
Soil temperature in top cell.
- *E_soil*: number [$\text{kg}/\text{m}^2/\text{s}$]
Evaporation of water from soil.
- *R_soil*: number [W/m^2]
Net longwave radiation from soil.
- *R_sun*: number [W/m^2]
Net longwave radiation from sunlit leaves.
- *R_shadow*: number [W/m^2]
Net longwave radiation from shadow leaves.
- *R_total*: number [W/m^2]
Net longwave radiation from soil and canopy.

88.3 none

Decoupling surface energy and water balance. Done using reference values and crop factors.

Used by bioclimate default svat (see 19.2, page 115) .

Chapter 89

tertiary

Transport of water and solute outside the matrix.

Used by movement component Tertiary (see 56, page 307) .

89.1 biopores

Tertiary domain divided into biopore classes.

```
< biopores  (active active ...)
              (pressure_initiate -3 [cm])
              (pressure_end -30 [cm])
              (pond_max 0.05 [cm])
              (classes classes ...)
              (pressure_limit pressure_limit)
              (pressure_barrier 5 [cm])
              (active_msg range)
              (old_dt 0 [cm])
              (deactivate_steps 3) >
```

- *active*: boolean (see section 4.1.2) soil cells
Optional state variable
Active biopores in cells.
- *pressure_initiate*: number [**cm**]
Parameter (default -3)
Pressure needed to activate biopore flow.
- *pressure_end*: number [**cm**]
Parameter (default -30)
Pressure below which biopore flow is deactivated.
- *pond_max*: number [**cm**]
Parameter (default 0.05)
Maximum height of ponding before spilling into biopores. After macropores are activated pond will have this height.
- *classes*: **biopore** component (see chapter 20) sequence
List of biopore classes.
- *pressure_limit*: number [**cm**]
Optional parameter
Limit to pressure difference for moving matrix water gradient to biopores.

The idea is that the water is extracted from the matrix by a hanging water column in the biopore, and that the suction is equal to the height of this water column. The pressure limit is then the maximal length of the column, or the point where the column breaks.

By default, this is equal to 'pressure_end'.

- *pressure_barrier*: number [**cm**]
Parameter (default 5)
Pressure barrier between matrix and biopore domain. If the pressure difference between the matrix and biopores is below this value, no water will transfer between the domains. If you specify a too small value for this parameter, the solution may be unstable.
- *active_msg*: string (see section 4.1.5)
Parameter (default 'range')
Control biopore activation and deactivation reports.
Possible values: cell: Report for each cell. range: Report for vertical range. none: No reports.
- *old_dt*: number [**cm**]
State variable (default 0)
Lowest dt imposed by weather during current event. In case of an event, the biopore module will suggest to keep a timestep that corresponds to the lowest suggestion from the weather module during the event.
- *deactivate_steps*: integer
State variable (default 3)
No matrix exchange for this number of timesteps. Automatically set when matrix pressure is in a disarray, such as after tillage operations, or calls to reserve models.

Log Variables

- *water_volume*: number [**cm**³]
Water volume.
- *water_height*: number [**cm**]
Water volume multiplied with surface area.
- *solute_mass*: submodel (see section 4.1.7) sequence
Total amount of solutes in biopores.

< <i>name</i> <i>value</i> >	Log Variables
– <i>value</i> : number [g]	Value for chemical.
– <i>name</i> : string (see section 4.1.5)	Name of chemical.
- *solute_storage*: submodel (see section 4.1.7) sequence
Total amount of solutes in biopores divided by surface area.

< <i>name</i> <i>value</i> >	Log Variables
– <i>value</i> : number [g/cm ²]	Value for chemical.

- *name*: string (see section 4.1.5)
Name of chemical.
- *ddt*: number [**h**]
Emulated timestep. Timestep scaled for available water.

89.2 none

No tertiary transport.

Used by movement rectangle Tertiary (see 56.2, page 307) .

89.3 old

Tertiary water and solute movement based on the obsolete 'macro' and 'mactrans' components. Provided for backward compatibility.

Used by movement vertical Tertiary (see 56.3, page 308) .

```
< old  (macro macro)
      (mactrans mactrans)  ; Default default value. >
```

- *macro*: **macro** component (see chapter 53)
Optional component
Preferential flow model. By default, preferential flow is enabled if and only if the combined amount of humus and clay in the top horizon is above 5%.
- *mactrans*: **mactrans** component (see chapter 54)
Component (default 'default')
Solute transport model in macropores.

Chapter 90

tortuosity

Solutes in the soil can't move the shortest way between two points. The tortuosity factor indicates how far the average solute have moved in absolute coordinates, when it has moved a given distance along the curved line. This component is responsible for calculating the soils tortuosity factor.

Used by horizon component tortuosity (see 47, page 231) .

90.1 M_Q

Millington-Quirk. $\Theta^{7/3} / \Theta_{sat}^2$.

90.2 linear

Linear Impedance factor. $a + b \Theta$. See also [Nye and Tinker, 1977]

Used by horizon component tortuosity (see 47, page 231) .

```
< linear (a a)
      (b 2 []) >
```

- a : number [cm^3/cm^3]
Optional parameter
Theta offset. By default, this corresponds to the wilting point.
- b : number (dimensionless)
Parameter (default 2)
Theta factor.

Chapter 91

transform

Generic transformations between soil components.

91.1 equilibrium

Two soil components reaching for equilibrium.

```
< equilibrium (equilibrium equilibrium)  
              (k_AB k_AB)  
              (k_BA k_BA) >
```

- *equilibrium*: **equilibrium** component (see chapter 39)
Function for calculating equilibrium between A and B.
- *k_AB*: **number** component (see chapter 59)
Transformation rate from soil component 'A' to 'B' [h⁻¹].
- *k_BA*: **number** component (see chapter 59)
Optional component
Transformation rate from soil component 'B' to 'A' [h⁻¹]. By default, this is identical to 'k_AB'.

Chapter 92

transport

Solute transport in primary domain.

92.1 Hansen

Solute transport using convection-dispersion.

Used by movement vertical matrix_solute (see 56.3, page 308) .

92.2 Mollerup

Coupled vertical and horizontal transport. See Mollerup 2007 for details.

Used by movement rectangle matrix_solute (see 56.2, page 307) .

```
< Mollerup (solver solver) ; Default cxsparse value.
      (debug 0)
      (stabilizing_method Timestep_reduction)
      (enable_boundary_diffusion true)
      (upstream_weight 1 [<fraction>]) >
```

- *solver*: **solver** component (see chapter 82)
Component (default 'cxsparse')
Model used for solving matrix equation system.
- *debug*: integer
Parameter (default 0)
Enable additional debug message. A value of 0 means no message, higher numbers means more messages.
- *stabilizing_method*: string (see section 4.1.5)
Optional parameter (default 'Timestep_reduction')
Method used for stabilizing results. Must be one of these:
None: No stabilizing method applied
Timestep_reduction: Reduce timesteps.
Streamline_diffusion: Increase diffusion.
- *enable_boundary_diffusion*: boolean (see section 4.1.2)
Parameter (default true)
If this is set, diffusion over boundaries is enabled.
- *upstream_weight*: number [<**fraction**>]
Parameter (default 1)
Upstream weighting factor: 1 = full upstream formulation, 0.5 = equal weight.

92.3 convection

Pure forward calculation of flow except through upper boundary. $J[\text{edge}] = q[\text{edge}] * C_{\text{old}}[\text{upstream}]$

Used by movement vertical matrix_solute (see 56.3, page 308) , and movement rectangle matrix_solute (see 56.2, page 307) .

92.4 none

Disable all transport except through boundaries.

Used by movement solute matrix_solid (see 56.1, page 307) , movement vertical matrix_solute (see 56.3, page 308) , and movement rectangle matrix_solute (see 56.2, page 307) .

Chapter 93

ui

Top level user interface.

Used by Toplevel @ ui (see 107.30, page 624) .

93.1 none

No user unterface.

This is useful when running from a batch program, or as a component in a larger system.

93.2 progress

Write progress on standard output (or standard error).

This is useful when starting the program from a text terminal, or from inside another program such as an editor that can capture the output.

Chapter 94

uifilter

Presentation of data in the user interface.

94.1 base

Base filter for the user interface.

94.2 raw

A ‘base’ model (see 94.1, page 491) build into Daisy.
Raw filter for the user interface.

94.3 simple

A ‘base’ model (see 94.1, page 491) build into Daisy.
Simple filter for the user interface.

```
< simple (default default)  
  (all all ...) >
```

- *default*: string (see section 4.1.5)
Parameter
Name of default component.
- *all*: submodel (see section 4.1.7) sequence
List of components to support.

```
< name all ... >
```

- *name*: string (see section 4.1.5)
Parameter
Name of this component.
- *all*: submodel (see section 4.1.7) sequence
List of models to support.

```
< name all ... >
```

- * *name*: string (see section 4.1.5)
Parameter
Name of this model.
- * *all*: string (see section 4.1.5) sequence
Parameter
List of parameters to support.

Chapter 95

uiitem

User interface information about a item.

< component (name *name*) >

- *name*: string (see section 4.1.5)
Parameter
Name of user interface unit.

95.1 raw

Raw item for the user interface.

Chapter 96

unit

Specify units of physical quantities.

Daisy will recognise both the build-in and user defined units, and convert between different units representing the same physical dimension. This is done by converting to and from the base (usually defined by SI) unit for that dimension.

96.1 SI

Base parameterization for all SI based units.

```
< SI  (time 0)
      (length 0)
      (mass 0)
      (electric_current 0)
      (thermodynamic_temperature 0)
      (amount_of_substance 0)
      (luminous_intensity 0) >
```

- *time*: integer
Parameter (default 0)
Dimension, base unit [s].
- *length*: integer
Parameter (default 0)
Dimension, base unit [m].
- *mass*: integer
Parameter (default 0)
Dimension, base unit [kg].
- *electric_current*: integer
Parameter (default 0)
Dimension, base unit [A].
- *thermodynamic_temperature*: integer
Parameter (default 0)
Dimension, base unit [K].
- *amount_of_substance*: integer
Parameter (default 0)
Dimension, base unit [mol].

- *luminous_intensity*: integer
Parameter (default 0)
Dimension, base unit [cd].

96.2 SIfactor

A ‘SI’ model (see 96.1, page 495) build into Daisy.

Convert to SI base units by multiplying with a factor.

```
< SIfactor (factor factor)
    ;; Shared parameters are described in section 96.1.
    (time 0)
    (length 0)
    (mass 0)
    (electric_current 0)
    (thermodynamic_temperature 0)
    (amount_of_substance 0)
    (luminous_intensity 0) >
```

- *factor*: number (dimensionless)
Parameter
Factor to multiply with to get base unit.

96.3

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Unitless.

```
< (factor 1 []) >
```

96.4 %

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Percent.

```
< (factor 0.01 []) >
```

96.5 <fraction>

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Unitless.

```
< (factor 1 []) >
```

96.6 <none>

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Unitless.

```
< (factor 1 []) >
```

96.7 A

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Ampere.


```
< (electric_current 1)
  (factor 1 []) >
```

96.8 J

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Joule.

```
< (time -2)
  (length 2)
  (mass 1)
  (factor 1 []) >
```

96.9 J/cm²/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Joule per square centimeter per hour.

```
< (time -3)
  (mass 1)
  (factor 2.77778 []) >
```

96.10 J/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Joule per square meter.

```
< (time -2)
  (length 2)
  (mass 1)
  (factor 1 []) >
```

96.11 K

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Kelvin.

```
< (thermodynamic_temperature 1)
  (factor 1 []) >
```

96.12 K⁻¹

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Per kelvin.

```
< (thermodynamic_temperature 1)
  (factor -1 []) >
```

96.13 K⁻²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Per kelvin squared.

```
< (thermodynamic_temperature 1)
  (factor -2 []) >
```

96.14 L/g

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 1 []) >
```

96.15 L/kg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 0.001 []) >
```

96.16 L/mg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 1000 []) >
```

96.17 MJ/d/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Megajoule per day per square meter.

```
< (time -3)
  (mass 1)
  (factor 11.5741 []) >
```

96.18 MJ/h/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Megajoule per hour per square meter.

```
< (time -3)
  (mass 1)
  (factor 277.778 []) >
```

96.19 MJ/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Megajoule per square meter.

```
< (time -2)
  (length 2)
  (mass 1)
  (factor 1e+06 []) >
```

96.20 MJ/m²/d

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Megajoule per square meter per day.

```
< (time -3)
  (mass 1)
  (factor 11.5741 []) >
```

96.21 MJ/m²/h

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Megajoule per square meter per hour.

```
< (time -3)
  (mass 1)
  (factor 277.778 []) >
```

96.22 MPa

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Megapascal.

```
< (time -2)
  (length -1)
  (mass 1)
  (factor 1e+06 []) >
```

96.23 MPa⁻¹

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Inverse pressure.

```
< (time 2)
  (length 1)
  (mass -1)
  (factor 1e-06 []) >
```

96.24 Mg DM/ha

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Ton dry matter per hectar.

```
< (length -2)
  (mass 1)
  (factor 0.1 []) >
```

96.25 Mg DM/ha/h

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Harvest and fertilizing.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-05 []) >
```

96.26 Mg w.w./ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Wet weight per area.

```
< (length -2)
    (mass 1)
    (factor 0.1 []) >
```

96.27 Mg/ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Ton per hectar.

```
< (length -2)
    (mass 1)
    (factor 0.1 []) >
```

96.28 Pa

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Pascal.

```
< (time -2)
    (length -1)
    (mass 1)
    (factor 1 []) >
```

96.29 Pa⁻¹

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base inverse pressure.

```
< (time 2)
    (length 1)
    (mass -1)
    (factor 1 []) >
```

96.30 T w.w./ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Wet weight per area.

```
< (length -2)
    (mass 1)
    (factor 0.1 []) >
```

96.31 W

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Watt.

```
< (time -3)
    (length 2)
    (mass 1)
    (factor 1 []) >
```

96.32 $W m^{-2}$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Watt per square meter.

```
< (time -3)
  (mass 1)
  (factor 1 []) >
```

96.33 W/cm^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Watt per square centimeter.

```
< (time -3)
  (mass 1)
  (factor 10000 []) >
```

96.34 W/m^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Watt per square meter.

```
< (time -3)
  (mass 1)
  (factor 1 []) >
```

96.35 Wh/m^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Watthours per square meter.

```
< (time -2)
  (length 2)
  (mass 1)
  (factor 3600 []) >
```

96.36 ag/L

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-18 []) >
```

96.37 cd

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Candela.

```
< (luminous_intensity 1)
  (factor 1 []) >
```

96.38 cm

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Centimeter.

```
< (length 1)
    (factor 0.01 []) >
```

96.39 cm R/cm³

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Root length per cubic centimeter.

```
< (length -2)
    (factor 10000 []) >
```

96.40 cm/d

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Soil water movement.

```
< (time -1)
    (length 1)
    (factor 1.15741e-07 []) >
```

96.41 cm/dg C/d

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Root growth.

```
< (time -1)
    (length 1)
    (thermodynamic_temperature -1)
    (factor 1.15741e-07 []) >
```

96.42 cm/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Soil water movement.

```
< (time -1)
    (length 1)
    (factor 2.77778e-06 []) >
```

96.43 cm⁻¹

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Per centimeter.

```
< (length -1)
    (factor 100 []) >
```

96.44 cm⁻²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Per square centimeter.

```
< (length -2)
    (factor 10000 []) >
```

96.45 cm^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Square centimeter.

```
< (length 2)
  (factor 0.0001 []) >
```

96.46 cm^2/h

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Square centimeters per hour.

```
< (time -1)
  (length 2)
  (factor 2.77778e-08 []) >
```

96.47 cm^2/s

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Square centimeters per second.

```
< (time -1)
  (length 2)
  (factor 0.0001 []) >
```

96.48 cm^3

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Cube centimeter.

```
< (length 3)
  (factor 1e-06 []) >
```

96.49 $cm^3 H_2O/cm^3$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Soil water fraction.

```
< (factor 1 []) >
```

96.50 cm^3/cm^3

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Soil water fraction.

```
< (factor 1 []) >
```

96.51 cm^3/g

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 0.001 []) >
```

96.52 cm^3/ng

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 1e+06 []) >
```

96.53 d

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Day.

```
< (time 1)
  (factor 86400 []) >
```

96.54 d^{-1}

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Day.

```
< (time -1)
  (factor 1.15741e-05 []) >
```

96.55 $\text{erg}/\text{cm}^2/\text{h}$

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

erg per square centimer per hour.

```
< (time -3)
  (mass 1)
  (factor 2.778e-07 []) >
```

96.56 fg/L

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-15 []) >
```

96.57 fraction

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Unitless.

```
< (factor 1 []) >
```

96.58 g

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Gram.

```
< (mass 1)
  (factor 0.001 []) >
```


96.59 g C/L

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 1 []) >
```

96.60 g C/cm²

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Soil scale carbon per area.

```
< (length -2)
  (mass 1)
  (factor 10 []) >
```

96.61 g C/cm³

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Carbon concentration.

```
< (length -3)
  (mass 1)
  (factor 1000 []) >
```

96.62 g C/m²

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Crop scale carbon per area.

```
< (length -2)
  (mass 1)
  (factor 0.001 []) >
```

96.63 g DM/cm²

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Gram dry matter per square centimeter.

```
< (length -2)
  (mass 1)
  (factor 10 []) >
```

96.64 g DM/ha

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Field scale crop mass per area.

```
< (length -2)
  (mass 1)
  (factor 1e-07 []) >
```

96.65 g DM/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Crop scale dry matter per area.

```
< (length -2)
    (mass 1)
    (factor 0.001 []) >
```

96.66 g N/cm²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Soil scale nitrogen per area.

```
< (length -2)
    (mass 1)
    (factor 10 []) >
```

96.67 g N/cm²/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Nitrogen per aquare centimeter per hour.

```
< (time -1)
    (length -2)
    (mass 1)
    (factor 0.00277778 []) >
```

96.68 g N/cm³

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Nitrogen concentration.

```
< (length -3)
    (mass 1)
    (factor 1000 []) >
```

96.69 g N/l

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Nitrogen concentration.

```
< (length -3)
    (mass 1)
    (factor 1 []) >
```

96.70 g N/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Crop scale nitrogen per area.

```
< (length -2)
    (mass 1)
    (factor 0.001 []) >
```

96.71 $g N/m^2/h$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Nitrogen per aquare meter per hour.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-07 []) >
```

96.72 $g w.w./m^2$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Wet weight per area.

```
< (length -2)
  (mass 1)
  (factor 0.001 []) >
```

96.73 g/L

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 1 []) >
```

96.74 $g/cm R/h$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Gram per centimeter root per hour.

```
< (time -1)
  (length -1)
  (mass 1)
  (factor 2.77778e-05 []) >
```

96.75 $g/cm/h$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Mass per length flux.

```
< (time -1)
  (length -1)
  (mass 1)
  (factor 2.77778e-05 []) >
```

96.76 g/cm^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Soil scale mass per area.

```
< (length -2)
  (mass 1)
  (factor 10 []) >
```

96.77 $\text{g}/\text{cm}^2/\text{h}$

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Mass per area flux.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 0.00277778 []) >
```

96.78 $\text{g}/\text{cm}^2/\text{mm}$

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Irrigation and percolation concentration.

```
< (length -3)
  (mass 1)
  (factor 10000 []) >
```

96.79 g/cm^3

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Solute concentration.

```
< (length -3)
  (mass 1)
  (factor 1000 []) >
```

96.80 $\text{g}/\text{cm}^3/\text{h}$

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Gram per cubic centimeter per hour.

```
< (time -1)
  (length -3)
  (mass 1)
  (factor 0.277778 []) >
```

96.81 g/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Gram per hour.

```
< (time -1)
  (mass 1)
  (factor 2.77778e-07 []) >
```

96.82 g/ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Field scale pesticide mass per area.

```
< (length -2)
  (mass 1)
  (factor 1e-07 []) >
```

96.83 $g/ha/d$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Pesticide application.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 1.15741e-12 []) >
```

96.84 $g/ha/h$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Pesticide application.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-11 []) >
```

96.85 $g/ha/mm$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

gram per hectare per millimeter.

```
< (length -3)
  (mass 1)
  (factor 0.0001 []) >
```

96.86 g/kg

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Gram per kilogram.

```
< (factor 0.001 []) >
```

96.87 g/l

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 1 []) >
```

96.88 g/m^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Crop scale mass per area.

```
< (length -2)
  (mass 1)
  (factor 0.001 []) >
```

96.89 $\text{g/m}^2/\text{h}$

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Gram per square meter per hour.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-07 []) >
```

96.90 g/mol

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Gram per mole.

```
< (mass 1)
  (amount_of_substance -1)
  (factor 0.001 []) >
```

96.91 h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Hour.

```
< (time 1)
  (factor 3600 []) >
```

96.92 hPa

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Hectopascal.

```
< (time -2)
  (length -1)
  (mass 1)
  (factor 100 []) >
```

96.93 h^{-1}

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Hour.

```
< (time -1)
  (factor 0.000277778 []) >
```

96.94 ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Hectare.

```
< (length 2)
  (factor 10000 []) >
```

96.95 hours

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Hour.

```
< (time 1)
  (factor 3600 []) >
```

96.96 kJ/m^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Kilojoule per square meter.

```
< (time -2)
  (length 2)
  (mass 1)
  (factor 1000 []) >
```

96.97 kPa

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Kilopascal.

```
< (time -2)
  (length -1)
  (mass 1)
  (factor 1000 []) >
```

96.98 kWh/m^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Kilowatthours per square meter.

```
< (time -2)
  (length 2)
  (mass 1)
  (factor 3.6e+06 []) >
```

96.99 kg

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Kilogram.

```
< (mass 1)
  (factor 1 []) >
```

96.100 $kg\ C/ha$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Field scale carbon per area.

```
< (length -2)
  (mass 1)
  (factor 0.0001 []) >
```

96.101 $kg\ C/ha/d$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Field scale carbon.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 1.15741e-09 []) >
```

96.102 kg C/ha/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Field scale application and removal of carbon.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-08 []) >
```

96.103 kg DM/ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Kilogram dry matter per hectar.

```
< (length -2)
  (mass 1)
  (factor 0.0001 []) >
```

96.104 kg DM/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per area.

```
< (length -2)
  (mass 1)
  (factor 1 []) >
```

96.105 kg DM/m³

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Mass per volume.

```
< (length -3)
  (mass 1)
  (factor 1 []) >
```

96.106 kg N/ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Field scale nitrogen per area.

```
< (length -2)
  (mass 1)
  (factor 0.0001 []) >
```

96.107 kg N/ha/d

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Field scale nitrogen.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 1.15741e-09 []) >
```


96.108 kg N/ha/h

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Field scale application and removal of nitrogen.

```
<  (time -1)
    (length -2)
    (mass 1)
    (factor 2.77778e-08 []) >
```

96.109 kg N/ha/mm

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Irrigation and percolation concentration.

```
<  (length -3)
    (mass 1)
    (factor 0.1 []) >
```

96.110 kg N/ha/y

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Deposition.

```
<  (time -1)
    (length -2)
    (mass 1)
    (factor 3.16887e-12 []) >
```

96.111 kg m⁻² s⁻¹

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Base mass per area flux.

```
<  (time -1)
    (length -2)
    (mass 1)
    (factor 1 []) >
```

96.112 kg w.w./ha

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Wet weight per area.

```
<  (length -2)
    (mass 1)
    (factor 0.0001 []) >
```

96.113 kg/Pa

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Base mass per pressure.

```
<  (time 2)
    (length 1)
    (factor 1 []) >
```

96.114 kg/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Kilogram per hour.

```
< (time -1)
  (mass 1)
  (factor 0.000277778 []) >
```

96.115 kg/ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Field scale mass per area.

```
< (length -2)
  (mass 1)
  (factor 0.0001 []) >
```

96.116 kg/ha/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Harvest and fertilizing.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-08 []) >
```

96.117 kg/ha/mm

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Irrigation and percolation concentration.

```
< (length -3)
  (mass 1)
  (factor 0.1 []) >
```

96.118 kg/ha/y

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Deposition.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 3.16887e-12 []) >
```

96.119 kg/m

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per length.

```
< (length -1)
  (mass 1)
  (factor 1 []) >
```

96.120 kg/m/s

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per length flux.

```
< (time -1)
  (length -1)
  (mass 1)
  (factor 1 []) >
```

96.121 kg/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per area.

```
< (length -2)
  (mass 1)
  (factor 1 []) >
```

96.122 kg/m²/s

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per area flux.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 1 []) >
```

96.123 kg/m³

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per volume.

```
< (length -3)
  (mass 1)
  (factor 1 []) >
```

96.124 kg/mol

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Kilogram per mole.

```
< (mass 1)
  (amount_of_substance -1)
  (factor 1 []) >
```

96.125 kg/s

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per time.

```
< (time -1)
  (mass 1)
  (factor 1 []) >
```

96.126 kgN/ha/year

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Deposition.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 3.16887e-12 []) >
```

96.127 l/g

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 1 []) >
```

96.128 l/ha

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Liter per hectar.

```
< (length 1)
  (factor 1e-07 []) >
```

96.129 l/ha/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Liter per hectar per hour.

```
< (time -1)
  (length 1)
  (factor 2.77778e-11 []) >
```

96.130 l/kg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 0.001 []) >
```

96.131 l/mg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
  (mass -1)
  (factor 1000 []) >
```

96.132 *m*

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Meter.

```
< (length 1)
   (factor 1 []) >
```

96.133 *m s⁻¹*

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Base speed.

```
< (time -1)
   (length 1)
   (factor 1 []) >
```

96.134 *m/s*

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Base speed.

```
< (time -1)
   (length 1)
   (factor 1 []) >
```

96.135 *mL/g*

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
   (mass -1)
   (factor 0.001 []) >
```

96.136 *mL/kg*

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```
< (length 3)
   (mass -1)
   (factor 1e-06 []) >
```

96.137 *m⁻¹*

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Per meter.

```
< (length -1)
   (factor 1 []) >
```

96.138 *m⁻²*

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Per square meter.

```
< (length -2)
   (factor 1 []) >
```

96.139 $\text{m}^{-2} \text{ kg s}^{-1}$

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base mass per area flux.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 1 []) >
```

96.140 m^2

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Square meter.

```
< (length 2)
  (factor 1 []) >
```

96.141 m^2/s

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base area per time unit.

```
< (time -1)
  (length 2)
  (factor 1 []) >
```

96.142 m^3

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Cube meter.

```
< (length 3)
  (factor 1 []) >
```

96.143 m^3/cm^3

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Kilo.

```
< (factor 1e+06 []) >
```

96.144 m^3/kg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Base volume per mass.

```
< (length 3)
  (mass -1)
  (factor 1 []) >
```

96.145 mg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Miligram.

```
< (mass 1)
  (factor 1e-06 []) >
```

96.146 mg N/kg dry soil

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Nitrogen concentration in dry soil.

```
< (factor 1e-06 []) >
```

96.147 mg N/l

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Nitrogen concentration.

```
< (length -3)
  (mass 1)
  (factor 0.001 []) >
```

96.148 mg/L

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 0.001 []) >
```

96.149 mg/g

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Milligram per gram.

```
< (factor 0.001 []) >
```

96.150 mg/ha

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Miligram per hectare

```
< (length -2)
  (mass 1)
  (factor 1e-10 []) >
```

96.151 mg/ha/d

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Pesticide application.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 1.15741e-15 []) >
```

96.152 mg/ha/h

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Pesticide application.

```

< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-14 []) >

```

96.153 mg/l

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```

< (length -3)
  (mass 1)
  (factor 0.001 []) >

```

96.154 mg/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Crop scale pesticide per area.

```

< (length -2)
  (mass 1)
  (factor 1e-06 []) >

```

96.155 minutes

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Minute.

```

< (time 1)
  (factor 60 []) >

```

96.156 ml/g

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```

< (length 3)
  (mass -1)
  (factor 0.001 []) >

```

96.157 ml/kg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Volume per mass.

```

< (length 3)
  (mass -1)
  (factor 1e-06 []) >

```

96.158 mm

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Millimeter.

```

< (length 1)
  (factor 0.001 []) >

```


96.159 mm/d

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Percolation intensity.

```
< (time -1)
  (length 1)
  (factor 1.15741e-08 []) >
```

96.160 mm/dg C/d

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Root growth.

```
< (time -1)
  (length 1)
  (thermodynamic_temperature -1)
  (factor 1.15741e-08 []) >
```

96.161 mm/h

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Percolation intensity.

```
< (time -1)
  (length 1)
  (factor 2.77778e-07 []) >
```

96.162 mm/s

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Percolation intensity.

```
< (time -1)
  (length 1)
  (factor 0.001 []) >
```

96.163 mm⁻¹

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Per millimeter.

```
< (length -1)
  (factor 100 []) >
```

96.164 mmol/cm³

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Millimole per cubic centimeter.

```
< (length -3)
  (amount_of_substance 1)
  (factor 1000 []) >
```

96.165 mmol/kg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Millimole per kilogram.

```
< (mass -1)
  (amount_of_substance 1)
  (factor 0.001 []) >
```

96.166 mmol/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Millimole per square meter.

```
< (length -2)
  (amount_of_substance 1)
  (factor 0.001 []) >
```

96.167 mmol/m²/s

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Millimole per square meter per second.

```
< (time -1)
  (length -2)
  (amount_of_substance 1)
  (factor 0.001 []) >
```

96.168 mol

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Mole.

```
< (amount_of_substance 1)
  (factor 1 []) >
```

96.169 mol/cm³

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Mole per cubic centimeter.

```
< (length -3)
  (amount_of_substance 1)
  (factor 1e+06 []) >
```

96.170 mol/kg

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Mole per kilogram.

```
< (mass -1)
  (amount_of_substance 1)
  (factor 1 []) >
```

96.171 mol/m^2

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Mole per square meter.

```
< (length -2)
  (amount_of_substance 1)
  (factor 1 []) >
```

96.172 $mol/m^2/s$

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Mole per square meter per second.

```
< (time -1)
  (length -2)
  (amount_of_substance 1)
  (factor 1 []) >
```

96.173 mol/m^3

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Mole per cubic meter.

```
< (length -3)
  (amount_of_substance 1)
  (factor 1 []) >
```

96.174 ng/L

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-09 []) >
```

96.175 ng/MPa

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Nanogram per megapascal.

```
< (time 2)
  (length 1)
  (factor 1e-18 []) >
```

96.176 ng/cm^3

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Low solute concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-06 []) >
```

96.177 ng/kPa

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Nanogram per kilopascal.

```
< (time 2)
  (length 1)
  (factor 1e-15 []) >
```

96.178 ng/l

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Lower solute concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-09 []) >
```

96.179 ng/mm/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Mass per length flux.

```
< (time -1)
  (length -1)
  (mass 1)
  (factor 2.77778e-13 []) >
```

96.180 none

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Unitless.

```
< (factor 1 []) >
```

96.181 pg/L

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-12 []) >
```

96.182 ppm

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Part per million.

```
< (factor 1e-06 []) >
```

96.183 s

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.
Second.

```
< (time 1)
  (factor 1 []) >
```

96.184 s^{-1}

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Second.

```
< (time -1)
  (factor 1 []) >
```

96.185 seconds

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Second.

```
< (time 1)
  (factor 1 []) >
```

96.186 t/ha

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Ton per hectare.

```
< (length -2)
  (mass 1)
  (factor 0.1 []) >
```

96.187 ug

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Microgram.

```
< (mass 1)
  (factor 1e-09 []) >
```

96.188 ug/L

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-06 []) >
```

96.189 ug/L/MPa

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Microgram per liter per megapascal.

```
< (time 2)
  (length -2)
  (factor 1e-09 []) >
```

96.190 ug/ha

A ‘Sifactor’ model (see 96.2, page 496) build into Daisy.

Microgram per hectare

```
< (length -2)
  (mass 1)
  (factor 1e-13 []) >
```

96.191 ug/ha/d

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Pesticide application.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 1.15741e-18 []) >
```

96.192 ug/ha/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Pesticide application.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-17 []) >
```

96.193 ug/l

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Low solute concentration.

```
< (length -3)
  (mass 1)
  (factor 1e-06 []) >
```

96.194 ug/m²

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Microgram per square meter.

```
< (length -2)
  (mass 1)
  (factor 1e-09 []) >
```

96.195 ug/m²/h

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Microgram per square meter per hour.

```
< (time -1)
  (length -2)
  (mass 1)
  (factor 2.77778e-13 []) >
```

96.196 ug/m³

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Mass per volume.

```
< (length -3)
  (mass 1)
  (factor 1e-09 []) >
```

96.197 um

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Micrometer.

```
< (length 1)
  (factor 1e-06 []) >
```

96.198 umol/cm³

A ‘SIfactor’ model (see 96.2, page 496) build into Daisy.

Micromole per cubic centimeter.

```
< (length -3)
  (amount_of_substance 1)
  (factor 1 []) >
```

96.199 base

A base unit.

96.200 <error>

A ‘base’ model (see 96.199, page 527) build into Daisy.

Bogus unit.

96.201 <unknown>

A ‘base’ model (see 96.199, page 527) build into Daisy.

Nothing is known about the dimension of this unit.

96.202 dgEast

A ‘base’ model (see 96.199, page 527) build into Daisy.

Degrees East of Greenwich.

96.203 dgNorth

A ‘base’ model (see 96.199, page 527) build into Daisy.

Degrees North of Equator.

96.204 dry soil fraction

A ‘base’ model (see 96.199, page 527) build into Daisy.

Fraction of dry soil.

96.205 rad

A ‘base’ model (see 96.199, page 527) build into Daisy.

Radians

96.206 factor

Convert to base units by multiplying with a factor.

```
< factor (base base)
      (factor factor) >
```

- *base*: string (see section 4.1.5)
Parameter
Base unit to convert to and from.
- *factor*: number (dimensionless)
Parameter
Factor to multiply with to get base unit.

96.207 dg

A ‘factor’ model (see 96.206, page 528) build into Daisy.

```
Degrees
< (base rad)
  (factor 0.0174533 []) >
```

96.208 dgSouth

A ‘factor’ model (see 96.206, page 528) build into Daisy.

```
Degrees North of Equator.
< (base dgNorth)
  (factor -1 []) >
```

96.209 dgWest

A ‘factor’ model (see 96.206, page 528) build into Daisy.

```
Degrees West of Greenwich.
< (base dgEast)
  (factor -1 []) >
```

96.210 new dg

A ‘factor’ model (see 96.206, page 528) build into Daisy.

```
New degrees
< (base rad)
  (factor 0.015708 []) >
```

96.211 ppm dry soil

A ‘factor’ model (see 96.206, page 528) build into Daisy.

```
Part per million in dry soil.
< (base "dry soil fraction")
  (factor 1e-06 []) >
```


96.212 offset

Convert to base units by multiplying factor, then subtracting offset.

```
< offset (base base)
      (factor 1 [])
      (offset 0 []) >
```

- *base*: string (see section 4.1.5)
Parameter
Base unit to convert to and from.
- *factor*: number (dimensionless)
Parameter (default 1)
Factor to multiply with to get base unit.
- *offset*: number (dimensionless)
Parameter (default 0)
Offset to add after multiplying with factor to get base unit.

96.213 dg C

A ‘offset’ model (see 96.212, page 529) build into Daisy.
degree Celcius.

```
< (base K)
  (offset 273.15 []) >
```

96.214 dg F

A ‘offset’ model (see 96.212, page 529) build into Daisy.
degree Fahrenheit.

```
< (base K)
  (factor 0.555556 [])
  (offset 255.372 []) >
```

96.215 pF

log10 (- cmH2O).

Chapter 97

uz1d

The 'uz1d' component handles the horizontal water movement in the unsaturated zone soil matrix.

97.1 none

Disable transport

Used by uzrect v+h horizontal (see 99.3, page 539) .

97.2 richards

A numerical solution to Richard's Equation.

```
< richards  (max_iterations 25)
              (max_absolute_difference 0.02 [cm])
              (max_relative_difference 0.001 [])
              (max_time_step_reductions 4)
              (time_step_reduction 4)
              (K_average K_average) ; Default arithmetic value. >
```

- *max_iterations*: integer
Parameter (default 25)
Maximum number of iterations when seeking convergence before reducing the time step.
- *max_absolute_difference*: number [cm]
Parameter (default 0.02)
Maximum absolute difference in 'h' values for convergence.
- *max_relative_difference*: number (dimensionless)
Parameter (default 0.001)
Maximum relative difference in 'h' values for convergence.
- *max_time_step_reductions*: integer
Parameter (default 4)
Number of times we may reduce the time step before giving up
- *time_step_reduction*: integer
Parameter (default 4)
Divide the time step with this at each reduction.

- *K_average*: **average** component (see chapter 18)
Optional component (default 'arithmetic')
Model for calculating average K between cells.

Chapter 98

uzmodel

The 'uzmodel' component handles the vertical water movement in the unsaturated zone soil matrix.

98.1 lr

Use gravitational water movement for wet soil, where $h > h_{fc}$. There are no water movement when $h < h_{fc}$, except at the layers down to z_{top} , where there can be Darcy movement.

Used by movement vertical matrix_water (see 56.3, page 308) .

```
< lr (overflow_warn true)
    (h_fc -100 [cm])
    (z_top -10 [cm]) >
```

- *overflow_warn*: boolean (see section 4.1.2)
Parameter (default true)
If true, warn the first time the soil profile is oversaturated.
- *h_fc*: number [cm]
Parameter (default -100)
Field capacity.
- *z_top*: number [cm]
Parameter (default -10)
Depth of layer where upward water movement is possible.

98.2 none

No water movement, and no sink.

Used by uzrect v+h vertical (see 99.3, page 539) .

98.3 richards

A numerical solution to Richard's Equation.

Used by movement vertical matrix_water (see 56.3, page 308) , and uzrect v+h vertical (see 99.3, page 539) .

```

< richards (debug 0)
          (max_iterations 25)
          (max_absolute_difference 0.02 [cm])
          (max_relative_difference 0.001 [])
          (max_time_step_reductions 16)
          (time_step_reduction 4)
          (max_number_of_small_time_steps 200000)
          (msg_number_of_small_time_steps 5000)
          (require_positive_convergence true)
          (K_average K_average) >

```

- *debug*: integer
Parameter (default 0)
Print additional debug messages, higher numbers means more messages.
- *max_iterations*: integer
Parameter (default 25)
Maximum number of iterations when seeking convergence before reducing the time step.
- *max_absolute_difference*: number [**cm**]
Parameter (default 0.02)
Maximum absolute difference in 'h' values for convergence.
- *max_relative_difference*: number (dimensionless)
Parameter (default 0.001)
Maximum relative difference in 'h' values for convergence.
- *max_time_step_reductions*: integer
Parameter (default 16)
Number of times we may reduce the time step before giving up
- *time_step_reduction*: integer
Parameter (default 4)
Divide the time step with this at each reduction.
- *max_number_of_small_time_steps*: integer
Parameter (default 200000)
Maximum number of small time steps in a large time step.
- *msg_number_of_small_time_steps*: integer
Parameter (default 5000)
Number of small time steps in a large time step between message.
- *require_positive_convergence*: boolean (see section 4.1.2)
Parameter (default true)
True if convergence criteria is also used for saturated cells. Setting this to false may speed up simulation, but give worse estimate of pressure and flow in saturated soil.
- *K_average*: **average** component (see chapter 18)
Optional component
Model for calculating average K between cells. The default model is 'geometric' if there are macropores, and 'arithmetic' otherwise.

98.4 richards2

A numerical solution to Richard's Equation.

```
< richards2 (debug 0)
            (max_iterations 10)
            (max_absolute_difference 0.02 [cm])
            (max_relative_difference 0.001 [])
            (max_number_of_small_time_steps 200000)
            (msg_number_of_small_time_steps 5000)
            (K_average K_average)
            (timestep_factor 0.7 [])
            (min_timestep 1e-10 [])
            (low_iterations 3)
            (high_iterations 4) >
```

- *debug*: integer
Parameter (default 0)
Print additional debug messages, higher numbers means more messages.
- *max_iterations*: integer
Parameter (default 10)
Above this, give up convergence.
- *max_absolute_difference*: number [cm]
Parameter (default 0.02)
Maximum absolute difference in 'h' values for convergence.
- *max_relative_difference*: number (dimensionless)
Parameter (default 0.001)
Maximum relative difference in 'h' values for convergence.
- *max_number_of_small_time_steps*: integer
Parameter (default 200000)
Maximum number of small time steps in a large time step.
- *msg_number_of_small_time_steps*: integer
Parameter (default 5000)
Number of small time steps in a large time step between message.
- *K_average*: **average** component (see chapter 18)
Optional component
Model for calculating average K between cells. The default model is 'geometric' if there are macropores, and 'arithmetic' otherwise.
- *timestep_factor*: number (dimensionless)
Parameter (default 0.7)
Multiply timestep with this if less than 'low_iterations' is used. Divide timestep with this if more than 'high_iterations' is used.
- *min_timestep*: number (dimensionless)
Parameter (default 1e-10)
Smallest timestep before giving up.
- *low_iterations*: integer
Parameter (default 3)
Below this, increase timestep.

- *high_iterations*: integer
Parameter (default 4)
Above this, decrease timestep.

Chapter 99

uzrect

2D water movement in a rectangular grid.

99.1 Mollerup

A finite volume solution to matrix water transport. See Mollerup 2007 for details.

Used by movement rectangle matrix_water (see 56.2, page 307) .

```
< Mollerup (solver solver) ; Default cxsparse value.
      (debug 0)
      (max_iterations 12)
      (max_absolute_difference 0.02 [cm])
      (max_relative_difference 0.001 [])
      (max_time_step_reductions 16)
      (time_step_reduction 4)
      (max_number_of_small_time_steps 20000)
      (msg_number_of_small_time_steps 100)
      (K_average K_average) ; Default arithmetic value.
      (max_iterations_timestep_reduction_factor 0)
      (max_pressure_potential 1e+09 [])
      (min_pressure_potential -1e+09 [])
      (forced_T forced.T) >
```

- *solver*: **solver** component (see chapter 82)
Component (default 'cxsparse')
Model used for solving matrix equation system.
- *debug*: integer
Parameter (default 0)
Level of debug messages:
= 0: no debug messages. > 0: Initial h and Theta per time step. > 1: Same, per iteration. = 3: Upper boundary extra info. = 4: Drain extra info. = 5: Remaining water.
- *max_iterations*: integer
Parameter (default 12)
Maximum number of iterations when seeking convergence before reducing the time step.
- *max_absolute_difference*: number [cm]
Parameter (default 0.02)
Maximum absolute difference in 'h' values for convergence.

- *max_relative_difference*: number (dimensionless)
Parameter (default 0.001)
Maximum relative difference in 'h' values for convergence.
- *max_time_step_reductions*: integer
Parameter (default 16)
Number of times we may reduce the time step before giving up
- *time_step_reduction*: integer
Parameter (default 4)
Divide the time step with this at each reduction.
- *max_number_of_small_time_steps*: integer
Parameter (default 20000)
Maximum number of small time steps in a large time step.
- *msg_number_of_small_time_steps*: integer
Parameter (default 100)
Number of small time steps in a large time step between message.
- *K_average*: **condedge** component (see chapter 27)
Component (default 'arithmetic')
Model for calculating average vertical K between cells.
- *max_iterations_timestep_reduction_factor*: integer
Parameter (default 0)
Multiply 'max.iterations' with this factor for each timestep reduction.
- *max_pressure_potential*: number (dimensionless)
Parameter (default 1e+09)
Maximum pressure potential for convergence.
- *min_pressure_potential*: number (dimensionless)
Parameter (default -1e+09)
minimum pressure potential for convergence.
- *forced_T*: number [**dg C**]
Optional parameter
Force transport equations to use this water temperature.

Log Variables

- *Theta_error*: number (dimensionless) soil cells
Water mass balance error per cell.
- *Kedge*: number [**cm/h**] soil edges
Conductivity between cells. The value logged is the value used for the last small timestep in the previous large timestep.

99.2 const

Steady-state water flow.

Used by movement rectangle matrix_water (see 56.2, page 307) .

```
< const (q_x 0 [cm/h])
      (q_z 0 [cm/h]) >
```

- q_x : number [**cm/h**]
Parameter (default 0)
Horizontal flow.
- q_z : number [**cm/h**]
Parameter (default 0)
Vertical flow upwards.

99.3 $v+h$

Transport water in the matrix in two phases, first vertical, then horizontal.

Used by movement rectangle `matrix_water` (see 56.2, page 307) .

```
< "v+h" (vertical vertical ...) ; Has default value.
      (horizontal horizontal ...) ; Has default value. >
```

- *vertical*: **uzmodel** component (see chapter 98) sequence
Component (has default value with length 2)

```
(vertical richards
      none)
```

Parameter description:

Vertical matrix water transport models. Each model will be tried in turn, until one succeeds. If none succeeds, the simulation ends.

- *horizontal*: **uz1d** component (see chapter 97) sequence
Component (has default value with length 1)

```
(horizontal none)
```

Parameter description:

Horizontal matrix water transport models. Each model will be tried in turn, until one succeeds. If none succeeds, the simulation ends.

Chapter 100

vegetation

That green stuff.

```
< component (EpInterchange 0.6 [<fraction>]) >
```

- *EpInterchange*: number [**<fraction>**]
Parameter (default 0.6)
Canopy adsorbtion fraction of unreached potential soil evaporation.

Log Variables

- *N*: number [**kg N/ha**]
Current nitrogen content of vegetation.
- *LAI*: number [**m²/m²**]
Total LAI of all crops on this column
- *LAIvsH*: plf [**m²/m² → cm**]
Total canopy LAI below given height
- *height*: number [**cm**]
Max crop height in canopy
- *albedo*: number (dimensionless)
Another reflection factor
- *cover*: number [**m²/m²**]
Fraction of soil covered by crops
- *interception_capacity*: number [**mm**]
Canopy water storage capacity
- *N_fixated*: number [**kg N/ha/h**]
Nitrogen fixation rate.
- *HvsLAI*: plf [**cm → m²/m²**]
Height in which there is a given LAI below in total canopy
- *ACExt_PAR*: number (dimensionless)
Canopy extinction coefficient of PAR (how fast the light dim as a function of LAI passed)
- *ACRef_PAR*: number (dimensionless)
Canopy reflection coefficient of PAR

- *ACExt_NIR*: number (dimensionless)
Canopy extinction coefficient of NIR (how fast the light dim as a function of LAI passed)
- *ACRef_NIR*: number (dimensionless)
Canopy reflection coefficient of NIR
- *ARExt*: number (dimensionless)
Radiation Extinction coefficient (like *ACExt*, but for all radiation, not just light)
- *EpFactorDry*: number (dimensionless)
Reference to potential evapotranspiration
- *EpFactorWet*: number (dimensionless)
Reference to potential evapotranspiration
- *shadow_stomata_conductance*: number [m/s]
Field based stomata conductance of shadow leaves.
- *sunlit_stomata_conductance*: number [m/s]
Field based stomata conductance of sunlit leaves.

100.1 afforestation

A growing forest.

```
< afforestation (Root Root) ; Default classic va
                (Canopy Canopy) ; Has default value.
                (root_DM 2 [Mg DM/ha])
                (root_am root_am ...) ; Has default value.
                (N_actual N_actual)
                (N_per_LAI N_per_LAI)
                (litter_am litter_am ...) ; Has default value.
                (Albedo 0.2 [])
                (planting_time planting_time) ; Has partial value.
                (canopy_height canopy_height)
                (root_depth root_depth)
                (LAI_shape LAI_shape)
                (LAI_min LAI_min)
                (LAI_max LAI_max)
                (N_nonleaves N_nonleaves)
                (litterfall_shape litterfall_shape)
                (litterfall_total litterfall_total)
                (litterfall_C_per_DM 0.42 [])
                (litterfall_C_per_N litterfall_C_per_N)
                (rhizodeposition_shape rhizodeposition_shape)
                (rhizodeposition_total rhizodeposition_total)
                (rhizodeposition_C_per_DM 0.42 [])
                (rhizodeposition_C_per_N rhizodeposition_C_per_N)
                ;; Shared parameters are described in chapter 100.
                (EpInterchange 0.6 [<fraction>]) >
```

- *Root*: **root** component (see chapter 72)
Component (default ‘classic’)
Root system.

- *Canopy*: **CanopySimple** fixed component (see section 107.4)
Submodel (has fully specified default value)
Canopy.
- *root_DM*: number [**Mg DM/ha**]
Parameter (default 2)
Afforestation root drymatter.
- *root_am*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(root_am "AOM-SLOW"
        "AOM-FAST")
```

Parameter description:
Rhizodeposition AOM parameters.

- *N_actual*: number [**g N/m²**]
Optional state variable
N uptake until now (default: 'N_demand').
- *N_per_LAI*: number [**kg N/ha**]
Parameter
N content as function of LAI.
- *litter_am*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(litter_am "AOM-SLOW"
          "AOM-FAST")
```

Parameter description:
Litter AOM parameters.

- *Albedo*: number (dimensionless)
Parameter (default 0.2)
Reflection factor.
- *planting_time*: **Time** fixed component (see section 107.27)
Submodel (has partially specified default value)
Reference time for yearly growth parameters.
- *canopy_height*: plf [**y → cm**]
Parameter
Forest height as a function of years after planting.
- *root_depth*: plf [**y → cm**]
Parameter
Depth of effective root zone as a function of years after planting.
- *LAI_shape*: plf [**d → <none>**]
Parameter
LAI factor as a function of Julian day. The value '1' represents 'LAI_min' and the value '5' represents 'LAI_max' for the specific year.
- *LAI_min*: plf [**y → m²/m²**]
Parameter
Yearly minimum LAI as a function of years after planting.

- *LAI_max*: plf [$y \rightarrow m^2/m^2$]
Parameter
Yearly maximum LAI as a function of years after planting.
- *N_nonleaves*: plf [$y \rightarrow kg\ N/ha$]
Parameter
Nitrogen not accounted for by seasonal LAI variation. This is a function of years after planting.
- *litterfall_shape*: plf [$d \rightarrow <none>$]
Parameter
Relative speed of litterfall.

The total litterfall over a year is specified by 'litterfall_total'. This parameter specifies how the the litterfall is divided over a year. The function is integrated over a year. The litterfall in a specific period is then the integration of this parameter over the period, divided by the integration of this function over a year, multiplied with the total litterfall for that year.
- *litterfall_total*: plf [$y \rightarrow Mg\ DM/ha$]
Parameter
Yearly litterfall as function of years after planting.
- *litterfall_C_per_DM*: number (dimensionless)
Parameter (default 0.42)
Carbon content of litter dry matter.
- *litterfall_C_per_N*: number (dimensionless)
Parameter
C/N ratio in litter.
- *rhizodeposition_shape*: plf [$d \rightarrow <none>$]
Parameter
Relative speed of rhizodeposition.

The total rhizodeposition over a year is specified by 'rhizodeposition_total'. This parameter specifies how the the rhizodeposition is divided over a year. The function is integrated over a year. The rhizodeposition in a specific period is then the integration of this parameter over the period, divided by the integration of this function over a year, multiplied with the total rhizodeposition for that year.
- *rhizodeposition_total*: plf [$y \rightarrow Mg\ DM/ha$]
Parameter
Yearly rhizodeposition as function of years after planting.
- *rhizodeposition_C_per_DM*: number (dimensionless)
Parameter (default 0.42)
Carbon content of litter dry matter.
- *rhizodeposition_C_per_N*: number (dimensionless)
Parameter
C/N ratio in rhizodeposition.

Log Variables

- *N_demand*: number [$g\ N/m^2$]
Current potential N content.

- *N_uptake*: number [g N/m²/h]
Nitrogen uptake rate.
- *N_litter*: number [g N/m²/h]
Nitrogen in litterfall.
- *N_rhizo*: number [g N/m²/h]
Nitrogen in rhizodeposition.

100.2 crops

Keep track of all crops on the field.

Used by column default Vegetation (see 26.1, page 153) .

```
< crops (crops)
      (ForcedLAI)
      ;; Shared parameters are described in chapter 100.
      (EpInterchange 0.6 [<fraction>]) >
```

- *crops*: **crop** component (see chapter 29) sequence
Component (default: an empty sequence)
List of crops growing in the field

- *ForcedLAI*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
By default, the total LAI for the vegetation will be the sum of the simulated LAI for the individual crops. However, you can force the model to use a different values for LAI by setting this attribute. The specified LAI will be distributed among the crops on the field corresponding to their simulated LAI.
'ForcedLAI' can be useful if you have measured the total LAI on the field, and want to force the model to confirm to the measurements.
'ForcedDAY' will not affect the LAI for crops that have not yet emerged. If no crops have emerged on the field, it will be ignored.

```
< year LAIvsDAY >
```

- *year*: integer
Parameter
Year for which to use forced LAI.
- *LAIvsDAY*: plf [yday → m²/m²]
Parameter
LAI as a function of Julian day.

The simulated LAI will be used before the first day you specify and after the last specified day. Simulated LAI will also be used whenever 'LAIvsDAY' becomes negative.

Log Variables

- *harvest*: **Harvest** fixed component (see section 107.15) sequence
Harvest current timestep.

100.3 permanent

Permanent (non-crop) vegetation.

```

< permanent (Root Root) ; Default classic value.
              (Canopy Canopy) ; Has default value.
              (Height 80 [cm])
              (root_DM 2 [Mg DM/ha])
              (N_actual N_actual)
              (LAIvsDAY LAIvsDAY)
              (YearlyLAI)
              (LAIfactor 1 [])
              (N_per_LAI 10 [kg N/ha/LAI])
              (DM_per_LAI 0.5 [Mg DM/ha/LAI])
              (litter_am litter_am ...) ; Has default value.
              (Albedo 0.2 [])
              ;; Shared parameters are described in chapter 100.
              (EpInterchange 0.6 [<fraction>]) >

```

- *Root*: **root** component (see chapter 72)
Component (default ‘classic’)
Root system.
- *Canopy*: **CanopySimple** fixed component (see section 107.4)
Submodel (has fully specified default value)
Canopy.
- *Height*: number [cm]
Parameter (default 80)
Permanent height of vegetation.
- *root_DM*: number [Mg DM/ha]
Parameter (default 2)
Permanent root drymatter.
- *N_actual*: number [g N/m²]
Optional state variable
N uptake until now (default: ‘N.demand’).
- *LAIvsDAY*: plf [yday → m²/m²]
Parameter
LAI as a function of Julian day. These numbers are used when there are no yearly numbers (YearlyLAI).
- *YearlyLAI*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Yearly LAI measurements.

```

      < year LAIvsDAY >

```

 - *year*: integer
Parameter
Year for which to use yearly LAI measurements.
 - *LAIvsDAY*: plf [m²/m² → yday]
Optional parameter
LAI as a function of Julian day.
The default LAI will be used before the first day you specify and after the last specified day. Default LAI will also be used whenever ‘LAIvsDAY’ becomes negative.

- *LAIfactor*: number (dimensionless)
Parameter (default 1)
Multiply calculated LAI with this number for quick scaling.
- *N_per_LAI*: number [**kg N/ha/LAI**]
Parameter (default 10)
N content as function of LAI.
- *DM_per_LAI*: number [**Mg DM/ha/LAI**]
Parameter (default 0.5)
DM as function of LAI.
- *litter_am*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(litter_am "AOM-SLOW"
           "AOM-FAST")
```

Parameter description:
Litter AOM parameters.

- *Albedo*: number (dimensionless)
Parameter (default 0.2)
Reflection factor.

Log Variables

- *N_demand*: number [**g N/m²**]
Current potential N content.
- *N_uptake*: number [**g N/m²/h**]
Nitrogen uptake this hour.
- *N_litter*: number [**g N/m²/h**]
Nitrogen in litter this hour.

100.4 Bush

A ‘permanent’ model (see 100.3, page 545) defined in ‘vegetation.dai’.

100.5 Coniferous

A ‘permanent’ model (see 100.3, page 545) defined in ‘vegetation.dai’.

100.6 Grass

A ‘permanent’ model (see 100.3, page 545) defined in ‘vegetation.dai’.

100.7 Hardwood

A ‘permanent’ model (see 100.3, page 545) defined in ‘vegetation.dai’.

Chapter 101

vernalization

Requirement for a cold period before flowering.

101.1 default

Temperature sum dependent vernalization.

Used by crop Rye Vernal (see 29.29, page 176) , crop Winter Barley Vernal (see 29.40, page 177) , crop Winter Barley; Koge Vernal (see 29.42, page 177) , crop Winter Rape Vernal (see 29.43, page 177) , crop Winter Wheat Vernal (see 29.44, page 177) , crop Grass to grain Vernal (see 29.10, page 174) , crop Ryegrass Vernal (see 29.30, page 176) , and crop Wclover Vernal (see 29.36, page 176) .

```
< default (DSLim DSLim)  
          (TaLim TaLim)  
          (TaSum TaSum) >
```

- *DSLim*: number [DS]
Parameter
Development stage at vernalization.
- *TaLim*: number [dg C]
Parameter
Vernalization temperature threshold.
- *TaSum*: number [dg C d]
State variable
Vernalization temperature-sum requirement.

101.2 none

No vernalization.

Used by crop default Vernal (see 29.1, page 171) .

Chapter 102

volume

A subset of 3D space.

Used by Irrigation @ event volume (see 107.10, page 602) .

102.1 box

A volume defined by intervals on each axis. By default, the intervals fill the entire axis. You can modify this by setting the parameters.

Used by action table volume (see 15.54, page 92) , action extern_fertigation volume (see 15.9, page 78) , action extern_subsoil volume (see 15.10, page 79) , action irrigate_subsoil volume (see 15.20, page 83) , log select volume (see 52.3, page 283) , select flow volume (see 80.5, page 431) , and select volume_base volume (see 80.16, page 434) .

```
< box (top top) ; Default empty value.  
      (bottom bottom) ; Default empty value.  
      (left left) ; Default empty value.  
      (right right) ; Default empty value.  
      (front front) ; Default empty value.  
      (back back) ; Default empty value. >
```

- *top*: **bound** component (see chapter 22)
Component (default 'empty')
Upper boundary on the z-axis.
- *bottom*: **bound** component (see chapter 22)
Component (default 'empty')
Lower boundary on the z-axis.
- *left*: **bound** component (see chapter 22)
Component (default 'empty')
Lower boundary on the x-axis.
- *right*: **bound** component (see chapter 22)
Component (default 'empty')
Upper boundary on the x-axis.
- *front*: **bound** component (see chapter 22)
Component (default 'empty')
Lower boundary on the y-axis.

- *back*: **bound** component (see chapter 22)
Component (default ‘empty’)
Upper boundary on the y-axis.

Chapter 103

weather

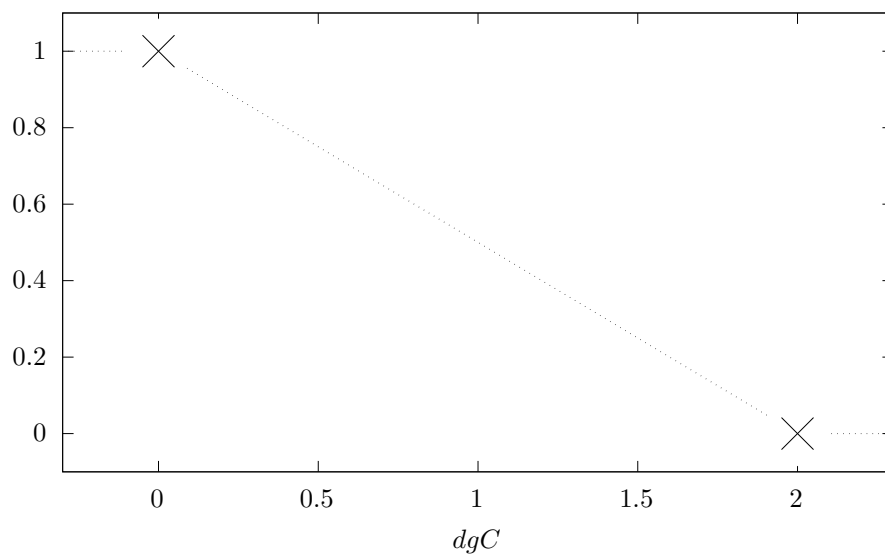
A 'wsources' is a source of raw weatherdata.

103.1 weather

Weather interface implementation.

```
< weather (snow_fraction snow_fraction) ; Has default value.  
          (max_rain max_rain) >
```

- *snow_fraction*: plf [dg C → <fraction>]
Parameter (has default value with 2 points)



Value description: This value is used by DMI as well as NMI. See also [Førland et al., 1996, Vejen et al., 2014]

```
(snow_fraction (0 [dg C] 1 []) (2 [dg C] 0 []))
```

Parameter description:

Fraction of precipitation that falls as snow as function of air temperature.

- *max_rain*: number [mm]
Optional parameter
Largest amount of rain in one timestep. By default, no limit on rain.

Log Variables

- *deposit*: submodel (see section 4.1.7) sequence
Total atmospheric deposition.
 < *name value* > **Log Variables**
 - *value*: number [**g/cm²/h**]
Value for chemical.
 - *name*: string (see section 4.1.5)
Name of chemical.
- *day_length*: number [**h**]
Number of light hours this day.
- *cloudiness_index*: number [<**fraction**>]
Fraction of sky covered by clouds, 1 = clear sky.
- *reference_evapotranspiration*: number [**mm/h**]
Reference evapotranspiration
- *air_temperature*: number [**dg C**]
Air temperature.
- *day_cycle*: number (dimensionless)
Current level extraterrestrial radiation reative to day average.
- *extraterrestrial_radiation*: number [**W/m²**]
Extraterrestrial radiation.
- *global_radiation*: number [**W/m²**]
Global radiation.
- *daily_air_temperature*: number [**dg C**]
Average temperature this day.
- *daily_min_air_temperature*: number [**dg C**]
Minumum temperature this day.
- *daily_max_air_temperature*: number [**dg C**]
Maximum temperature this day.
- *daily_global_radiation*: number [**W/m²**]
Average global radiation this day.
- *daily_extraterrestrial_radiation*: number [**W/m²**]
Average extraterrestrial radiation this day.
- *diffuse_radiation*: number [**W/m²**]
Diffuse radiation.
- *ground_heat_flux*: number [**W/m²**]
Ground heat flux.
- *rain*: number [**mm/h**]
Rain.
- *snow*: number [**mm/h**]
Snow.

- *precipitation*: number [**mm/h**]
Precipitation.
- *vapor_pressure*: number [**Pa**]
Humidity.
- *relative_humidity*: number [<**fraction**>]
Relative humidity.
- *h_atm*: number [**cm**]
Pressure head.
- *air_pressure*: number [**Pa**]
Air pressure.
- *wind*: number [**m/s**]
Wind speed.
- *co2*: number [**Pa**]
Atmospheric CO2 pressure.
- *sunrise*: number [**h**]
Time of sunrise.
- *sunset*: number [**h**]
Time of sunset.

103.2 base

A ‘weather’ base model (see 103.1, page 553) build into Daisy.

Weather that does not change during the simulation.

```

< base (T_min T_min)
      (GlobRad GlobRad)
      (AirTemp AirTemp)
      (T_max T_max)
      (Precip Precip)
      (RefEvap RefEvap)
      (VapPres VapPres)
      (DiffRad DiffRad)
      (GHF GHF)
      (CloudinessIndex CloudinessIndex)
      (RelHum RelHum)
      (Wind Wind)
      (CO2 CO2)
      (Latitude Latitude)
      (Longitude Longitude)
      (Elevation Elevation)
      (TimeZone TimeZone)
      (ScreenHeight ScreenHeight)
      (TAverage TAverage)
      (TAmplitude TAmplitude)
      (MaxTDay MaxTDay)
      (NH4WetDep NH4WetDep)
      (NH4DryDep NH4DryDep)
      (NO3WetDep NO3WetDep)
      (NO3DryDep NO3DryDep)
      (Deposition Deposition)
      (DepDry DepDry)
      (DepDryNH4 DepDryNH4)
      (DepWetNH4 DepWetNH4)
      (PAverage PAverage)
      (Timestep Timestep)
      (end end)
      (begin begin)
      (Station Station)
      (Note Note)
      (Surface Surface)
      (PrecipCorrect PrecipCorrect ...)
      (PrecipScale PrecipScale ...)
      (GlobRadScale GlobRadScale ...)
      (TempOffset TempOffset ...)
      (Begin Begin)
      (End End)
      ;; Shared parameters are described in section 103.1.
      (snow_fraction snow_fraction) ; Has default value.
      (max_rain max_rain) >

```

- *T_min*: number [dgC]
Optional parameter
Minimum air temperature.
- *GlobRad*: number [W/m²]
Optional parameter
Global radiation.
- *AirTemp*: number [dgC]

Optional parameter
Air temperature.

- *T_{max}*: number [**dgC**]
Optional parameter
Maximum air temperature.
- *Precip*: number [**mm/h**]
Optional parameter
Precipitation.
- *RefEvap*: number [**mm/h**]
Optional parameter
Reference evapotranspiration.
- *VapPres*: number [**Pa**]
Optional parameter
Vapor pressure.
- *DiffRad*: number [**W/m²**]
Optional parameter
Diffuse radiation.
- *GHF*: number [**W/m²**]
Optional parameter
Ground heat flux.
- *CloudinessIndex*: number [**fraction**]
Optional parameter
Cloudiness index, 1 = clear sky.
- *RelHum*: number [**fraction**]
Optional parameter
Relative humidity.
- *Wind*: number [**m/s**]
Optional parameter
Wind speed.
- *CO2*: number [**<fraction>**]
Optional parameter
Atmospheric CO2 level.
- *Latitude*: number [**dgNorth**]
Optional parameter
Location of station (north-south).
- *Longitude*: number [**dgEast**]
Optional parameter
Location of station (east-west).
- *Elevation*: number [**m**]
Optional parameter
Station altitude over sea level.
- *TimeZone*: number [**dgEast**]
Optional parameter
Time zone.

- *ScreenHeight*: number [**m**]
Optional parameter
Measurement altitude over ground level.
- *TAverage*: number [**dgC**]
Optional parameter
Yearly average temperature.
- *TAmplitude*: number [**dgC**]
Optional parameter
Typical temperature variation over the seasons. If you fit the daily average temperature over a year to a sinus curve, this would be the amplitude.
- *MaxTDay*: number [**yday**]
Optional parameter
Typical day where the temperature is highest. If you fit the daily average temperature over a year to a sinus curve, this would be maximum point.
- *NH4WetDep*: number [**ppm**]
Optional parameter
NH4 concentration in precipitation.
- *NH4DryDep*: number [**kg N/ha/y**]
Optional parameter
Dry deposition of NH4.
- *NO3WetDep*: number [**ppm**]
Optional parameter
NO3 concentration in precipitation.
- *NO3DryDep*: number [**kg N/ha/y**]
Optional parameter
Dry deposition of NO3.
- *Deposition*: number [**kg N/ha/y**]
Optional parameter
Total N deposition.
- *DepDry*: number [**<fraction>**]
Optional parameter
Fraction of total N deposition that is dry.
- *DepDryNH4*: number [**<fraction>**]
Optional parameter
NH4 fraction of dry deposition.
- *DepWetNH4*: number [**<fraction>**]
Optional parameter
NH4 fraction of wet deposition.
- *PAverage*: number [**mm**]
Optional parameter
Average precipitation. Used for deviding precipitation into dry and wet.
- *Timestep*: number [**h**]
Optional parameter
Timestep for weather data.

- *end*: **Time** fixed component (see section 107.27)
Optional submodel
Only use data before this date.
- *begin*: **Time** fixed component (see section 107.27)
Optional submodel
Only use data after this date.
- *Station*: string (see section 4.1.5)
Optional parameter
Name of weather station.
- *Note*: string (see section 4.1.5)
Optional parameter
Note regarding this set of data.
- *Surface*: string (see section 4.1.5)
Optional parameter
Surface type. Either 'reference' for a weather station standard of short grass, or 'field' for measurements directly at the field.
- *PrecipCorrect*: number (dimensionless) sequence
Optional parameter
Correction factors for precipitation. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *PrecipScale*: number (dimensionless) sequence
Optional parameter
Scale factors for precipitation. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *GlobRadScale*: number (dimensionless) sequence
Optional parameter
Scale factors for global radiation. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *TempOffset*: number [dg C] sequence
Optional parameter
Temperature offset. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *Begin*: **Time** fixed component (see section 107.27)
Optional submodel
Beginning of weather data.
- *End*: **Time** fixed component (see section 107.27)
Optional submodel
End of weather data.

103.3 const

A 'base' model (see 103.2, page 555) build into Daisy.

103.4 none

A ‘const’ model (see 103.3, page 559) build into Daisy.

No weather. No precipitation or global radiation, temperature is 10 dg Celcius. The static information is taken from a climate station operated by Copenhagen University, and located in Taastrup, Denmark. You can overwrite any parameters to simulate an experiment under controled, constant weather conditions.

```
< (GlobRad 0 [W/m2])  
  (AirTemp 10 [dgC])  
  (Precip 0 [mm/h])  
  (Latitude 56 [dgNorth])  
  (Longitude 12 [dgEast])  
  (Elevation 30 [m])  
  (TimeZone 15 [dgEast])  
  (ScreenHeight 2 [m])  
  (TAverage 7.8 [dgC])  
  (TAmplitude 8.5 [dgC])  
  (MaxTDay 209 [yday])  
  (Station Taastrup)  
  (Surface reference) >
```

103.5 null

A ‘const’ model (see 103.3, page 559) build into Daisy.

Missing weather data. Use for indicating no weather data available.

Used by weather combine reserve (see 103.8, page 564) .

103.6 table

A ‘base’ model (see 103.2, page 555) build into Daisy.

Read weather data from a file.


```

< table (file file)
      (missing missing ...) ; Has default value.
      (filter)
      (original original ...)
      (dim_line dim_line)
      ;; Shared parameters are described in section 103.2.
      (T_min T_min)
      (GlobRad GlobRad)
      (AirTemp AirTemp)
      (T_max T_max)
      (Precip Precip)
      (RefEvap RefEvap)
      (VapPres VapPres)
      (DiffRad DiffRad)
      (GHF GHF)
      (CloudinessIndex CloudinessIndex)
      (RelHum RelHum)
      (Wind Wind)
      (CO2 CO2)
      (Latitude Latitude)
      (Longitude Longitude)
      (Elevation Elevation)
      (TimeZone TimeZone)
      (ScreenHeight ScreenHeight)
      (TAverage TAverage)
      (TAmplitude TAmplitude)
      (MaxTDay MaxTDay)
      (NH4WetDep NH4WetDep)
      (NH4DryDep NH4DryDep)
      (NO3WetDep NO3WetDep)
      (NO3DryDep NO3DryDep)
      (Deposition Deposition)
      (DepDry DepDry)
      (DepDryNH4 DepDryNH4)
      (DepWetNH4 DepWetNH4)
      (PAverage PAverage)
      (Timestep Timestep)
      (end end)
      (begin begin)
      (snow_fraction snow_fraction) ; Has default value.
      (max_rain max_rain)
      (Station Station)
      (Note Note)
      (Surface Surface)
      (PrecipCorrect PrecipCorrect ...)
      (PrecipScale PrecipScale ...)
      (GlobRadScale GlobRadScale ...)
      (TempOffset TempOffset ...)
      (Begin Begin)
      (End End) >

```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.

- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

- *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.

103.7 default

A 'table' model (see 103.6, page 560) build into Daisy.

Read weather data from specific file.

```

< default  file
    (missing_years)
    ;; Shared parameters are described in section 103.6.
    (T_min T_min)
    (GlobRad GlobRad)
    (AirTemp AirTemp)
    (T_max T_max)
    (Precip Precip)
    (RefEvap RefEvap)
    (VapPres VapPres)
    (DiffRad DiffRad)
    (GHF GHF)
    (CloudinessIndex CloudinessIndex)
    (RelHum RelHum)
    (Wind Wind)
    (CO2 CO2)
    (Latitude Latitude)
    (Longitude Longitude)
    (Elevation Elevation)
    (TimeZone TimeZone)
    (ScreenHeight ScreenHeight)
    (TAverage TAverage)
    (TAmplitude TAmplitude)
    (MaxTDay MaxTDay)
    (NH4WetDep NH4WetDep)
    (NH4DryDep NH4DryDep)
    (NO3WetDep NO3WetDep)
    (NO3DryDep NO3DryDep)
    (Deposition Deposition)
    (DepDry DepDry)
    (DepDryNH4 DepDryNH4)
    (DepWetNH4 DepWetNH4)
    (PAverage PAverage)
    (Timestep Timestep)
    (end end)
    (begin begin)
    (missing missing ...) ; Has default value.
    (filter)
    (original original ...)
    (dim_line dim_line)
    (snow_fraction snow_fraction) ; Has default value.
    (max_rain max_rain)
    (Station Station)
    (Note Note)
    (Surface Surface)
    (PrecipCorrect PrecipCorrect ...)
    (PrecipScale PrecipScale ...)
    (GlobRadScale GlobRadScale ...)
    (TempOffset TempOffset ...)
    (Begin Begin)
    (End End) >

```

- *file*: string (see section 4.1.5)
Parameter

Name of Daisy file where data is found.

- *missing_years*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)

How to get data for dates outside the range of the weather file.

The value is a list of maps. Each map consist of two intervals, and indicates that missing data from the first interval should be read from the second interval instead. Each interval consists of two years, the first and last year of that interval.

When the simulation requests weather data from a date outside the range covered by the weather file, the model will look up each member of the list, to see if the year is covered by the first interval. If so, it will use weather data from the same day in the corresponding year in the second interval.

If a given year is covered by multiple intervals in the list, the first one will be used.

< *from to* >

- *from*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(from <missing from> <missing to>)

Parameter description:

Interval of years to map from.

< *from to* >

- * *from*: integer
Parameter
First year of interval.
- * *to*: integer
Parameter
First year of interval.

- *to*: submodel (see section 4.1.7)
Submodel (has partially specified default value)

(to <missing from> <missing to>)

Parameter description:

Interval of years to map to.

< *from to* >

- * *from*: integer
Parameter
First year of interval.
- * *to*: integer
Parameter
First year of interval.

103.8 combine

A ‘weather’ model (see 103.1, page 553) build into Daisy.

Combine multiple weather sources.

```
< combine (entry entry ...)
  (reserve reserve) ; Default null value.
  ;; Shared parameters are described in section 103.1.
  (snow_fraction snow_fraction) ; Has default value.
  (max_rain max_rain) >
```

- *entry*: submodel (see section 4.1.7) sequence
List of weather sources.


```

      < (source source)
        (end end)
        (begin begin)
        (use use ...) ; Has default value. >
      
```

 - *source*: **weather** component (see chapter 103)
Source of weather data.
 - *end*: **Time** fixed component (see section 107.27)
Optional submodel
Use weather data from source until this time. By default, use data until end of source.
 - *begin*: **Time** fixed component (see section 107.27)
Optional submodel
Use weather data from source after this time. By default, use data from start of source.
 - *use*: string (see section 4.1.5) sequence
Parameter (has default value with length 1)

(use Any)

Parameter description:
List of weather data to use from source. Specify 'Any' to use all present weather data.
- *reserve*: **weather** component (see chapter 103)
Component (default 'null')
Reserve weather model to use when no source match.

103.9 indirect

A 'weather' model (see 103.1, page 553) build into Daisy.

Delegate to another weather source.

```

< indirect (source source)
  ;; Shared parameters are described in section 103.1.
  (snow_fraction snow_fraction) ; Has default value.
  (max_rain max_rain) >

```

- *source*: **weather** component (see chapter 103)
Use this weather source.

103.10 time

A 'indirect' model (see 103.9, page 565) build into Daisy.

Change time period for weather data.

You can either specify 'offset' directly, or indirectly through 'from' and 'to'. If you chose the later, offset will be calculated January the first in the year specified with 'from' will be mapped to January the first in the year specified with 'to'. Leap years mean that other dates may not be mapped exactly to corresponding dates in other years.

```
< time (offset offset)  
      (from from)  
      (to to)  
      ;; Shared parameters are described in section 103.9.  
      (source source)  
      (snow_fraction snow_fraction) ; Has default value.  
      (max_rain max_rain) >
```

- *offset*: **Timestep** fixed component (see section 107.25)
Optional submodel
Time offset. Positive mean move time ahead.
- *from*: integer
Optional state variable
Year in weather data.
- *to*: integer
Optional state variable
Year in simulation.

Chapter 104

WSe

The water stress effect on crop growth.

104.1 full

Water stress has full effect on crop growth. This means that if there is 50% water stress, assimilate production will be cut into half.

104.2 none

Water stress has no effect on plant growth.

104.3 partial

Water stress has partial effect on crop growth.

With this model, there will be full production when there is enough available soil water to cover the potential evapotranspiration, and no production when there is no soil water available. In between production is controlled by the 'y_half' parameter.

See SH:REFERENCE for more explanation.

Used by crop Potato; Triada water_stress_effect (see 29.26, page 175) .

< **partial** (y_half *y_half*) >

- *y_half*: number [<**fraction**>]

Parameter

Effect on assimilate production of water stress. This parameter specifies the effect on assimilate production (compared to potential) when the amount of available soil water is enough to cover exactly half the potential evapotranspiration.

Chapter 105

xysource

XY data series.

105.1 arithmetic

Read a daisy log, weather or data file. Calculate an x and an y value for each time step, based on the value in the various columns.

```
< arithmetic (x x)
              (file file)
              (with with)
              (missing missing ...) ; Has default value.
              (filter)
              (original original ...)
              (dim_line dim_line)
              (style style)
              (title title)
              (valid valid) ; Default true value.
              (y y)
              (xbar xbar)
              (ybar ybar) >
```

- *x*: **number** component (see chapter 59)
Expression for calculating the x value for this source for each row. The expression can refer to the value in a specific column by the tag for that column.
- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.

By default, data from dwf and dlf files will be drawn with lines, and data from ddf files will be drawn with points.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:

List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
 Submodel (default: an empty sequence)
 Only include data from rows that passes all these filters.

```
< tag allowed... >
```

 - *tag*: string (see section 4.1.5)
 Parameter
 Name of column in Daisy log file to filter for.
 - *allowed*: string (see section 4.1.5) sequence
 Parameter
 List of allowable values in filter.
- *original*: string (see section 4.1.5) sequence
 Optional parameter
 List of dimensions of the data in the data file.
 If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.
 By default Daisy will use the names specified in data file.
- *dim_line*: boolean (see section 4.1.2)
 Optional parameter
 If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *style*: integer
 Optional parameter
 Style to use for this dataset.
 By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.
 Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.
 The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
 Optional parameter
 Name of data series for the legend on the graph.
 By default the name of the 'x' and 'y' objects.
- *valid*: **boolean** component (see chapter 21)
 Component (default 'true')
 Ignore entries if this boolean expression is false.
- *y*: **number** component (see chapter 59)
 Expression for calculating the y value for this source for each row. The expression can refer to the value in a specific column by the tag for that column.
- *xbar*: **number** component (see chapter 59)
 Optional component
 Expression for calculating x errorbar for this source for each row. The expression can refer to the value in a specific column by the tag for that column.

- *ybar*: **number** component (see chapter 59)
Optional component
Expression for calculating y errorbar for this source for each row. The expression can refer to the value in a specific column by the tag for that column.

105.2 combine

Combine data from multiple time series with a single expression. Data from times series are matched by date.

```
< combine  (x x)
           (source source ...)
           (with with)
           (style style)
           (title title)
           (y y) >
```

- *x*: **number** component (see chapter 59)
Expression for calculating the x value for this source for each row. A row is any date found in any of the member of 'source'. The expression may refer to the value of each source by its title.
- *source*: **source** component (see chapter 83) sequence
List of sources for data. The style information for the sources is ignored, but the dates, title and value is used as specified by 'expr' to calculate the combined date and value pairs.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.

By default, let the first source decide.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.

By default a combination of the x and y objects.
- *y*: **number** component (see chapter 59)
Expression for calculating the y value for this source for each row. A row is any date found in any of the member of 'source'. The expression may refer to the value of each source by its title.

105.3 flux

Read a daisy 2d log file, extract flux through line. Values between 'begin' and 'end' will be accumulated, unless 'when' is set.

```
< flux  (x x)
        (file file)
        (with with)
        (dimension dimension)
        (end end)
        (z z)
        (begin begin)
        (missing missing ...) ; Has default value.
        (filter)
        (original original ...)
        (dim_line dim_line)
        (style style)
        (title title)
        (when when)
        (pos_dim cm) >
```

- *x*: number [**cm**]
Optional parameter
Plot flux through this position.
- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
By default, data will be drawn with linespoints.
- *dimension*: string (see section 4.1.5)
Optional parameter
Dimension for data. By default, use dimension from file.
- *end*: **Time** fixed component (see section 107.27)
Optional submodel
Ignore values after this time.
- *z*: number [**cm**]
Optional parameter
Plot flux through this depth.
- *begin*: **Time** fixed component (see section 107.27)
Optional submodel
Ignore values before this time.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
 Submodel (default: an empty sequence)
 Only include data from rows that passes all these filters.

```
< tag allowed... >
```

 - *tag*: string (see section 4.1.5)
 Parameter
 Name of column in Daisy log file to filter for.
 - *allowed*: string (see section 4.1.5) sequence
 Parameter
 List of allowable values in filter.
- *original*: string (see section 4.1.5) sequence
 Optional parameter
 List of dimensions of the data in the data file.
 If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.
 By default Daisy will use the names specified in data file.
- *dim_line*: boolean (see section 4.1.2)
 Optional parameter
 If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *style*: integer
 Optional parameter
 Style to use for this dataset.
 By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.
 Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.
 The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
 Optional parameter
 Name of data series for the legend on the graph.
 By default the specified 'z' or 'x' value, time, and tag.
- *when*: **Time** fixed component (see section 107.27)
 Optional submodel
 Use value closest to this time.
- *pos_dim*: string (see section 4.1.5)
 Parameter (default 'cm')
 Dimension for soil position.

105.4 frequency

Read a daisy log, weather or data file. Plot the frequency of a specific column.

```

< frequency (file file)
  (with with)
  (missing missing ...) ; Has default value.
  (filter)
  (original original ...)
  (dim_line dim_line)
  (style style)
  (title title)
  (valid valid) ; Default true value.
  (sort sort)
  (plot plot)
  (percent "%")
  (use_sort_value false)
  (plot_on_x true) >

```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.

By default, data will be drawn with lines.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

```
(missing "" "00.00")
```

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

```
< tag allowed... >
```

- *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.
By default the name of the tag.
- *valid*: **boolean** component (see chapter 21)
Component (default 'true')
Ignore entries if this boolean expression is false.
- *sort*: **number** component (see chapter 59)
Expression for calculating the sorting value for this source for each row. The expression can refer to the value in a specific column by the tag for that column.
- *plot*: **number** component (see chapter 59)
Optional component
Expression for calculating the plot value for this source for each row. By default, this will be identical to 'sort'.
- *percent*: string (see section 4.1.5)
Parameter (default '%')
Symbol to use for percent.
- *use_sort_value*: boolean (see section 4.1.2)
Parameter (default false)
Show 'sort' value instead of accumulated frequency.
- *plot_on_x*: boolean (see section 4.1.2)
Parameter (default true)
Show plot values on x axis.

105.5 inline

A list of x, y pairs.

```
< inline (with with)
          (points points)
          (style style)
          (title title)
          (x_dimension x_dimension)
          (y_dimension y_dimension) >
```

- *with*: string (see section 4.1.5)
Parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
- *points*: plf [<unknown> → <unknown>]
Parameter
List of (x y) pairs.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.

By default the name of the 'x' and 'y' objects.
- *x_dimension*: string (see section 4.1.5)
Parameter
Dimension for x points.
- *y_dimension*: string (see section 4.1.5)
Parameter
Dimension for y points.

105.6 loop

Calculate x and y pairs based on a single variable.

The variable cover an interval from 'begin' to 'end' in fixed steps 'step'. The name of the variable is specified by 'tag'. The x and y expressions may refer to the variable.

```
< loop  (x x)                ; Default x value.
        (with lines)
        (end end)
        (begin begin)
        (style style)
        (title title)
        (tag x)
        (y y)
        (step step) >
```

- *x*: **number** component (see chapter 59)
Component (default 'x')
Expression for calculating the x value.

- *with*: string (see section 4.1.5)
Parameter (default 'lines')
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
- *end*: number [<user>]
Parameter
End of interval.
- *begin*: number [<user>]
Parameter
Start of interval.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.

By default the name of the 'x' and 'y' objects.
- *tag*: string (see section 4.1.5)
Parameter (default 'x')
Name of free variable to calculate the 'x' and 'y' expressions from.
- *y*: **number** component (see chapter 59)
Expression for calculating the y value.
- *step*: number [<user>]
Parameter
Disretization within interval.

105.7 merge

Merge multiple xy data series into one.

```
< merge  (source source ...)
         (with with)
         (style style)
         (title title)
         (x_dimension x_dimension)
         (y_dimension y_dimension) >
```

- *source*: **xysource** component (see chapter 105) sequence
XY data series to merge.

- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
By default, let the first source decide.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Parameter
Name of data series for the legend on the graph.
- *x_dimension*: string (see section 4.1.5)
Parameter
Dimension for x points.
- *y_dimension*: string (see section 4.1.5)
Parameter
Dimension for y points.

105.8 profile

Plot soil profile.

```
< profile (file file)
          (with with)
          (dimension dimension)
          (missing missing ...) ; Has default value.
          (filter)
          (original original ...)
          (dim_line dim_line)
          (style style)
          (title title)
          (to to)
          (when when) ; Has partial value.
          (pos_dim cm) >
```

- *file*: string (see section 4.1.5)
Parameter
Name of Daisy file where data is found.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
By default, data will be drawn with linespoints.

- *dimension*: string (see section 4.1.5)
Optional parameter
Dimension for data. By default, use dimension from file.
- *missing*: string (see section 4.1.5) sequence
Parameter (has default value with length 2)

(missing "" "00.00")

Parameter description:
List of strings indicating missing values.

- *filter*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Only include data from rows that passes all these filters.

< tag allowed... >

- *tag*: string (see section 4.1.5)
Parameter
Name of column in Daisy log file to filter for.
- *allowed*: string (see section 4.1.5) sequence
Parameter
List of allowable values in filter.

- *original*: string (see section 4.1.5) sequence
Optional parameter
List of dimensions of the data in the data file.

If the list has only one element, that element is used as the dimension for all columns in the file. Otherwise, the list must have one element for each column.

By default Daisy will use the names specified in data file.

- *dim_line*: boolean (see section 4.1.2)
Optional parameter
If true, assume the line after the tags contain dimensions. By default this will be true iff 'original' is not specified.

- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.

- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.
By default the specified 'z' value, time, and tag.

- *to*: number [**cm**]
Optional parameter
Plot profile to this depth. By default, plot full profile.
- *when*: **Time** fixed component (see section 107.27)
Submodel (has partially specified default value)
Use value closest to this time.
- *pos_dim*: string (see section 4.1.5)
Parameter (default 'cm')
Dimension for soil position.

105.9 xycombine

Combine data from multiple time series with a single expression. Data from times series are matched by date.

```
< xycombine (expr expr)
            (source source ...)
            (with with)
            (style style)
            (title title) >
```

- *expr*: **number** component (see chapter 59)
Expression for calculating the value for this source. The expression may refer to the value of each source by its title.
- *source*: **xysource** component (see chapter 105) sequence
List of sources for data. The style information for the sources is ignored, but the title and value is used as specified by 'expr' to calculate the combined x and y pairs. Either the x or y for all sources must be identical.
- *with*: string (see section 4.1.5)
Optional parameter
Specify style used for the data series on the graph. use 'points' to plot each point individually, or 'lines' to draw lines between them.
By default, let the first source decide.
- *style*: integer
Optional parameter
Style to use for this dataset.

By default, gnuplot will use style 1 for the first source to plot with lines, style 2 for the second, and so forth until it runs out of styles and has to start over. Points work similar, but with its own style counter. For color plots, points and lines with the same style number also have the same color.

Set style to 0 to reuse the style of the previous series, or to a negative number to explicitly request the default behaviour.

The 'style' parameter is only used if 'with' is either 'points' or 'lines'.
- *title*: string (see section 4.1.5)
Optional parameter
Name of data series for the legend on the graph.
By default use the expression.

Chapter 106

zone

A subset of 3D space.

Used by SoilRegion @ volume (see 107.20, page 611) .

106.1 box

A zone defined by intervals on each axis. By default, the intervals fill the entire axis. You can modify this by setting the parameters.

```
< box  (top top)           ; Default empty value.  
        (bottom bottom)    ; Default empty value.  
        (left left)         ; Default empty value.  
        (right right)       ; Default empty value.  
        (front front)       ; Default empty value.  
        (back back)         ; Default empty value. >
```

- *top*: **bound** component (see chapter 22)
Component (default 'empty')
Upper boundary on the z-axis.
- *bottom*: **bound** component (see chapter 22)
Component (default 'empty')
Lower boundary on the z-axis.
- *left*: **bound** component (see chapter 22)
Component (default 'empty')
Lower boundary on the x-axis.
- *right*: **bound** component (see chapter 22)
Component (default 'empty')
Upper boundary on the x-axis.
- *front*: **bound** component (see chapter 22)
Component (default 'empty')
Lower boundary on the y-axis.
- *back*: **bound** component (see chapter 22)
Component (default 'empty')
Upper boundary on the y-axis.

106.2 polygon

A 2D region defined by points connected by lines. The object is assumed to be infinite in the third dimension.

`< polygon (outer outer ...) >`

- *outer*: submodel (see section 4.1.7) sequence
Points surrounding the polygon in clockwise order.

`< z x >`

- *z*: number [**cm**]
Parameter
Vertical position.
- *x*: number [**cm**]
Parameter
Horizontal position.

Chapter 107

Fixed Components

Fixed components are similar to ordinary component, with the exceptions that there can only be one model, that is, only a single implementation of the component, and that it is not possible to define libraries of standard parameterizations for the model.

107.1 DOM

A single Dissolved Organic Matter pool.

```
< DOM (N N) ; Has partial value.
      (C C) ; Has partial value.
      (diffusion_coefficient diffusion_coefficient)
      (heat_factor heat_factor)
      (water_factor water_factor)
      (efficiency efficiency ...)
      (fractions fractions ...)
      (turnover_rate turnover_rate)
      (turnover_halftime turnover_halftime) >
```

- *N*: **DOM-Element** fixed component (see section 107.12)
Submodel (has partially specified default value)
Nitrogen content of DOM pool.
- *C*: **DOM-Element** fixed component (see section 107.12)
Submodel (has partially specified default value)
Carbon content of DOM pool.
- *diffusion_coefficient*: number [**cm²/s**]
Parameter
Diffusion coefficient.
- *heat_factor*: plf [**dg C** → **<none>**]
Optional parameter
Heat factor. If empty, use default from 'OrganicMatter'.
- *water_factor*: plf [**cm** → **<none>**]
Optional parameter
Water potential factor. If empty, use default from 'OrganicMatter'.
- *efficiency*: number [**<fraction>**] sequence
Parameter
the efficiency this pool can be digested by each of the SMB pools.

- *fractions*: number [**<fraction>**] sequence
Parameter
Fraction of this pool that ends up in each SMB pools
- *turnover_rate*: number [**h⁻¹**]
Optional parameter
Fraction converted to other pools each hour. You must specify either this or 'turnover_halftime'.
- *turnover_halftime*: number [**h**]
Optional parameter
Time until half had been converted to other pools. You must specify either this or 'turnover_rate'.

107.2 SoluteUptakeResult

Result of the 'solute_uptake' number model.

```
< SoluteUptakeResult  (uptake uptake)
                      (alpha alpha)
                      (D D)
                      (beta beta)
                      (C_root C_root) >
```

- *uptake*: number [**g/cm³/h**]
Parameter
Root uptake.
- *alpha*: number (dimensionless)
Parameter
Thingy.
- *D*: number [**cm²/h**]
Parameter
Diffusion thingy.
- *beta*: number (dimensionless)
Parameter
Other thingy.
- *C_root*: number [**g/cm³**]
Parameter
Concentration at root surface.

107.3 SoluteUptake2Result

Result of the 'solute_uptake2' number model.

```
< SoluteUptake2Result  (uptake uptake)
                      (alpha alpha)
                      (A A)
                      (B B)
                      (D D)
                      (beta beta)
                      (C1 C1)
                      (C2 C2)
                      (C_root C_root) >
```


- *uptake*: number [$\text{g}/\text{cm}^3/\text{h}$]
Parameter
Root uptake.
- *alpha*: number (dimensionless)
Parameter
Thingy.
- *A*: number [$\text{g}/\text{cm R}/\text{h}$]
Parameter
Uptake thingy.
- *B*: number [$\text{cm}^3 \text{ W}/\text{cm R}/\text{h}$]
Parameter
'C_root' dependent uptake thingy.
- *D*: number [cm^2/h]
Parameter
Diffusion thingy.
- *beta*: number (dimensionless)
Parameter
Other thingy.
- *C1*: number [g/cm^3]
Parameter
Concentration at root surface (mechanism 1 dominating).
- *C2*: number [g/cm^3]
Parameter
Concentration at root surface (mechanism 2 dominating).
- *C_root*: number [g/cm^3]
Parameter
Concentration at root surface.

107.4 CanopySimple

Simple canopy model.

```
< CanopySimple (PARref 0.06 [])
                (PARext 0.6 [])
                (NIRref 0.51 [])
                (NIRext 0.18 [])
                (EPext 0.5 [])
                (IntcpCap 0.5 [mm])
                (EpFac 1.2 [])
                (EpFacWet EpFacWet)
                (EpFacDS EpFacDS)           ; Has default value.
                (rs_max 100000 [s/m])
                (rs_min 100 [s/m])
                (Height 0 [cm])
                (leaf_width leaf_width)     ; Has default value. >
```

- *PARref*: number (dimensionless)
Parameter (default 0.06)
PAR reflectance.

- *PARext*: number (dimensionless)
Parameter (default 0.6)
PAR extinction coefficient.
- *NIRref*: number (dimensionless)
Parameter (default 0.51)
NIR reflectance. NIRref = 0.51 (Ross, 1975)
- *NIRext*: number (dimensionless)
Parameter (default 0.18)
NIR extinction coefficient. NIRext = 0.18 (Jones, 1983)
- *EPext*: number (dimensionless)
Parameter (default 0.5)
EP extinction coefficient.
- *IntcpCap*: number [**mm**]
Parameter (default 0.5)
Interception capacity.
- *EpFac*: number (dimensionless)
Parameter (default 1.2)
Value description: See figure 4 in the cited paper.
With a bare soil factor of 0.6 a combined Kc of 1.15 is reached at LAI=5. See also [Kjaersgaard et al., 2008]
Potential evapotranspiration factor.
- *EpFacWet*: number (dimensionless)
Optional parameter
Potential evapotranspiration factor for wet surface. By default this is identical to EpFac.
- *EpFacDS*: plf [**DS** → **<none>**]
Parameter (has default const value 1 [])
DS dependent potential evapotranspiration factor.
- *rs_max*: number [**s/m**]
Parameter (default 100000)
Maximum transpiration resistance.
- *rs_min*: number [**s/m**]
Parameter (default 100)
Minimum transpiration resistance.
- *Height*: number [**cm**]
State variable (default 0)
Crop height.
- *leaf_width*: plf [**DS** → **cm**]
Parameter (has default const value 3 [cm])
Leaf width.

Log Variables

- *CAI*: number [**m²/m²**]
Crop Area Index.
- *LAIvsH*: plf [**cm** → **m²/m²**]
Accumulated Leaf Area Index at Height.

107.5 CanopyStandard

Standard canopy model.

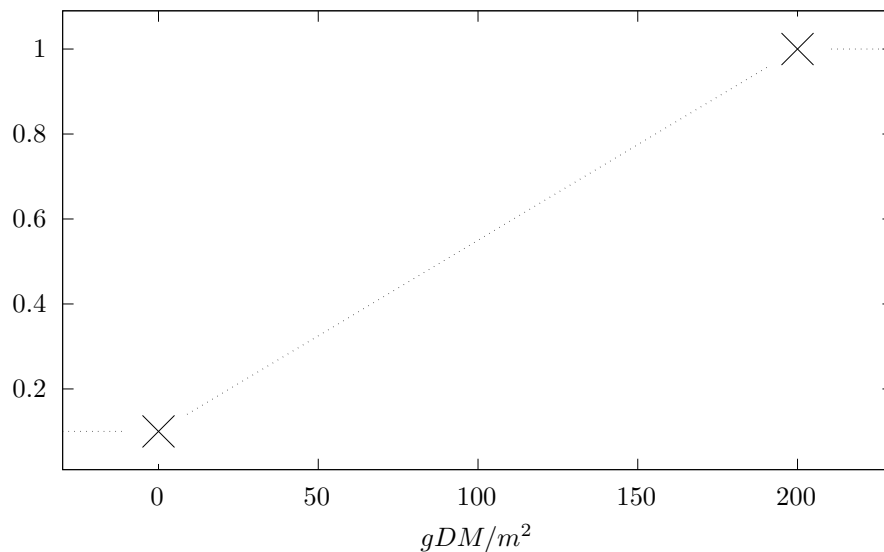
```
< CanopyStandard (PARref 0.06 [])
  (PARext 0.6 [])
  (NIRref 0.51 [])
  (NIRext 0.18 [])
  (EPext 0.5 [])
  (IntcpCap 0.5 [mm])
  (EpFac 1.2 [])
  (EpFacWet EpFacWet)
  (EpFacDS EpFacDS) ; Has default value.
  (rs_max 100000 [s/m])
  (rs_min 100 [s/m])
  (Height 0 [cm])
  (leaf_width leaf_width) ; Has default value.
  (SpLAI SpLAI)
  (LeafAImod LeafAImod) ; Has default value.
  (SpSOrgAI 0 [(m2/m2)/(g DM/m2)])
  (SOrgAImod SOrgAImod) ; Has default value.
  (SOrgPhotEff 1 [])
  (SOrgEffvsDS SOrgEffvsDS) ; Has default value.
  (SpStemAI 0 [(m2/m2)/(g DM/m2)])
  (StemAImod StemAImod) ; Has default value.
  (StemPhotEff 1 [])
  (StemEffvsDS StemEffvsDS) ; Has default value.
  (HvsDS HvsDS)
  (HvsWStem HvsWStem) ; Has default value.
  (LAIDist0 LAIDist0 ...)
  (LAIDist1 LAIDist1 ...)
  (PARrel 0.05 [])
  (Offset 0 [cm])
  (LeafAI 0 [m2/m2])
  (StemAI 0 [m2/m2])
  (SOrgAI 0 [m2/m2])
  (LADm -9999.99 [cm2/cm3]) >
```

- *PARref*: number (dimensionless)
Parameter (default 0.06)
PAR reflectance.
- *PARext*: number (dimensionless)
Parameter (default 0.6)
PAR extinction coefficient.
- *NIRref*: number (dimensionless)
Parameter (default 0.51)
NIR reflectance. NIRref = 0.51 (Ross, 1975)
- *NIRext*: number (dimensionless)
Parameter (default 0.18)
NIR extinction coefficient. NIRext = 0.18 (Jones, 1983)
- *EPext*: number (dimensionless)
Parameter (default 0.5)
EP extinction coefficient.

- *IntcpCap*: number [**mm**]
Parameter (default 0.5)
Interception capacity.
- *EpFac*: number (dimensionless)
Parameter (default 1.2)
Value description: See figure 4 in the cited paper.

With a bare soil factor of 0.6 a combined Kc of 1.15 is reached at LAI=5. See also [Kjaersgaard et al., 2008]
Potential evapotranspiration factor.
- *EpFacWet*: number (dimensionless)
Optional parameter
Potential evapotranspiration factor for wet surface. By default this is identical to EpFac.
- *EpFacDS*: plf [**DS** → **<none>**]
Parameter (has default const value 1 [])
DS dependent potential evapotranspiration factor.
- *rs_max*: number [**s/m**]
Parameter (default 100000)
Maximum transpiration resistance.
- *rs_min*: number [**s/m**]
Parameter (default 100)
Minimum transpiration resistance.
- *Height*: number [**cm**]
State variable (default 0)
Crop height.
- *leaf_width*: plf [**DS** → **cm**]
Parameter (has default const value 3 [cm])
Leaf width.
- *SpLAI*: number [**(m²/m²)/(g DM/m²)**]
Parameter
Specific leaf weight.
- *LeafAIMod*: plf [**DS** → **<none>**]
Parameter (has default const value 1 [])
Specific leaf weight modifier. Used only after the intital phase.
- *SpSOrgAI*: number [**(m²/m²)/(g DM/m²)**]
Parameter (default 0)
Specific storage organ weight. Used only after the intital phase.
- *SOrgAIMod*: plf [**DS** → **<none>**]
Parameter (has default const value 1 [])
Specific storage organ weight modifier. Used only after the intital phase.
- *SOrgPhotEff*: number (dimensionless)
Parameter (default 1)
Relative photosynthetic efficiency of storage organ. Used only after the intital phase.

- *SOrgEffvsDS*: plf [**DS** → <none>]
Parameter (has default const value 1 [])
SOrg photosynthetic efficiency as function of DS.
- *SpStemAI*: number [(m²/m²)/(g DM/m²)]
Parameter (default 0)
Specific stem weight. Used only after the intital phase.
- *StemAIMod*: plf [**DS** → <none>]
Parameter (has default const value 1 [])
Specific stem weight modifier. Used only after the intital phase.
- *StemPhotEff*: number (dimensionless)
Parameter (default 1)
Relative photosynthetic efficiency of stem. Used only after the intital phase.
- *StemEffvsDS*: plf [**DS** → <none>]
Parameter (has default const value 1 [])
Stem photosynthetic efficiency as function of DS.
- *HvsDS*: plf [**DS** → cm]
Parameter
Crop height as function of DS.
- *HvsWStem*: plf [g DM/m² → <fraction>]
Parameter (has default value with 2 points)



```
(HvsWStem (0 [g DM/m^2] 0.1 []) (200 [g DM/m^2] 1 []))
```

Parameter description:

Relative crop height as function of stem weight. By default, it needs 200 g DM/m² to reach full height.

- *LAI Dist0*: number (dimensionless) array of length 3
Parameter
Relative CAI distribution at DS=0.

- *LAI**Dist1*: number (dimensionless) array of length 3
Parameter
Relative CAI distribution at DS=1.
- *PAR**rel*: number (dimensionless)
Parameter (default 0.05)
Relative PAR below the canopy. If the relative PAR get below this, the bottom leaves will start dying.
- *Offset*: number [**cm**]
State variable (default 0)
Extra height after harvest.
- *LeafAI*: number [**m**²/**m**²]
State variable (default 0)
Leaf Area Index.
- *StemAI*: number [**m**²/**m**²]
State variable (default 0)
Stem Area Index.
- *SOrgAI*: number [**m**²/**m**²]
State variable (default 0)
Storage Organ Area Index.
- *LAD**m*: number [**cm**²/**cm**³]
State variable (default -9999.99)
Maximal Leaf Area Density.

Log Variables

- *CAI*: number [**m**²/**m**²]
Crop Area Index.
- *LAI**vsH*: plf [**cm** → **m**²/**m**²]
Accumulated Leaf Area Index at Height.
- *ForcedCAI*: number [**m**²/**m**²]
CAI forced upon us by vegetation module.
- *SimCAI*: number [**m**²/**m**²]
CAI simulated by crop model.
- *CAI**m**Rat*: number (dimensionless)
(CAIm - CAI) / CAIm.

107.6 CrpN

Default crop nitrogen parameters.

```

< CrpN  (PtLeafCnc PtLeafCnc)
        (CrLeafCnc CrLeafCnc)
        (NfLeafCnc NfLeafCnc)
        (PtStemCnc PtStemCnc)
        (CrStemCnc CrStemCnc)
        (NfStemCnc NfStemCnc)
        (PtSOrgCnc PtSOrgCnc)
        (CrSOrgCnc CrSOrgCnc)
        (NfSOrgCnc NfSOrgCnc)
        (PtRootCnc PtRootCnc)
        (CrRootCnc CrRootCnc)
        (NfRootCnc NfRootCnc)
        (TLLeafEff TLLeafEff) ; Has default value.
        (TLRootEff TLRootEff) ; Has default value.
        (NO3_root_min 0 [g N/cm3])
        (NH4_root_min 0 [g N/cm3])
        (NO3_root_min_luxury NO3_root_min_luxury)
        (NH4_root_min_luxury NH4_root_min_luxury)
        (nitrogen_stress_days 0 [d])
        (DS_fixate 42000 [])
        (DS_cut_fixate 0 [])
        (fixate_factor 0.8 [h-1])
        (DS_start_fixate DS_start_fixate) >

```

- *PtLeafCnc*: plf [DS → g N/g DM]
Parameter
Upper limit for N-concentration in leaves.
- *CrLeafCnc*: plf [DS → g N/g DM]
Parameter
Critical limit for N-concentration in leaves.
- *NfLeafCnc*: plf [DS → g N/g DM]
Parameter
Non-functional limit for N-concentration in leaves.
- *PtStemCnc*: plf [DS → g N/g DM]
Parameter
Upper limit for N-concentration in stem.
- *CrStemCnc*: plf [DS → g N/g DM]
Parameter
Critical limit for N-concentration in stem.
- *NfStemCnc*: plf [DS → g N/g DM]
Parameter
Non-functional limit for N-concentration in stem.
- *PtSOrgCnc*: plf [DS → g N/g DM]
Parameter
Upper limit for N-concentration in storage organ.
- *CrSOrgCnc*: plf [DS → g N/g DM]
Parameter
Critical limit for N-concentration in storage organ.

- *NfSOrgCnc*: plf [**DS** → **g N/g DM**]
Parameter
Non-functional limit for N-concentration in storage organ.
- *PtRootCnc*: plf [**DS** → **g N/g DM**]
Parameter
Upper limit for N-concentration in roots.
- *CrRootCnc*: plf [**DS** → **g N/g DM**]
Parameter
Critical limit for N-concentration in roots.
- *NfRootCnc*: plf [**DS** → **g N/g DM**]
Parameter
Non-functional lim for N-concentration in roots.
- *TLLefEff*: plf [**DS** → **<fraction>**]
Parameter (has default const value 0.9 [])
Translocation efficiency, Leaf.
- *TLLRootEff*: plf [**DS** → **<fraction>**]
Parameter (has default const value 0.1 [])
Translocation efficiency, Root.
- *NO3_root_min*: number [**g N/cm³**]
Parameter (default 0)
Minimum nitrate concentration near roots for uptake.
- *NH4_root_min*: number [**g N/cm³**]
Parameter (default 0)
Minimum ammonium concentration near roots for uptake.
- *NO3_root_min_luxury*: number [**g N/cm³**]
Optional parameter
Minimum nitrate concentration near roots for luxury uptake. Only used in reproductive phase. By default identical to 'NO3_root_min'.
- *NH4_root_min_luxury*: number [**g N/cm³**]
Optional parameter
Minimum ammonium concentration near roots for luxury uptake. Only used in reproductive phase. By default identical to 'NH4_root_min'.
- *nitrogen_stress_days*: number [**d**]
State variable (default 0)
Number of days production has halted due to nitrogen stress. This is the sum of nitrogen stress for each hour, multiplied with the action of the radiation of that day that was received that hour.
- *DS_fixate*: number (dimensionless)
Parameter (default 42000)
DS at which to start fixation of atmospheric N.
- *DS_cut_fixate*: number (dimensionless)
Parameter (default 0)
Restore fixation this DS after cut.
- *fixate_factor*: number [**h⁻¹**]
Parameter (default 0.8)
Fraction rate relative to potential N uptake.

- *DS_start_fixate*: number (dimensionless)
Optional state variable
Development stage at which to restart fixation after a cut.

Log Variables

- *PtNCnt*: number [g/m²]
Potential nitrogen content in crop.
- *CrNCnt*: number [g/m²]
Critical nitrogen content in crop.
- *NfNCnt*: number [g/m²]
Non-functional nitrogen content in crop.
- *nitrogen_stress*: number (dimensionless)
Nitrogen stress factor.
- *NNI*: number (dimensionless)
Nitrogen nutrition index. Actual nitrogen content of shoot divided by critical nitrogen content. Will be less than one for a stressed plant. For now, the storage organ is counted as part of the shoot.
- *Fixated*: number [g N/m²/h]
N fixation from air.
- *AccFixated*: number [g N/m²]
Accumuated N fixation from air.

107.7 Harvesting

Information about what happens to the crop at harvest and cut.

```
< Harvesting (Root Root ...) ; Has default value.
              (Stem Stem ...) ; Has default value.
              (Leaf Leaf ...) ; Has default value.
              (Dead Dead ...) ; Has default value.
              (SOrg SOrg ...) ; Has default value.
              (EconomicYield_W 1 [])
              (EconomicYield_N EconomicYield_N)
              (DSmax 0.8 [])
              (DSnew DSnew)
              (last_cut last_cut)
              (production_delay 0 [d])
              (cut_delay cut_delay) ; Has default value.
              (total_water_use 0 [kg H2O])
              (sorg_height sorg_height) >
```

- *Root*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(Root "AOM-SLOW"
      "AOM-FAST")
```

Parameter description:
Root AM parameters.

- *Stem*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(Stem "AOM-SLOW"
      "AOM-FAST")
```

Parameter description:
Stem AM parameters.

- *Leaf*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(Leaf "AOM-SLOW"
      "AOM-FAST")
```

Parameter description:
Leaf AM parameters.

- *Dead*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(Dead "AOM-SLOW"
      "AOM-FAST")
```

Parameter description:
Dead leaves AM parameters.

- *SOrg*: **AOM** component (see chapter 8) sequence
Component (has default value with length 2)

```
(SOrg "AOM-SLOW"
      "AOM-FAST")
```

Parameter description:
Storage organ AM parameters.

- *EconomicYield_W*: number (dimensionless)
Parameter (default 1)
Valuable fraction of storage organ (DM), e.g. grain or tuber.
- *EconomicYield_N*: number (dimensionless)
Optional parameter
Valuable fraction of storage organ (N). By default the value for DM is used.
- *DSmax*: number (dimensionless)
Parameter (default 0.8)
Maximal development stage for which the crop survives harvest.
- *DSnew*: number (dimensionless)
Optional parameter
New development stage after harvest. If not specified, use the DS where an uncut crop would first reach the height it now has after the cut. I.e. it uses the inverse function of the HvsDS Canopy parameter to find the new DS.
- *last_cut*: **Time** fixed component (see section 107.27)
Optional submodel
Date of last cut. Used for calculating cut delay.

- *production_delay*: number [d]
State variable (default 0)
production delay caused by last cut
- *cut_delay*: plf [kg DM/ha → d]
Parameter (has default const value 0 [d])
Production and development delay in days as a function of the shoot DM removed by harvest. By default, there is no delay.
- *total_water_use*: number [kg H₂O]
State variable (default 0)
Total evapotranspiration since emergence.
- *sorg_height*: number [cm]
Optional parameter
Vertical location of storage organ. Set this to a negative number for root fruits, this will cause harvesting to imply a suitable tillage operation, and guarantee that harvest will kill the plant. By default, the storage organ is assumed to be located far above ground.

Log Variables

- *cut_stress*: number [<fraction>]
Stress induced due to last cut.

107.8 Partition

Assimilate partitioning in the default crop model. The 'Root' parameter determine what fraction of the assimilate for growth goes to roots at a given development stage. The remaining assimilate goes to the shoot. The 'Leaf' and 'Stem' parameters determine what fraction of the shoot assimilate goes to the leaf and stem respectively. The remaining shoot assimilate will go to the storage organ.

```
< Partition (Root Root)
              (Stem Stem)
              (Leaf Leaf)
              (RSR RSR)
              (max_WRoot max_WRoot) ; Has default value.
              (nitrogen_stress_limit 1 [])
              (NNI_crit 0 [])
              (NNI_inc 0 []) >
```

- *Root*: plf [DS → <fraction>]
Parameter
Fraction of assimilate for growth that goes to the roots, as a function of the crop development stage. The remaining growth assimilate goes to the shoot.
- *Stem*: plf [DS → <fraction>]
Parameter
Fraction of shoot assimilate that goes to the stem.
- *Leaf*: plf [DS → <fraction>]
Parameter
Fraction of shoot assimilate that goes to the leaves.
- *RSR*: plf [DS → <none>]
Parameter

Maximal root/shoot ratio as a function of development state. If the root/shoot ratio is above this, the roots will start dying.

- *max_WRoot*: plf [$\mathbf{DS} \rightarrow \mathbf{g\ DM/m^2}$]
 Parameter (has default const value $1\text{e}+06$ [g DM/m²])
 Maximal root DM as a function of development state. If the root DM is above this, no assimilate will be allocated to them.

- *nitrogen_stress_limit*: number (dimensionless)
 Parameter (default 1)
 If nitrogen stress is above this number and DS is above 1, allocate all assimilate to the storage organ.

- *NNI_crit*: number (dimensionless)
 Parameter (default 0)
 When NNI is below this value, modify Stem/Leaf partitioning. See also [Gyldengren et al., 2020]

- *NNI_inc*: number (dimensionless)
 Parameter (default 0)
 Stem/Leaf partitioning modifier for low NNI. See also [Gyldengren et al., 2020]

Log Variables

- *cf*: number (dimensionless)
 Stem/Leaf partitioning modifier for low NNI when $\mathbf{DS} < 1.0$. $\mathbf{cf} = 1 + (\mathbf{NNI_crit} - \mathbf{NNI}) * \mathbf{NNI_inc}$. Stem assimilate partitioning is increased with cf, which is taken from the amount allocated to leaves. See also [Gyldengren et al., 2020]

107.9 Production

Crop production in the default crop model.

```

< Production  (WRoot 0.001 [g DM/m2])
               (ShldResC 0 [<fraction>])
               (ReMobilEff 0.85 [<fraction>])
               (ReMobilDS 1.2 [DS])
               (ReMobilRt 0.1 [d-1])
               (ReMobilMod ReMobilMod) ; Has default value.
               (StemRes 0 [g DM/m2])
               (CH20ReleaseRate 0.04 [h-1])
               (E_Root 0.69 [g DM-C/g Ass-C])
               (E_Leaf 0.68 [g DM-C/g Ass-C])
               (E_Stem 0.66 [g DM-C/g Ass-C])
               (E_SOrg E_SOrg)
               (MaintResp_DS MaintResp_DS) ; Has default value.
               (r_Root 0.015 [d-1])
               (r_Leaf r_Leaf)
               (r_Stem r_Stem)
               (r_SOrg r_SOrg)
               (ExfoliationFac 1 [h-1])
               (LfDR LfDR)
               (RtDR RtDR)
               (Large_RtDR 0.05 [d-1])
               (RtDR.T_factor RtDR.T_factor) ; Has default value.
               (water_log_h_limit 0 [cm])
               (water_log_root_limit 0.1 [cm/cm3])
               (RtDR_water_log_factor RtDR_water_log_factor) ; Has default value.
               (RtDR_water_log_addend RtDR_water_log_addend) ; Has default value.
               (IntDSRelRtRes 0.8 [])
               (EndDSRelRtRes 0.8 [])
               (RelRateRtRes 0.05 [d-1])
               (LfRtRelRtRes 0.8 [])
               (CH20Pool 0.001 [g CH20/m2])
               (WLeaf 0.001 [g DM/m2])
               (WStem 0 [g DM/m2])
               (WSOrg 0 [g DM/m2])
               (WDead 0 [g DM/m2])
               (NLeaf 0 [g N/m2])
               (NStem 0 [g N/m2])
               (NRoot 0 [g N/m2])
               (NSOrg 0 [g N/m2])
               (NDead 0 [g N/m2])
               (NCrop NCrop)
               (C_AM 0 [g C/m2])
               (N_AM 0 [g N/m2])
               (DailyNetRoot 0 [g DM/m2])
               (DailyNetShoot 0 [g DM/m2]) >

```

- *WRoot*: number [g DM/m²]
State variable (default 0.001)
Root dry matter weight.
- *ShldResC*: number [<fraction>]
Parameter (default 0)
Capacity of shielded reserves (fraction of stem DM).
- *ReMobilEff*: number [<fraction>]

Parameter (default 0.85)
Efficiency of reserve mobilization

- *ReMobilDS*: number [**DS**]
Parameter (default 1.2)
Remobilization, Initial DS.
- *ReMobilRt*: number [**d**⁻¹]
Parameter (default 0.1)
Remobilization, release rate.
- *ReMobilMod*: plf [**DS** → <**none**>]
Parameter (has default const value 1 [])
Modifier of remobilization rate depending on DS.
- *StemRes*: number [**g DM/m**²]
State variable (default 0)
Shielded reserves in stems.
- *CH2OReleaseRate*: number [**h**⁻¹]
Parameter (default 0.04)
CH2O Release Rate constant.
- *E_Root*: number [**g DM-C/g Ass-C**]
Parameter (default 0.69)
Conversion efficiency, root.
- *E_Leaf*: number [**g DM-C/g Ass-C**]
Parameter (default 0.68)
Conversion efficiency, leaf.
- *E_Stem*: number [**g DM-C/g Ass-C**]
Parameter (default 0.66)
Conversion efficiency, stem.
- *E_SOrg*: number [**g DM-C/g Ass-C**]
Parameter
Conversion efficiency, storage organ.
- *MaintResp_DS*: plf [**DS** → <**none**>]
Parameter (has default const value 1 [])
Reduction of maintenance respiration with crop senescence
- *r_Root*: number [**d**⁻¹]
Parameter (default 0.015)
Maintenance respiration coefficient, root.
- *r_Leaf*: number [**d**⁻¹]
Parameter
Maintenance respiration coefficient, leaf.
- *r_Stem*: number [**d**⁻¹]
Parameter
Maintenance respiration coefficient, stem.
- *r_SOrg*: number [**d**⁻¹]
Parameter
Maintenance respiration coefficient, storage organ.

- *ExfoliationFac*: number [$\mathbf{h}^{-1}$]
Parameter (default 1)
Exfoliation factor, 0-1.
- *LfDR*: plf [$\mathbf{DS} \rightarrow \mathbf{d}^{-1}$]
Parameter
Death rate of Leafs.
- *RtDR*: plf [$\mathbf{DS} \rightarrow \mathbf{d}^{-1}$]
Parameter
Death rate of Roots.
- *Large_RtDR*: number [$\mathbf{d}^{-1}$]
Parameter (default 0.05)
Extra death rate for large root/shoot.
- *RtDR_T_factor*: plf [$\mathbf{dg} \mathbf{C} \rightarrow \langle \mathbf{none} \rangle$]
Parameter (has default const value 1 [])
Temperature dependent factor for root death rate.
- *water_log_h_limit*: number [$\mathbf{cm}$]
Parameter (default 0)
Roots are water logged if the soil water potential exceed this threshold.
- *water_log_root_limit*: number [$\mathbf{cm}/\mathbf{cm}^3$]
Parameter (default 0.1)
Soil is part of the root zone if root density is above this threshold. Set it to zero to use the calculated root depth instead.
- *RtDR_water_log_factor*: plf [$\langle \mathbf{fraction} \rangle \rightarrow \langle \mathbf{none} \rangle$]
Parameter (has default const value 1 [])
Water logging dependent factor for root death rate. This is a function of the fraction of the root zone being water logged.
- *RtDR_water_log_addend*: plf [$\langle \mathbf{fraction} \rangle \rightarrow \mathbf{d}^{-1}$]
Parameter (has default const value 0 [$\mathbf{d}^{-1}$])
Water logging dependent addend for root death rate. This is a function of the fraction of the root zone being water logged.
- *IntDSRelRtRes*: number (dimensionless)
Parameter (default 0.8)
Initial DS for the release of root reserves.
- *EndDSRelRtRes*: number (dimensionless)
Parameter (default 0.8)
End DS for the release of root reserves.
- *RelRateRtRes*: number [$\mathbf{d}^{-1}$]
Parameter (default 0.05)
Release rate of root reserves.
- *LfRtRelRtRes*: number (dimensionless)
Parameter (default 0.8)
Max Leaf:Root for the release of root res.
- *CH2OPool*: number [$\mathbf{g} \mathbf{CH2O}/\mathbf{m}^2$]
State variable (default 0.001)
CH2O Pool.

- *WLeaf*: number [g DM/m²]
State variable (default 0.001)
Leaf dry matter weight.
- *WStem*: number [g DM/m²]
State variable (default 0)
Stem dry matter weight.
- *WSOrg*: number [g DM/m²]
State variable (default 0)
Storage organ dry matter weight.
- *WDead*: number [g DM/m²]
State variable (default 0)
Dead leaves dry matter weight.
- *NLeaf*: number [g N/m²]
State variable (default 0)
Nitrogen stored in the leaves.
- *NStem*: number [g N/m²]
State variable (default 0)
Nitrogen stored in the stem.
- *NRoot*: number [g N/m²]
State variable (default 0)
Nitrogen stored in the roots.
- *NSOrg*: number [g N/m²]
State variable (default 0)
Nitrogen stored in the storage organ.
- *NDead*: number [g N/m²]
State variable (default 0)
Nitrogen stored in dead leaves.
- *NCrop*: number [g N/m²]
Optional state variable
Total crop nitrogen content. By default, this will start as the amount of N in the seed.
- *C_{AM}*: number [g C/m²]
State variable (default 0)
Added C in plant material.
- *N_{AM}*: number [g N/m²]
State variable (default 0)
Added N in plant material.
- *DailyNetRoot*: number [g DM/m²]
State variable (default 0)
Root growth minus root respiration so far this day.
- *DailyNetShoot*: number [g DM/m²]
State variable (default 0)
Leaf growth minus leaf respiration so far this day.

Log Variables

- *CRoot*: number [g C/m²]
Root C weight.
- *WCrop*: number [g DM/m²]
Total crop dry matter, including roots and dead leaves.
- *CLeaf*: number [g C/m²]
Leaf C weight.
- *CStem*: number [g C/m²]
Stem C weight.
- *CSOrg*: number [g C/m²]
Storage organ C weight.
- *CDead*: number [g C/m²]
Dead leaves C weight.
- *CCrop*: number [g C/m²]
Crop C weight.
- *PotCanopyAss*: number [g CH₂O/m²/h]
Potential canopy assimilation, i.e. stressfree production.
- *CanopyAss*: number [g CH₂O/m²/h]
Canopy assimilation.
- *NetPhotosynthesis*: number [g CO₂/m²/h]
Net Photosynthesis.
- *AccNetPhotosynthesis*: number [g CO₂/m²]
Accumulated Net Photosynthesis.
- *Respiration*: number [g CH₂O/m²/h]
Crop Respiration.
- *MaintRespiration*: number [g CH₂O/m²/h]
Maintenance Respiration.
- *GrowthRespiration*: number [g CH₂O/m²/h]
Growth Respiration.
- *LeafRespiration*: number [g CO₂/m²/h]
Total Leaf Respiration.
- *StemRespiration*: number [g CO₂/m²/h]
Total Stem Respiration.
- *SOrgRespiration*: number [g CO₂/m²/h]
Total SOrg Respiration.
- *RootRespiration*: number [g CO₂/m²/h]
Total Root Respiration.
- *LeafMaintRespiration*: number [g CO₂/m²/h]
Leaf Maintenance Respiration.
- *StemMaintRespiration*: number [g CO₂/m²/h]
Stem Maintenance Respiration.

- *SOrgMaintRespiration*: number [g CO₂/m²/h]
SOrg Maintenance Respiration.
- *RootMaintRespiration*: number [g CO₂/m²/h]
Root Maintenance Respiration.
- *LeafGrowthRespiration*: number [g CO₂/m²/h]
Leaf Growth Respiration.
- *StemGrowthRespiration*: number [g CO₂/m²/h]
Stem Growth Respiration.
- *SOrgGrowthRespiration*: number [g CO₂/m²/h]
SOrg Growth Respiration.
- *RootGrowthRespiration*: number [g CO₂/m²/h]
Root Growth Respiration.
- *IncWLeaf*: number [g DM/m²/h]
Leaf growth.
- *IncWStem*: number [g DM/m²/h]
Stem growth.
- *IncWSOrg*: number [g DM/m²/h]
Storage organ growth.
- *IncWRoot*: number [g DM/m²/h]
Root growth.
- *DeadWLeaf*: number [g DM/m²/h]
Leaf DM removed.
- *DeadNLeaf*: number [g N/m²/h]
Leaf N removed.
- *DeadWRoot*: number [g DM/m²/h]
Root DM removed.
- *DeadNRoot*: number [g N/m²/h]
Root N removed.
- *C_Loss*: number [g C/m²/h]
C lost from the crop
- *water_logged*: number [<fraction>]
Fraction of root zone water logged.

107.10 Irrigation

Keep track of active irrigation events. Usually not set explicitly, but may be found in a checkpoint.

< Irrigation (event) >

- *event*: submodel (see section 4.1.7) sequence
Submodel (default: an empty sequence)
Currently active irrigation events.

- ```

< (solute solute ...)
 (flux flux)
 (volume volume)
 (temperature temperature)
 (time_left time_left)
 (target target)
 (silence silence) >

```
- *solute*: submodel (see section 4.1.7) sequence  
Solute in irrigation water.  

```

< name value >

```

    - \* *name*: string (see section 4.1.5)  
State variable  
Name of chemical.
    - \* *value*: number [g/cm<sup>2</sup>/mm]  
State variable  
Value for chemical.
  - *flux*: number [mm/h]  
State variable  
Water applied.
  - *volume*: **volume** component (see chapter 102)  
Optional component  
Soil volume to apply for subsoil irrigation. Ignored for overhead and surface irrigation.
  - *temperature*: number [dg C]  
Optional state variable  
Irrigation temperature. By default, use daily air temperature. Ignored for subsoil irrigation.
  - *time\_left*: number [h]  
State variable  
Time left of this irrigation event.
  - *target*: string (see section 4.1.5)  
State variable  
Where to apply the irrigation.  
overhead: Above crop canopy. surface: On soil surface, below crop canopy. subsoil: In the soil. The 'volume' parameter will specify where.
  - *silence*: boolean (see section 4.1.2)  
State variable  
True if event should not declare when it is over.

## 107.11 Bioincorporation

Biological incorporation of organic matter in soil.

```

< Bioincorporation (AOM AOM ...) ; Has default value.
 (T_factor T_factor) ; Has default value.
 (R_max 0.5 [g DM/m2/h])
 (k_half 1 [g DM/m2])
 (C_per_N_factor C_per_N_factor) ; Has default value.
 (respiration 0 [<fraction>])
 (distribution distribution) ; Has default value. >

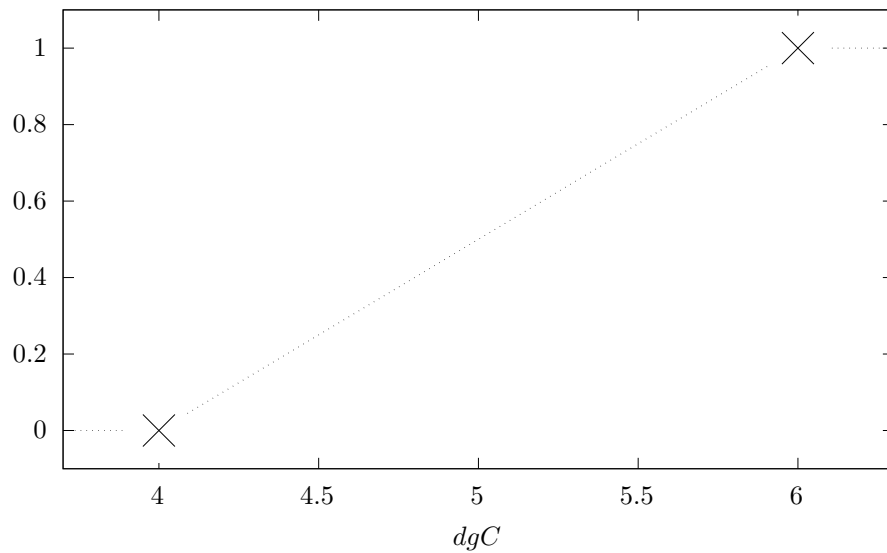
```

- *AOM*: **AOM** component (see chapter 8) sequence  
Component (has default value with length 2)

```
(AOM "AOM-SLOW-BIOINCORPORATION"
 "AOM-FAST")
```

Parameter description:  
Incorporated AM parameters.

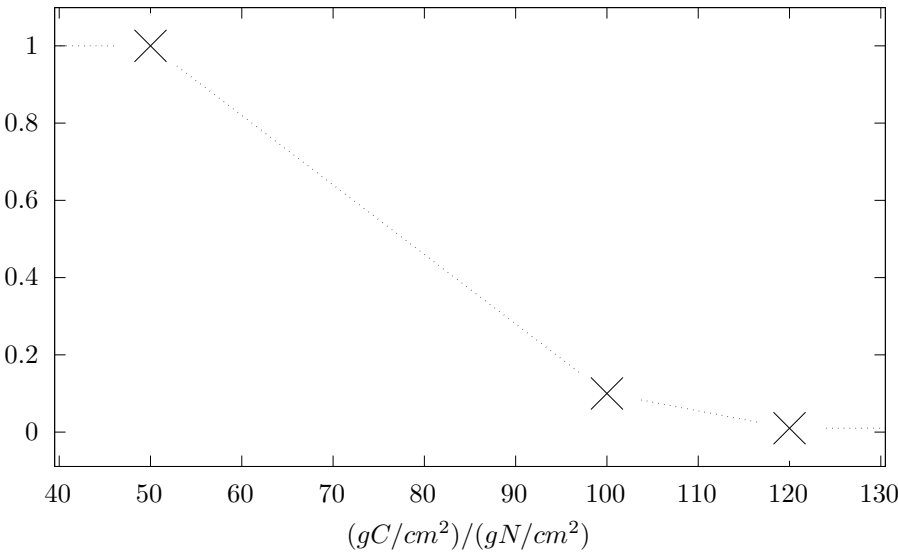
- *T\_factor*: plf [**dg C**  $\rightarrow$  **<none>**]  
Parameter (has default value with 2 points)



```
(T_factor (4 [dg C] 0 []) (6 [dg C] 1 []))
```

Parameter description:  
Limiting factor for low temperature.

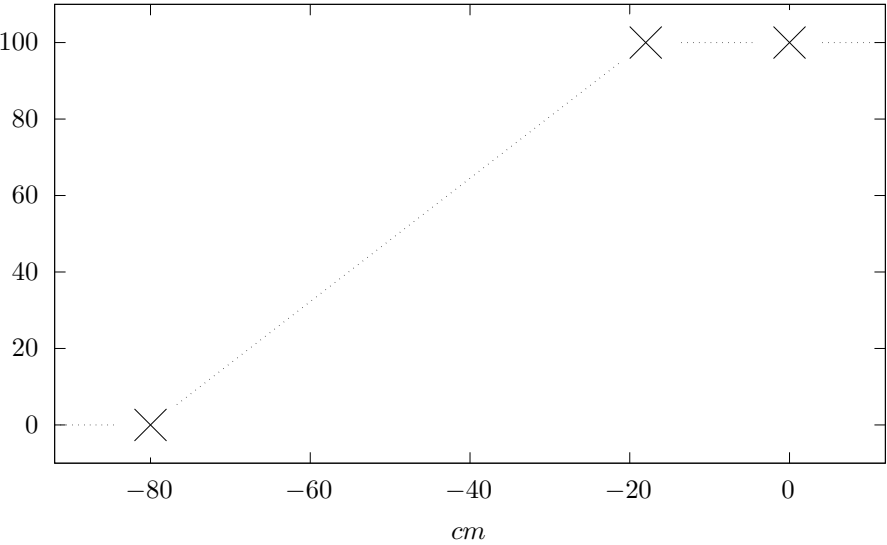
- *R\_max*: number [**g DM/m<sup>2</sup>/h**]  
Parameter (default 0.5)  
Maximal speed of incorporation.
- *k\_half*: number [**g DM/m<sup>2</sup>**]  
Parameter (default 1)  
Halflife constant.
- *C\_per\_N\_factor*: plf [**(g C/cm<sup>2</sup>)/(g N/cm<sup>2</sup>)**  $\rightarrow$  **<none>**]  
Parameter (has default value with 3 points)



```
(C_per_N_factor (50 [(g C/cm^2)/(g N/cm^2)] 1 [])
 (100 [(g C/cm^2)/(g N/cm^2)] 0.1 [])
 (120 [(g C/cm^2)/(g N/cm^2)] 0.01 []))
```

Parameter description:  
Limiting factor for high C/N ratio.

- *respiration*: number [**<fraction>**]  
Parameter (default 0)  
Fraction of C lost in respiration.
- *distribution*: plf [**cm** → **<none>**]  
Parameter (has default value with 3 points)



```
(distribution (-80 [cm] 0 []) (-18 [cm] 100 []) (0 [cm] 100 []))
```

Parameter description:

Distribution of incorporated matter in the soil. (X, Y), where X is the depth (negative numbers), and Y is the relative weight in that depth. To get the fraction in a specific interval [a:b], we integrate the plf over that interval, and divide by the integration over the whole profile.

## Log Variables

- *CO2*: number [g C/m<sup>2</sup>/h]  
C respired.
- *speed*: number [g DM/m<sup>2</sup>/h]  
Fraction of litter incorporated this hour. The formula is  $\text{speed} = (\text{R\_max} * \text{litter}) / (\text{k\_half} + \text{litter})$ .
- *DM*: number [g DM/m<sup>2</sup>/h]  
DM removed from surface.
- *C\_removed*: number [g C/m<sup>2</sup>/h]  
C removed from surface.
- *N\_removed*: number [g N/m<sup>2</sup>/h]  
N removed from surface.
- *C\_added*: number [g C/cm<sup>3</sup>/h] soil cells  
C added to soil.
- *N\_added*: number [g N/cm<sup>3</sup>/h] soil cells  
N added to soil.

## 107.12 DOM-Element

A single element in a Dissolved Organic Matter pool.

Used by DOM @ N (see 107.1, page 583) .

< "DOM-Element" (M M ...) >

- *M*: number [g/cm<sup>3</sup>] soil cells  
State variable  
Mass in water and soil.

## Log Variables

- *C*: number [g/cm<sup>3</sup>] soil cells  
Concentration in water.
- *S*: number [g/cm<sup>3</sup>/h] soil cells  
Combined source term.
- *J\_matrix*: number [g/cm<sup>2</sup>/h] soil edges  
Transportation in matrix (positive up).
- *J\_tertiary*: number [g/cm<sup>2</sup>/h] soil edges  
Transportation outside matrix (positive up).
- *S\_p*: number [g/cm<sup>3</sup>/h] soil cells  
Source term (macropore transport only).
- *S\_drain*: number [g/cm<sup>3</sup>/h] soil cells  
Source term (soil drainage only).

## 107.13 Fetch

A summary file line.

```
< Fetch tag >
```

- *tag*: string (see section 4.1.5)  
Parameter  
The tag of a column in the log file to summarize in this line.

## 107.14 FetchPretty

A summary file line.

```
< FetchPretty tag
 (factor 1 [])
 (name name) >
```

- *tag*: string (see section 4.1.5)  
Parameter  
The tag of a column in the log file to summarize in this line.
- *factor*: number (dimensionless)  
Parameter (default 1)  
Factor to multiply with to get the sum. Typically 1.0 to add this line, or -1.0 to subtract it.
- *name*: string (see section 4.1.5)  
Optional parameter  
Name to use for this line. By default use the tag.

## 107.15 Harvest

Log of all harvests during the simulation.

```
< Harvest (crop crop)
 (column column)
 (water_stress_days water_stress_days)
 (sow_time sow_time)
 (emerge_time emerge_time)
 (flowering_time flowering_time)
 (ripe_time ripe_time)
 (nitrogen_stress_days nitrogen_stress_days)
 (harvest_time harvest_time)
 (stem_DM stem_DM)
 (stem_N stem_N)
 (stem_C stem_C)
 (dead_DM dead_DM)
 (dead_N dead_N)
 (dead_C dead_C)
 (leaf_DM leaf_DM)
 (leaf_N leaf_N)
 (leaf_C leaf_C)
 (sorg_DM sorg_DM)
 (sorg_N sorg_N)
 (sorg_C sorg_C)
 (water_productivity water_productivity)
 (harvest_index harvest_index) >
```

- *crop*: string (see section 4.1.5)  
State variable  
Name of crop that was harvested.
- *column*: string (see section 4.1.5)  
State variable  
Name of column where the yield were harvested.
- *water\_stress\_days*: number [d]  
State variable  
Production days lost due to water stress.
- *sow\_time*: **Time** fixed component (see section 107.27)  
Optional submodel  
Time of sowing or planting of crop.
- *emerge\_time*: **Time** fixed component (see section 107.27)  
Optional submodel  
Time the crop emerge on the field.
- *flowering\_time*: **Time** fixed component (see section 107.27)  
Optional submodel  
Time of flowering.
- *ripe\_time*: **Time** fixed component (see section 107.27)  
Optional submodel  
Time the crop became ripe.
- *nitrogen\_stress\_days*: number [d]  
State variable  
Production days lost due to water stress.
- *harvest\_time*: **Time** fixed component (see section 107.27)  
Optional submodel  
Time of the harvest operation.
- *stem\_DM*: number [g/m<sup>2</sup>]  
State variable  
Total stem dry matter in harvest.
- *stem\_N*: number [g/m<sup>2</sup>]  
State variable  
Total stem nitrogen in harvest.
- *stem\_C*: number [g/m<sup>2</sup>]  
State variable  
Total stem carbon in harvest.
- *dead\_DM*: number [g/m<sup>2</sup>]  
State variable  
Total dead leaf dry matter in harvest.
- *dead\_N*: number [g/m<sup>2</sup>]  
State variable  
Total dead leaf nitrogen in harvest.
- *dead\_C*: number [g/m<sup>2</sup>]  
State variable  
Total dead leaf carbon in harvest.



- *leaf\_DM*: number [g/m<sup>2</sup>]  
State variable  
Total leaf dry matter in harvest.
- *leaf\_N*: number [g/m<sup>2</sup>]  
State variable  
Total leaf nitrogen in harvest.
- *leaf\_C*: number [g/m<sup>2</sup>]  
State variable  
Total leaf carbon in harvest.
- *sorg\_DM*: number [g/m<sup>2</sup>]  
State variable  
Total storage organ dry matter in harvest.
- *sorg\_N*: number [g/m<sup>2</sup>]  
State variable  
Total storage organ nitrogen in harvest.
- *sorg\_C*: number [g/m<sup>2</sup>]  
State variable  
Total storage organ carbon in harvest.
- *water\_productivity*: number [kg DM/m<sup>3</sup> H<sub>2</sub>O]  
State variable  
Storage organ harvested per evapotranspiration.
- *harvest\_index*: number [<fraction>]  
State variable  
Economic yield as fraction of total shoot.

## Log Variables

- *sorg\_N\_per\_DM*: number [<fraction>]  
Nitrogen fraction is storage dy matter.

## 107.16 Geometry1D

A one dimensional discretization of the soil.

< Geometry1D (zplus *zplus* ...) >

- *zplus*: number [cm] soil cells  
Optional parameter  
Depth of each numeric layer (a negative number). The end points are listed descending from the surface to the bottom.

## 107.17 GeometryRect

A rectangular discretization of the soil.

< GeometryRect (zplus *zplus* ...) (xplus *xplus* ...) >

- *zplus*: number [cm] sequence  
Parameter  
Depth of each numeric layer (a negative number). The end points are listed descending from the surface to the bottom.

- *xplus*: number [**cm**] sequence  
Parameter  
Horizontal end of each numeric layer (a positive number). The end points are listed ascending from left (0.0) to right.

## 107.18 HorHeat

Heat capacity and conductivity per horizon.

Used by horizon component HorHeat (see 47, page 231) .

```
< HorHeat (intervals 100)
 (quarts_form_factor 2 [])
 (mineral_form_factor 4 [])
 (C_soil C_soil)
 (K_water K_water ...)
 (K_ice K_ice ...) >
```

- *intervals*: integer  
Parameter (default 100)  
Number of numeric intervals to use in the heat conductivity table.
- *quarts\_form\_factor*: number (dimensionless)  
Parameter (default 2)  
Geometry factor used for conductivity calculation.
- *mineral\_form\_factor*: number (dimensionless)  
Parameter (default 4)  
Geometry factor used for conductivity calculation.
- *C\_soil*: number [**erg/cm<sup>3</sup>/dg C**]  
Optional parameter  
The soils heat capacity. By default, this is calculated from the soil constituents.
- *K\_water*: number [**erg/s/cm/dg C**] sequence  
Optional parameter  
Heat conductivity table for water in soil. By default, this is calculated from the soil constituents.
- *K\_ice*: number [**erg/s/cm/dg C**] sequence  
Optional parameter  
Heat conductivity table for solid frozen soil. By default, this is calculated from the soil constituents.

## 107.19 SoilLayer

A location and content of a soil layer. The layers apply to the soil section not covered by the 'zones' parameter.

Used by Soil @ horizons (see 107.21, page 611) .

```
< SoilLayer end horizon >
```

- *end*: number [**cm**]  
Parameter  
End point of this layer (a negative number).
- *horizon*: **horizon** component (see chapter 47)  
Soil properties of this layer.

## 107.20 SoilRegion

A location and content of a soil zone. If several zones cover the same soil, the first one listed is used. If no zones cover the soil, the 'horizons' parameter is used.

With regard to the numeric discretization, the whole cell is assumed to be of the soil found in the cell center.

Used by Soil @ zones (see 107.21, page 611) .

< SoilRegion *volume horizon* >

- *volume*: **zone** component (see chapter 106)  
Volume covered by this zone.
- *horizon*: **horizon** component (see chapter 47)  
Soil properties of this zone.

## 107.21 Soil

The soil submodel provides the numeric and physical properties of the soil.

```
< Soil (horizons horizons ...)
 (zones)
 (Hydraulic Hydraulic ...)
 (MaxRootingDepth MaxRootingDepth)
 (dispersivity 5 [cm])
 (dispersivity_transversal dispersivity_transversal)
 (border border ...)
 (frozen_water_Kfactor 0.561 []) >
```

- *horizons*: **SoilLayer** fixed component (see section 107.19) sequence  
Layered description of the soil properties. The horizons can be overlapped by the 'zones' parameter. Some groundwater models, specifically 'pipe', may cause an extra horizon to be added below the one specified here if you do not also specify an explicit geometry.
- *zones*: **SoilRegion** fixed component (see section 107.20) sequence  
Submodel (default: an empty sequence)  
Regions with special soil properties. This overrules the 'horizons' parameter.
- *Hydraulic*: **hydraulic** component (see chapter 48) soil cells  
Optional component  
Hydraulic properties for each cell. Use when the hydraulic properties are dynamic and cells specific. For example when implementing hysteresis.
- *MaxRootingDepth*: number [**cm**]  
Parameter  
Depth at the end of the root zone (a positive number).
- *dispersivity*: number [**cm**]  
Parameter (default 5)  
Dispersion length.
- *dispersivity\_transversal*: number [**cm**]  
Optional parameter  
Transversal dispersion length. By default, this is 0.1 times the dispersivity.
- *border*: number [**cm**] sequence  
Optional parameter  
List of flux depths where a mass balance should be possible when logging. By default, this is the endpoint of each horizon.

- *frozen\_water\_K\_factor*: number (dimensionless)  
Optional parameter (default 0.561)  
Hydraulic conductivity for water below 0 dg C compared to 20 dg C. The default value, 0.561, corresponds to 0 dg C water viscosity.

## 107.22 SoilHeat

Temperature and heat flux in soil.

```
< SoilHeat (T T ...)
 (q_lim 2.69978e+08 [erg/cm2/h])
 (initial_T initial_T ...)
 (initial_T_plf initial_T_plf)
 (h_frozen -15000 [cm-1])
 (enable_ice false)
 (ice_dt 0.05 [h])
 (T_limited T_limited)
 (T_thawing_epsilon 0.001 [dg C]) >
```

- *T*: number [dg C] soil cells  
Optional state variable  
Soil temperature.
- *q\_lim*: number [erg/cm<sup>2</sup>/h]  
Parameter (default 2.69978e+08)  
Limit surface flux to this value. Negative for no limit.
- *initial\_T*: submodel (see section 4.1.7) sequence  
Optional submodel  
Initial value of the 'T' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'T' is unspecified.  

```
< end value >
```

  - *end*: number [cm]  
Parameter  
End point of this layer (a negative number).
  - *value*: number [dg C]  
Parameter  
Soil temperature.
- *initial\_T\_plf*: plf [cm → dg C]  
Optional parameter  
Initial value of the 'T' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'T' and 'initial\_T' are unspecified.
- *h\_frozen*: number [cm<sup>-1</sup>]  
Parameter (default -15000)  
Pressure below which no more water will freeze.
- *enable\_ice*: string (see section 4.1.5)  
Parameter (default 'false')  
Set this to 'false' or 'true'

- *ice\_dt*: number [**h**]  
Parameter (default 0.05)  
Timestep to use when ice is enabled. This applies when there are subzero temperatures in the profile.
- *T\_limited*: number [**dg C**]  
Optional state variable  
Surface temperature limited by 'q\_lim'.
- *T\_thawing\_epsilon*: number [**dg C**]  
Parameter (default 0.001)  
Subtract this from T\_thawing for stability.

## Log Variables

- *state*: number (dimension not specified) soil cells  
Current freezing/melting state.
- *q*: number [**erg/cm<sup>2</sup>/h**] soil edges  
Heat flux.
- *conductivity*: number [**erg/cm/dg C/h**]  
Heat conductivity.
- *capacity*: number [**erg/cm<sup>3</sup>/dg C**]  
Heat capacity.
- *capacity\_apparent*: number [**erg/cm<sup>3</sup>/dg C**]  
Apparent heat capacity, including freezing.
- *T\_freezing*: number [**dg C**] soil cells  
Freezing point depression for freezing.
- *T\_thawing*: number [**dg C**] soil cells  
Freezing point depression for thawing.

## 107.23 SoilWater

Keep track of water and pressure in the soil matrix.

```
< SoilWater (h h ...)
 (Theta Theta ...)
 (S_permanent)
 (default_h default_h)
 (max_sink_change 0.1 [])
 (use_last true)
 (initial_h initial_h ...)
 (initial_h_plf initial_h_plf)
 (initial_Theta initial_Theta ...)
 (initial_Theta_plf initial_Theta_plf)
 (X_ice X_ice ...)
 (X_ice_buffer X_ice_buffer ...) >
```

- *h*: number [**cm**] soil cells  
Optional state variable  
Soil water pressure.

- *Theta*: number [**<fraction>**] soil cells  
Optional state variable  
Soil water content.
- *S\_permanent*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
State variable (default: an empty sequence)  
Permanent water sink, e.g. subsoil irrigation.
- *default\_h*: number [**cm**]  
Optional parameter  
Default h at surface if neither it nor Theta is specified. The depth will be added to the pressure. By default, this will be based on either distance to groundwater or field capacity (pF 2), whichever is lower.
- *max\_sink\_change*: number (dimensionless)  
Parameter (default 0.1)  
Largest change to available water within a timestep. This is used for calculating the suggested timestep. The suggested timestep will be small enough that the change water due to forward calculated sinks (S\_forward) alone is less than the specified value.  
  
Plant available water is defined as the difference between saturation and wilting point.
- *use\_last*: boolean (see section 4.1.2)  
Parameter (default true)  
Use last value specified for 'h' or 'Theta' for the rest of the profile.
- *initial\_h*: submodel (see section 4.1.7) sequence  
Optional submodel  
Initial value of the 'h' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'h' is unspecified.  

```
< end value >
```

  - *end*: number [**cm**]  
Parameter  
End point of this layer (a negative number).
  - *value*: number [**cm**]  
Parameter  
Soil water pressure.
- *initial\_h\_plf*: plf [**cm → cm**]  
Optional parameter  
Initial value of the 'h' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the the value of the PLF at the center of the cell. Only used if 'h' and 'initial\_h' are unspecified.
- *initial\_Theta*: submodel (see section 4.1.7) sequence  
Optional submodel  
Initial value of the 'Theta' parameter. The initial value is given as a sequence of (END VALUE) pairs, starting from the top and going down. The parameter will be initialized to VALUE from the END of the previous layer, to the END of the current layer. Only used if 'Theta' is unspecified.  

```
< end value >
```

- *end*: number [**cm**]  
Parameter  
End point of this layer (a negative number).
- *value*: number [<**fraction**>]  
Parameter  
Soil water content.
- *initial\_Theta\_plf*: plf [**cm** → <**fraction**>]  
Optional parameter  
Initial value of the 'Theta' parameter. The initial value is given as a sequence of (DEPTH VALUE) pairs, starting from the bottom and going up. The parameter will be initialized to the value of the PLF at the center of the cell. Only used if 'Theta' and 'initial\_Theta' are unspecified.
- *X\_ice*: number [<**fraction**>] soil cells  
Optional state variable  
Ice volume fraction in soil.
- *X\_ice\_buffer*: number (dimensionless) soil cells  
Optional state variable  
Ice volume that didn't fit the soil during freezing.

## Log Variables

- *K*: number [**cm/h**] soil cells  
Hydraulic conductivity.
- *S\_indirect\_drain*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
Matrix water sink due to soil drainage, only counting biopores.
- *S\_soil\_drain*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
Matrix water sink due to soil drainage, not counting biopores.
- *S\_p\_drain*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
Water moving from biopores to drain. This only reflects water in the tertiary domain, not in the matrix.
- *S\_B2M*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
Water flowing from biopores to matrix.
- *S\_M2B*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
Water flowing from matrix to biopores. This does not include drain or drain connected biopores.
- *S\_root*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
Water sink due to root uptake.
- *tillage*: number [**cm<sup>3</sup>/cm<sup>3</sup>/h**] soil cells  
Changes in water content due to tillage operations.
- *dt*: number [**h**]  
Suggested timestep length (based on S\_forward). The absolute value is used, negative numbers indicate source based limits.
- *sink\_cell*: integer  
Cell with largest forward sink compared to available water.

- *Theta\_primary*: number [ $\text{cm}^3/\text{cm}^3$ ] soil cells  
Water content in primary matrix system. Conventionally, this is the intra-aggregate pores.
- *Theta\_secondary*: number [ $\text{cm}^3/\text{cm}^3$ ] soil cells  
Water content in secondary matrix system. Conventionally, this is the inter-aggregate pores.
- *q*: number [ $\text{cm}/\text{h}$ ] soil edges  
Matrix water flux (positive numbers mean upward).
- *h\_ice*: number (dimensionless) soil cells  
Pressure at which all air is out of the matrix. When there are no ice, this is 0.0. When there are ice, the ice is presumed to occupy the large pores, so it is  $h$  ( $\text{Theta\_sat} - X_{\text{ice}}$ ).
- *Theta\_p*: number [ $\text{cm}^3/\text{cm}^3$ ] soil cells  
Water content in tertiary (biopore) system.
- *S\_sum*: number [ $\text{cm}^3/\text{cm}^3/\text{h}$ ] soil cells  
Total water sink (due to root uptake and macropores).
- *S\_drain*: number [ $\text{cm}^3/\text{cm}^3/\text{h}$ ] soil cells  
Matrix water sink due to soil drainage.
- *S\_incorp*: number [ $\text{cm}^3/\text{cm}^3/\text{h}$ ] soil cells  
Incorporated water sink, typically from subsoil irrigation.
- *S\_ice*: number [ $\text{cm}^3/\text{cm}^3/\text{h}$ ] soil cells  
Ice sink due to thawing or freezing.
- *S\_ice\_water*: number [ $\text{cm}^3/\text{cm}^3/\text{h}$ ] soil cells  
Water sink due to thawing or freezing.
- *S\_forward\_total*: number [ $\text{cm}^3/\text{cm}^3/\text{h}$ ] soil cells  
Sink (including source terms) at beginning of timestep. Used for limiting size of timestep. Currently this includes drain and tertiary domain (biopores).
- *S\_forward\_sink*: number [ $\text{cm}^3/\text{cm}^3/\text{h}$ ] soil cells  
Sink (excluding source terms) at beginning of timestep. Used for limiting size of timestep. Currently this includes drain and tertiary domain (biopores).
- *q\_primary*: number [ $\text{cm}/\text{h}$ ] soil edges  
Primary domain water flux (positive numbers mean upward).
- *q\_secondary*: number [ $\text{cm}/\text{h}$ ] soil edges  
Secondary domain water flux (positive numbers mean upward).
- *q\_p*: number [ $\text{cm}/\text{h}$ ] soil edges  
Water flux in macro pores (positive numbers mean upward).
- *Cw2*: number [ $\text{cm}^{-1}$ ] soil cells  
Specific water capacity.
- *table\_low*: number [ $\text{cm}$ ]  
Groundwater table estimated by pressure in lowest unsaturated cell. If there are multiple unsaturated cells in the same depth, the one with the lowest pressure will be used.



- *table\_high*: number [**cm**]  
Groundwater table estimated by pressure in highest saturated cell. If there are multiple saturated cells in the same depth, the one with the highest pressure will be used.

## 107.24 Snow

Simulate snow pack on surface. \_Snow Hydrology\_, U.S. Corps of Engineers, 1956.

```
< Snow (partial_day 0 [h])
 (Ssnow 0 [mm])
 (Swater 0 [mm])
 (age 0 [h])
 (dZs 0 [m])
 (mf 10 [m-1])
 (mtprime 0.0833333 [kg/m2/h/dg C])
 (mrprime 1.5e-07 [kg/J])
 (m1 2 [])
 (m2 0.00416667 [h-1])
 (rho_s 100 [kg/m3])
 (f_c 0.07 [])
 (rho_1 200 [kg/m3])
 (rho_2 0.5 [m-1])
 (Psa 5 [mm])
 (fsa 0.9 [<fraction>])
 (K_snow_factor 2.86e-06 [W m5/kg2/dg C])
 (partial_snow 0 [mm])
 (partial_precipitation 0 [mm]) >
```

- *partial\_day*: number [**h**]  
State variable (default 0)  
Time since last day. Snow age is reset when at least 'Psa' is received within a day.
- *Ssnow*: number [**mm**]  
State variable (default 0)  
Snow storage expressed as water.
- *Swater*: number [**mm**]  
State variable (default 0)  
Water in snow storage.
- *age*: number [**h**]  
State variable (default 0)  
Time since last snow.
- *dZs*: number [**m**]  
State variable (default 0)  
Depth of snow layer.
- *mf*: number [**m<sup>-1</sup>**]  
Parameter (default 10)  
Snow pack depth melting factor.
- *mtprime*: number [**kg/m<sup>2</sup>/h/dg C**]  
Parameter (default 0.0833333)  
Air temperature melting factor.

- *mrprime*: number [**kg/J**]  
Parameter (default 1.5e-07)  
Radiation melting factor.
- *m1*: number (dimensionless)  
Parameter (default 2)  
Radiation melting linear.
- *m2*: number [**h<sup>-1</sup>**]  
Parameter (default 0.00416667)  
Radiation melting exponential factor.
- *rho\_s*: number [**kg/m<sup>3</sup>**]  
Parameter (default 100)  
Density of newly fallen snow.
- *f\_c*: number (dimensionless)  
Parameter (default 0.07)  
Water capacity in snow factor.
- *rho\_1*: number [**kg/m<sup>3</sup>**]  
Parameter (default 200)  
Water collapse factor.
- *rho\_2*: number [**m<sup>-1</sup>**]  
Parameter (default 0.5)  
Snow collapse factor.
- *Psa*: number [**mm**]  
Parameter (default 5)  
Amount of snow in a day required for snow to become new.
- *fsa*: number [**<fraction>**]  
Parameter (default 0.9)  
Fraction of snow in precipitation required for snow to become new.
- *K\_snow\_factor*: number [**W m<sup>5</sup>/kg<sup>2</sup>/dg C**]  
Parameter (default 2.86e-06)  
Factor related to thermal conductivity for snow water mix.
- *partial\_snow*: number [**mm**]  
State variable (default 0)  
Amount of snow received this day so far. Snow age is reset when at least 'Psa'  
is received within a day.
- *partial\_precipitation*: number [**mm**]  
State variable (default 0)  
Amount of precipitation received this day so far. At least 'fsa' of this must  
be snow for snow age to be reset.

## Log Variables

- *EvapSnowPack*: number [**mm/h**]  
Evaporation from snowpack.
- *q\_s*: number [**mm/h**]  
Leaking water.

## 107.25 Timestep

Relative time.

```
< Timestep (seconds 0)
 (minutes 0)
 (hours 0)
 (days 0)
 (microseconds 0) >
```

- *seconds*: integer  
State variable (default 0)  
Number of seconds.
- *minutes*: integer  
State variable (default 0)  
Number of minutes.
- *hours*: integer  
State variable (default 0)  
Number of hours.
- *days*: integer  
State variable (default 0)  
Number of days.
- *microseconds*: integer  
State variable (default 0)  
Number of microseconds.

## 107.26 Weatherdata

Weather data.

```

< Weatherdata (T_min T_min)
 (GlobRad GlobRad)
 (AirTemp AirTemp)
 (T_max T_max)
 (Precip Precip)
 (RefEvap RefEvap)
 (VapPres VapPres)
 (DiffRad DiffRad)
 (GHF GHF)
 (CloudinessIndex CloudinessIndex)
 (RelHum RelHum)
 (Wind Wind)
 (CO2 CO2)
 (Latitude Latitude)
 (Longitude Longitude)
 (Elevation Elevation)
 (TimeZone TimeZone)
 (ScreenHeight ScreenHeight)
 (TAverage TAverage)
 (TAmplitude TAmplitude)
 (MaxTDay MaxTDay)
 (NH4WetDep NH4WetDep)
 (NH4DryDep NH4DryDep)
 (NO3WetDep NO3WetDep)
 (NO3DryDep NO3DryDep)
 (Deposition Deposition)
 (DepDry DepDry)
 (DepDryNH4 DepDryNH4)
 (DepWetNH4 DepWetNH4)
 (PAverage PAverage)
 (Timestep Timestep)
 (Station Station)
 (Note Note)
 (Surface Surface)
 (PrecipCorrect PrecipCorrect ...)
 (PrecipScale PrecipScale ...)
 (GlobRadScale GlobRadScale ...)
 (TempOffset TempOffset ...)
 (Begin Begin)
 (End End) >

```

- *T\_min*: number [dgC]  
Optional parameter  
Minimum air temperature.
- *GlobRad*: number [W/m<sup>2</sup>]  
Optional parameter  
Global radiation.
- *AirTemp*: number [dgC]  
Optional parameter  
Air temperature.
- *T\_max*: number [dgC]  
Optional parameter  
Maximum air temperature.

- *Precip*: number [**mm/h**]  
Optional parameter  
Precipitation.
- *RefEvap*: number [**mm/h**]  
Optional parameter  
Reference evapotranspiration.
- *VapPres*: number [**Pa**]  
Optional parameter  
Vapor pressure.
- *DiffRad*: number [**W/m<sup>2</sup>**]  
Optional parameter  
Diffuse radiation.
- *GHF*: number [**W/m<sup>2</sup>**]  
Optional parameter  
Ground heat flux.
- *CloudinessIndex*: number [**fraction**]  
Optional parameter  
Cloudiness index, 1 = clear sky.
- *RelHum*: number [**fraction**]  
Optional parameter  
Relative humidity.
- *Wind*: number [**m/s**]  
Optional parameter  
Wind speed.
- *CO2*: number [**<fraction>**]  
Optional parameter  
Atmospheric CO2 level.
- *Latitude*: number [**dgNorth**]  
Optional parameter  
Location of station (north-south).
- *Longitude*: number [**dgEast**]  
Optional parameter  
Location of station (east-west).
- *Elevation*: number [**m**]  
Optional parameter  
Station altitude over sea level.
- *TimeZone*: number [**dgEast**]  
Optional parameter  
Time zone.
- *ScreenHeight*: number [**m**]  
Optional parameter  
Measurement altitude over ground level.
- *TAverage*: number [**dgC**]  
Optional parameter  
Yearly average temperature.

- *TA*mplitude: number [**dgC**]  
Optional parameter  
Typical temperature variation over the seasons. If you fit the daily average temperature over a year to a sinus curve, this would be the amplitude.
- *MaxTDay*: number [**yday**]  
Optional parameter  
Typical day where the temperature is highest. If you fit the daily average temperature over a year to a sinus curve, this would be maximum point.
- *NH<sub>4</sub>WetDep*: number [**ppm**]  
Optional parameter  
NH<sub>4</sub> concentration in precipitation.
- *NH<sub>4</sub>DryDep*: number [**kg N/ha/y**]  
Optional parameter  
Dry deposition of NH<sub>4</sub>.
- *NO<sub>3</sub>WetDep*: number [**ppm**]  
Optional parameter  
NO<sub>3</sub> concentration in precipitation.
- *NO<sub>3</sub>DryDep*: number [**kg N/ha/y**]  
Optional parameter  
Dry deposition of NO<sub>3</sub>.
- *Deposition*: number [**kg N/ha/y**]  
Optional parameter  
Total N deposition.
- *DepDry*: number [**<fraction>**]  
Optional parameter  
Fraction of total N deposition that is dry.
- *DepDryNH<sub>4</sub>*: number [**<fraction>**]  
Optional parameter  
NH<sub>4</sub> fraction of dry deposition.
- *DepWetNH<sub>4</sub>*: number [**<fraction>**]  
Optional parameter  
NH<sub>4</sub> fraction of wet deposition.
- *PAverage*: number [**mm**]  
Optional parameter  
Average precipitation. Used for deviding precipitation into dry and wet.
- *Timestep*: number [**h**]  
Optional parameter  
Timestep for weather data.
- *Station*: string (see section 4.1.5)  
Optional parameter  
Name of weather station.
- *Note*: string (see section 4.1.5)  
Optional parameter  
Note regarding this set of data.

- *Surface*: string (see section 4.1.5)  
Optional parameter  
Surface type. Either 'reference' for a weather station standard of short grass, or 'field' for measurements directly at the field.
- *PrecipCorrect*: number (dimensionless) sequence  
Optional parameter  
Correction factors for precipitation. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *PrecipScale*: number (dimensionless) sequence  
Optional parameter  
Scale factors for precipitation. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *GlobRadScale*: number (dimensionless) sequence  
Optional parameter  
Scale factors for global radiation. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *TempOffset*: number [dg C] sequence  
Optional parameter  
Temperature offset. Can contain one or twelve numbers, in the later case the numbers corresponds to months.
- *Begin*: **Time** fixed component (see section 107.27)  
Optional submodel  
Beginning of weather data.
- *End*: **Time** fixed component (see section 107.27)  
Optional submodel  
End of weather data.

## 107.27 Time

Year, month, day and hour, minute, second and microsecond.

Used by Harvesting @ last\_cut (see 107.7, page 593) , Harvest @ sow\_time (see 107.15, page 607) , and Weatherdata @ Begin (see 107.26, page 619) .

```
< Time year month mday hour
 (microsecond 0)
 (second 0)
 (minute 0) >
```

- *year*: integer  
State variable  
Current year.
- *month*: integer  
State variable  
Current month.
- *mday*: integer  
State variable  
Current day in the month.
- *hour*: integer  
State variable (default 0)  
Current hour.

- *microsecond*: integer  
State variable (default 0)  
Current microsecond.
- *second*: integer  
State variable (default 0)  
Current second.
- *minute*: integer  
State variable (default 0)  
Current minute.

## Log Variables

- *yday*: integer  
Current Julian day.
- *week*: integer  
Current week.
- *wday*: string (see section 4.1.5)  
Current weekday. Monday is 1, Sunday is 7.

## 107.28 IntegerCondClause

If condition is true, return value.

- ```
< IntegerCondClause  condition value >
```
- *condition*: **boolean** component (see chapter 21)
Condition to test for.
 - *value*: integer
Parameter
Value to return.

107.29 StringerCondClause

If condition is true, return value.

- ```
< StringerCondClause condition value >
```
- *condition*: **boolean** component (see chapter 21)  
Condition to test for.
  - *value*: string (see section 4.1.5)  
Parameter  
Value to return.

## 107.30 Toplevel

The top level syntax for a Daisy setup file.

- ```
< Toplevel  (ui ui)
              (input input)
              (directory directory)
              (run run)
              (path path ...) ; Has default value.
              (allow_old_units true)
              (install_directory install_directory) ; Has default value. >
```


- *ui*: **ui** component (see chapter 93)
Optional component
Top level user interface.
- *input*: **parser** component (see chapter 61)
Optional component
Command to add more information about the simulation.
- *directory*: string (see section 4.1.5)
Optional parameter
Run program in this directory. This can affect both where input files are found and where log files are generated.
- *run*: **program** component (see chapter 66)
Optional component
Program to run.

If this option is specified, all the 'Daisy' specific top-level attributes will be ignored. If unspecified, run 'Daisy' on the current top-level attributes.
- *path*: string (see section 4.1.5) sequence
Parameter (has default value with length 3)

```
(path "." "/home/runner/work/daisy/daisy/lib" "/home/runner/work/daisy/daisy/sample")
```

Parameter description:

List of directories to search for input files in. The special value "." means the current directory.

By default, this variable will be initialised from the DAISYPATH environment variable if it exists. The value of the variable should be a list of directories to search for input files in, separated by semicolon on MS Windows, or colon on other systems. If the DAISYPATH environment variable is not set, the path will be initialized to the working directory followed by the standard parameter libraries.

- *allow_old_units*: boolean (see section 4.1.2)
Parameter (default true)
OBSOLETE: Set this to true to enable the old system of build-in unit conversation.
- *install_directory*: string (see section 4.1.5)
Parameter (default '/home/runner/work/daisy/daisy')
Directory where Daisy has been installed.

This is used for looking up files that came with the installation, in particular the parameter library. By default, the value of the DAISYHOME environment variable is used. If DAISYHOME is not set, and the program is running under MS Windows, the value of the "Install Directory" registry key is used. If that is not set either (or we are not running MS Windows), a hardcoded value is used. This is "C:/daisy" under MS Windows, or "/usr/local/daisy" on other systems.

The value found in the manual corresponds to the system where the manual was generated.

Part III

Appendix

Version

Daisy version 7.1.12. LaTeX manual generated: Tue Jun 30 10:01:21 2026

Appendix A

Daisy Weather File Format

The Daisy Weather File format (*dwf* for short) is a flexible format for specifying weather data. The easiest way to create a *dwf* file is to edit an existing file. Here, we will describe the format for reference purposes.

First some general syntax. Empty lines and lines beginning with ‘#’ are ignored, and can occur anywhere except at the very first line. Line starting with ‘#’ are called *comment lines*. All words are case-sensitive, and must be written exactly as specified.

The first line *must* begin with the string ‘*dwf-0.0*’, followed by whitespace. The rest of the line is ignored.

After the first line, the keyword section follows. Each line (except blank lines and comment lines) in the keyword section have the general format:

keyword: value

or

keyword: value dimension

All keywords, with the exception of ‘*Note*’ and ‘*Timestep*’, must occur exactly once. The sequence doesn’t matter.

Here is a list of the recognized keywords, and the dimensions allowed for them:

Station The rest of the line (after the colon) contains the name of the weather station.

Elevation Height above sea level of station, given in [m].

Longitude Longitude of station, given in either [dgEast] or [dgWest].

Latitude Latitude of station, given in either [dgNorth] or [dgSouth].

TimeZone Time zone used for the data, given in either [dgEast] or [dgWest].

Surface Measurement conditions at surface, the value should either be ‘*reference*’ (short grass) or ‘*field*’ if the data have been measured at the simulated field.

ScreenHeight Measurement height above surface, given in [m].

PrecipCorrect Factors to multiply to the precipitation for each month. Should have the form of twelve numbers, separated by spaces, the first number representing the factor for January, and the last for December. So if the first number is 1.20 and the weather data for a given day in January specifies 10 mm precipitation, Daisy will calculate with 12 mm precipitation for that day.

Begin First data point, given in the format *yyyy-mm-dd* or *yyyy-mm-dd:hh*.

End Last data point, given in the format `yyyy-mm-dd` or `yyyy-mm-dd:hh`. Note that the date specified by **End** should be a few days before the actual end of the weather data, to prevent Daisy from reading past the end of the file. Also, make sure the simulation ends before the weather data, or the results may be lost.

Timestep Time between data points, given in `[hours]`. You can leave out this keyword, in that case the time for each data point must be specified. Leaving it out allows for varying timesteps.

Note You can have any number of these, but they must come in sequence. That is, no other keywords between two notes. You can have any text after the keyword.

TAverage Average temperature for the location, given in `[dgC]`.

TAmplitude Amplitude of yearly temperature variation, given in `[dgC]`.

MaxTDay The Julian day with the highest temperature (on average), given in `[yday]`.

There are two ways of specifying deposition. Directly

NH4WetDep NH_4 deposition in precipitation, given in `[ppm]`.

NH4DryDep NH_4 deposition from air, given in `[kgN/ha/year]`.

NO3WetDep NO_3 deposition in precipitation, given in `[ppm]`.

NO3DryDep NO_3 deposition from air, given in `[kgN/ha/year]`.

Or alternatively, indirectly

Deposition Mean yearly deposition, given in `[kgN/ha/year]`.

PAverage Total yearly precipitation, given in `[mm]`.

DepDry Dry fraction of total deposition (from air). By default 40%. The remainder will fall with the precipitation.

DepDryNH4 Fraction of NH_4 in deposition from air. By default 60%. The remainder will fall in the form of NO_3 .

DepWetNH4 Fraction of NH_4 in deposition from rain and snow. By default 50%. The remainder will fall in the form of NO_3 .

You cannot specify both. If you use the indirect specification, these numbers will be used for calculating the equivalents of the direct specification of deposition.

After the keywords there should be a line consisting solely of hyphens. It marks the beginning of the data section. The lines in the data section are divided into columns by whitespace. All lines must have the same number of columns. The first line contains the name of the data type specified in each column, the second line the dimension of that data, and the following lines the actual measurement data in chronological order. The following is a list of the possible data type names, as well as the recognized dimensions.

Year Given in `[year]`.

Month Given in `[month]`.

Day Given in `[mday]`.

Hour Given in [hour].

GlobRad Global radiation, given in $[W/m^2]$.

AirTemp Air temperature, given in $[dgC]$.

T_min Daily minimum air temperature, given in $[dgC]$.

T_max Daily maximum air temperature, given in $[dgC]$.

Precip Precipitation, given in $[mm/h]$ or $[mm/d]$.

RefEvap Reference evapotranspiration, given in $[mm/h]$ or $[mm/d]$.

VapPres Vapor pressure, given in $[Pa]$.

RelHum Relative humidity, given in [fraction] or [%].

Wind Wind speed, given in $[m/s]$

The columns can be arranged in any sequence. GlobRad is mandatory, Year, Month, and Day are mandatory if no timestep have been specified. You cannot specify both VapPres and relHum. All other columns are optional.

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